

**The data file AMMONIAEX
(based on AMUJEL format):
Additional Atomic and Molecular Data for
EIRENE for AMMONIA**

**S. Touchard
LSPM – 99 avenue J.B. Clement
93430 Villetaneuse
FR**

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Available via D-mail from s.touchard@lspm.cnrs.fr

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4 Appendix

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Introduction

Record:

- April 17: First version

to be done:

- To be done

Additional atomic data fits, read by EIRENE Specific reaction for AMMONIAC
Format as HYDHEL DATEN [2] or METHANE DATEN [1] or AMJUEL DATEN
SPECIFIC BIBLIOGRAPHY IS AVIABLE

End of preface:

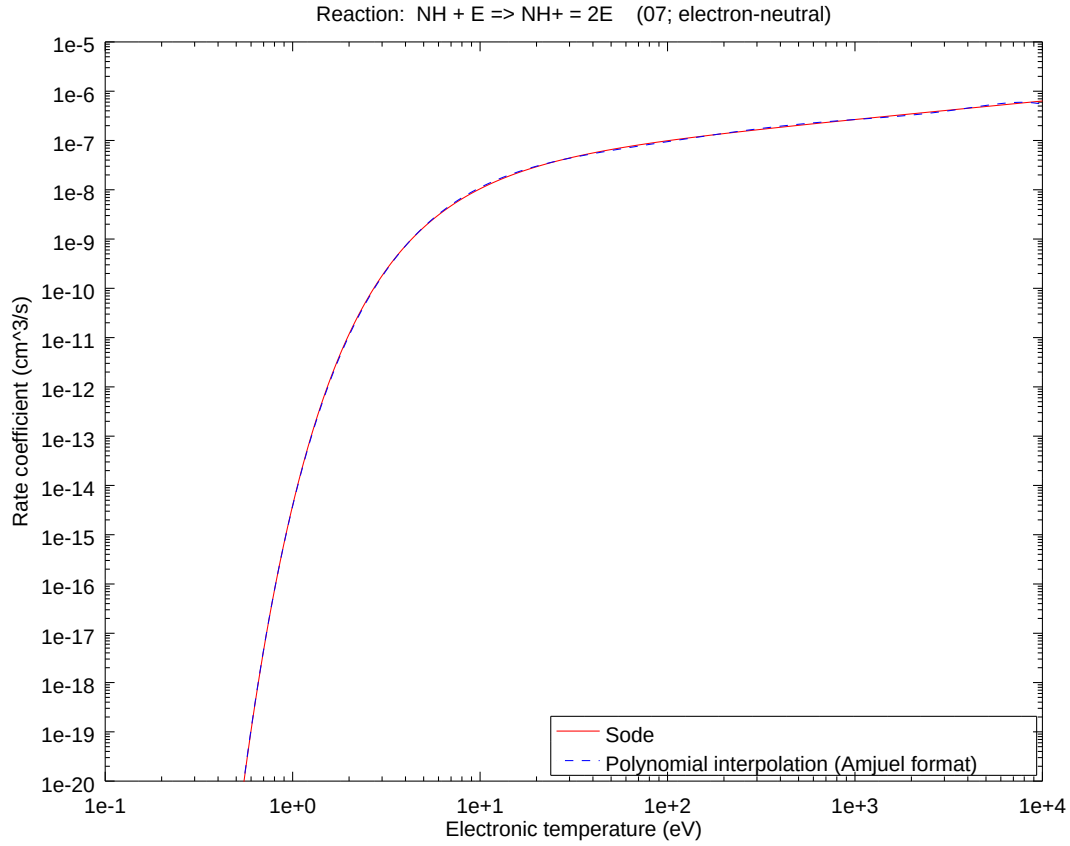
This next string is searched by EIRENE in subroutine SLREAC to initialize search for a particular set of fit coefficients. From here on, a character string ‘**H.n**’, n an integer, must only appear in the section title, but not in the text. Likewise: identifiers p0, a0, b0, ..., h0, k0 are used in SLREAC and must not appear in the text elsewhere, from here on.

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.....  
.  
.      ##BEGIN DATA HERE##      .  
.  
.....
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0 H.0 : Fits for electron-neutral

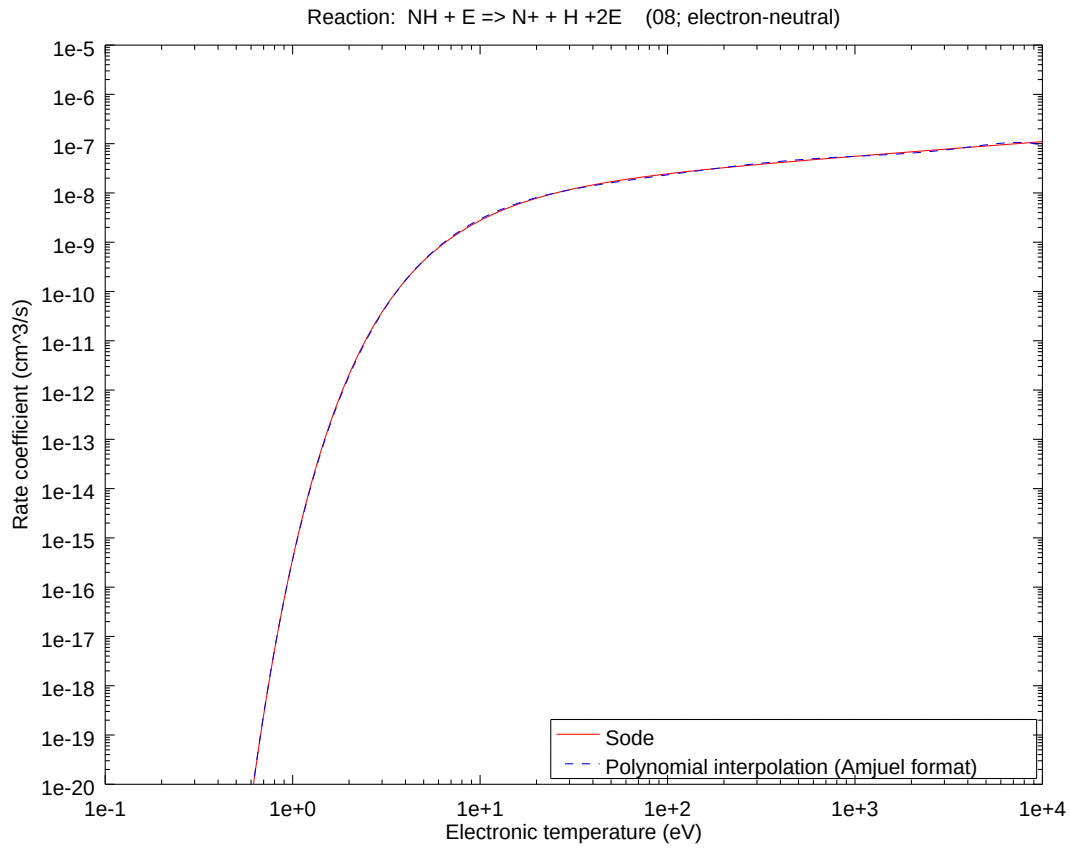
0.1 Reaction 07 $\text{NH} + \text{E} = \text{NH}^+ + 2\text{E}$

b0	-3.319843294396D+01	b1	1.572589154276D+01	b2	-7.650034869943D+00
b3	2.652640889075D+00	b4	-6.985494189552D-01	b5	1.309333127369D-01
b6	-1.558126112726D-02	b7	1.026447081867D-03	b8	-2.820271950207D-05



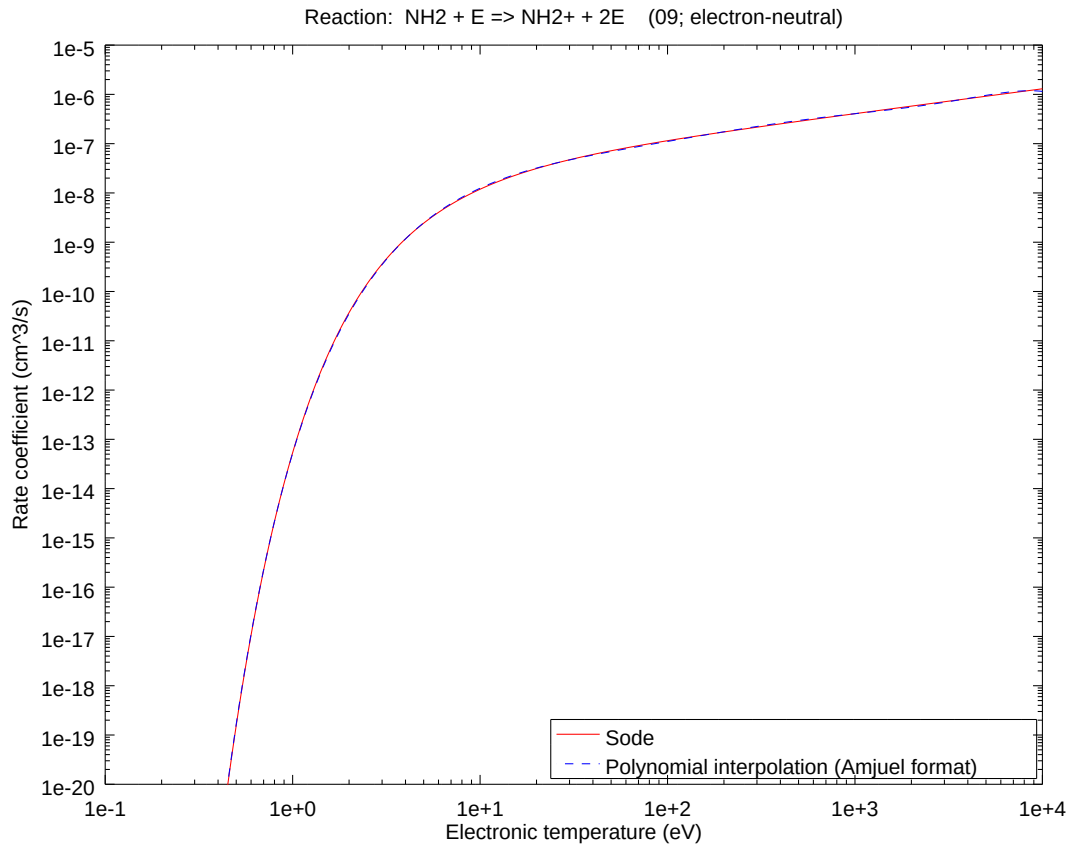
0.2 Reaction 08 $\text{NH} + \text{E} = \text{N}^+ + \text{H} + 2\text{E}$

b0	-3.554735633352D+01	b1	1.696437674301D+01	b2	-8.306880988537D+00
b3	2.880401533522D+00	b4	-7.585281618352D-01	b5	1.421754887175D-01
b6	-1.691909697615D-02	b7	1.114579723499D-03	b8	-3.062425707068D-05



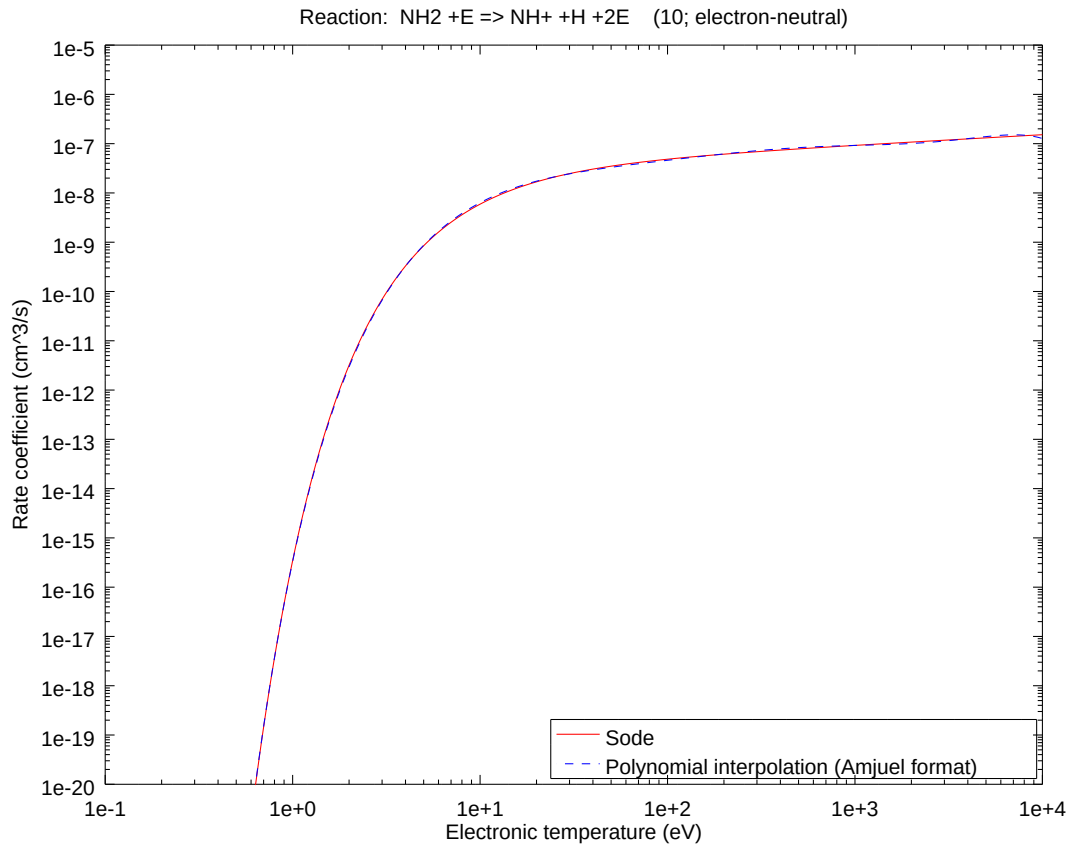
0.3 Reaction 09 $\text{NH}_2 + \text{E} = \text{NH}_2^+ + 2\text{E}$

b0	-3.058208344880D+01	b1	1.279264397226D+01	b2	-6.123978850051D+00
b3	2.123482700099D+00	b4	-5.592003095574D-01	b5	1.048142723007D-01
b6	-1.247305603473D-02	b7	8.216877866463D-04	b8	-2.257674124117D-05



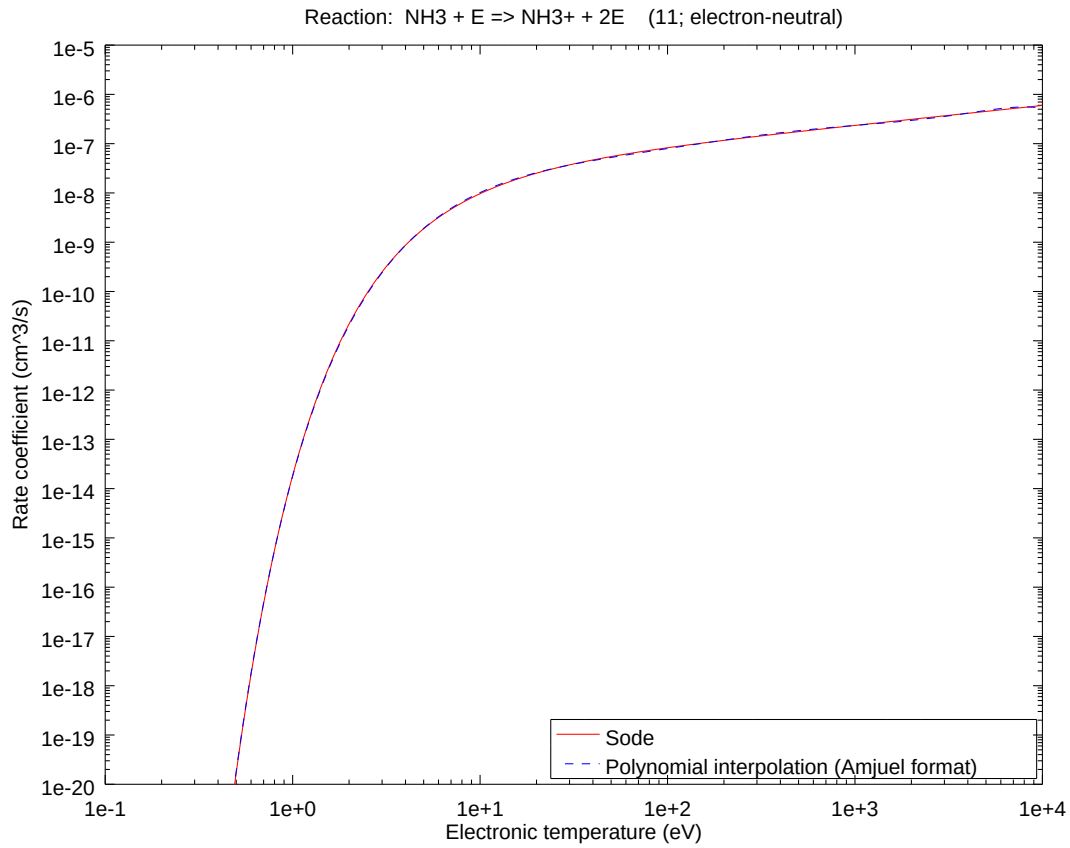
0.4 Reaction 10 $\text{NH}_2 + \text{E} = \text{NH}^+ + \text{H} + 2\text{E}$

b0	-3.563666632219D+01	b1	1.802442033722D+01	b2	-8.874830639953D+00
b3	3.077337429095D+00	b4	-8.103894808667D-01	b5	1.518961671970D-01
b6	-1.807587233421D-02	b7	1.190784639196D-03	b8	-3.271806775030D-05



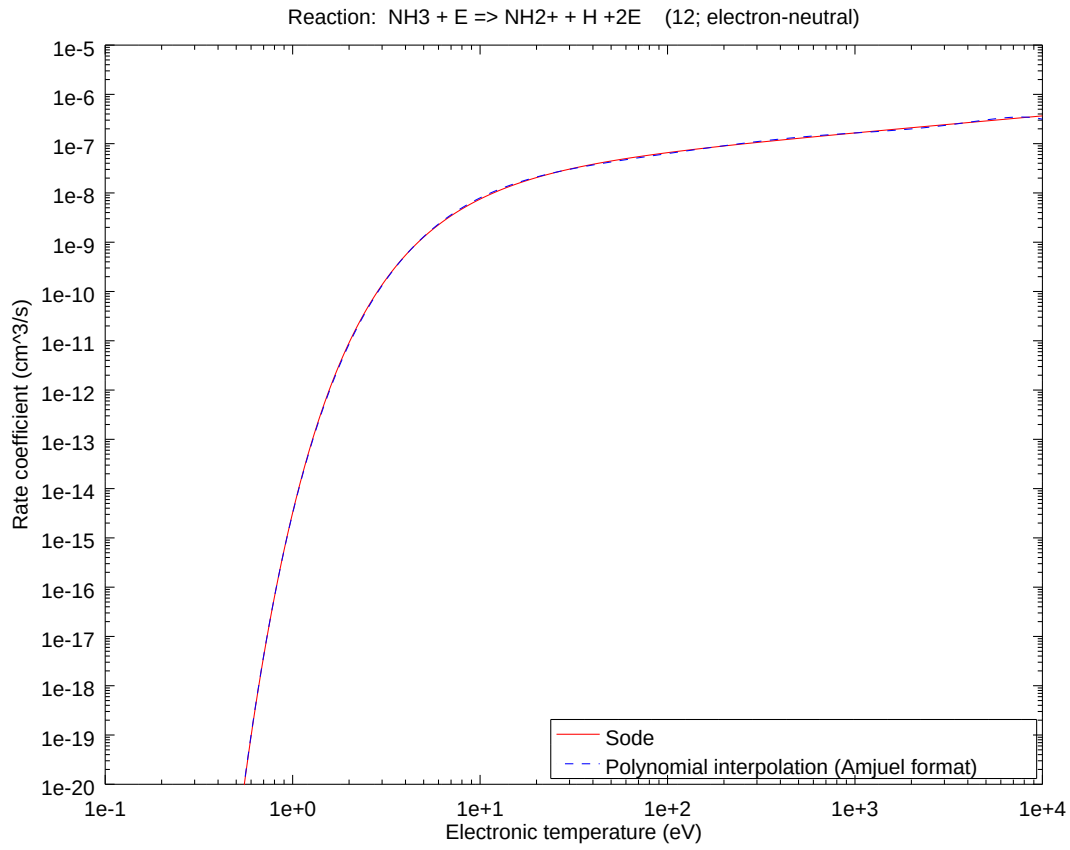
0.5 Reaction 11 $\text{NH}_3 + \text{E} = \text{NH}_3^+ + 2\text{E}$

b0	-3.165130180136D+01	b1	1.389216810181D+01	b2	-6.721560657193D+00
b3	2.330693511964D+00	b4	-6.137674365384D-01	b5	1.150421166139D-01
b6	-1.369018488974D-02	b7	9.018686109884D-04	b8	-2.477979421712D-05



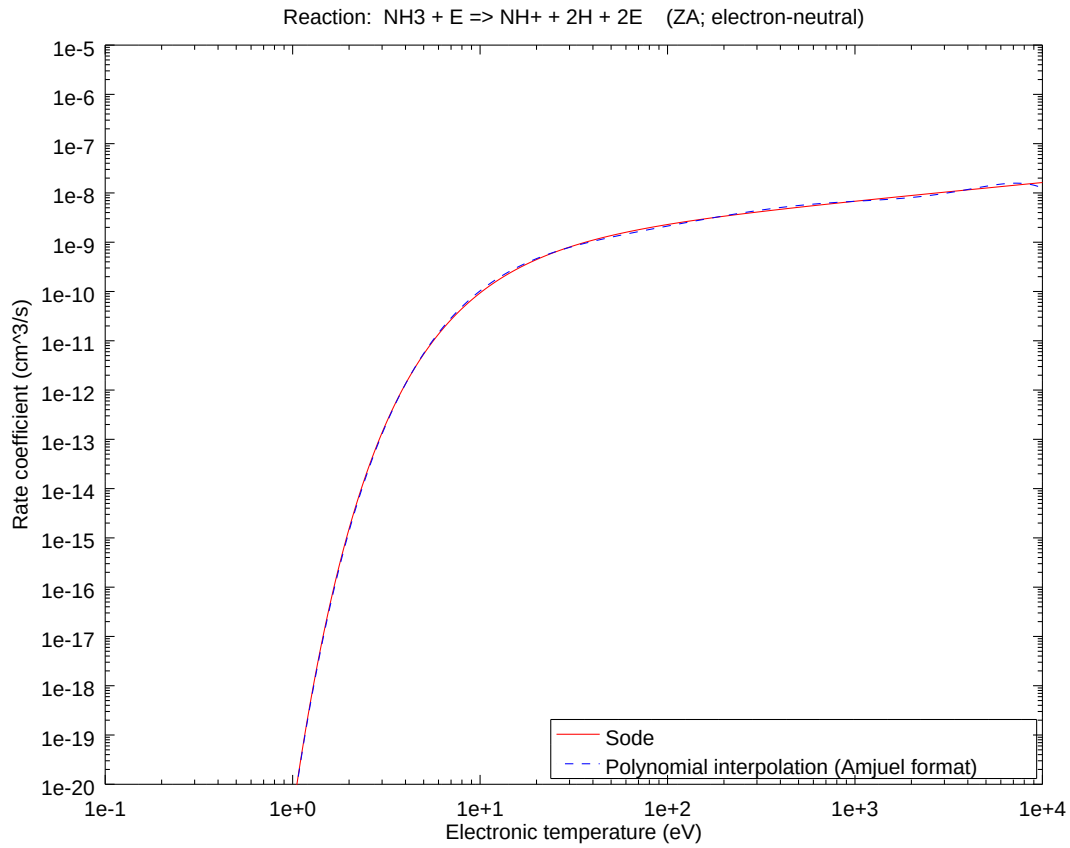
0.6 Reaction 12 $\text{NH}_3 + \text{E} = \text{NH}_2^+ + \text{H} + 2\text{E}$

b0	-3.339021332416D+01	b1	1.561658416230D+01	b2	-7.610525328975D+00
b3	2.638941000688D+00	b4	-6.949416750226D-01	b5	1.302570916252D-01
b6	-1.550078979800D-02	b7	1.021145870340D-03	b8	-2.805706310696D-05



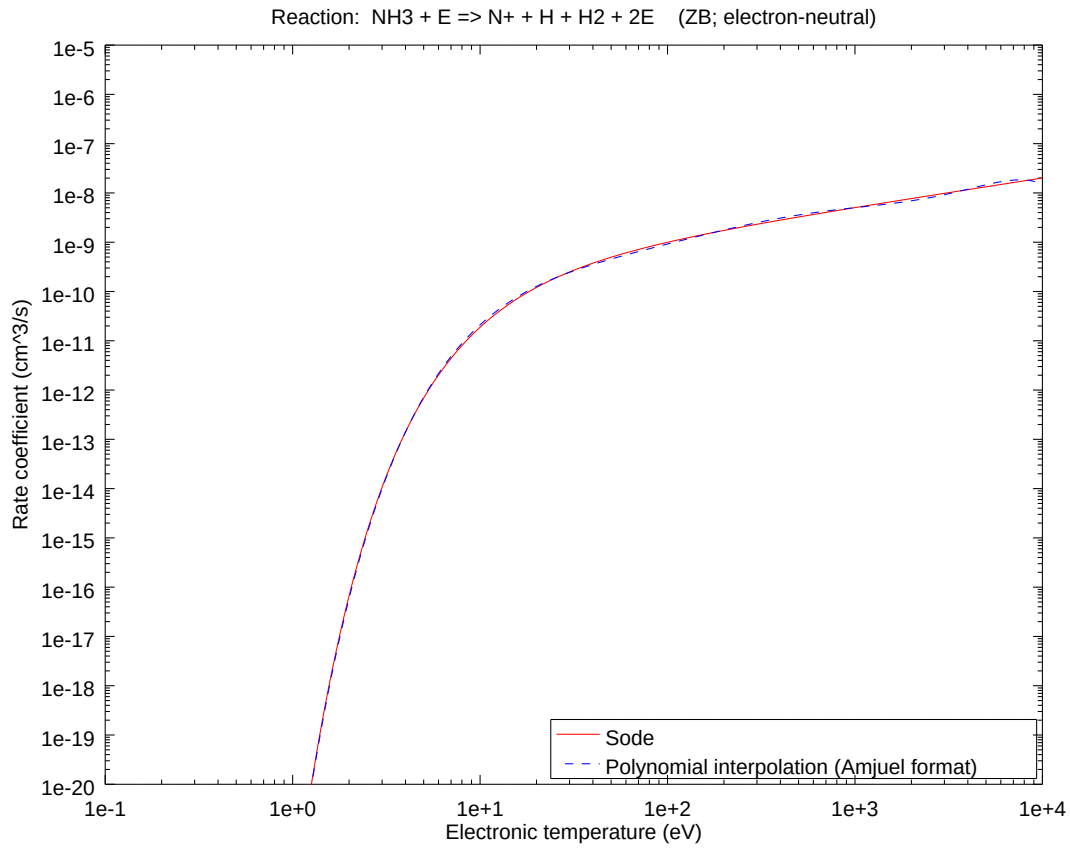
0.7 Reaction ZA $\text{NH}_3 + \text{E} = \text{NH}^+ + 2\text{H} + 2\text{E}$

b0	-4.744940094576D+01	b1	2.620437918686D+01	b2	-1.287023297035D+01
b3	4.462738642305D+00	b4	-1.175222586054D+00	b5	2.202790271093D-01
b6	-2.621353550525D-02	b7	1.726869654839D-03	b8	-4.744757070522D-05



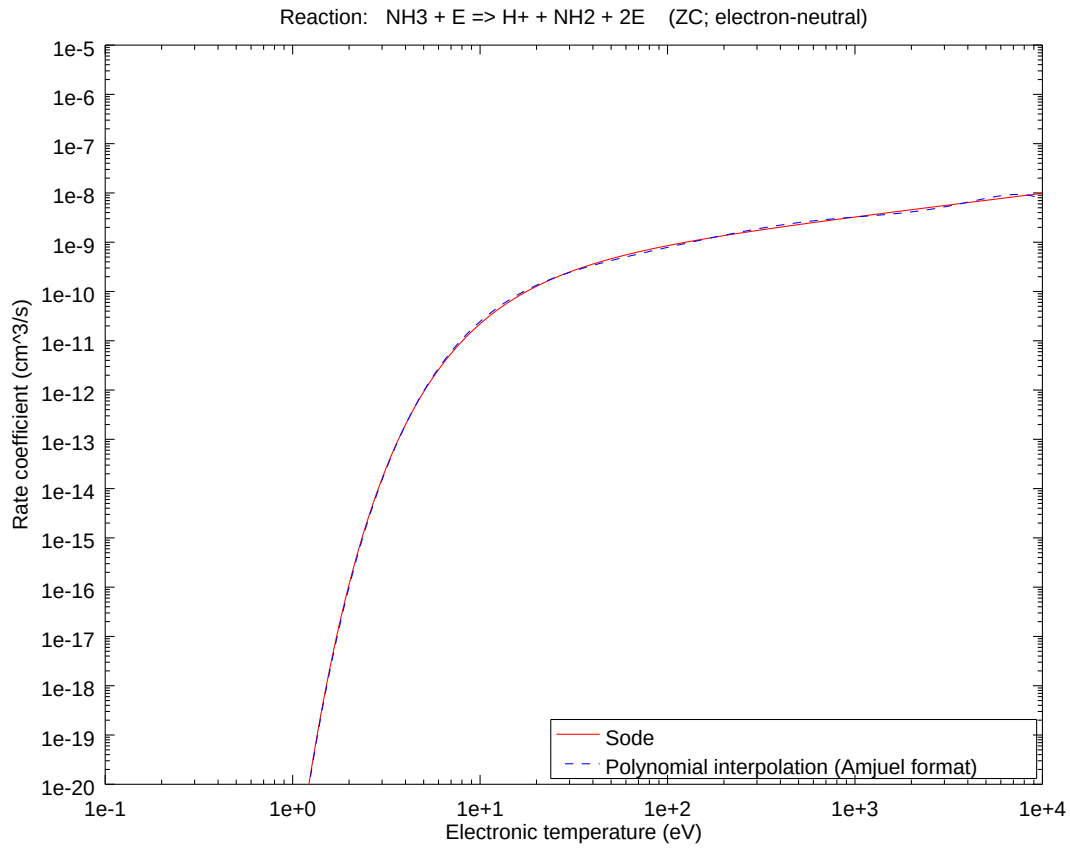
0.8 Reaction ZB $\text{NH}_3 + \text{E} = \text{N} + \text{H} + \text{H}_2 + 2\text{E}$

b0	-5.220926834260D+01	b1	2.933892541899D+01	b2	-1.432220860093D+01
b3	4.966209540554D+00	b4	-1.307807175578D+00	b5	2.451301529612D-01
b6	-2.917085685543D-02	b7	1.921689178447D-03	b8	-5.280044322530D-05



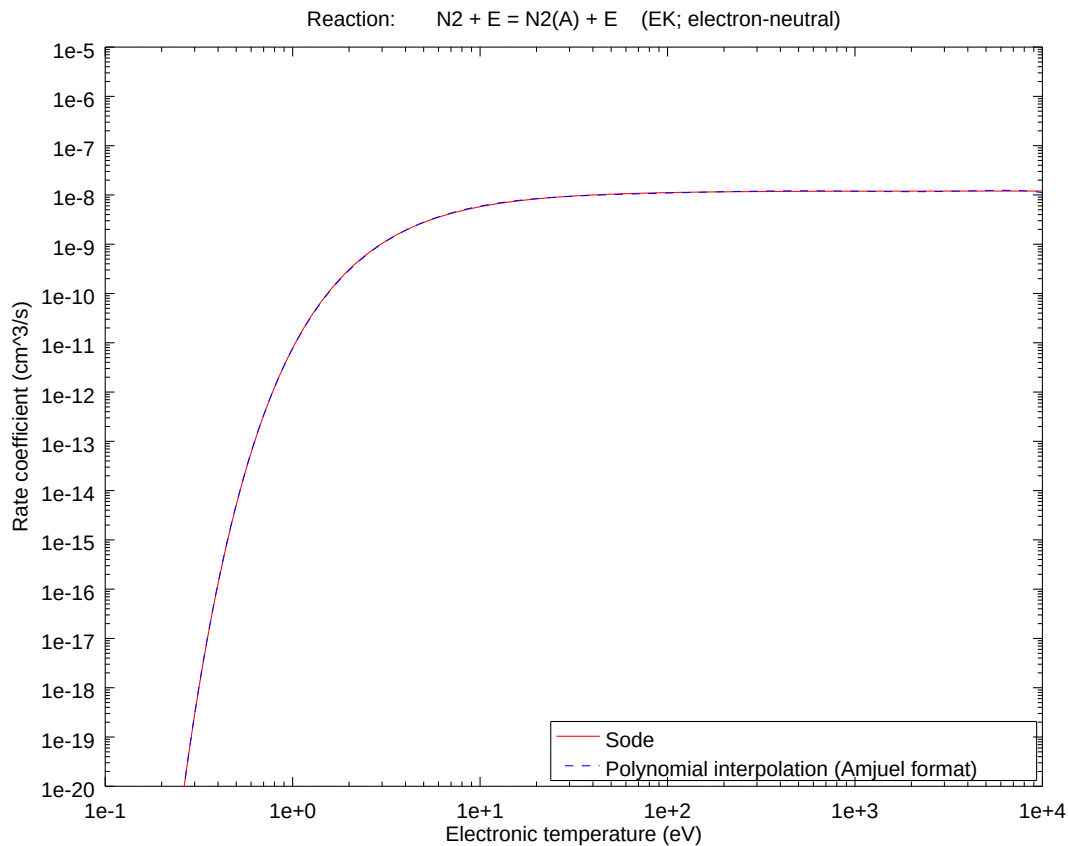
0.9 Reaction ZC $\text{NH}_3 + \text{E} = \text{H} + \text{NH}_2 + 2\text{E}$

b0	-5.136820819564D+01	b1	2.877282140387D+01	b2	-1.409996743298D+01
b3	4.889147668373D+00	b4	-1.287513615957D+00	b5	2.413264092084D-01
b6	-2.871820562836D-02	b7	1.891869863609D-03	b8	-5.198112600285D-05



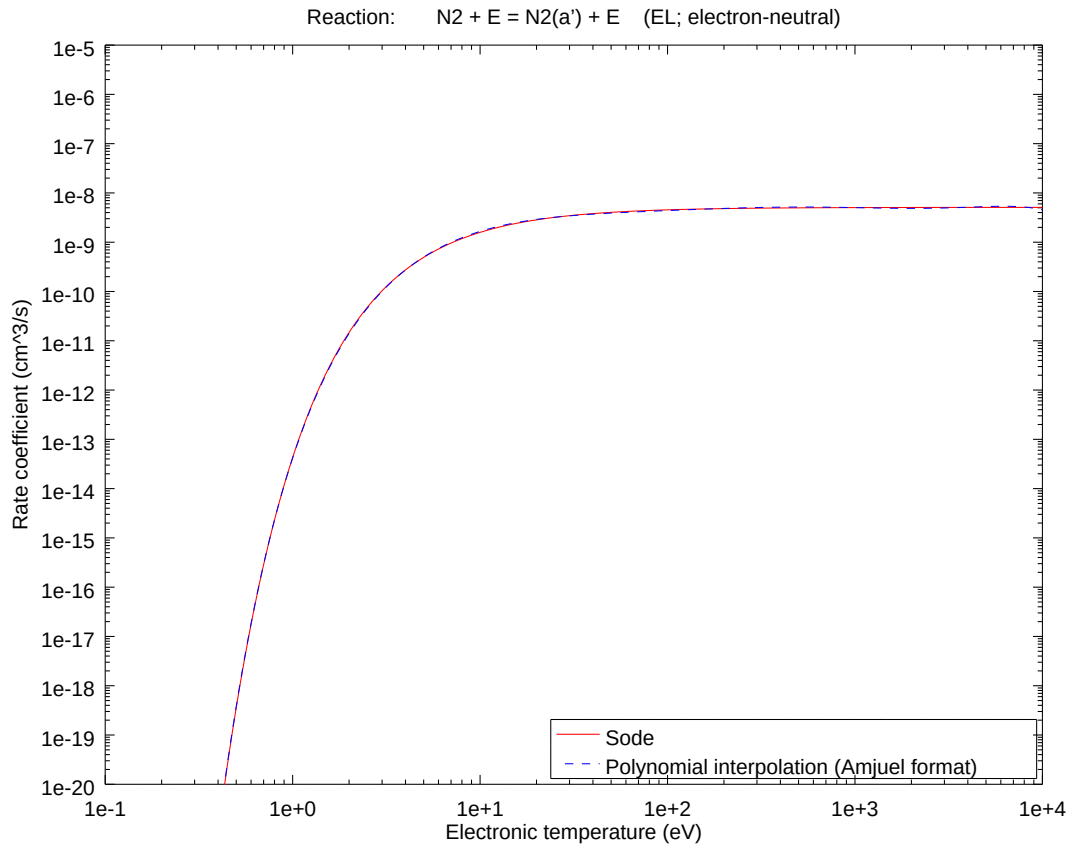
0.10 Reaction EK N2 + E = N2(A) + E

b0	-2.559242769964D+01	b1	7.276452157772D+00	b2	-3.625000383821D+00
b3	1.256964759575D+00	b4	-3.310105058187D-01	b5	6.204328699087D-02
b6	-7.383244459270D-03	b7	4.863861575793D-04	b8	-1.336397425082D-05



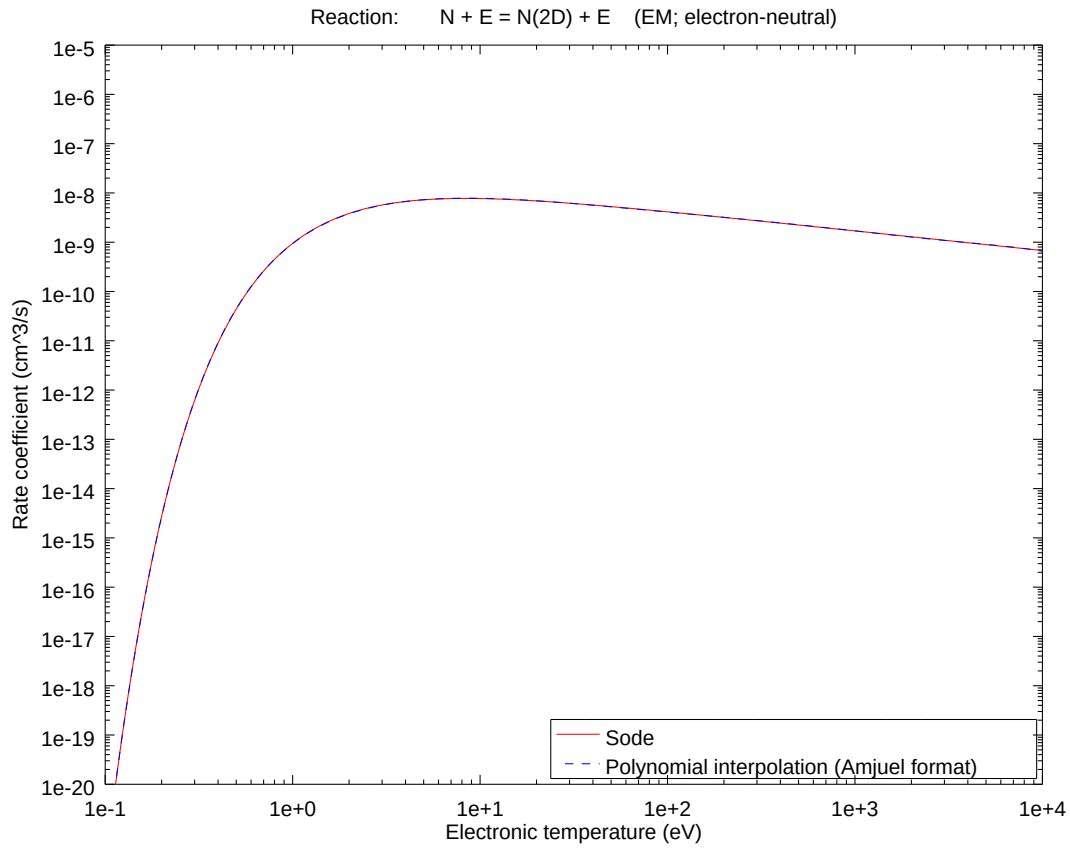
0.11 Reaction EL N2 + E = N2(a') + E

b0	-3.080643141587D+01	b1	1.158879097062D+01	b2	-5.773331673960D+00
b3	2.001896190658D+00	b4	-5.271815821553D-01	b5	9.881280993505D-02
b6	-1.175887298758D-02	b7	7.746395343463D-04	b8	-2.128404073461D-05



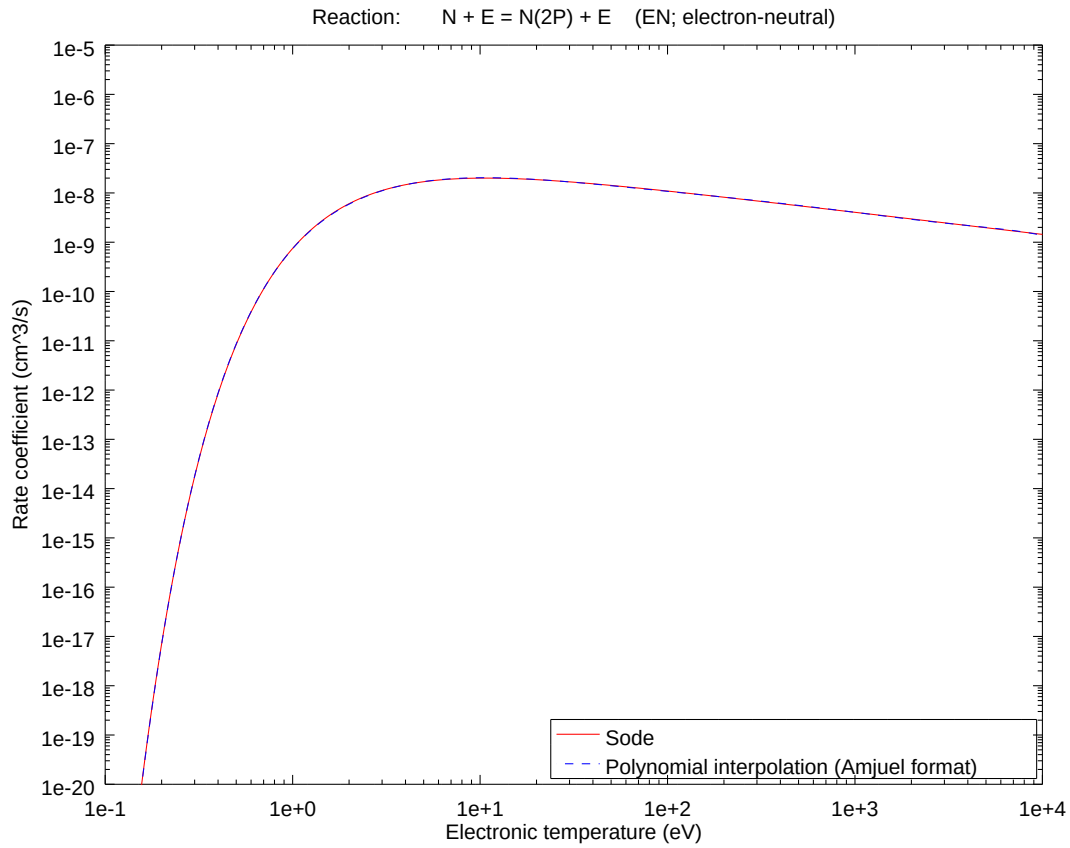
0.12 Reaction EM $N + E = N(2D) + E$

b0	-2.078384988604D+01	b1	2.920996557021D+00	b2	-1.654462028038D+00
b3	5.736828262365D-01	b4	-1.510742771788D-01	b5	2.831675904897D-02
b6	-3.369736912610D-03	b7	2.219882326827D-04	b8	-6.099361544993D-06



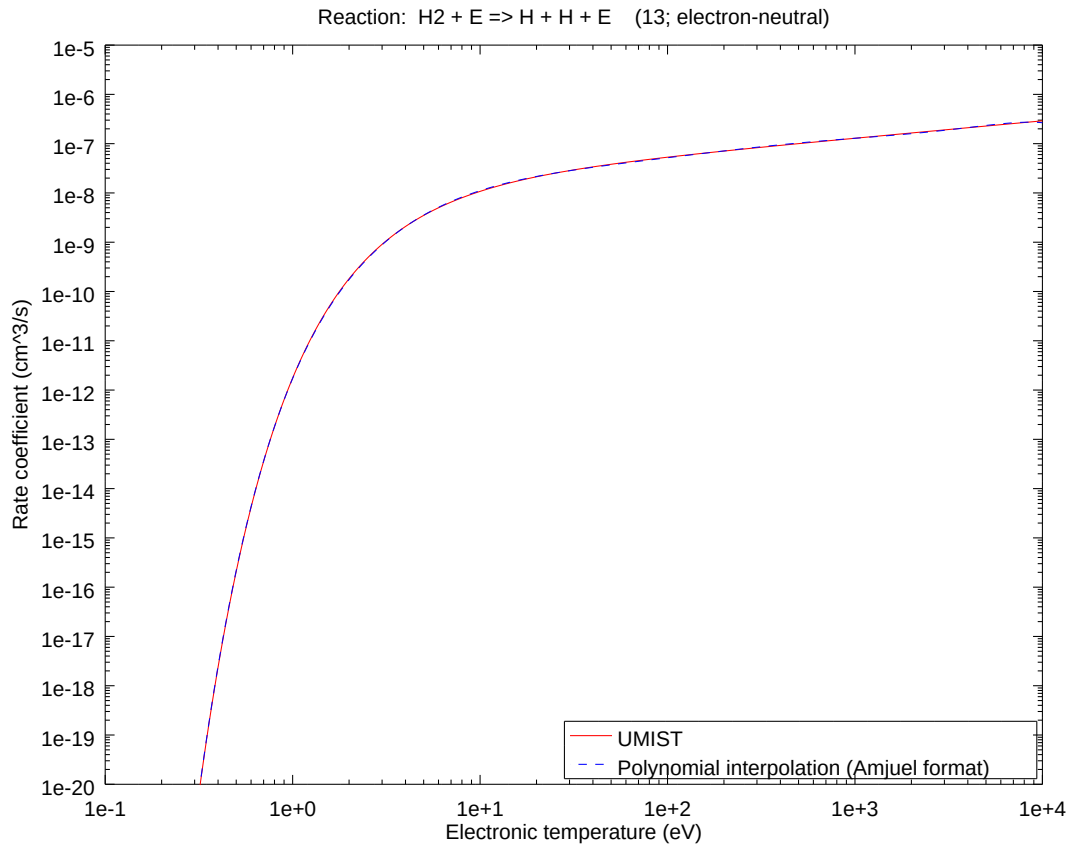
0.13 Reaction EN $N + E = N(2P) + E$

b0	-2.102160644758D+01	b1	4.308442827971D+00	b2	-2.370572458084D+00
b3	8.219933032642D-01	b4	-2.164646359577D-01	b5	4.057326669703D-02
b6	-4.828279755381D-03	b7	3.180726916050D-04	b8	-8.739383706258D-06



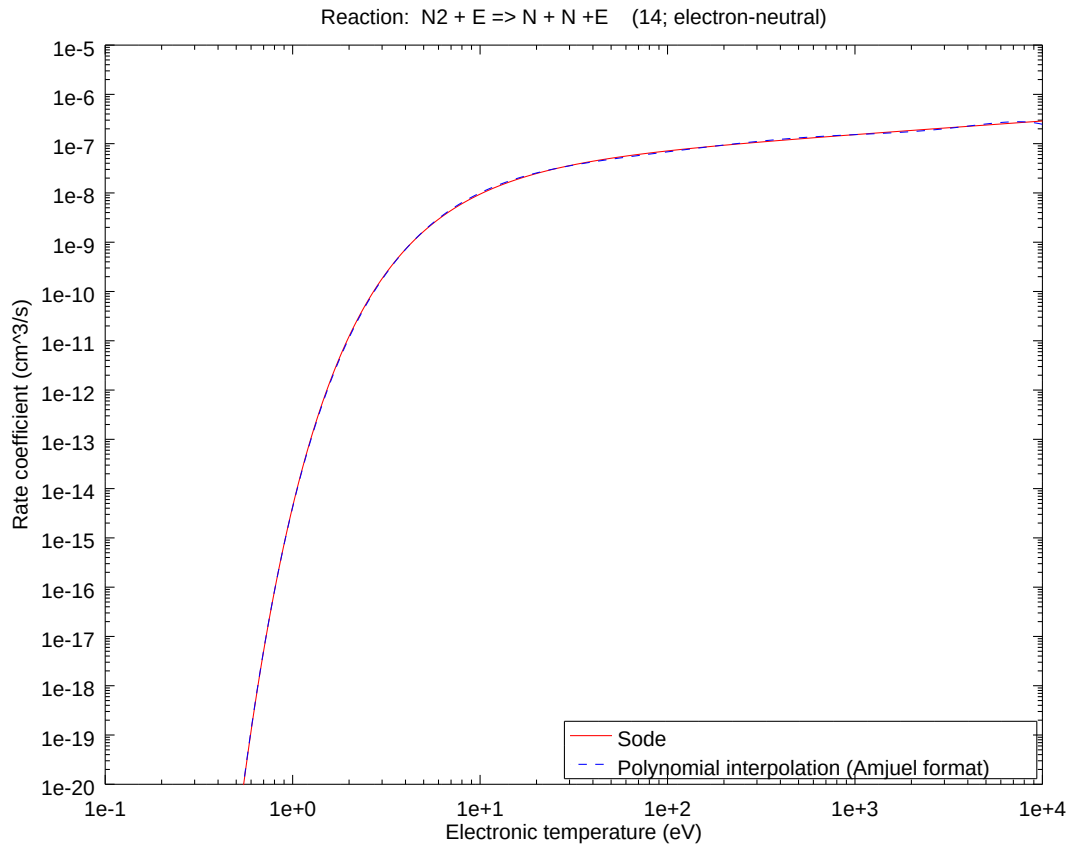
0.14 Reaction 13 $\text{H}_2 + \text{E} = \text{H} + \text{H} + \text{E}$

b0	-2.065501354542D+04	b1	1.199383411831D+04	b2	-3.061947982450D+03
b3	4.473020579061D+02	b4	-4.081227556395D+01	b5	2.377835412190D+00
b6	-8.628814122052D-02	b7	1.781520903670D-03	b8	-1.601159377178D-05



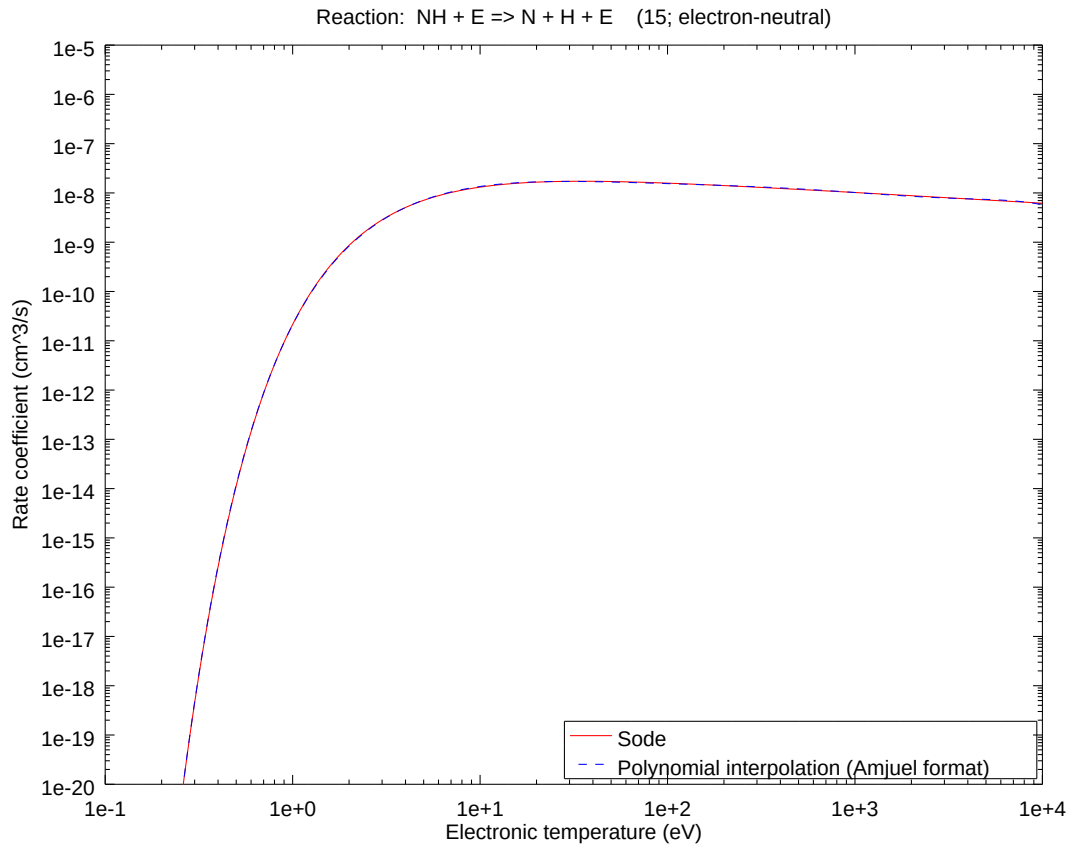
0.15 Reaction 14 $\text{N}_2 + \text{E} = \text{N} + \text{N} + \text{E}$

b0	-3.310497821898D+01	b1	1.566554523300D+01	b2	-7.669789640427D+00
b3	2.659490833269D+00	b4	-7.003532909215D-01	b5	1.312714232927D-01
b6	-1.562149679189D-02	b7	1.029097687630D-03	b8	-2.827554769962D-05



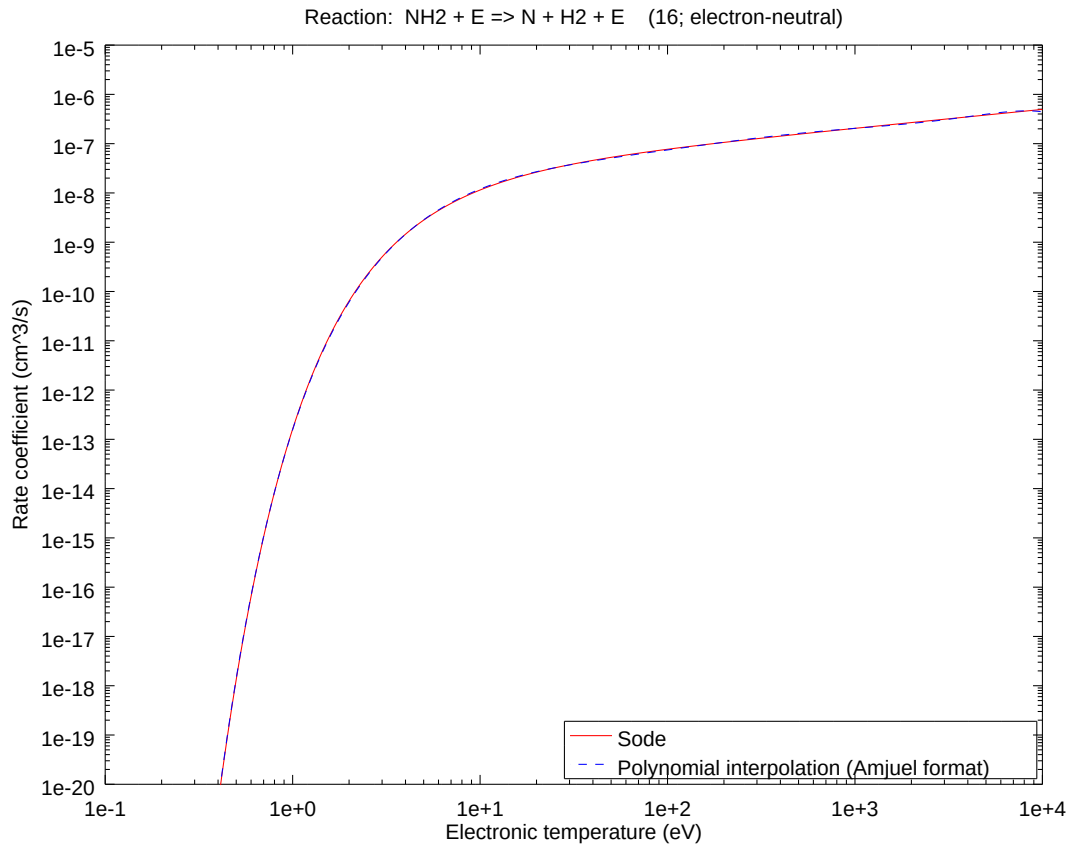
0.16 Reaction 15 $\text{NH} + \text{E} = \text{N} + \text{H} + \text{E}$

b0	-2.457785758960D+01	b1	7.403421947312D+00	b2	-3.797854625556D+00
b3	1.316901771271D+00	b4	-3.467943855239D-01	b5	6.500175435420D-02
b6	-7.735306524768D-03	b7	5.095789580089D-04	b8	-1.400122097940D-05



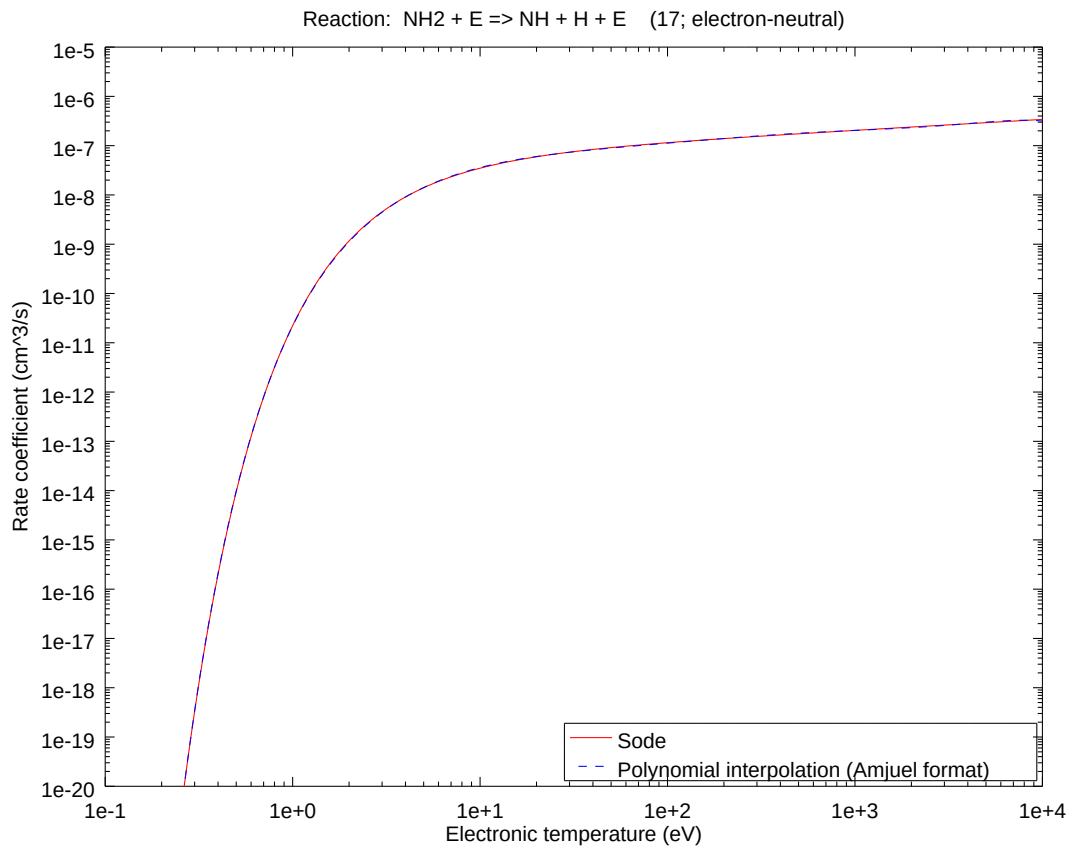
0.17 Reaction 16 $\text{NH}_2 + \text{E} = \text{N} + \text{H}_2 + \text{E}$

b0	-2.947714258173D+01	b1	1.172095540666D+01	b2	-5.649864358434D+00
b3	1.959084039446D+00	b4	-5.159073823659D-01	b5	9.669961896125D-02
b6	-1.150740008366D-02	b7	7.580732483252D-04	b8	-2.082886449991D-05



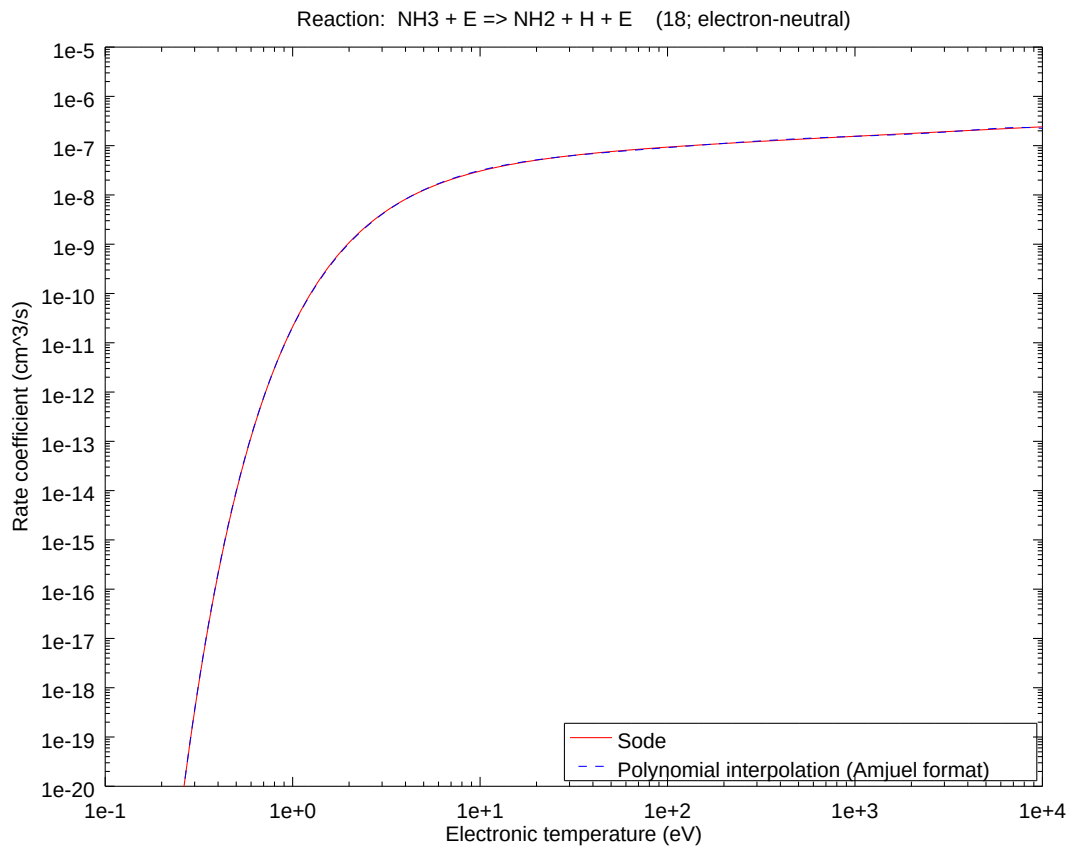
0.18 Reaction 17 $\text{NH}_2 + \text{E} = \text{NH} + \text{H} + \text{E}$

b0	-2.454118936625D+01	b1	7.764114566845D+00	b2	-3.758345084588D+00
b3	1.303201882883D+00	b4	-3.431866415913D-01	b5	6.432553324258D-02
b6	-7.654835195510D-03	b7	5.042777464821D-04	b8	-1.385556458430D-05



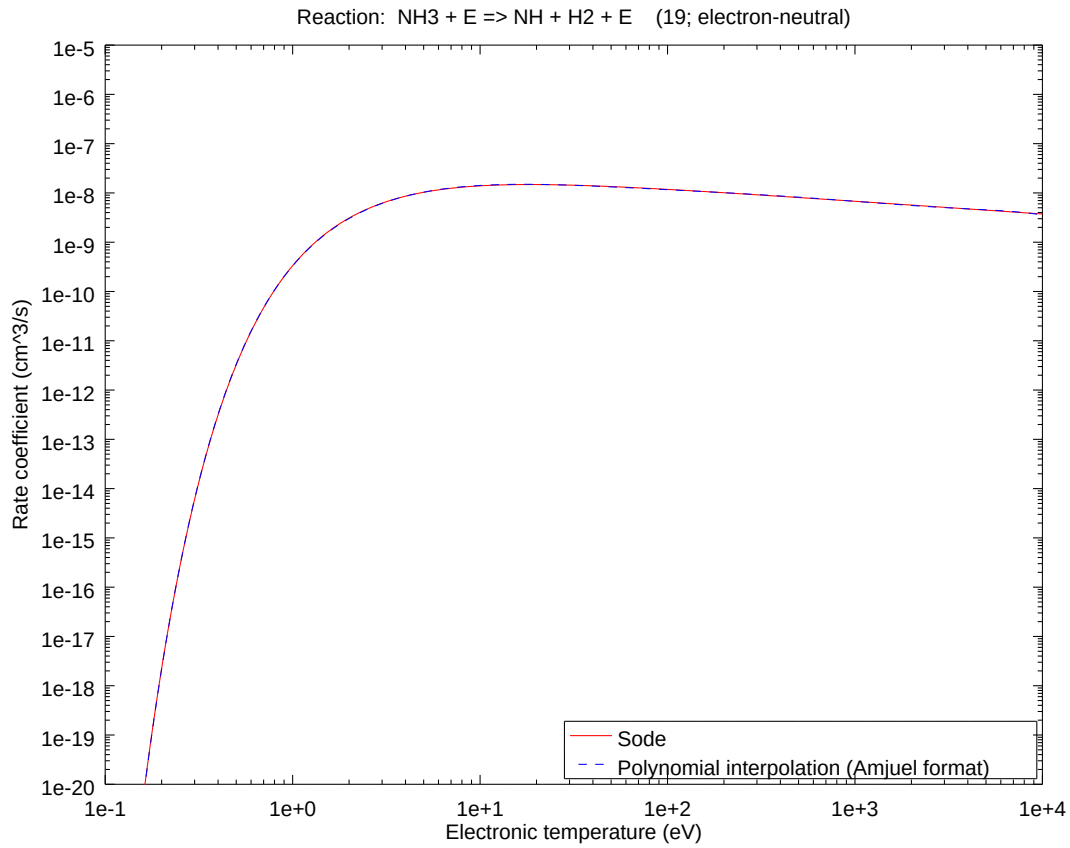
0.19 Reaction 18 $\text{NH}_3 + \text{E} = \text{NH}_2 + \text{H} + \text{E}$

b0	-2.459014390392D+01	b1	7.714287721729D+00	b2	-3.748467699346D+00
b3	1.299776910786D+00	b4	-3.422847056081D-01	b5	6.415647796467D-02
b6	-7.634717363196D-03	b7	5.029524436004D-04	b8	-1.381915048552D-05



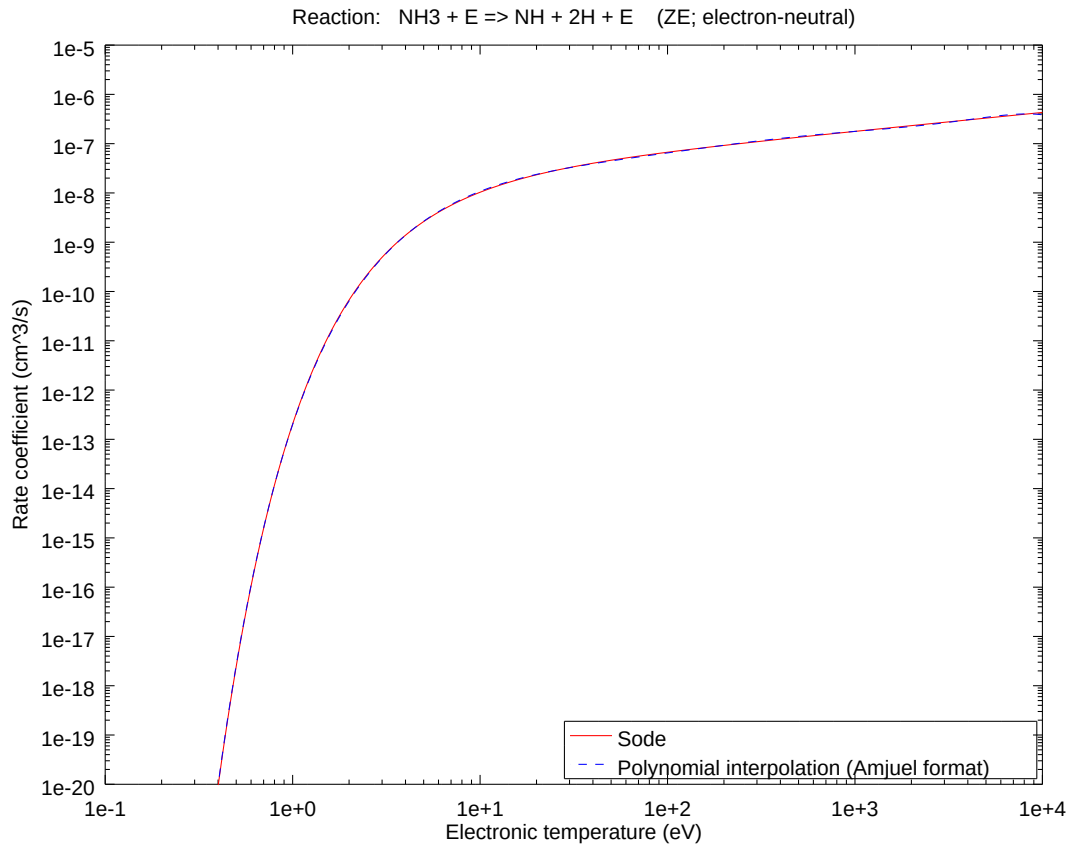
0.20 Reaction 19 $\text{NH}_3 + \text{E} = \text{NH} + \text{H}_2 + \text{E}$

b0	-2.182891305431D+01	b1	4.508356250529D+00	b2	-2.375511150705D+00
b3	8.237057893126D-01	b4	-2.169156039493D-01	b5	4.065779433598D-02
b6	-4.838338671538D-03	b7	3.187353430458D-04	b8	-8.757590755646D-06



0.21 Reaction ZE NH3 + E = NH + 2H + E

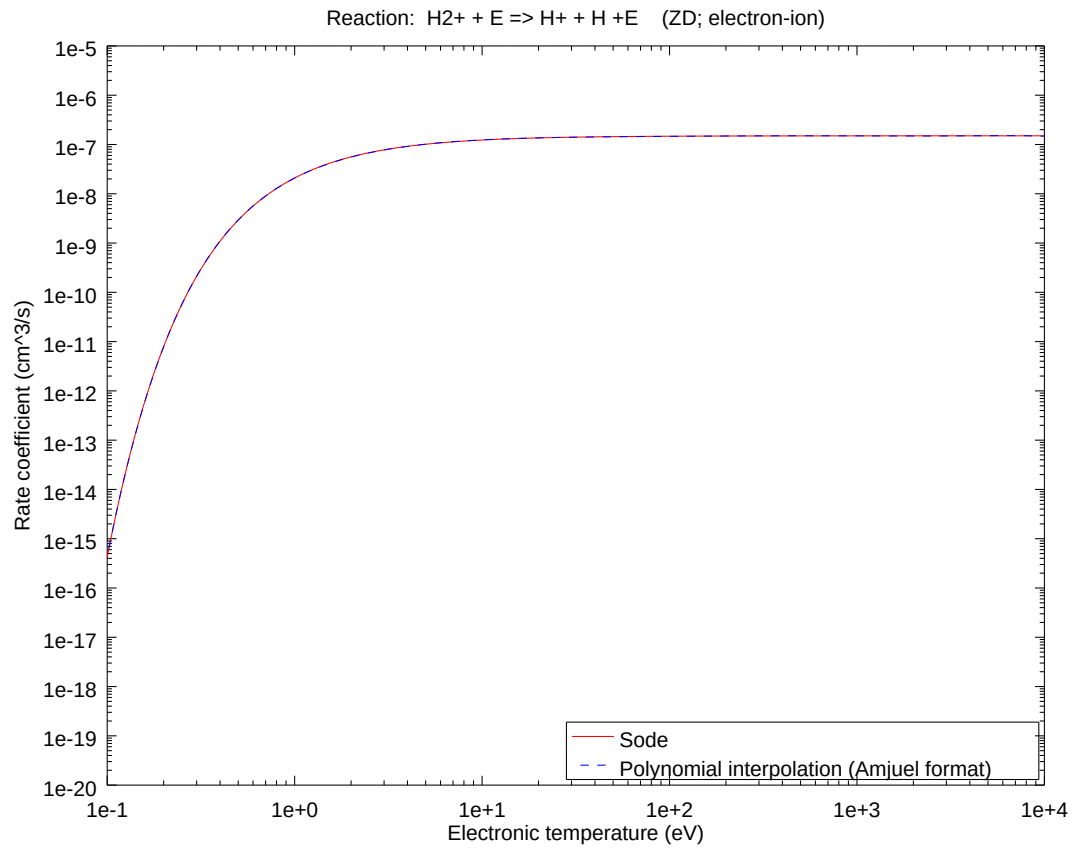
b0	-2.923951508276D+01	b1	1.134424534945D+01	b2	-5.462194038836D+00
b3	1.894009569605D+00	b4	-4.987705986859D-01	b5	9.348756868107D-02
b6	-1.112516126969D-02	b7	7.328924935732D-04	b8	-2.013699662317D-05



1 H.1 : Fits for electron-ion

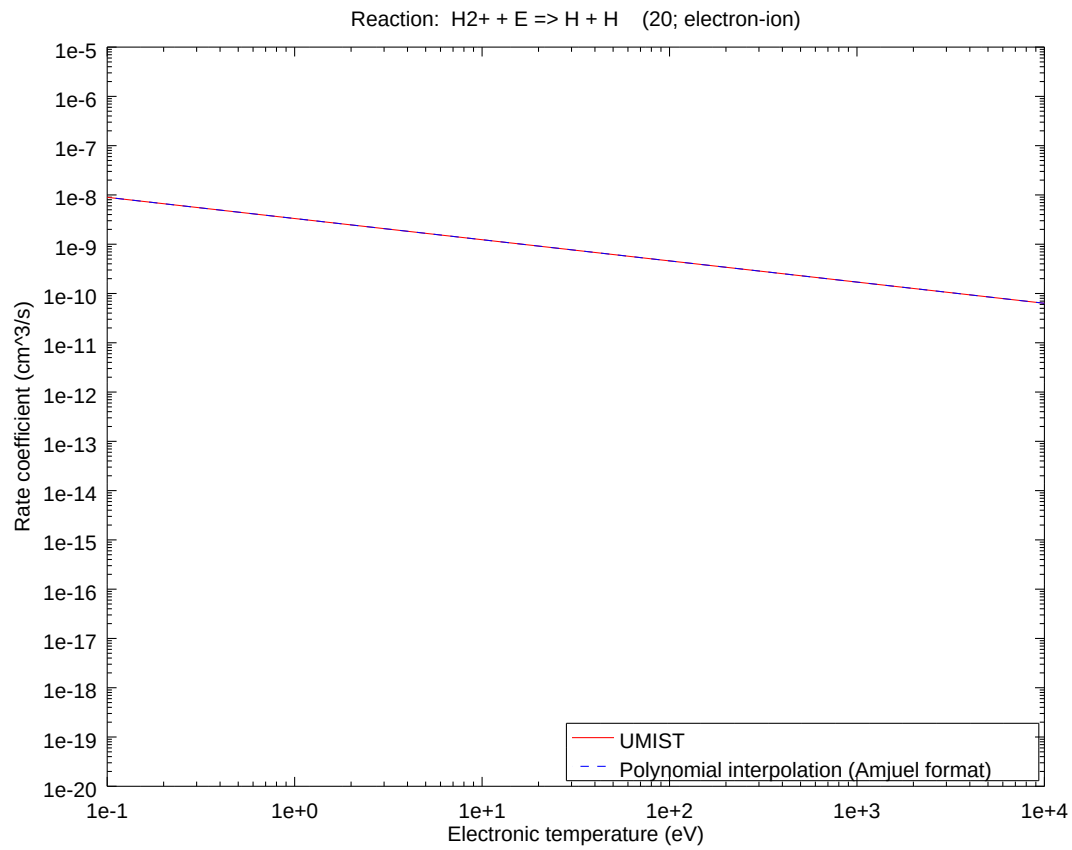
1.1 Reaction ZD $\text{H}_2^+ + \text{E} = \text{H}^+ + \text{H} + \text{E}$

b0	-1.768640642427D+01	b1	1.952944243980D+00	b2	-9.729224463388D-01
b3	3.373597515480D-01	b4	-8.884069434098D-02	b5	1.665194487357D-02
b6	-1.981606482938D-03	b7	1.305423338462D-04	b8	-3.586788729444D-06



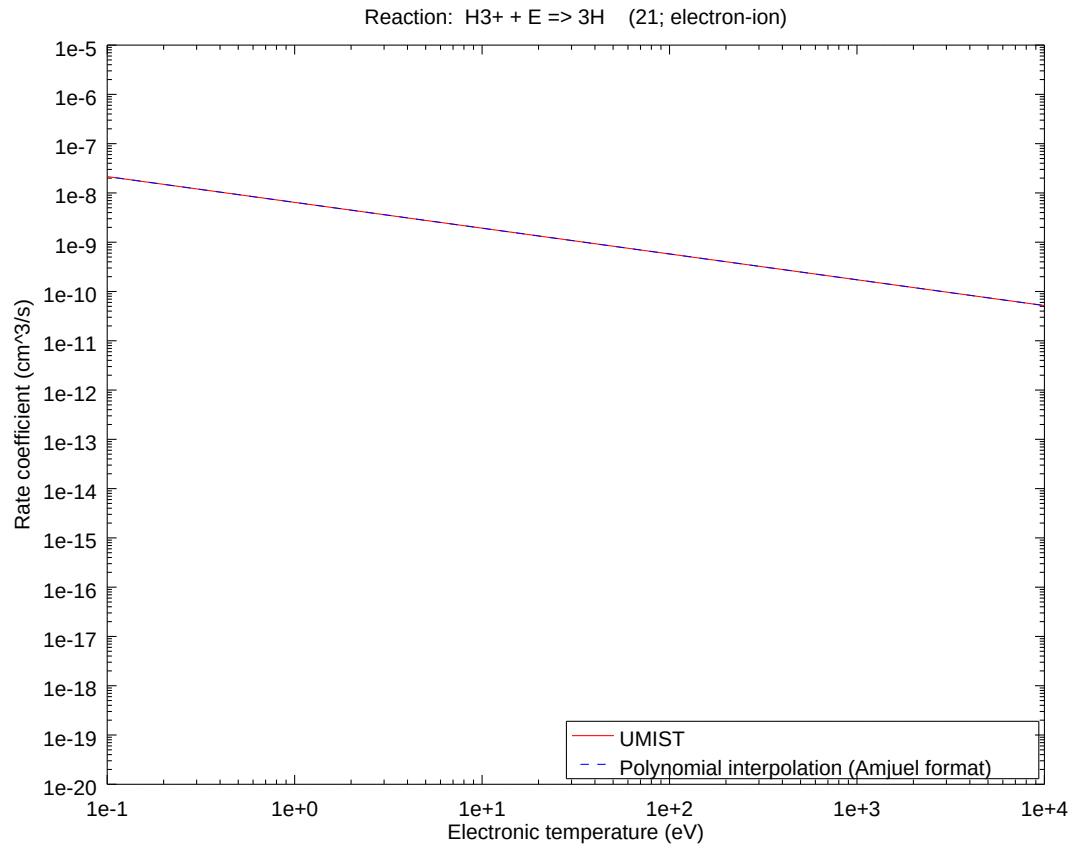
1.2 Reaction 20 $\text{H}_2^+ + \text{E} = \text{H} + \text{H}$

b0	-1.549805065050D+01	b1	-4.300000000721D-01	b2	2.121207859872D-11
b3	-3.525233366392D-12	b4	3.619396884458D-13	b5	-2.350039868387D-14
b6	9.422902459647D-16	b7	-2.133892806437D-17	b8	2.090784587778D-19



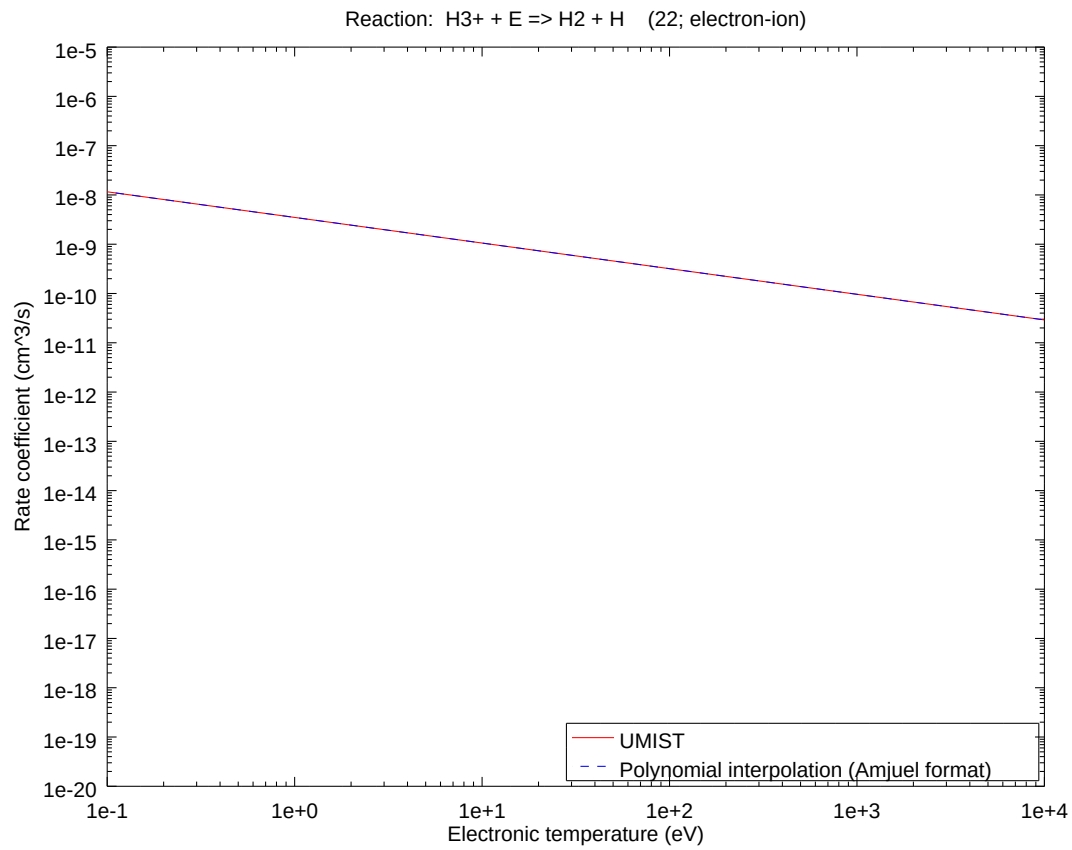
1.3 Reaction 21 $\text{H3}^+ + \text{E} = 3\text{H}$

b0	-1.396227856104D+01	b1	-5.235000000456D-01	b2	1.377323514977D-11
b3	-2.364345055224D-12	b4	2.514131485131D-13	b5	-1.691918639573D-14
b6	7.023760130112D-16	b7	-1.642171432285D-17	b8	1.653920297816D-19



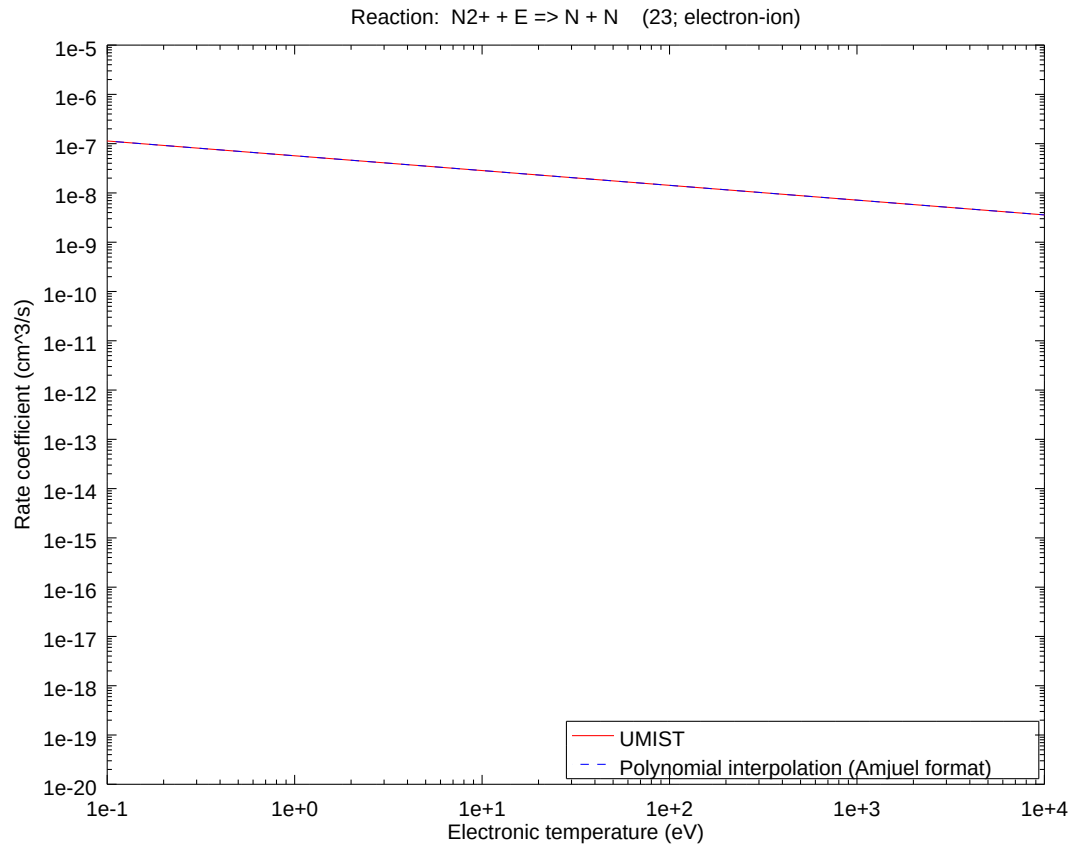
1.4 Reaction 22 $\text{H3}^+ + \text{E} = \text{H2} + \text{H}$

b0	-1.460456292784D+01	b1	-5.199999999415D-01	b2	-1.797642964283D-11
b3	3.106425910878D-12	b4	-3.308321264474D-13	b5	2.226194511892D-14
b6	-9.250457320960D-16	b7	2.171052866419D-17	b8	-2.203798900785D-19



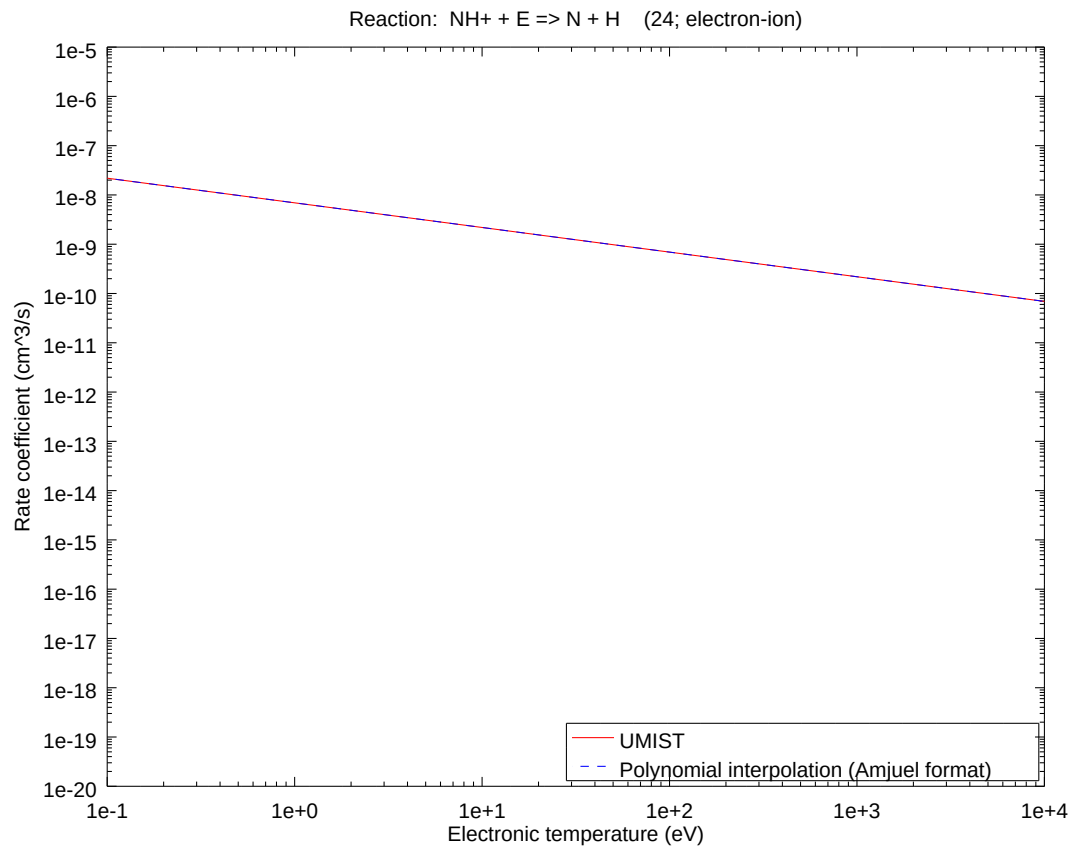
1.5 Reaction 23 $\text{N}_2^+ + \text{E} = \text{N} + \text{N}$

b0	-1.387633265741D+01	b1	-3.000000000638D-01	b2	2.042553247897D-11
b3	-3.686395140958D-12	b4	4.096234141478D-13	b5	-2.867492878980D-14
b6	1.234826176798D-15	b7	-2.991603590027D-17	b8	3.123447038938D-19



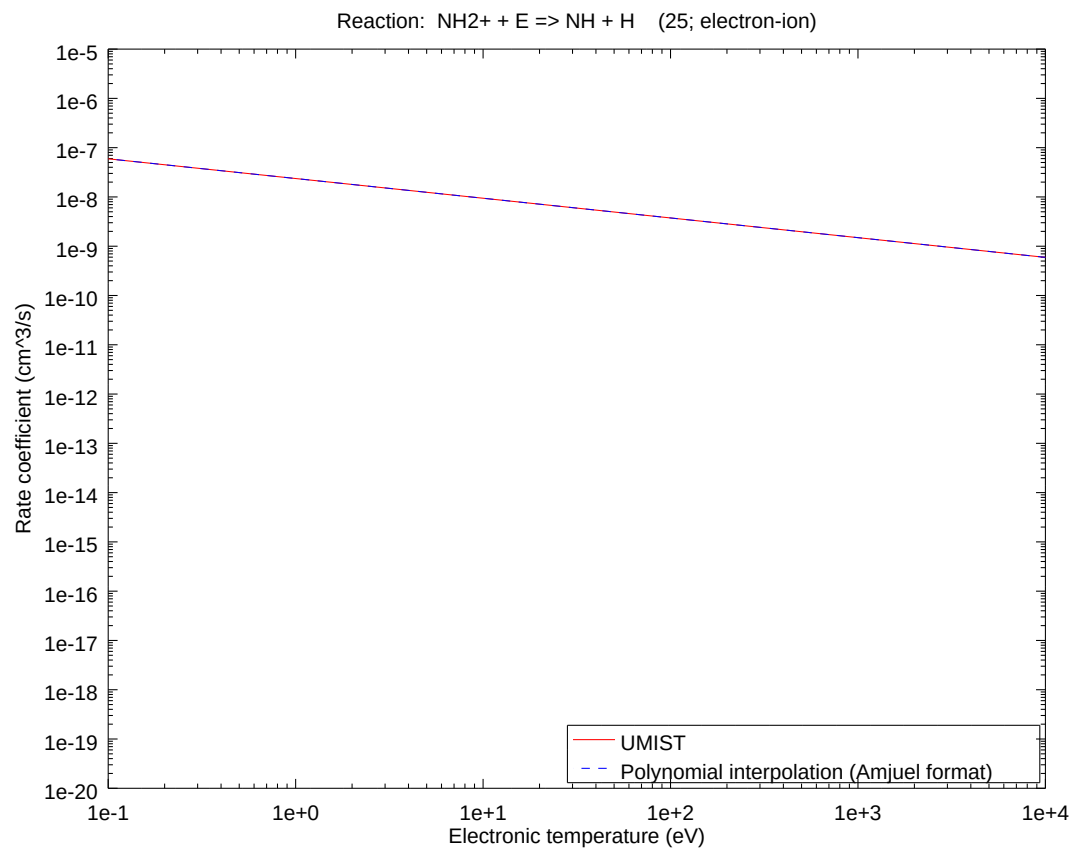
1.6 Reaction 24 $\text{NH}^+ + \text{E} = \text{N} + \text{H}$

b0	-1.411017448387D+01	b1	-5.000000000414D-01	b2	1.345152089079D-11
b3	-2.452613591631D-12	b4	2.740123797404D-13	b5	-1.919475414462D-14
b6	8.233565854067D-16	b7	-1.978385863314D-17	b8	2.040557824651D-19



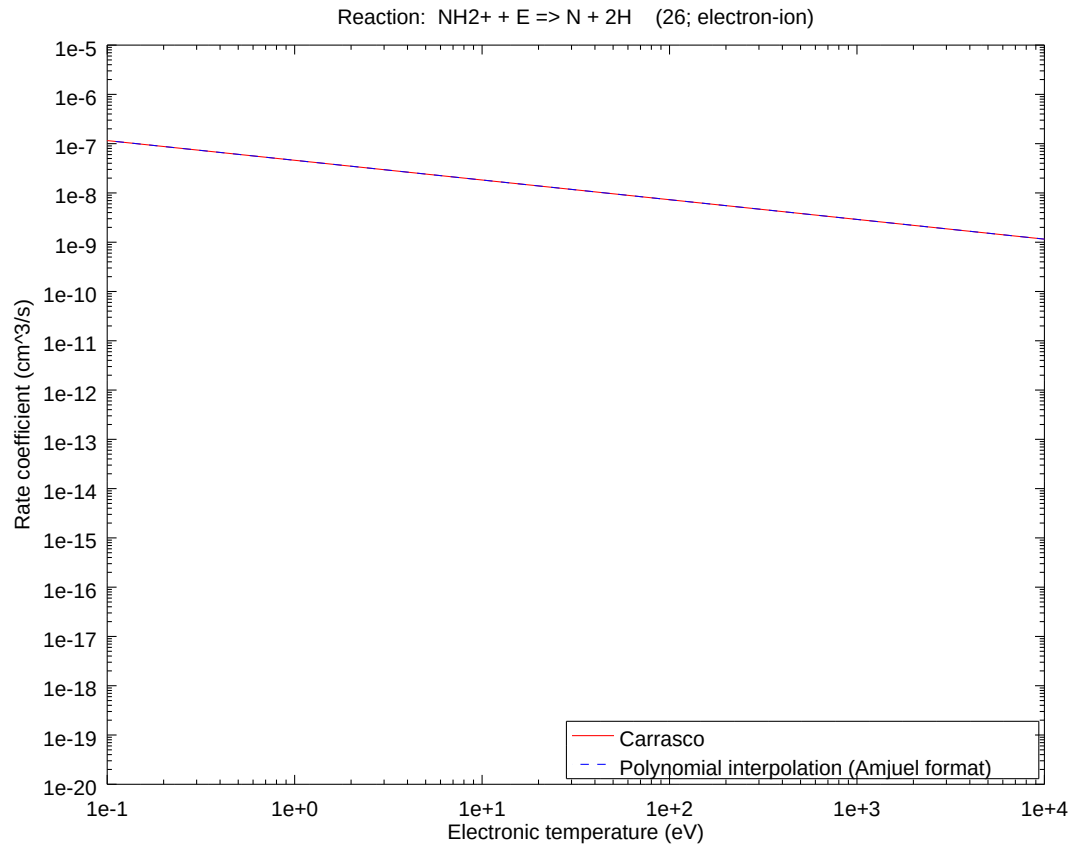
1.7 Reaction 25 $\text{NH}_2^+ + \text{E} = \text{NH} + \text{H}$

b0	-1.381678003369D+01	b1	-4.000000000763D-01	b2	2.281875835896D-11
b3	-3.855397636026D-12	b4	4.021080079869D-13	b5	-2.649928985121D-14
b6	1.077536689384D-15	b7	-2.472359944128D-17	b8	2.451558560319D-19



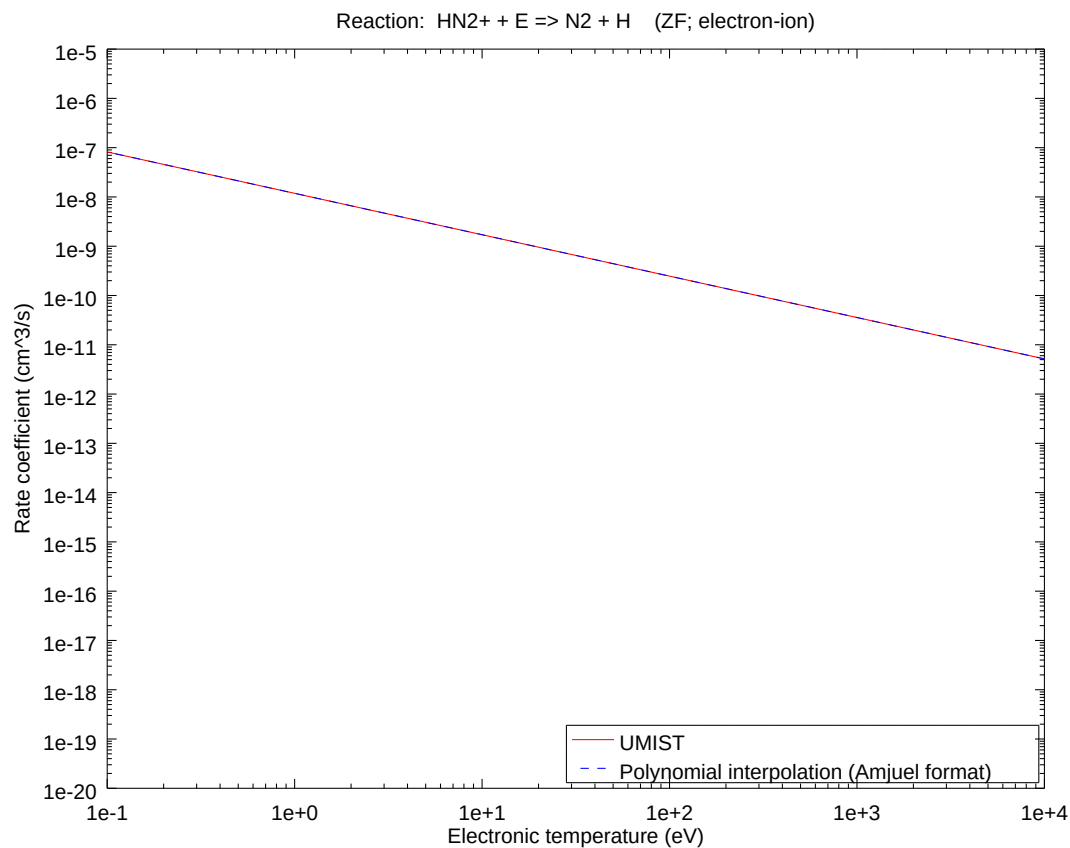
1.8 Reaction 26 $\text{NH}_2^+ + \text{E} = \text{N} + 2\text{H}$

b0	-1.689486230264D+01	b1	-4.000000000000D-01	b2	3.922332100276D-15
b3	4.933121613073D-15	b4	-2.597700526548D-15	b5	1.035377292669D-16
b6	1.120423733007D-16	b7	-1.903463952049D-17	b8	8.964767471941D-19



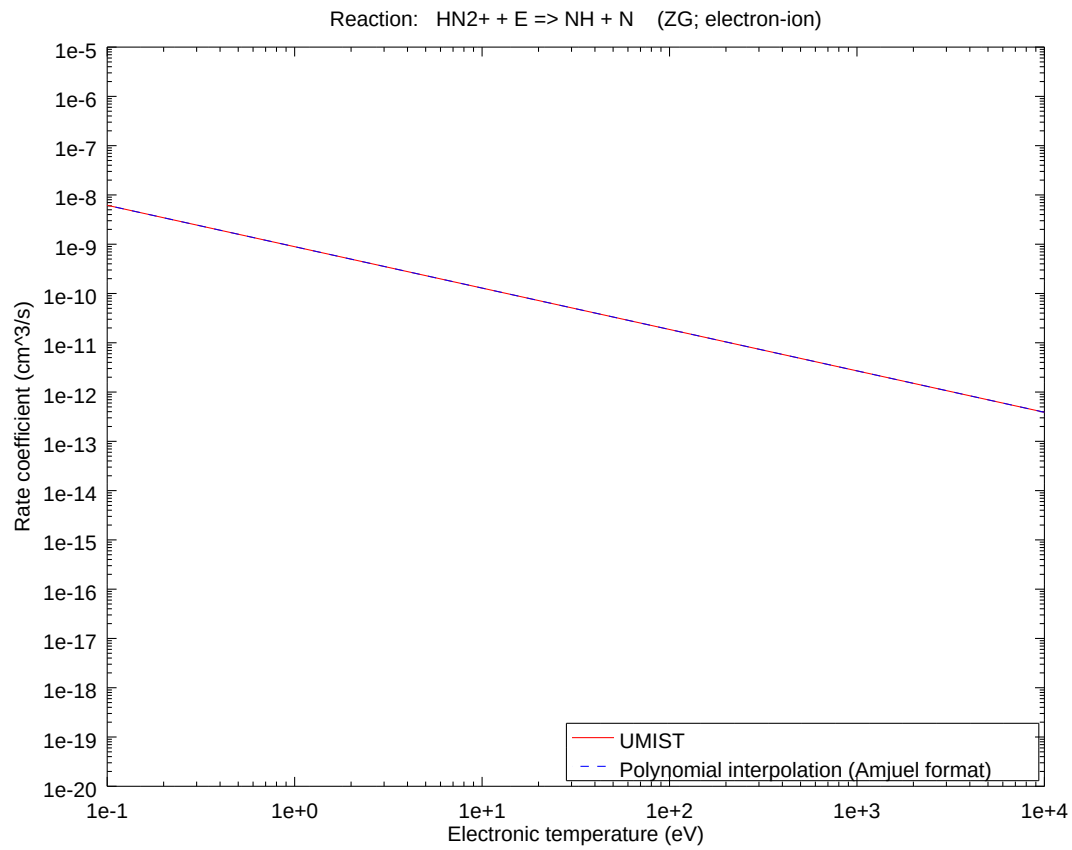
1.9 Reaction ZF $\text{HN}_2^+ + \text{E} = \text{N}_2 + \text{H}$

b0	-1.039082501294D+01	b1	-8.400000000933D-01	b2	2.798050081896D-11
b3	-4.729451154959D-12	b4	4.924607470924D-13	b5	-3.234789685169D-14
b6	1.309240010427D-15	b7	-2.985976716106D-17	b8	2.938852736197D-19



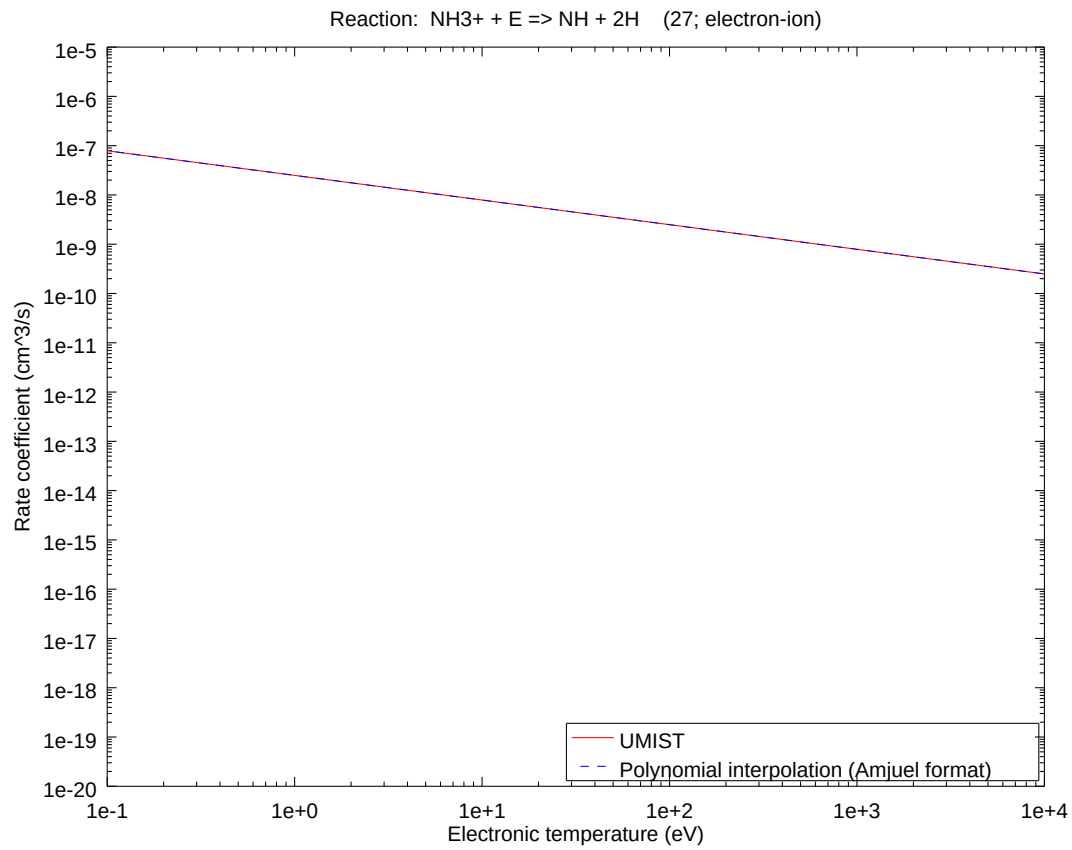
1.10 Reaction ZG $\text{HN}_2^+ + \text{E} = \text{NH} + \text{N}$

b0	-1.297717827918D+01	b1	-8.400000000205D-01	b2	7.952829649458D-12
b3	-1.672429370773D-12	b4	2.099308337076D-13	b5	-1.619696727516D-14
b6	7.533144016107D-16	b7	-1.937737405572D-17	b8	2.116902016248D-19



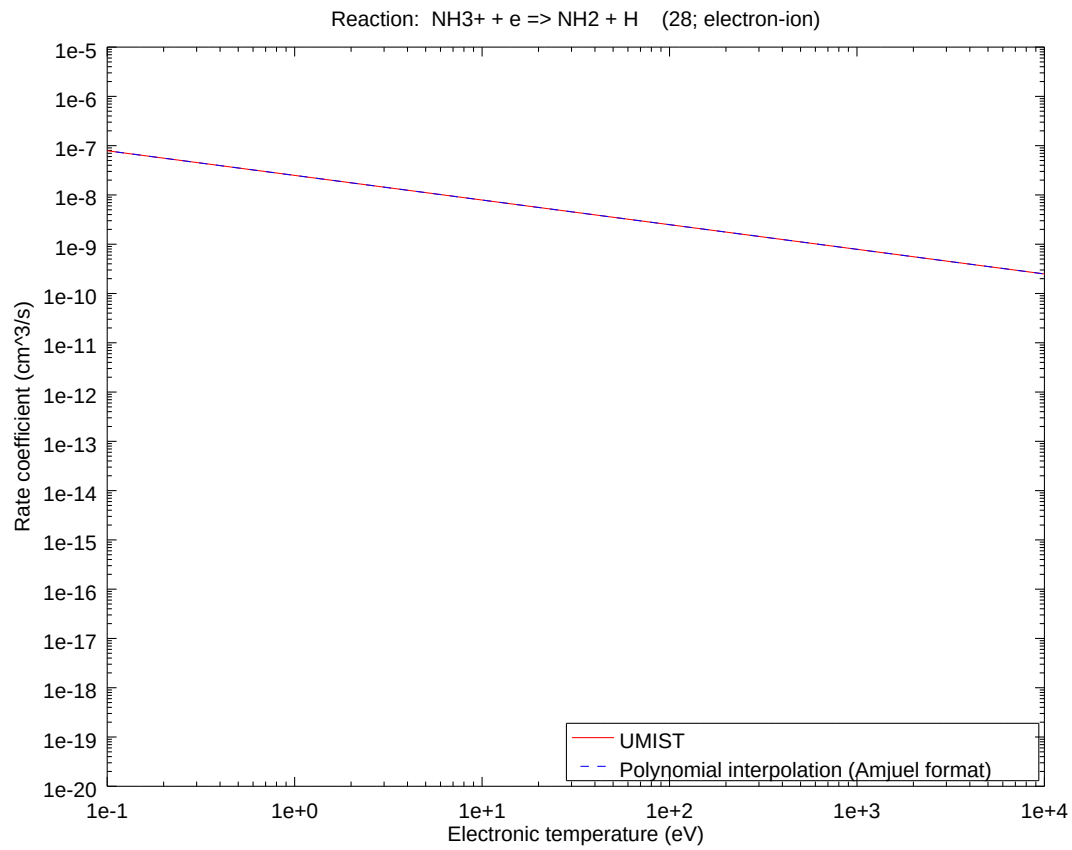
1.11 Reaction 27 $\text{NH}_3^+ + \text{E} = \text{NH} + 2\text{H}$

b0	-1.282794948273D+01	b1	-4.999999999830D-01	b2	-4.349711744970D-12
b3	5.810946018121D-13	b4	-4.249268933396D-14	b5	1.519714196652D-15
b6	-7.943027281034D-18	b7	-1.062362698928D-18	b8	2.274013726615D-20



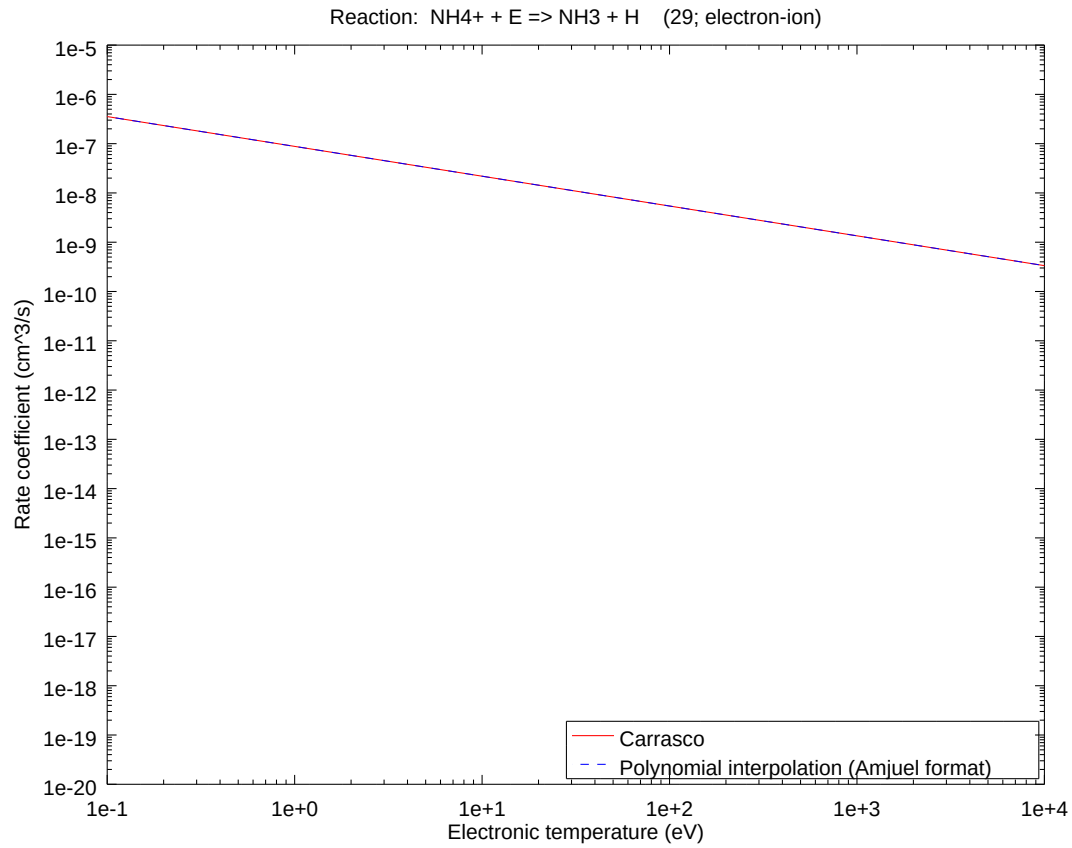
1.12 Reaction 28 $\text{NH}_3^+ + e = \text{NH}_2 + \text{H}$

b0	-1.282794948273D+01	b1	-4.999999999830D-01	b2	-4.349711744970D-12
b3	5.810946018121D-13	b4	-4.249268933396D-14	b5	1.519714196652D-15
b6	-7.943027281034D-18	b7	-1.062362698928D-18	b8	2.274013726615D-20



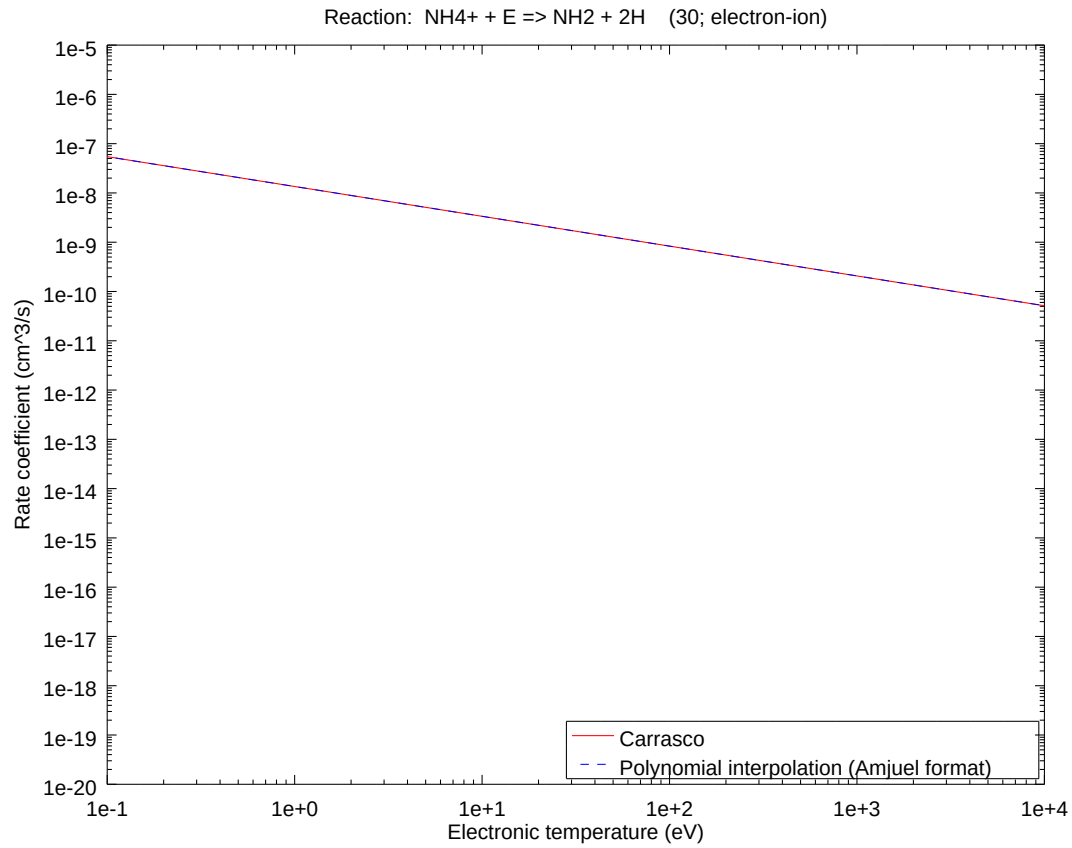
1.13 Reaction 29 $\text{NH}_4^+ + \text{E} = \text{NH}_3 + \text{H}$

b0	-1.624544842816D+01	b1	-6.050000000000D-01	b2	0.000000000000D+00
b3	8.306736808840D-16	b4	-4.686048483672D-17	b5	-2.950510937902D-16
b6	1.050373201078D-16	b7	-1.304766889864D-17	b8	5.502683070338D-19



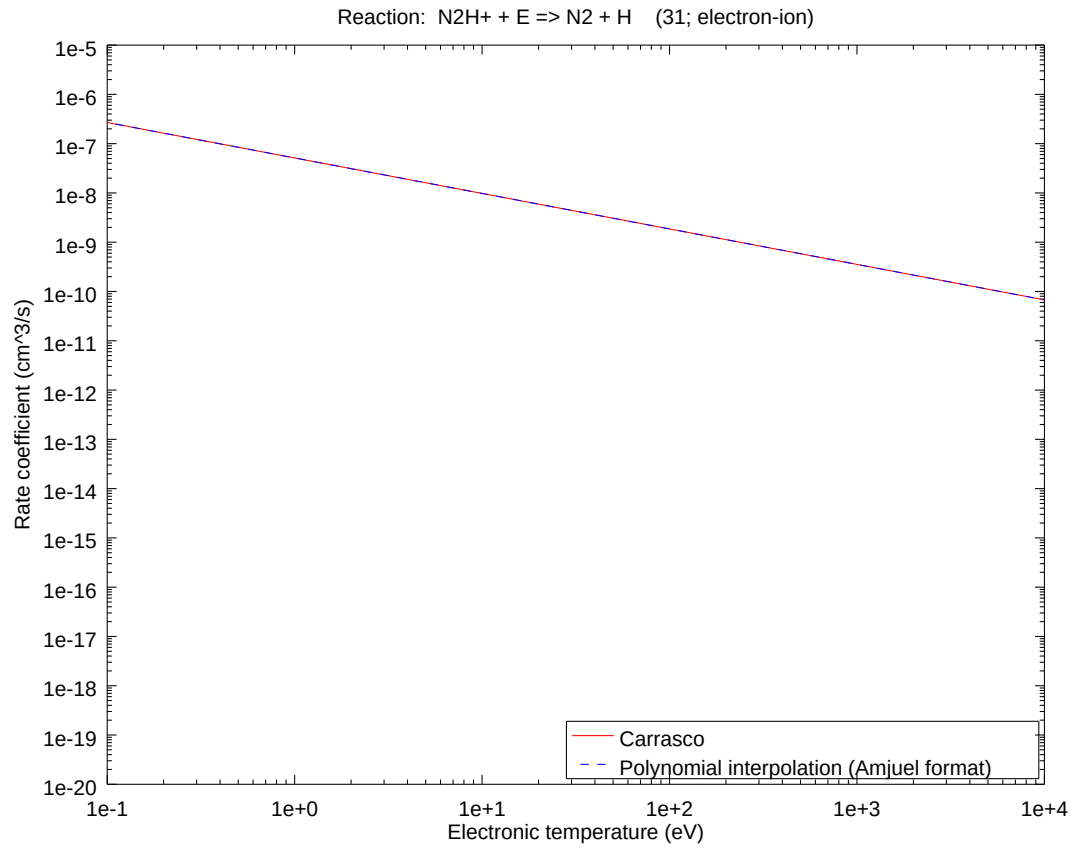
1.14 Reaction 30 $\text{NH}_4^+ + \text{E} = \text{NH}_2 + 2\text{H}$

b0	-1.811912501986D+01	b1	-6.050000000000D-01	b2	2.941749075207D-14
b3	-9.380868417975D-15	b4	-3.748190362625D-15	b5	2.200689446104D-15
b6	-3.784562783185D-16	b7	2.695309824768D-17	b8	-6.649703260841D-19



1.15 Reaction 31 $\text{N}_2\text{H}^+ + \text{E} = \text{N}_2 + \text{H}$

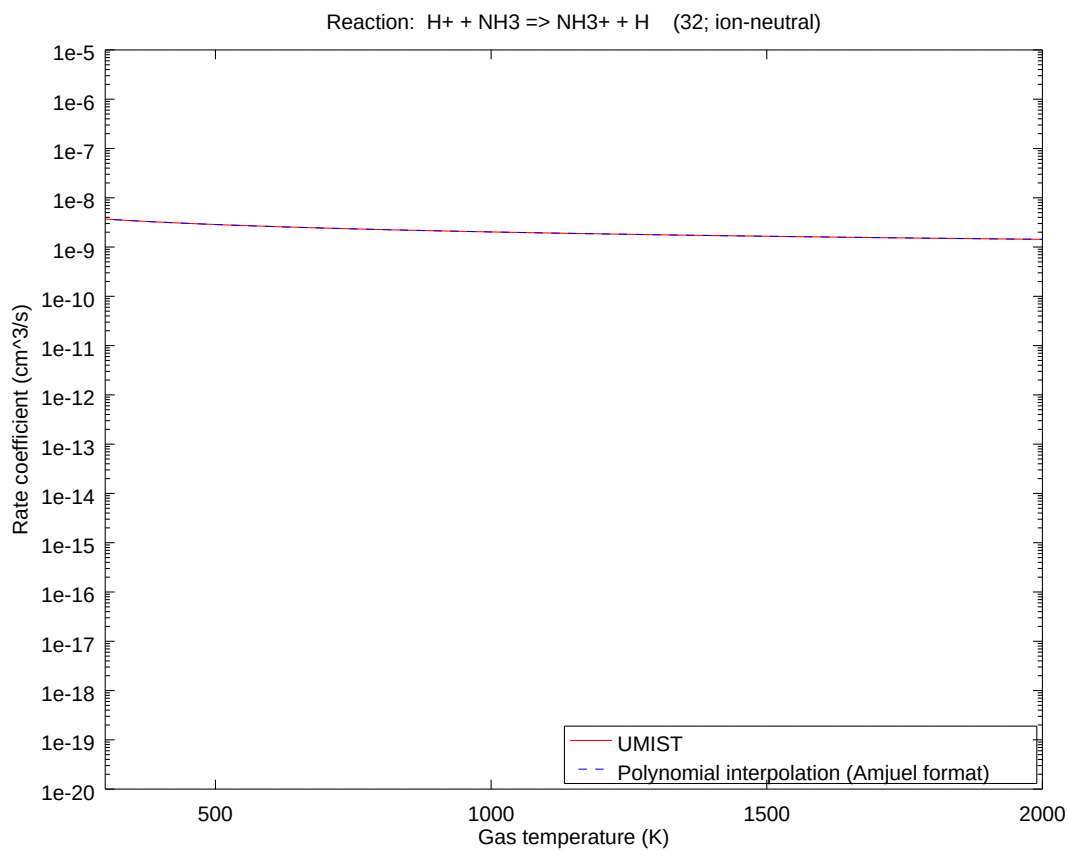
b0	-1.678575516040D+01	b1	-7.200000000000D-01	b2	2.941749075207D-14
b3	-1.766359676538D-15	b4	-4.297937615358D-15	b5	1.044560914165D-15
b6	-3.956040344312D-17	b7	-8.226431945192D-18	b8	6.016644534124D-19



2 H.2 : Fits for ion-neutral

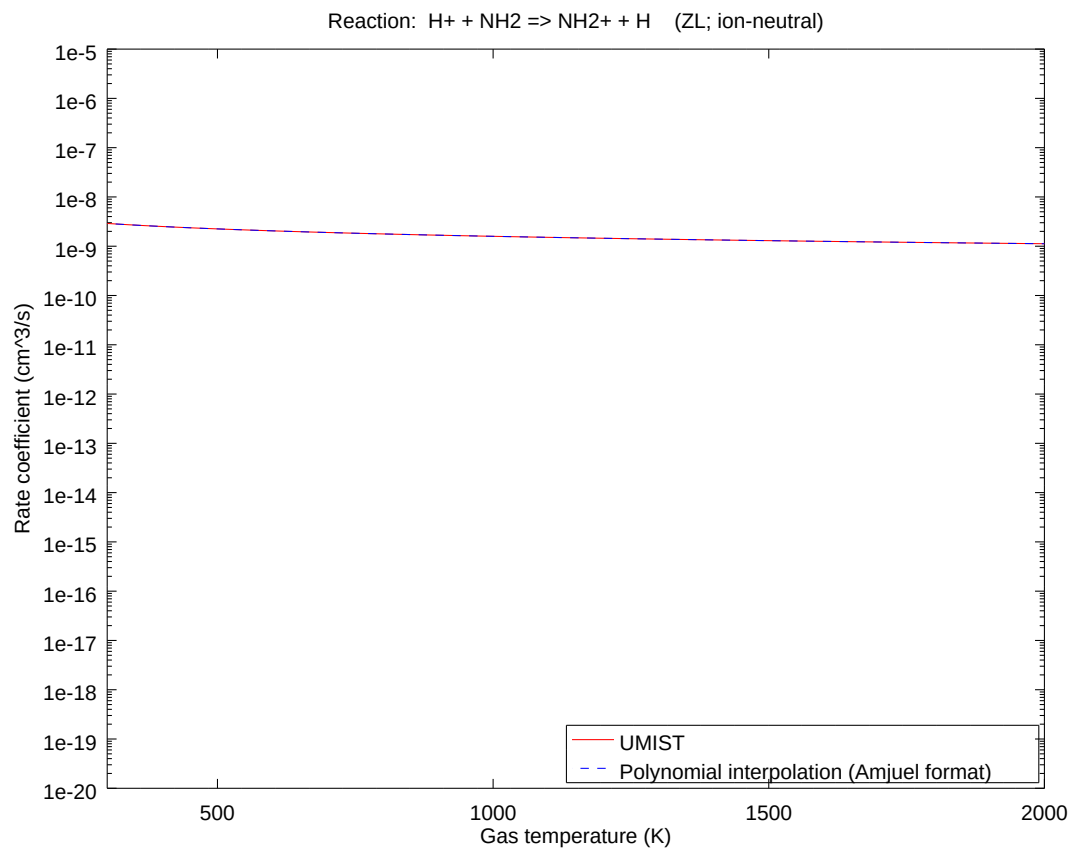
2.1 Reaction 32 $\text{H}^+ + \text{NH}_3 = \text{NH}_3^+ + \text{H}$

b0	-1.656303543225D+01	b1	-5.000077079232D-01	b2	4.088471613298D-06
b3	-1.237295628666D-06	b4	2.336655020426D-07	b5	-2.819853205399D-08
b6	2.123596137960D-09	b7	-9.124645179622D-11	b8	1.712688566827D-12



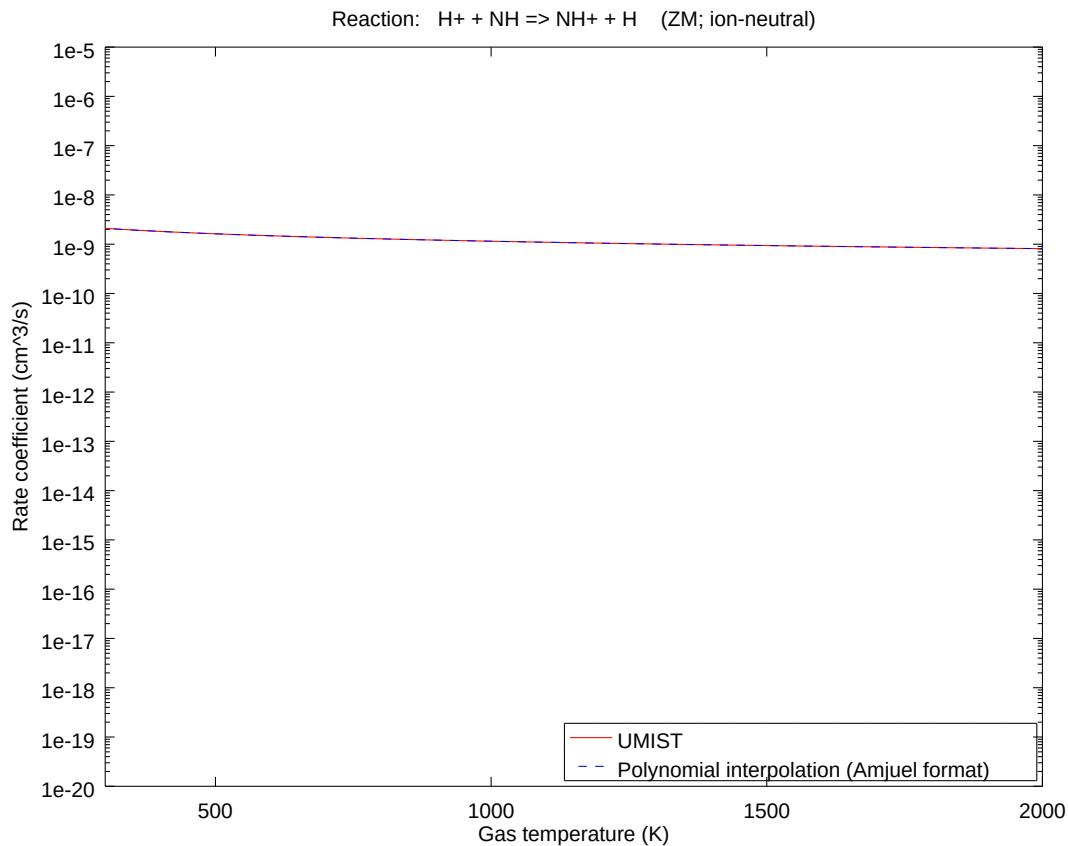
2.2 Reaction ZL $\text{H}^+ + \text{NH}_2 = \text{NH}_2^+ + \text{H}$

b0	-1.680665785521D+01	b1	-5.000073125322D-01	b2	3.888134643719D-06
b3	-1.179489359870D-06	b4	2.232783051952D-07	b5	-2.700857065920D-08
b6	2.038743138113D-09	b7	-8.780409578528D-11	b8	1.651879750166D-12



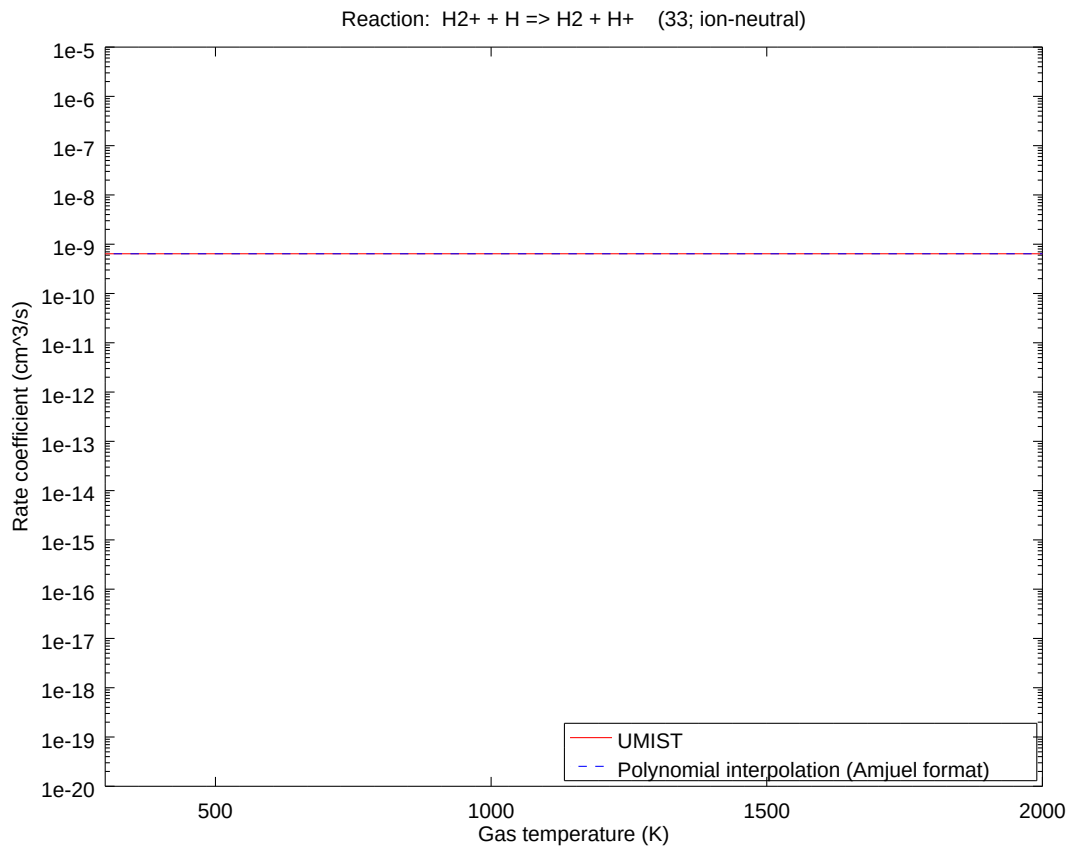
2.3 Reaction ZM $\text{H}^+ + \text{NH} = \text{NH}^+ + \text{H}$

b0	-1.712943118603D+01	b1	-5.000073751283D-01	b2	3.915015979476D-06
b3	-1.185728681478D-06	b4	2.241022714450D-07	b5	-2.706574339404D-08
b6	2.039907770134D-09	b7	-8.772115525199D-11	b8	1.647867472525D-12



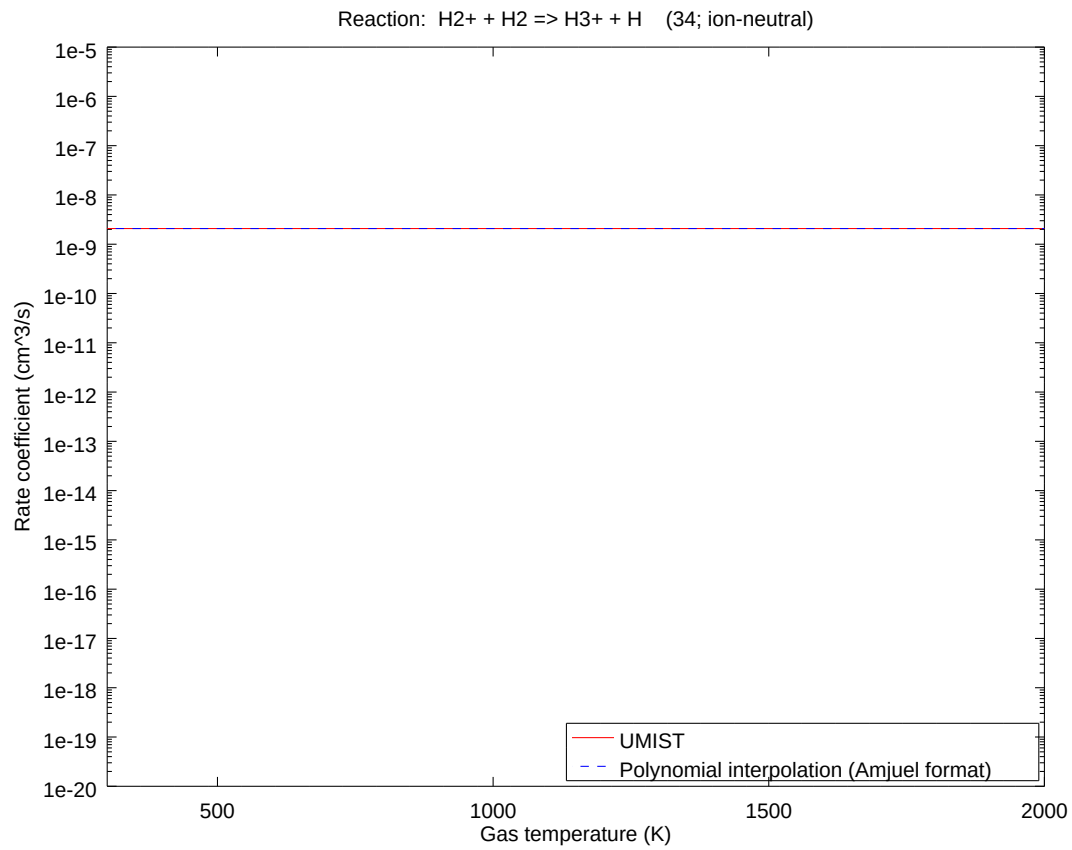
2.4 Reaction 33 $\text{H}_2^+ + \text{H} = \text{H}_2 + \text{H}^+$

b0	-2.116954665470D+01	b1	-7.634682401144D-06	b2	4.051497619569D-06
b3	-1.226749575953D-06	b4	2.318097332644D-07	b5	-2.799270224514D-08
b6	2.109588923865D-09	b7	-9.071422773444D-11	b8	1.704106649575D-12



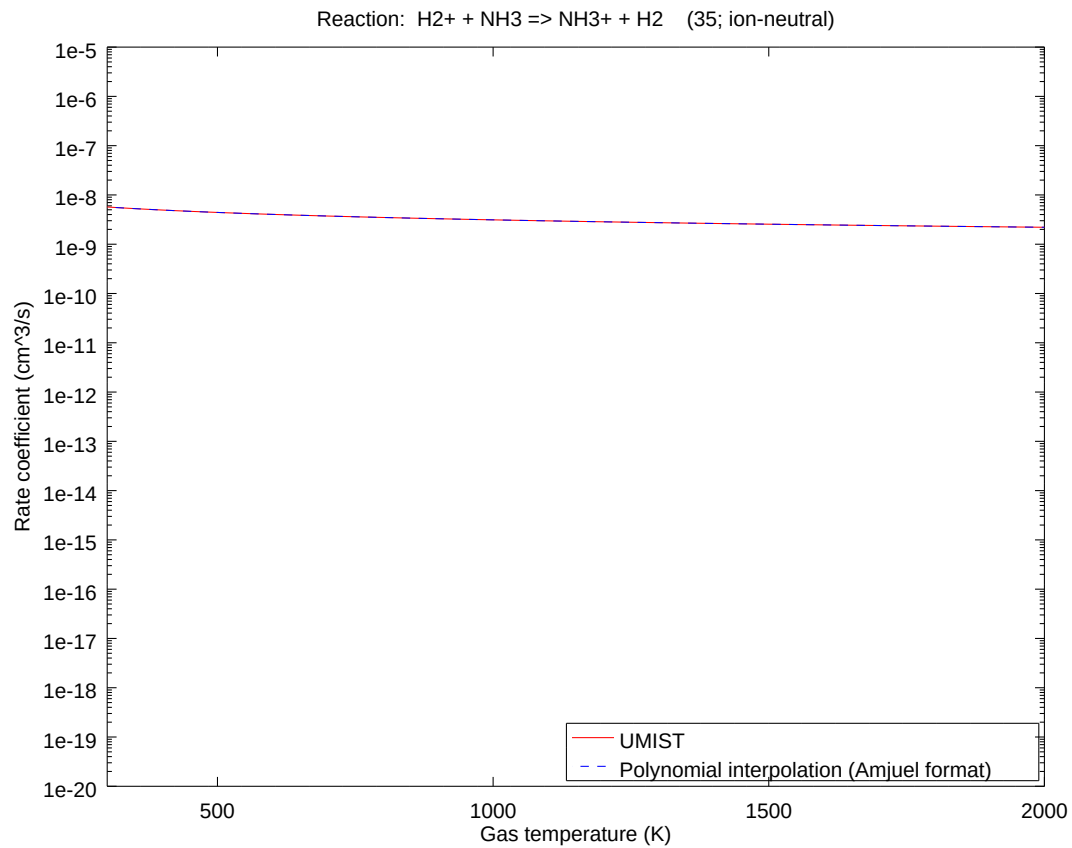
2.5 Reaction 34 $\text{H}_2^+ + \text{H}_2 = \text{H}_3^+ + \text{H}$

b0	-1.999089004022D+01	b1	-9.577084393834D-06	b2	5.069944528382D-06
b3	-1.531399950189D-06	b4	2.886758489133D-07	b5	-3.477536837052D-08
b6	2.614423463825D-09	b7	-1.121524204190D-10	b8	2.101793115336D-12



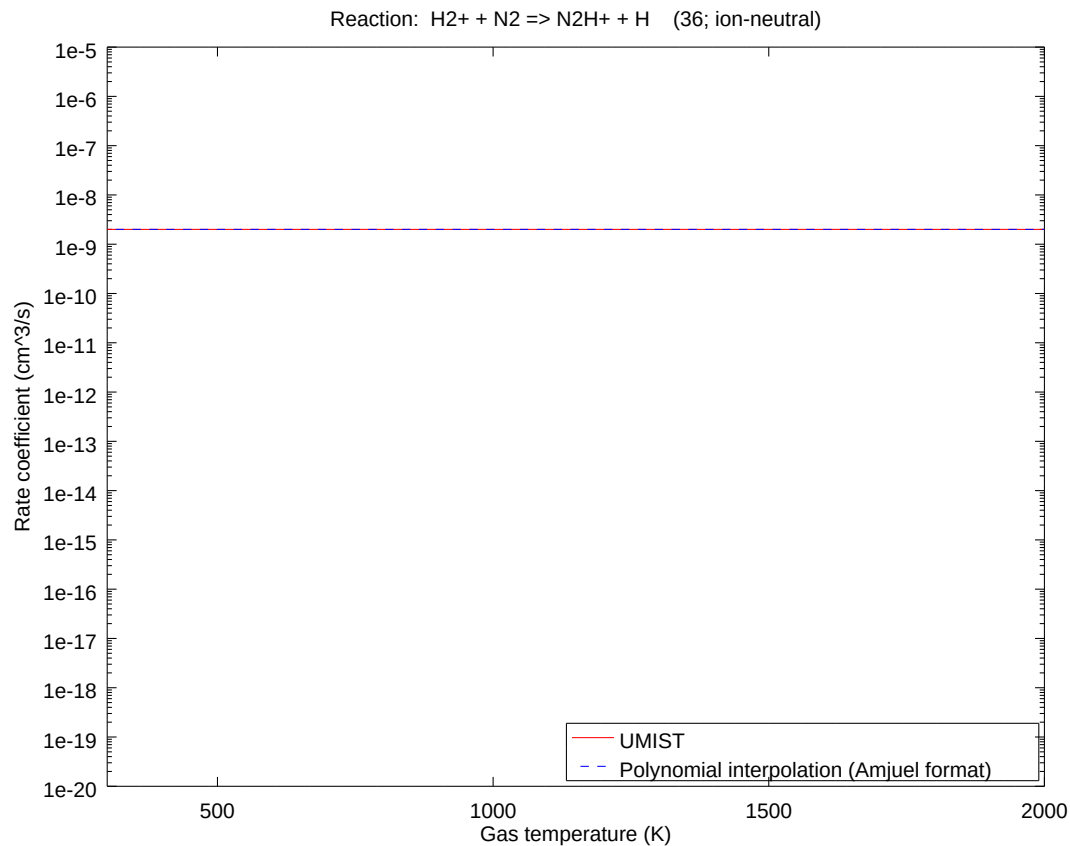
2.6 Reaction 35 $\text{H}_2^+ + \text{NH}_3 = \text{NH}_3^+ + \text{H}_2$

b0	-1.613090326241D+01	b1	-5.000062741049D-01	b2	3.331122736185D-06
b3	-1.009136852705D-06	b4	1.907871341007D-07	b5	-2.305095617096D-08
b6	1.738086303688D-09	b7	-7.477908987689D-11	b8	1.405509750146D-12



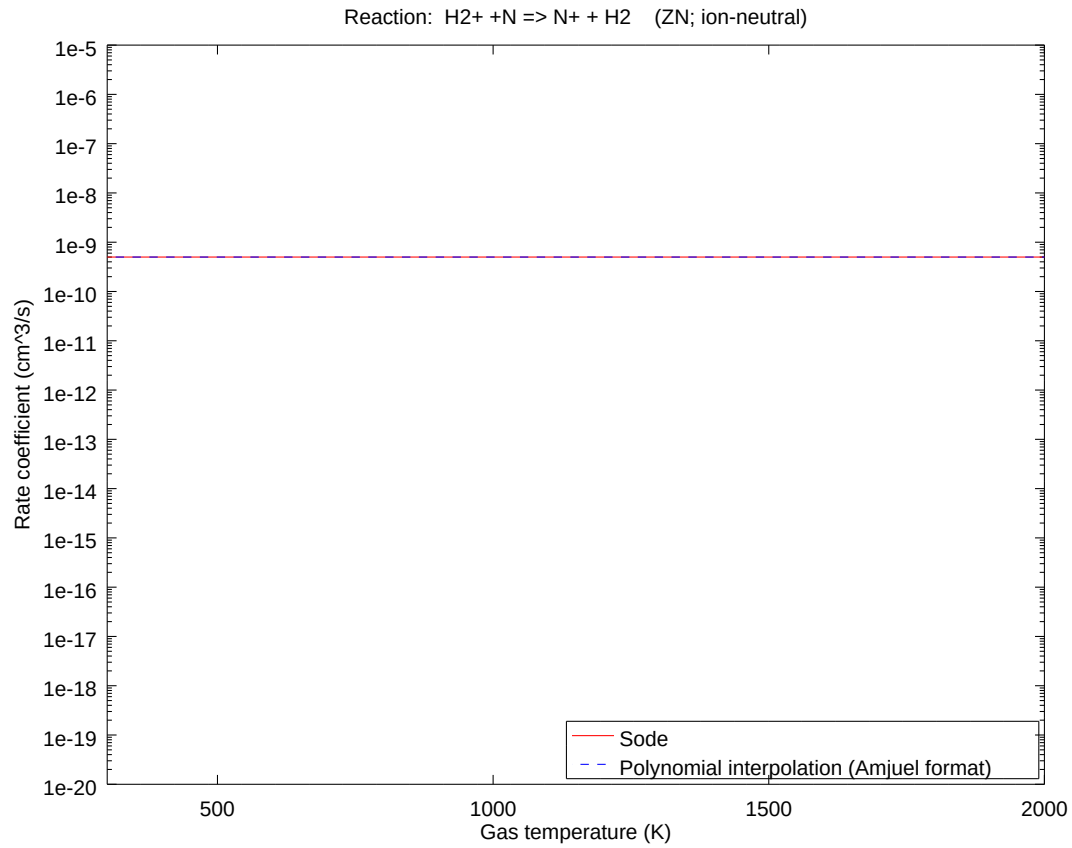
2.7 Reaction 36 $\text{H}_2^+ + \text{N}_2 = \text{N}_2\text{H}^+ + \text{H}$

b0	-2.003011395895D+01	b1	-5.708629727668D-06	b2	3.030475015499D-06
b3	-9.178726557271D-07	b4	1.734868129724D-07	b5	-2.095388099415D-08
b6	1.579350557928D-09	b7	-6.791926991658D-11	b8	1.275929759107D-12



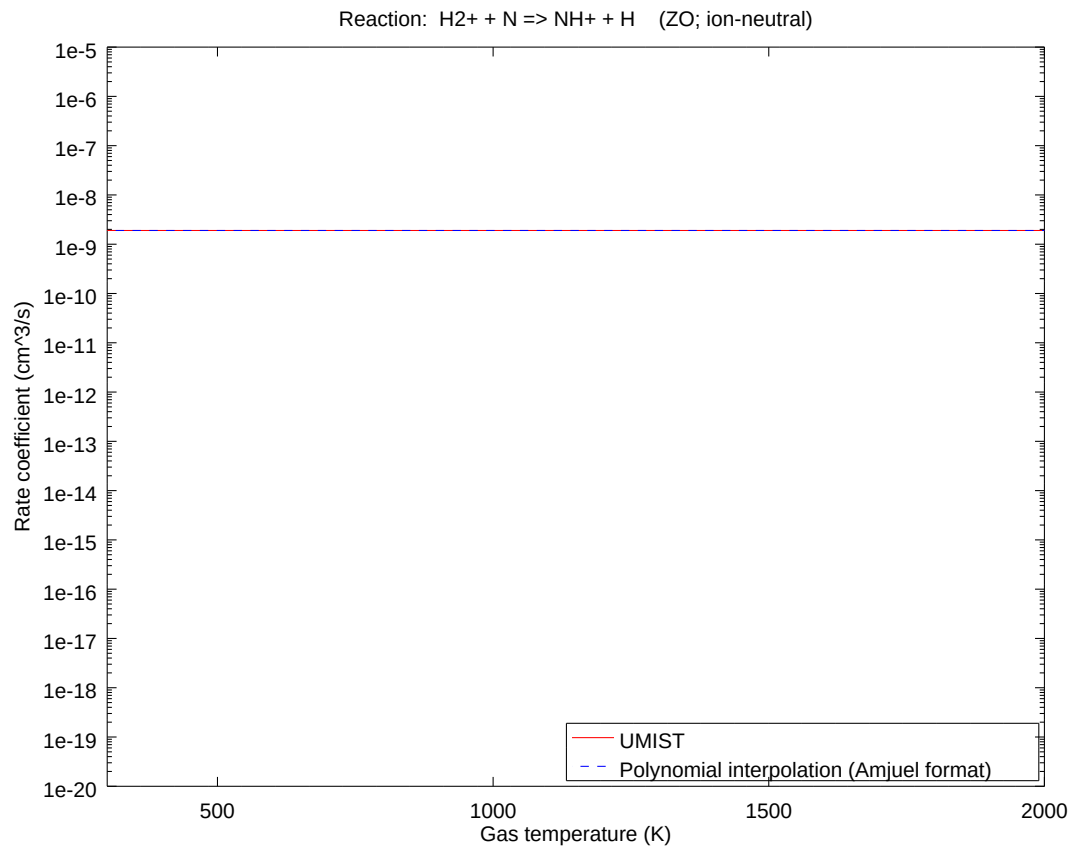
2.8 Reaction ZN $\text{H}_2^+ + \text{N} = \text{N}^+ + \text{H}_2$

b0	-2.141641302012D+01	b1	-8.131359156625D-09	b2	-1.095575028514D-08
b3	-8.351628081406D-09	b4	-3.940087662388D-09	b5	-1.178197239087D-09
b6	-2.181281550989D-10	b7	-2.286666084442D-11	b8	-1.039602156424D-12



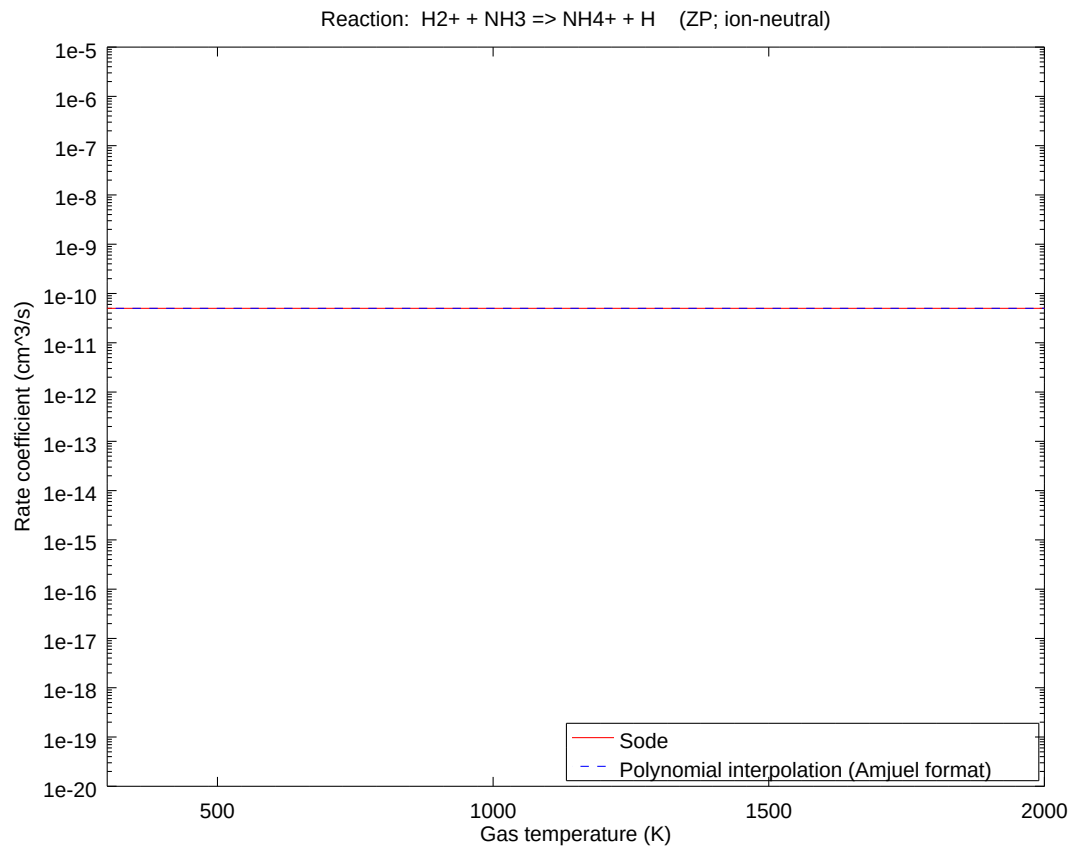
2.9 Reaction ZO H2+ + N = NH+ + H

b0	-2.008140404183D+01	b1	-9.585254955163D-06	b2	5.074660572956D-06
b3	-1.532904255055D-06	b4	2.889674760882D-07	b5	-3.481066367425D-08
b6	2.617031891671D-09	b7	-1.122600731671D-10	b8	2.103691015112D-12



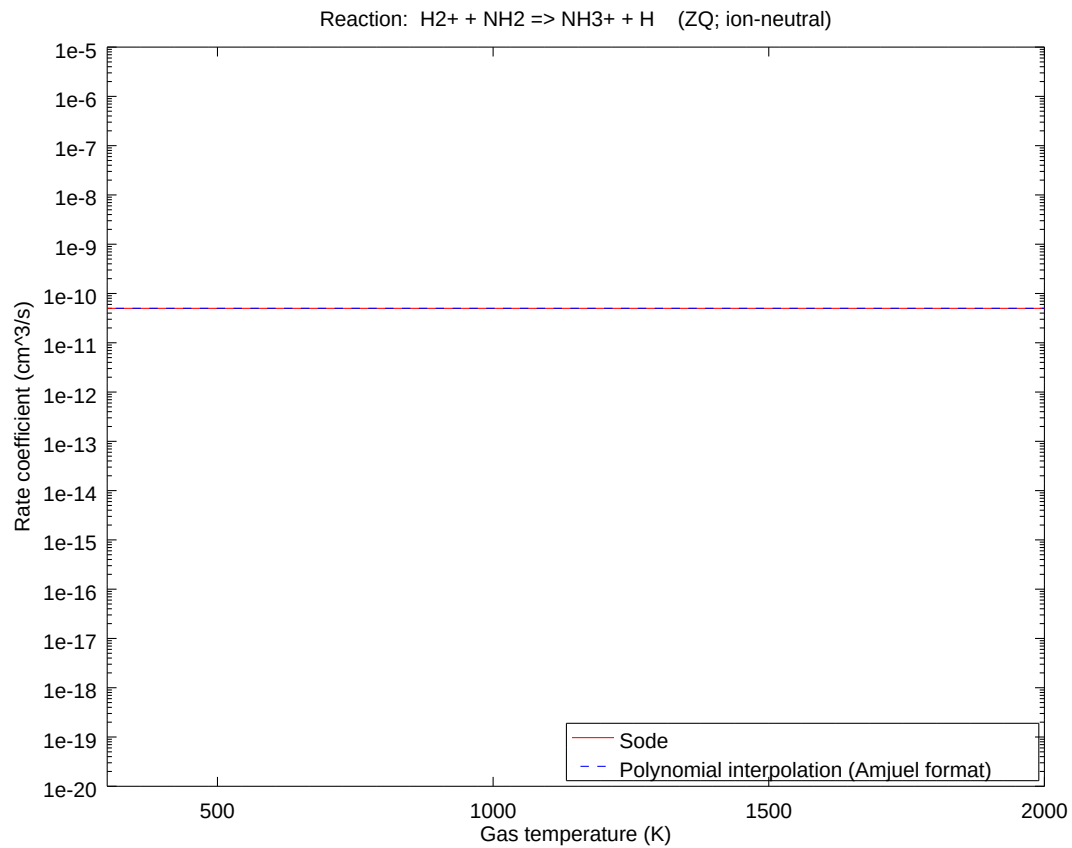
2.10 Reaction ZP $\text{H}_2^+ + \text{NH}_3 = \text{NH}_4^+ + \text{H}$

b0 -2.371899811251D+01	b1 -6.363672383446D-09	b2 -8.725644566069D-09
b3 -6.759829558708D-09	b4 -3.236074176671D-09	b5 -9.803478896226D-10
b6 -1.835703774470D-10	b7 -1.943046114932D-11	b8 -8.903776913581D-13



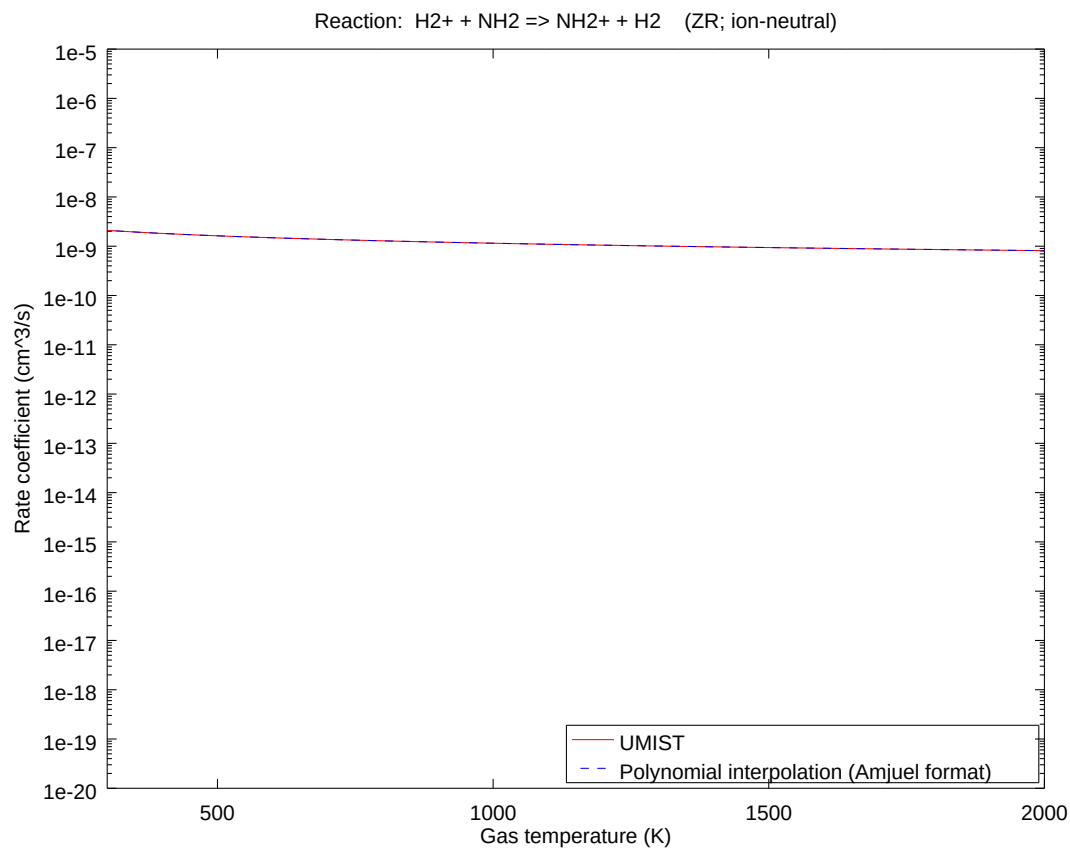
2.11 Reaction ZQ $\text{H}_2^+ + \text{NH}_2 = \text{NH}_3^+ + \text{H}$

b0	-2.371899811251D+01	b1	-6.363672383446D-09	b2	-8.725644566069D-09
b3	-6.759829558708D-09	b4	-3.236074176671D-09	b5	-9.803478896226D-10
b6	-1.835703774470D-10	b7	-1.943046114932D-11	b8	-8.903776913581D-13



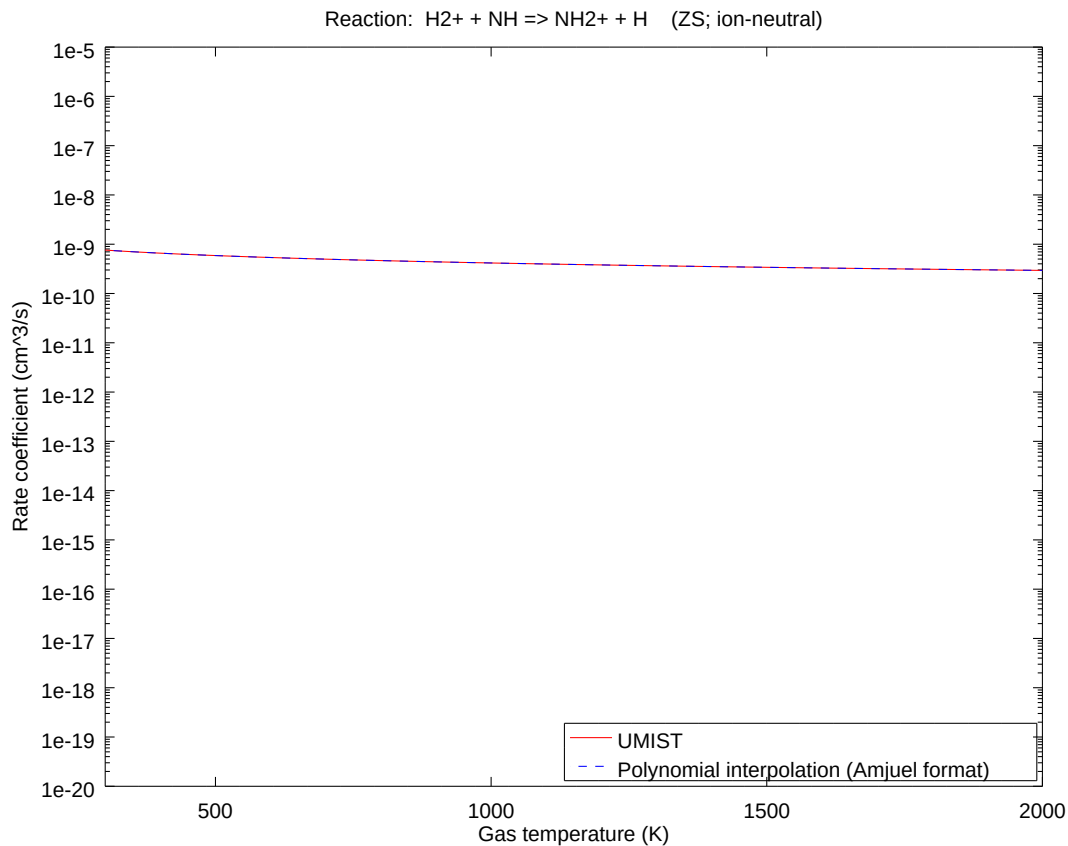
2.12 Reaction ZR $\text{H}_2^+ + \text{NH}_2 = \text{NH}_2^+ + \text{H}_2$

b0	-1.712943118603D+01	b1	-5.000073751283D-01	b2	3.915015979476D-06
b3	-1.185728681478D-06	b4	2.241022714450D-07	b5	-2.706574339404D-08
b6	2.039907770134D-09	b7	-8.772115525199D-11	b8	1.647867472525D-12



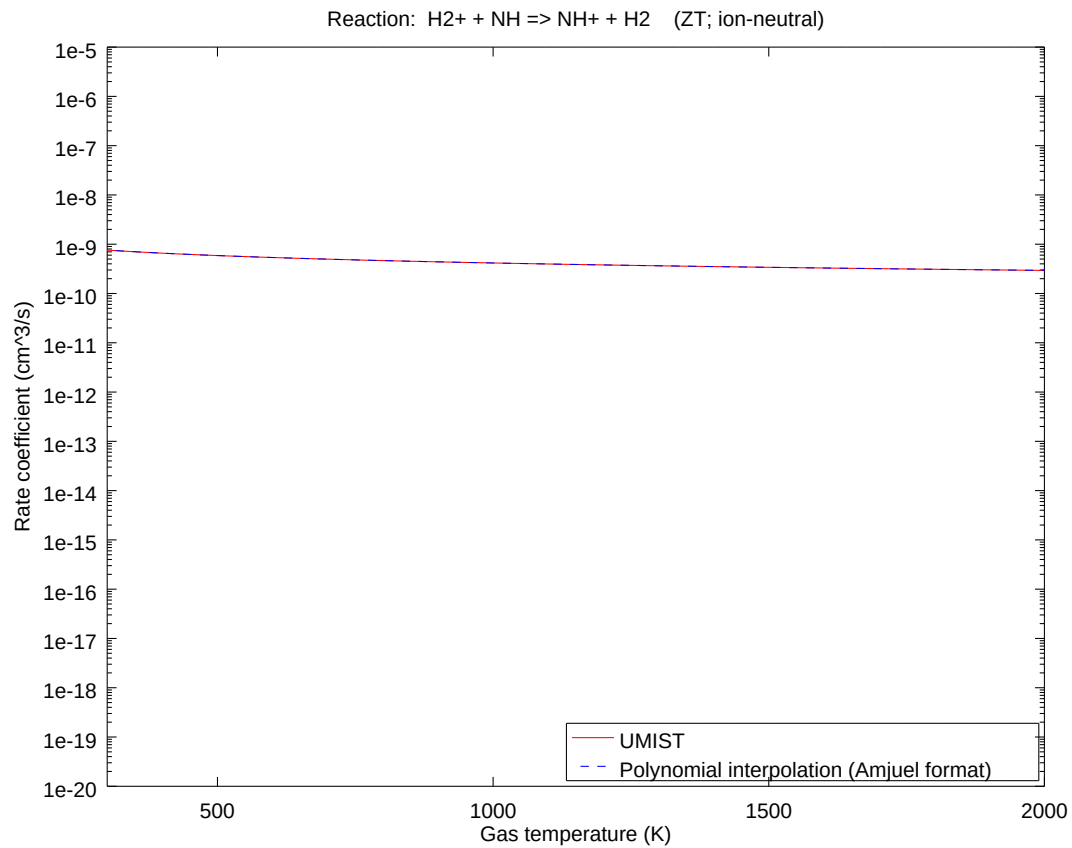
2.13 Reaction ZS $\text{H}_2^+ + \text{NH} = \text{NH}_2^+ + \text{H}$

b0	-1.814580427512D+01	b1	-5.000087064857D-01	b2	4.618164018360D-06
b3	-1.397643488903D-06	b4	2.639638767822D-07	b5	-3.185784069788D-08
b6	2.399468556864D-09	b7	-1.031159964675D-10	b8	1.935838480628D-12



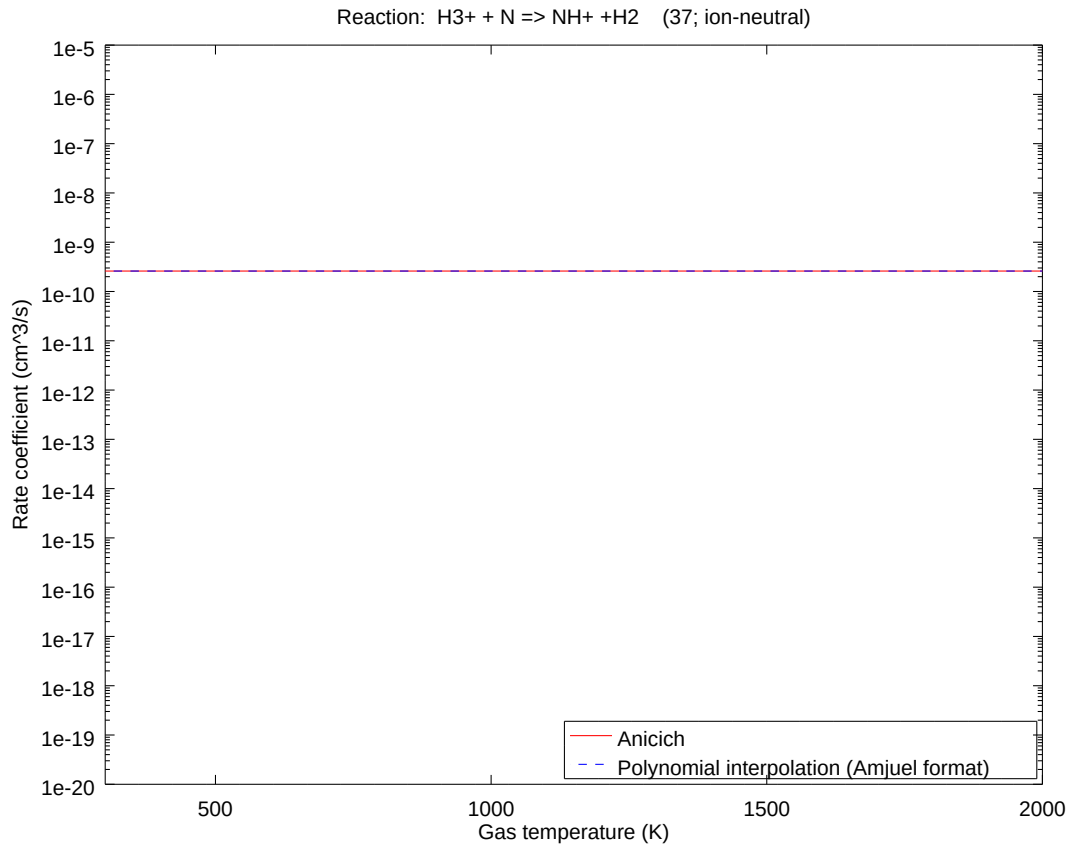
2.14 Reaction ZT $\text{H}_2^+ + \text{NH} = \text{NH}^+ + \text{H}_2$

b0	-1.814580427512D+01	b1	-5.000087064857D-01	b2	4.618164018360D-06
b3	-1.397643488903D-06	b4	2.639638767822D-07	b5	-3.185784069788D-08
b6	2.399468556864D-09	b7	-1.031159964675D-10	b8	1.935838480628D-12



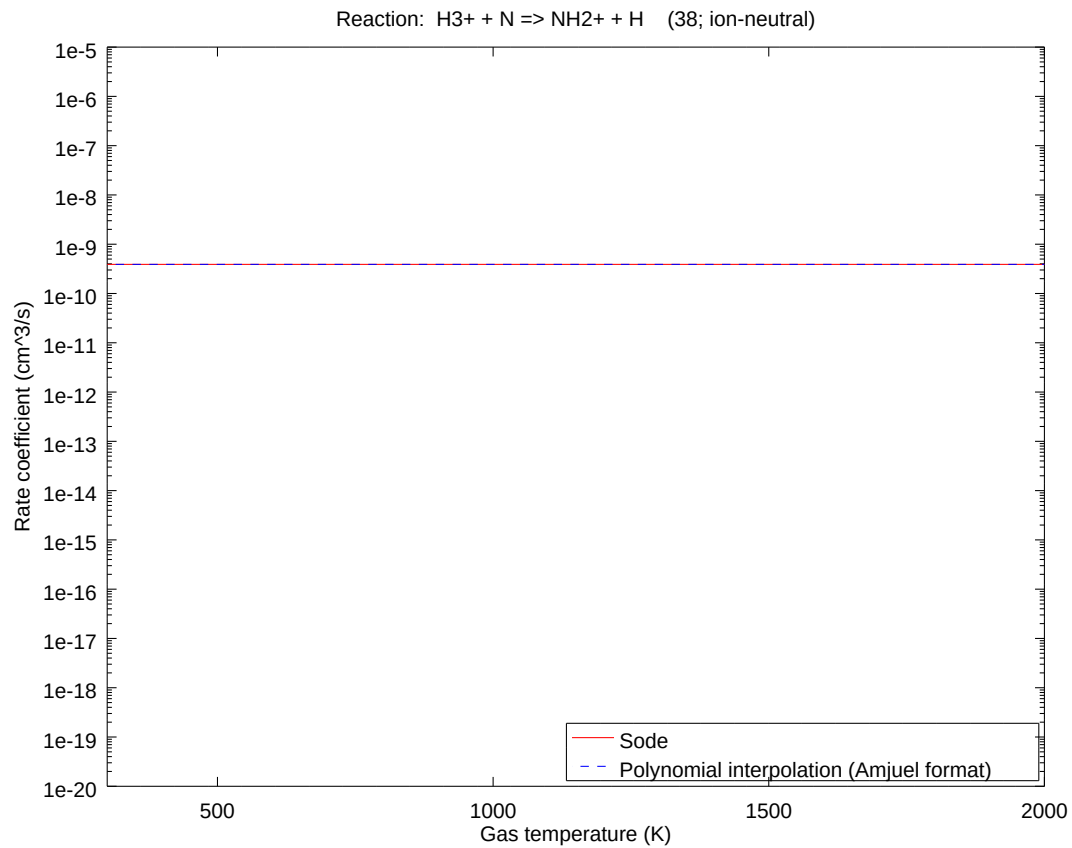
2.15 Reaction 37 $\text{H}_3^+ + \text{N} = \text{NH}^+ + \text{H}_2$

b0	-2.207033948612D+01	b1	-3.623757885018D-09	b2	-4.668986273792D-09
b3	-3.379118748538D-09	b4	-1.500903112887D-09	b5	-4.184475810622D-10
b6	-7.140512382280D-11	b7	-6.805668193736D-12	b8	-2.766550189438D-13



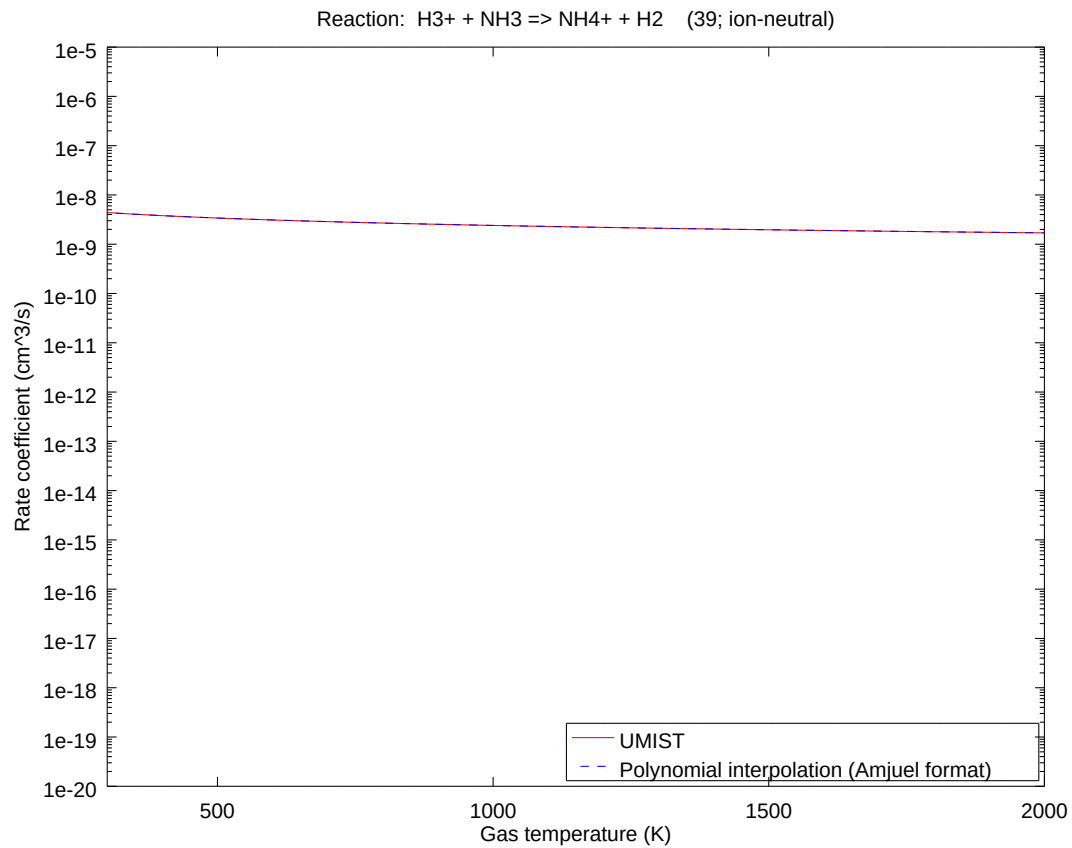
2.16 Reaction 38 $\text{H}_3^+ + \text{N} = \text{NH}_2^+ + \text{H}$

b0 -2.166487437743D+01	b1 -1.988647619827D-09	b2 -2.741337813120D-09
b3 -2.124900712044D-09	b4 -1.012364016074D-09	b5 -3.034255125527D-10
b6 -5.584840449730D-11	b7 -5.769232918336D-12	b8 -2.559637643636D-13



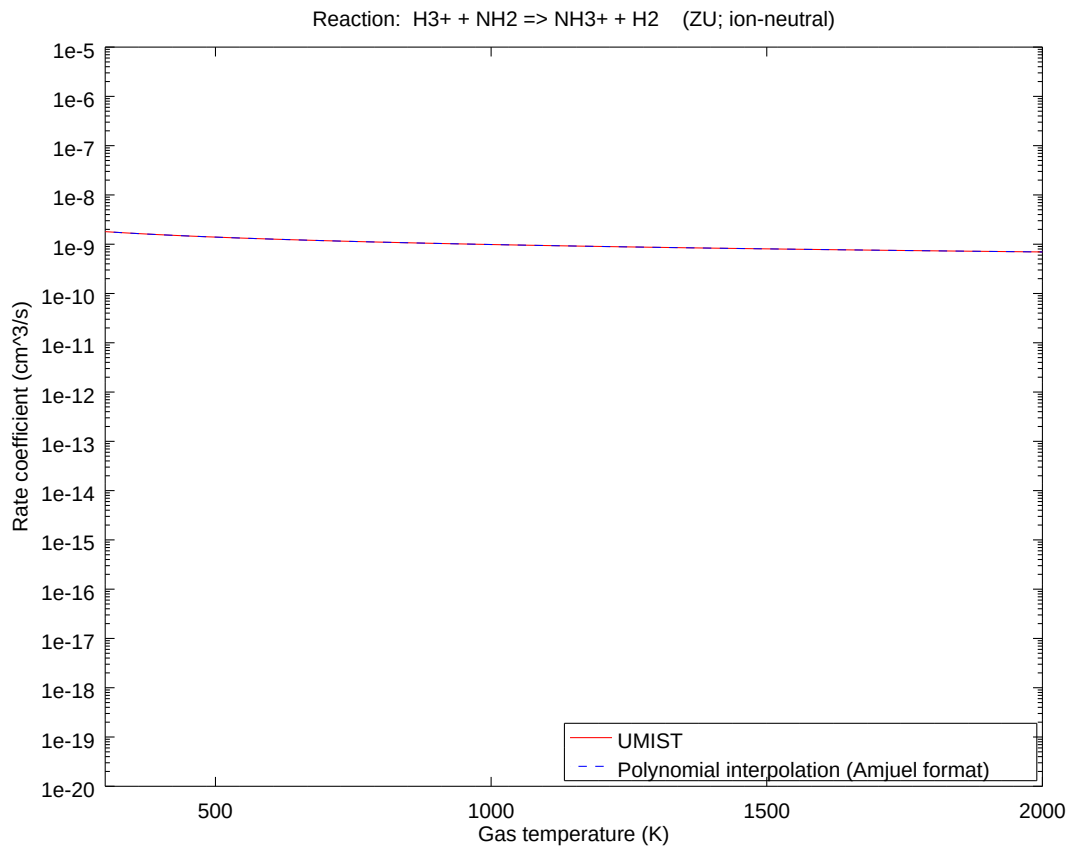
2.17 Reaction 39 $\text{H3}^+ + \text{NH3} = \text{NH4}^+ + \text{H2}$

b0	-1.639204044396D+01	b1	-5.000059513628D-01	b2	3.139138526953D-06
b3	-9.446899057460D-07	b4	1.774081872835D-07	b5	-2.128956082351D-08
b6	1.594306718881D-09	b7	-6.812002941964D-11	b8	1.271439302053D-12



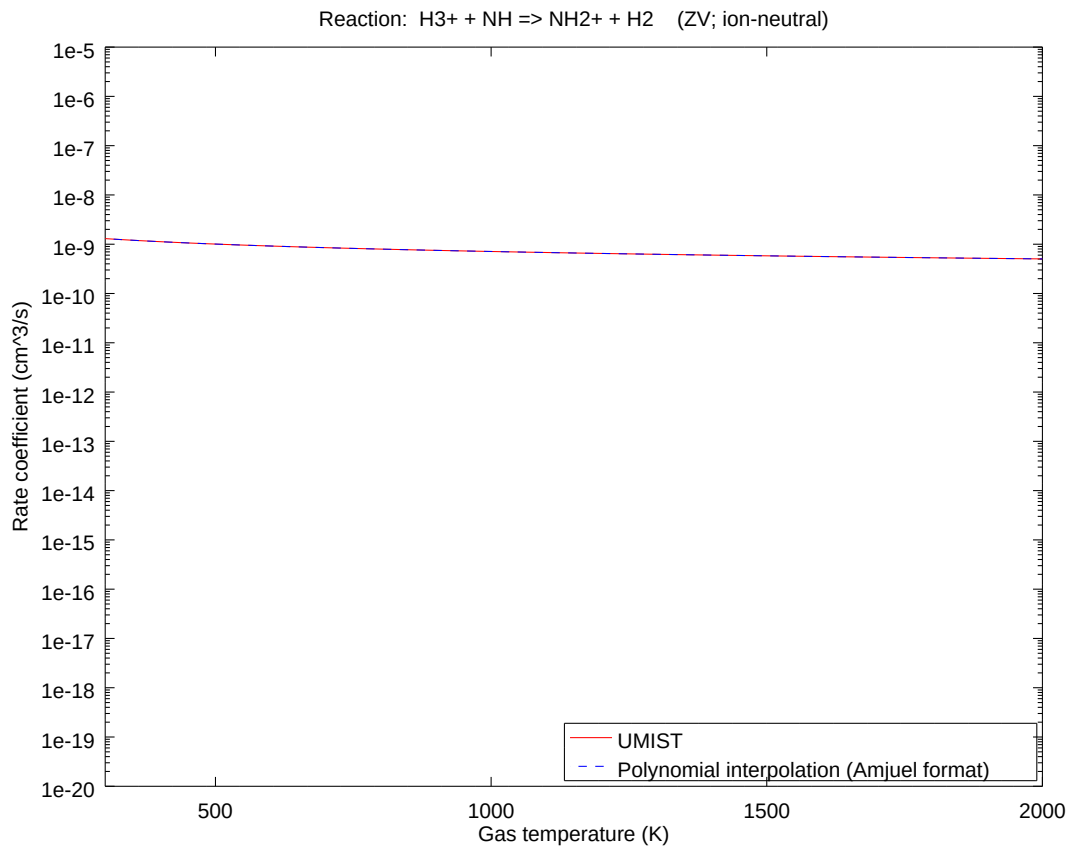
2.18 Reaction ZU $\text{H}_3^+ + \text{NH}_2 = \text{NH}_3^+ + \text{H}_2$

b0	-1.728358060535D+01	b1	-5.000088937553D-01	b2	4.714037353071D-06
b3	-1.425531498388D-06	b4	2.690034692978D-07	b5	-3.243693609527D-08
b6	2.440769381003D-09	b7	-1.047864198345D-10	b8	1.965148931997D-12



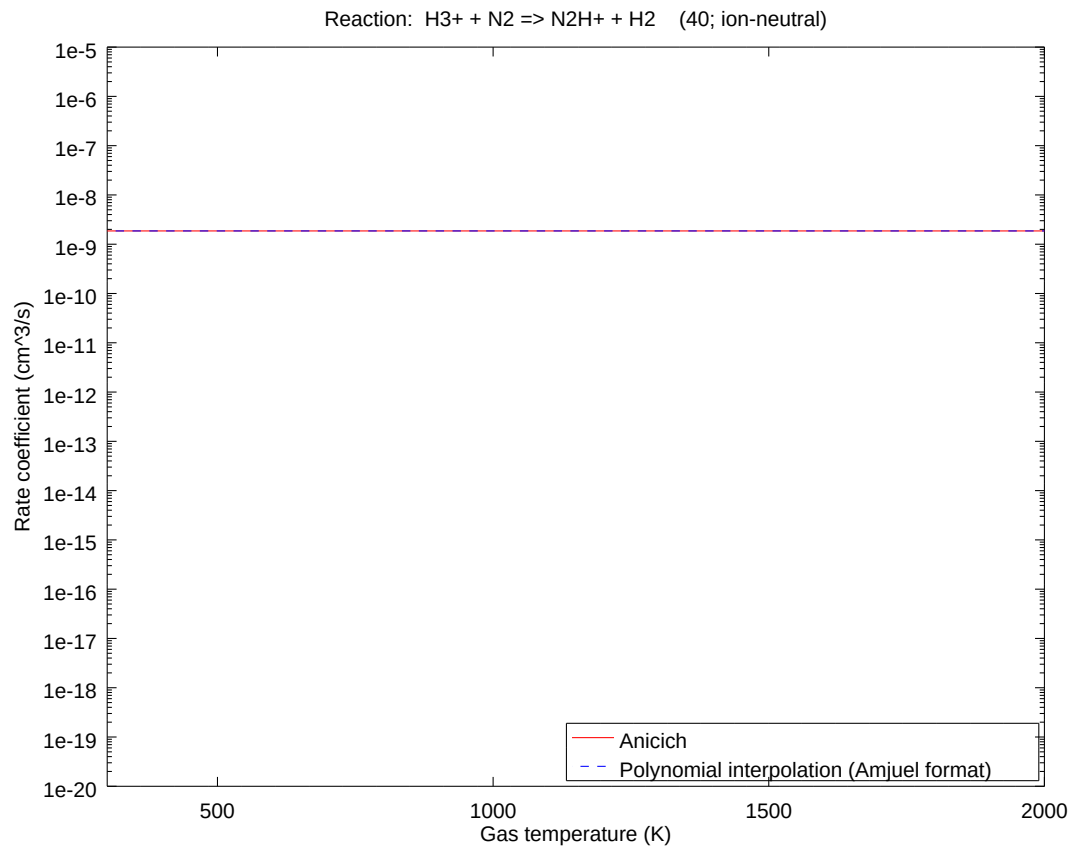
2.19 Reaction ZV $\text{H3}^+ + \text{NH} = \text{NH2}^+ + \text{H2}$

b0	-1.760900272475D+01	b1	-5.000092198705D-01	b2	4.879305705195D-06
b3	-1.473306590622D-06	b4	2.776200337504D-07	b5	-3.342981786956D-08
b6	2.512153628768D-09	b7	-1.077142229298D-10	b8	2.017598767023D-12



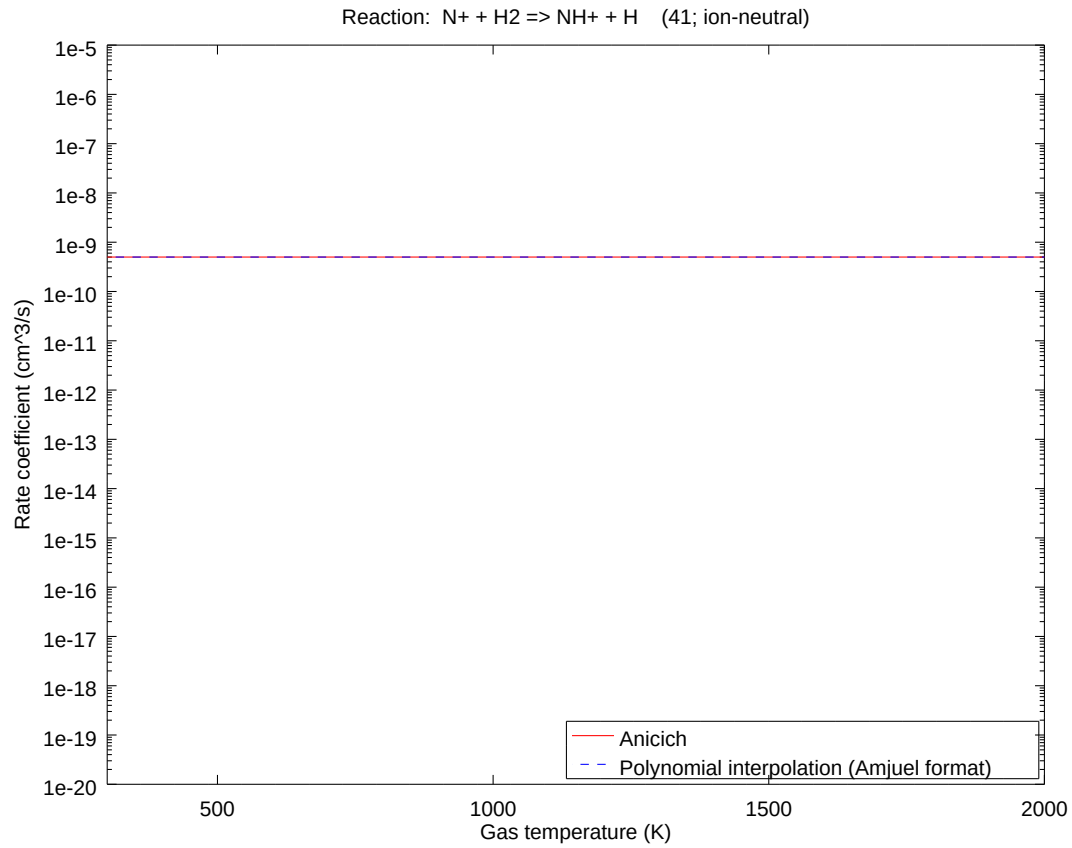
2.20 Reaction 40 $\text{H3}^+ + \text{N2} = \text{N2H}^+ + \text{H2}$

b0 -2.010268935009D+01	b1 -2.695722329098D-09	b2 -3.594896348717D-09
b3 -2.700687664952D-09	b4 -1.249225306535D-09	b5 -3.640588946150D-10
b6 -6.522819114944D-11	b7 -6.563569168921D-12	b8 -2.836978713469D-13



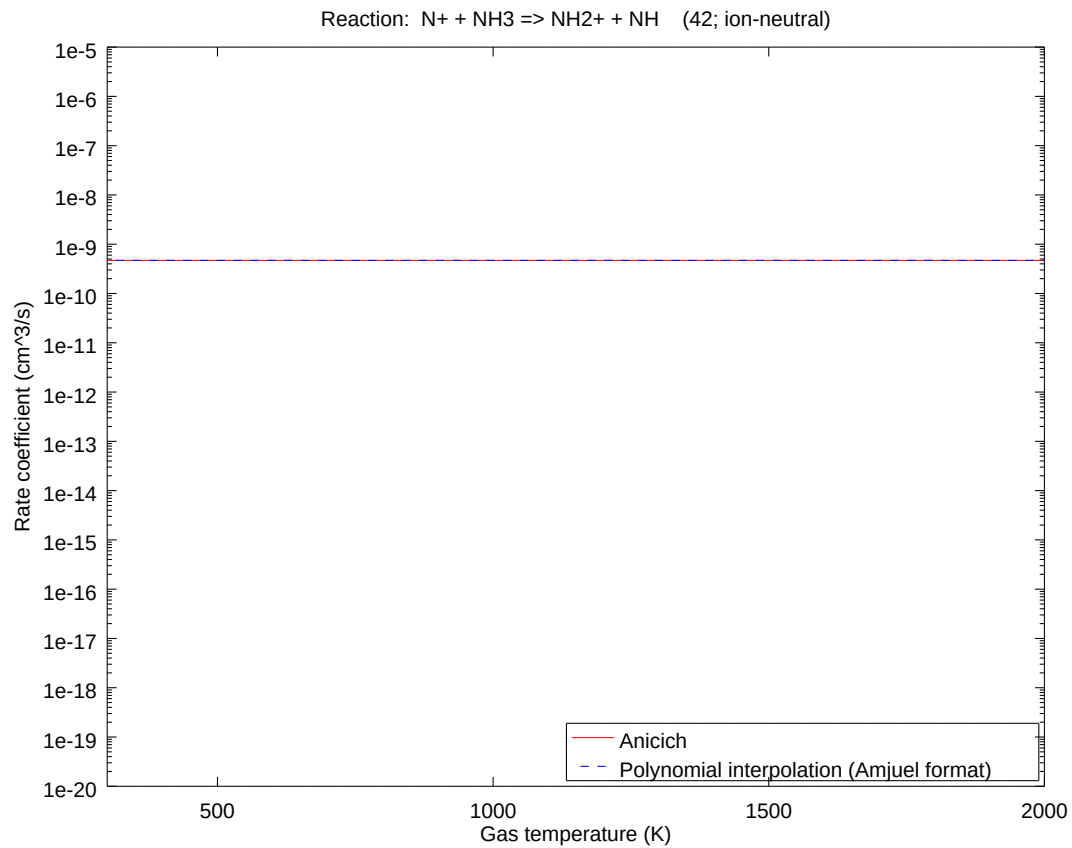
2.21 Reaction 41 $\text{N}^+ + \text{H}_2 = \text{NH}^+ + \text{H}$

b0	-2.141641302012D+01	b1	-8.131359156625D-09	b2	-1.095575028514D-08
b3	-8.351628081406D-09	b4	-3.940087662388D-09	b5	-1.178197239087D-09
b6	-2.181281550989D-10	b7	-2.286666084442D-11	b8	-1.039602156424D-12



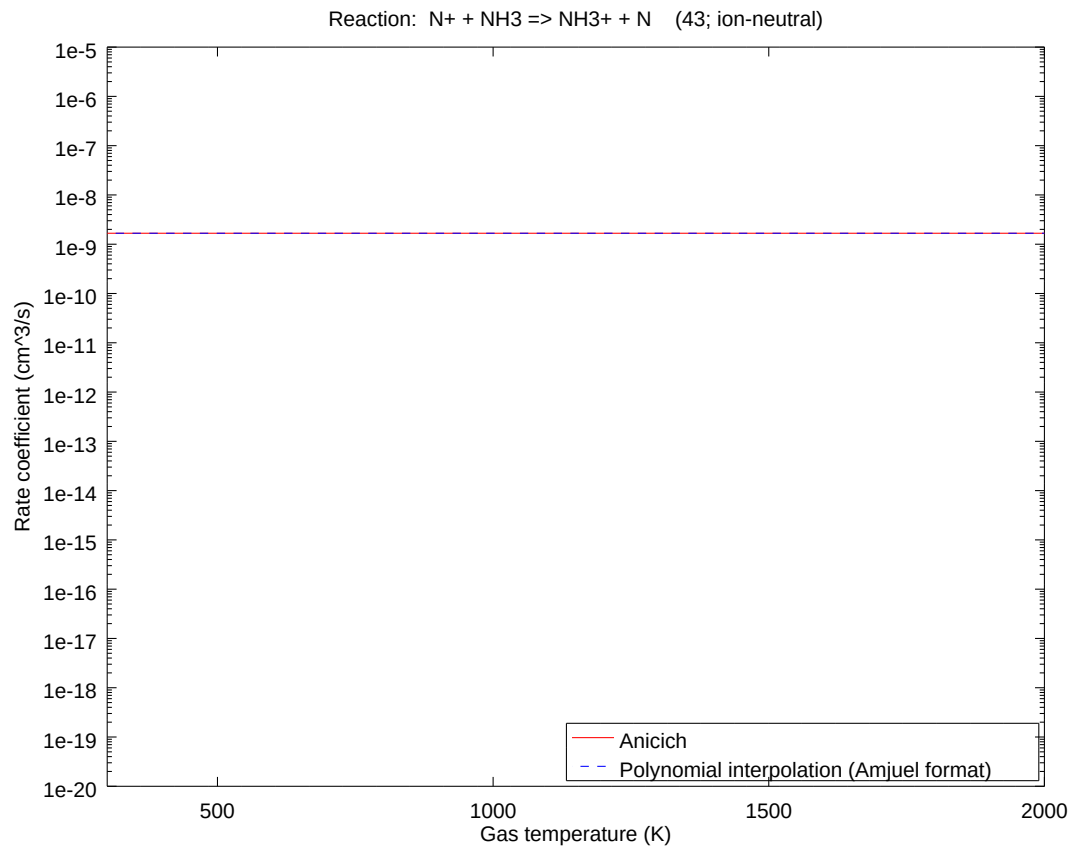
2.22 Reaction 42 $\text{N}^+ + \text{NH}_3 = \text{NH}_2^+ + \text{NH}$

b0	-2.147828842200D+01	b1	-2.430569313122D-09	b2	-3.283908649853D-09
b3	-2.506457922747D-09	b4	-1.181624038523D-09	b5	-3.522528278959D-10
b6	-6.484145197853D-11	b7	-6.738684678419D-12	b8	-3.027606809019D-13



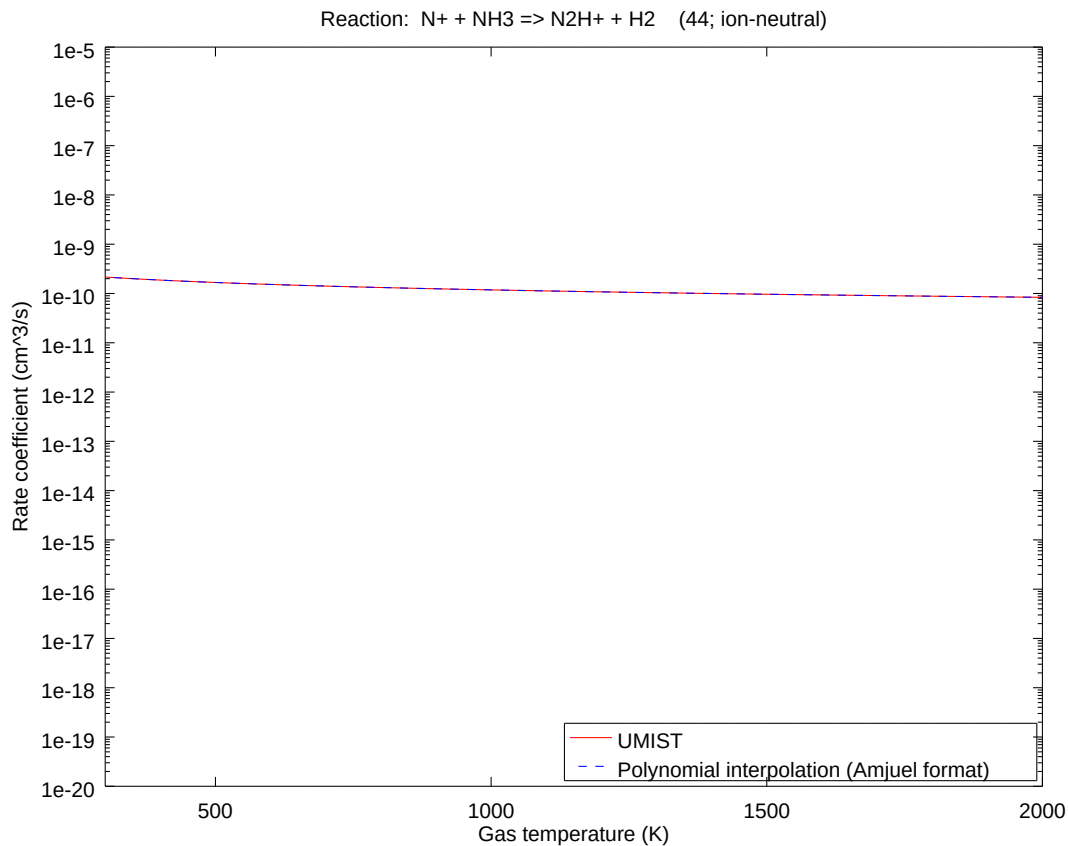
2.23 Reaction 43 $\text{N}^+ + \text{NH}_3 = \text{NH}_3^+ + \text{N}$

b0 -2.021044221088D+01	b1 -1.060612063908D-09	b2 -1.337145314168D-09
b3 -9.351499390680D-10	b4 -3.941653592280D-10	b5 -1.014622139491D-10
b6 -1.528981482726D-11	b7 -1.186867806159D-12	b8 -3.275609035177D-14



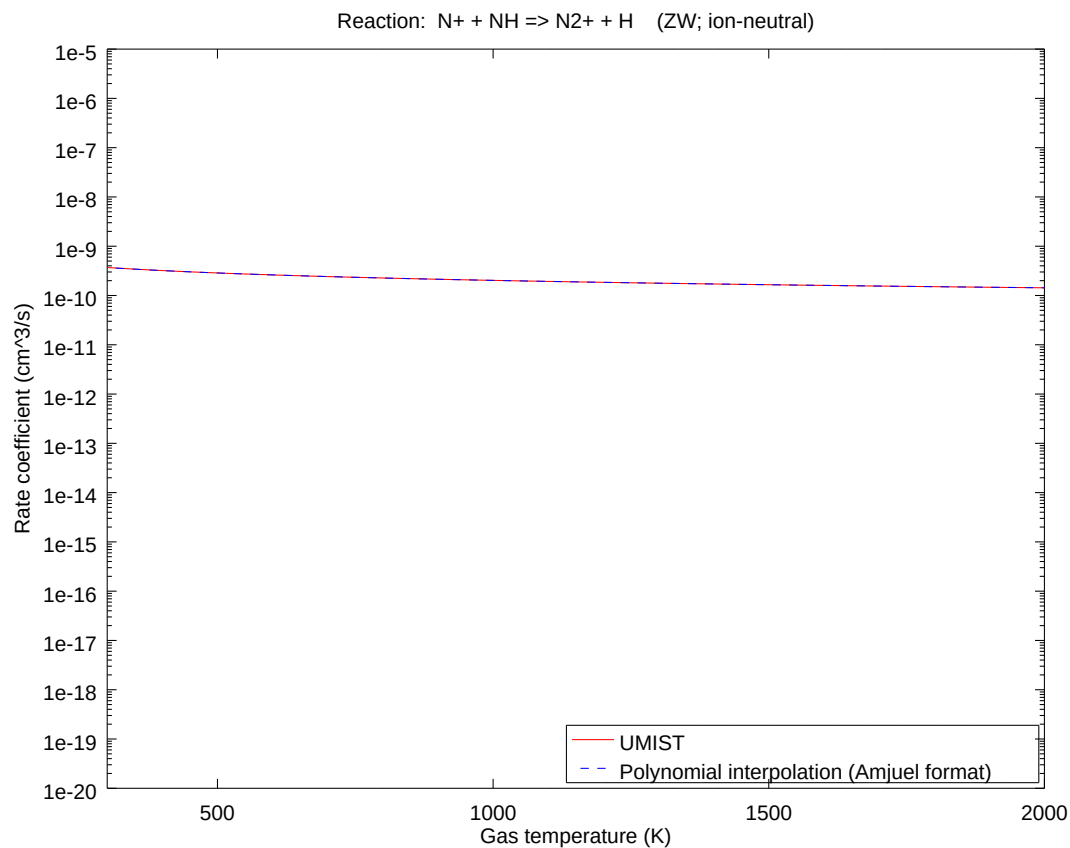
2.24 Reaction 44 $\text{N}^+ + \text{NH}_3 = \text{N}_2\text{H}^+ + \text{H}_2$

b0	-1.940384516491D+01	b1	-5.000076572078D-01	b2	4.061769492417D-06
b3	-1.229348581878D-06	b4	2.322044984655D-07	b5	-2.802883522007D-08
b6	2.111454182423D-09	b7	-9.075821457031D-11	b8	1.704267576556D-12



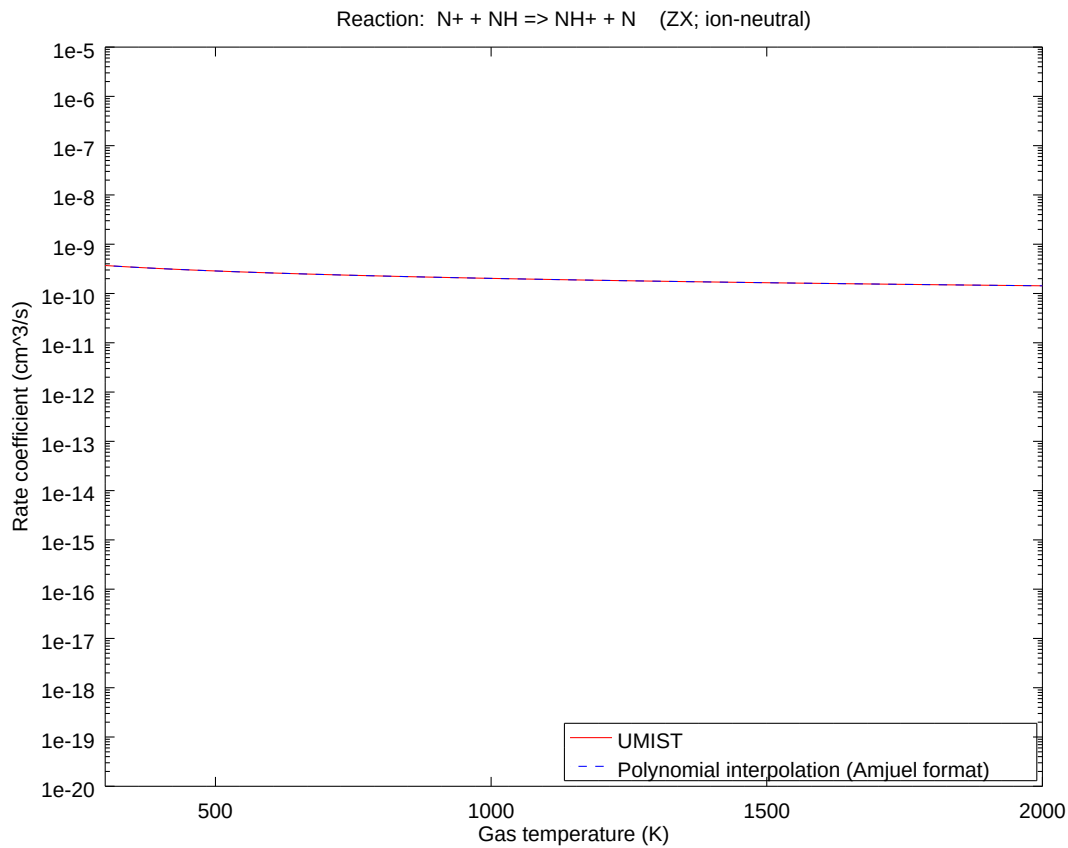
2.25 Reaction ZW $\text{N}^+ + \text{NH} = \text{N}_2^+ + \text{H}$

b0	-1.886562001770D+01	b1	-5.000083254610D-01	b2	4.416728021362D-06
b3	-1.336868267588D-06	b4	2.525191120122D-07	b5	-3.048048278980D-08
b6	2.296021843252D-09	b7	-9.868327746585D-11	b8	1.852873176705D-12



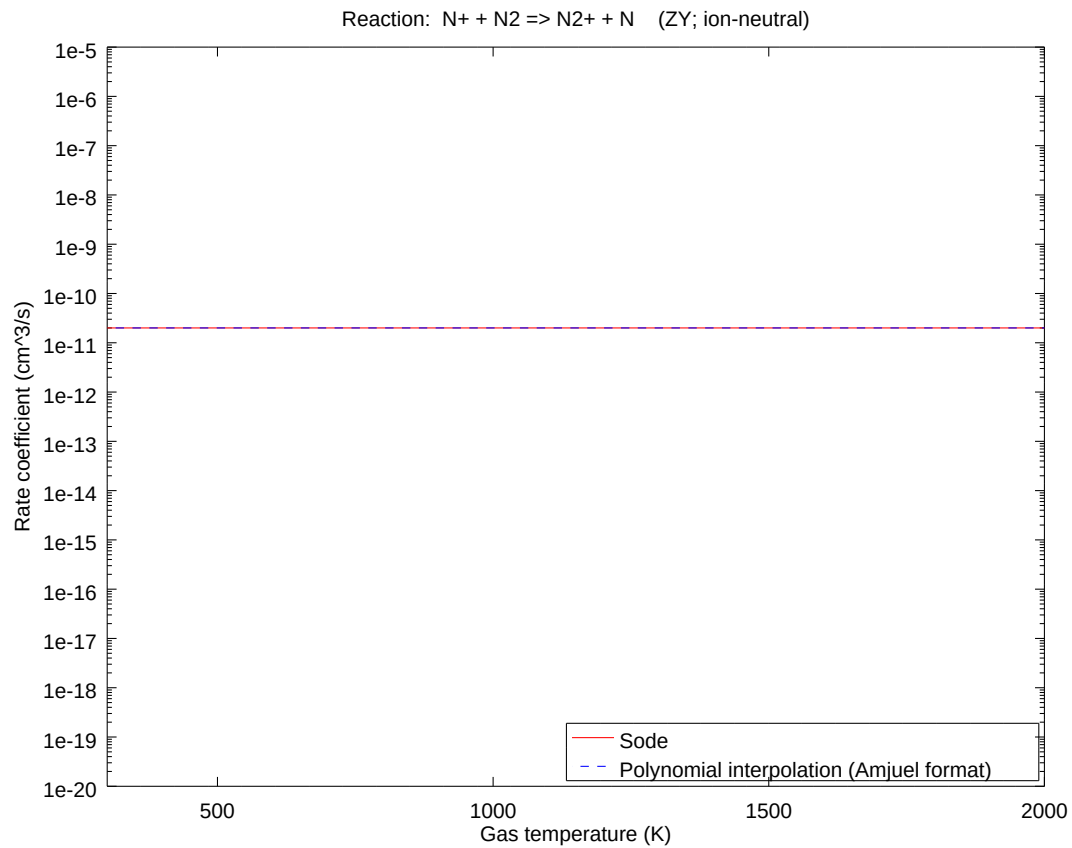
2.26 Reaction ZX $\text{N}^+ + \text{NH} = \text{NH}^+ + \text{N}$

b0	-1.886562001770D+01	b1	-5.000083254610D-01	b2	4.416728021362D-06
b3	-1.336868267588D-06	b4	2.525191120122D-07	b5	-3.048048278980D-08
b6	2.296021843252D-09	b7	-9.868327746585D-11	b8	1.852873176705D-12



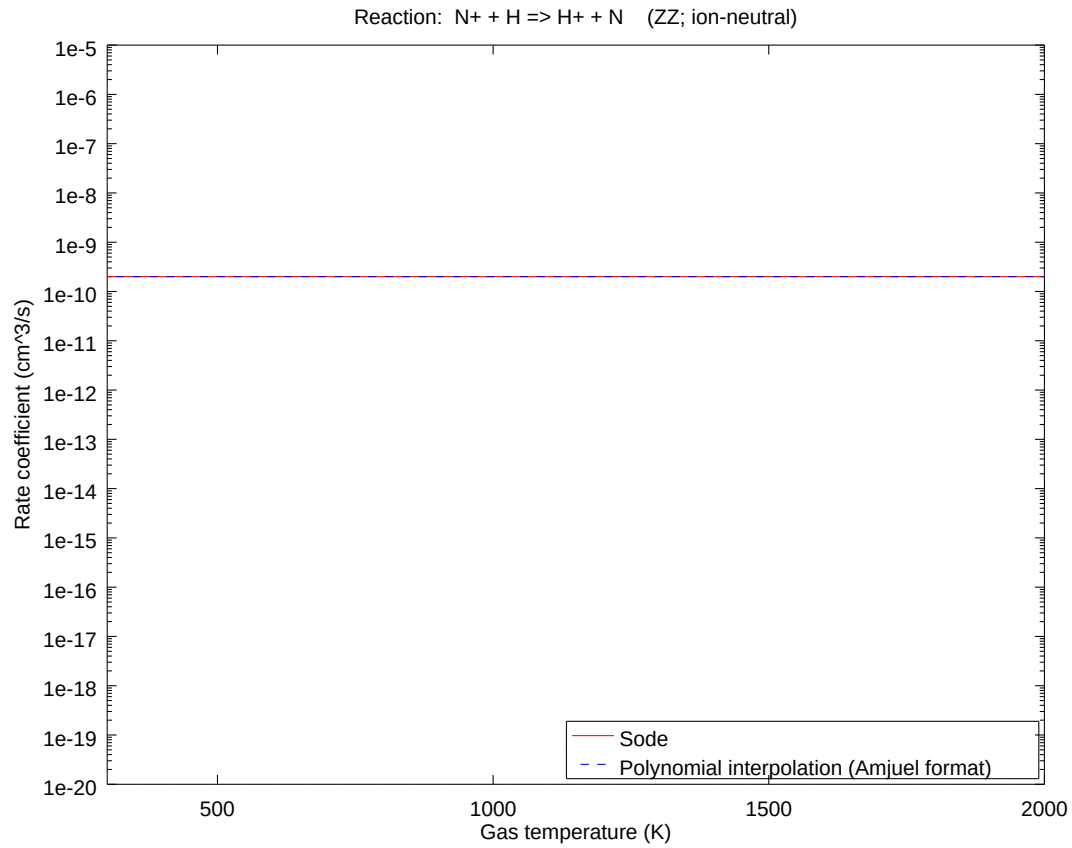
2.27 Reaction ZY $\text{N}^+ + \text{N}_2 = \text{N}_2^+ + \text{N}$

b0 -2.463528884331D+01	b1 -2.872491006416D-09	b2 -3.785933285449D-09
b3 -2.806324618315D-09	b4 -1.277897224942D-09	b5 -3.654923868938D-10
b6 -6.400079359295D-11	b7 -6.258974726443D-12	b8 -2.609437550917D-13



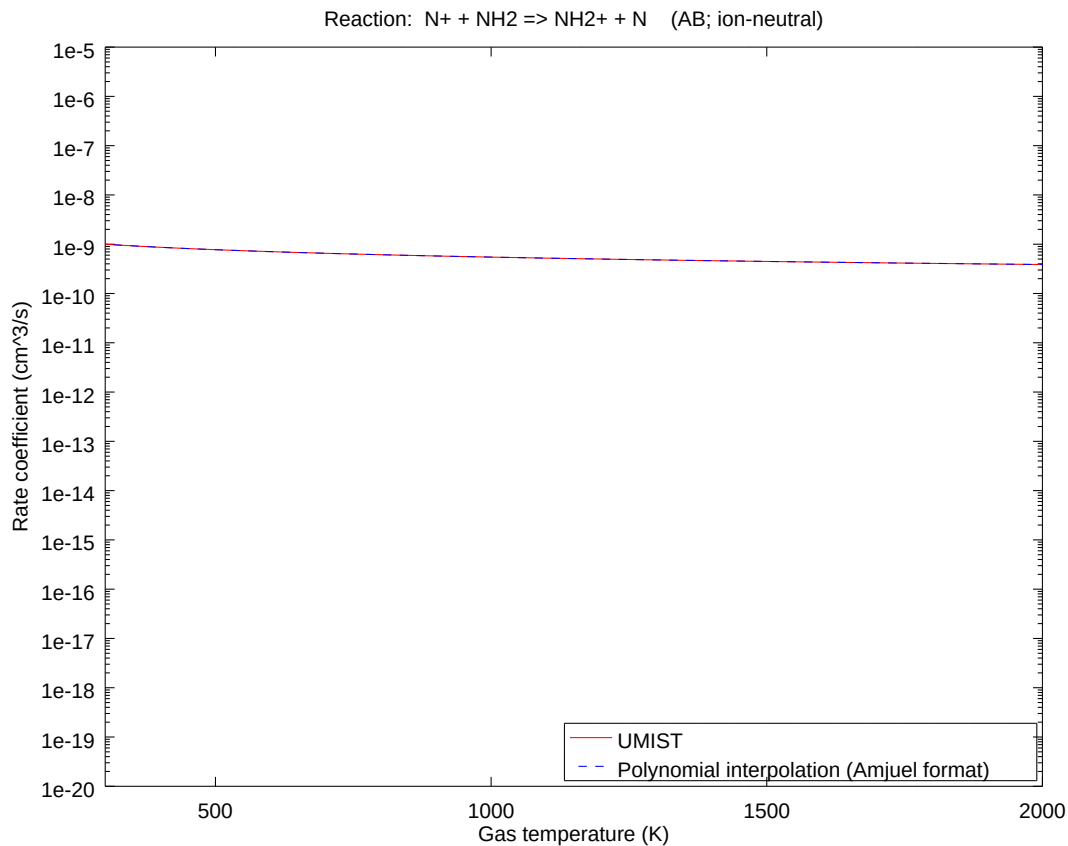
2.28 Reaction ZZ N+ + H = H+ + N

b0 -2.233270375134D+01	b1 -6.142711536798D-09	b2 -8.333989980676D-09
b3 -6.392983540575D-09	b4 -3.032443889312D-09	b5 -9.108067599946D-10
b6 -1.691793226648D-10	b7 -1.777090335237D-11	b8 -8.083911312511D-13



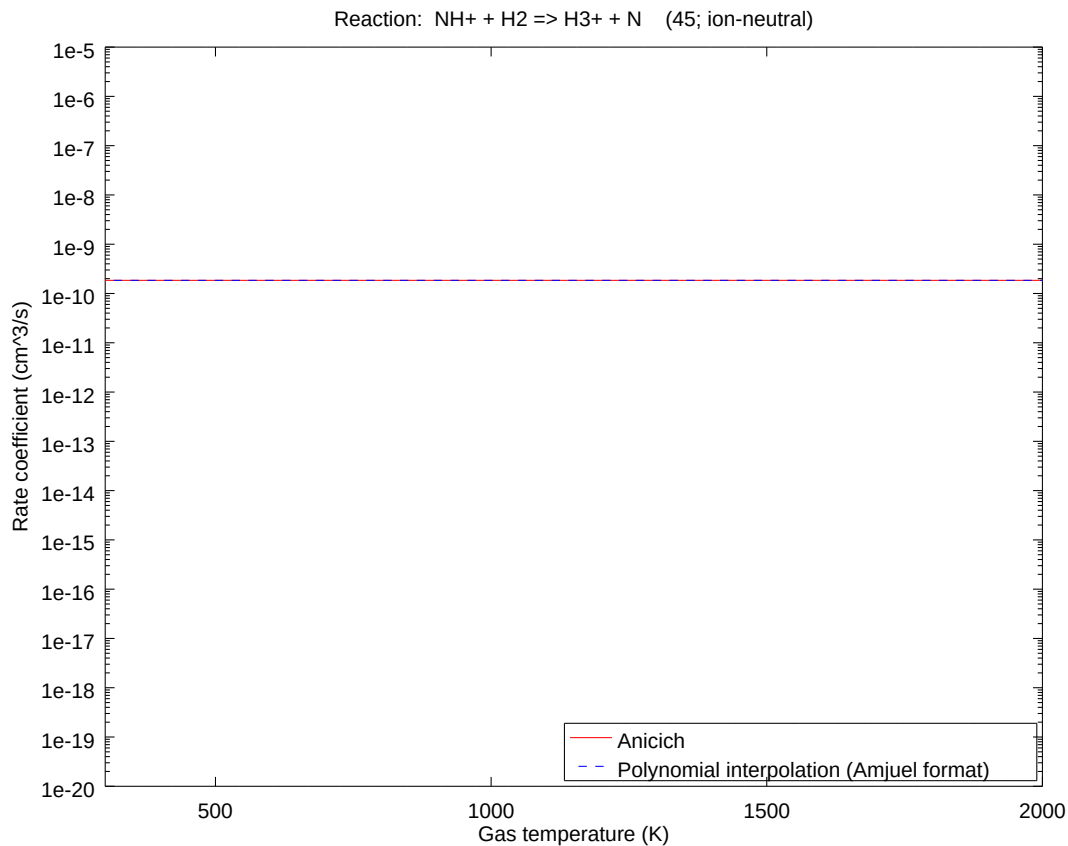
2.29 Reaction AB $\text{N}^+ + \text{NH}_2 = \text{NH}_2^+ + \text{N}$

b0	-1.787136826730D+01	b1	-5.000076950241D-01	b2	4.084561632382D-06
b3	-1.236959245409D-06	b4	2.337546976851D-07	b5	-2.822689445576D-08
b6	2.126996754399D-09	b7	-9.144435480296D-11	b8	1.717327548274D-12



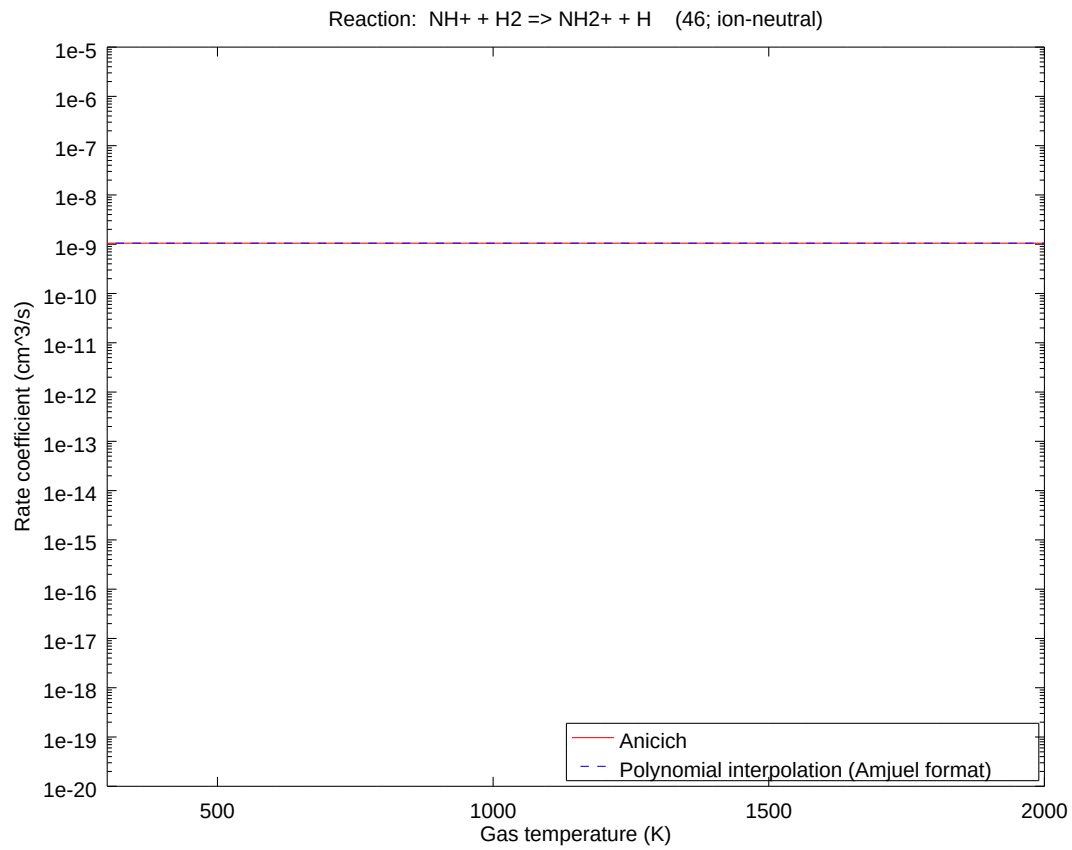
2.30 Reaction 45 $\text{NH}^+ + \text{H}_2 = \text{H}_3^+ + \text{N}$

b0	-2.241066529185D+01	b1	-3.049259683734D-09	b2	-4.012682428600D-09
b3	-2.974461675562D-09	b4	-1.357342594500D-09	b5	-3.901757428464D-10
b6	-6.894669735610D-11	b7	-6.842366892766D-12	b8	-2.917263387613D-13



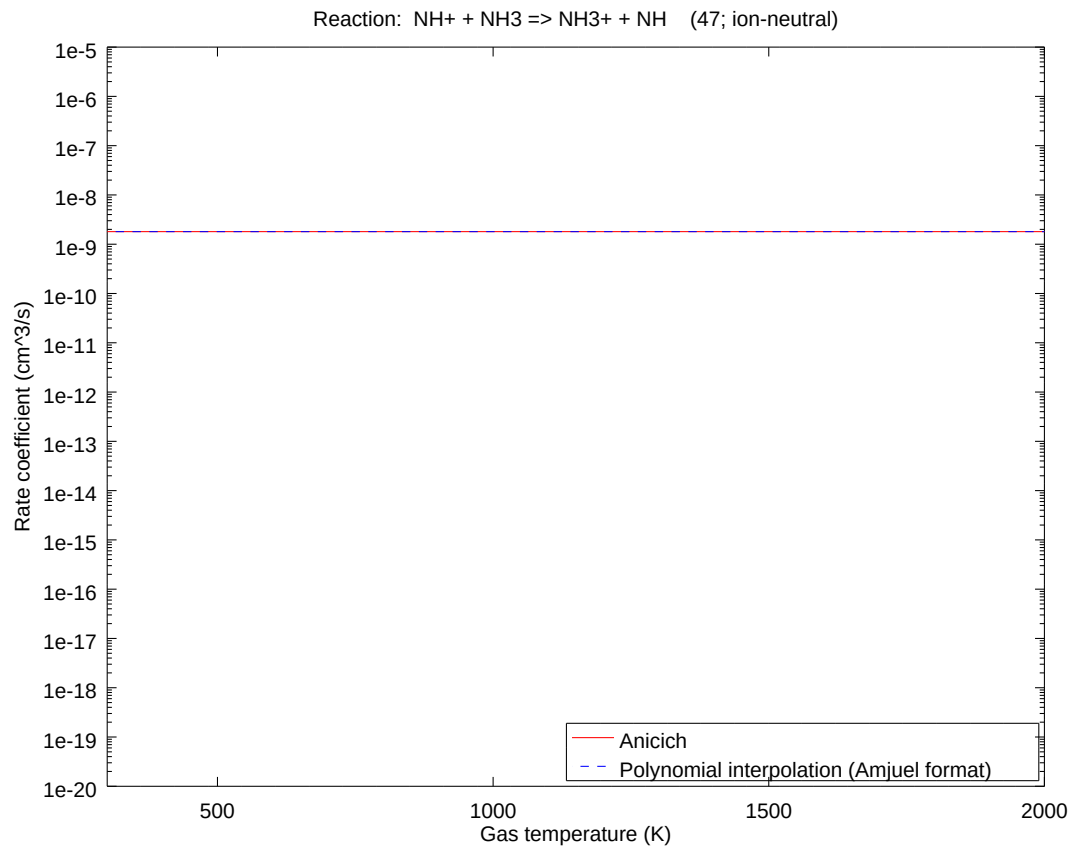
2.31 Reaction 46 $\text{NH}^+ + \text{H}_2 = \text{NH}_2^+ + \text{H}$

b0	-2.067447567583D+01	b1	-9.633892913827D-09	b2	-1.317039007359D-08
b3	-1.018519707762D-08	b4	-4.873207235255D-09	b5	-1.477236209958D-09
b6	-2.770904454089D-10	b7	-2.940953542269D-11	b8	-1.352598756840D-12



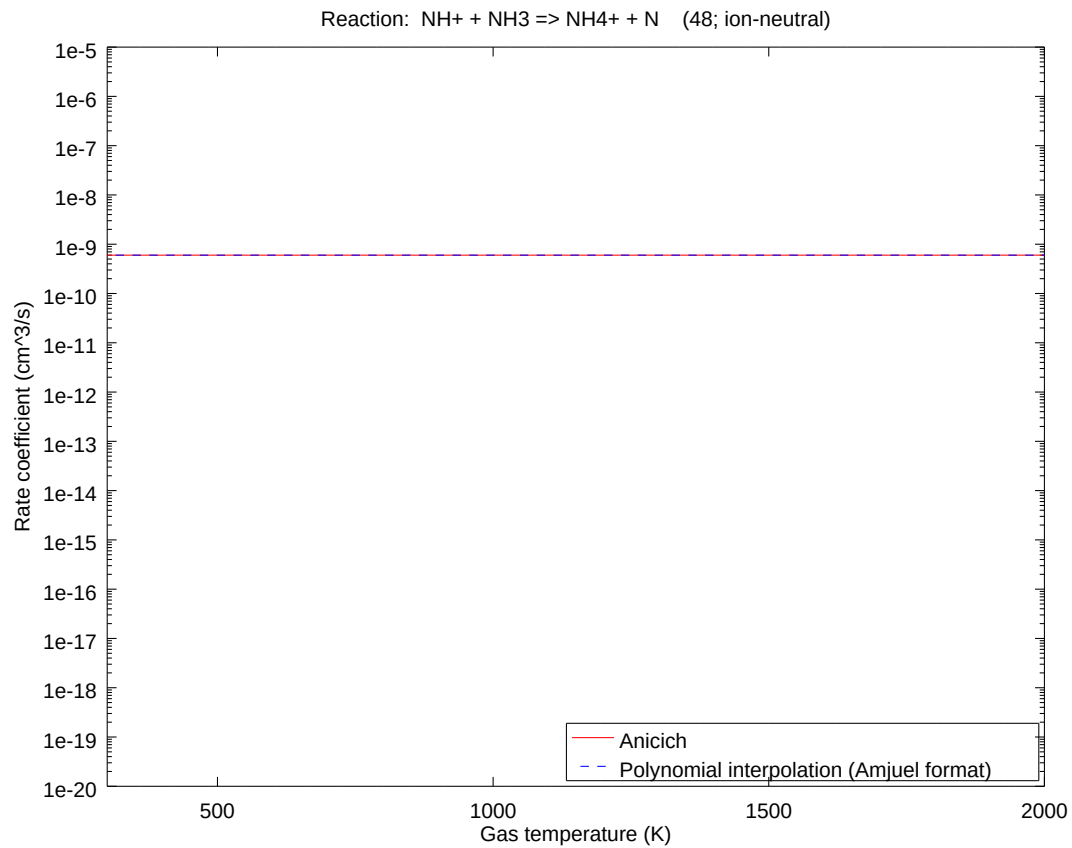
2.32 Reaction 47 $\text{NH}^+ + \text{NH}_3 = \text{NH}_3^+ + \text{NH}$

b0 -2.013547917424D+01	b1 -6.805594076740D-09	b2 -9.136011450841D-09
b3 -6.926777430142D-09	b4 -3.243493823577D-09	b5 -9.603967624261D-10
b6 -1.756031873909D-10	b7 -1.812846346319D-11	b8 -8.090929294731D-13



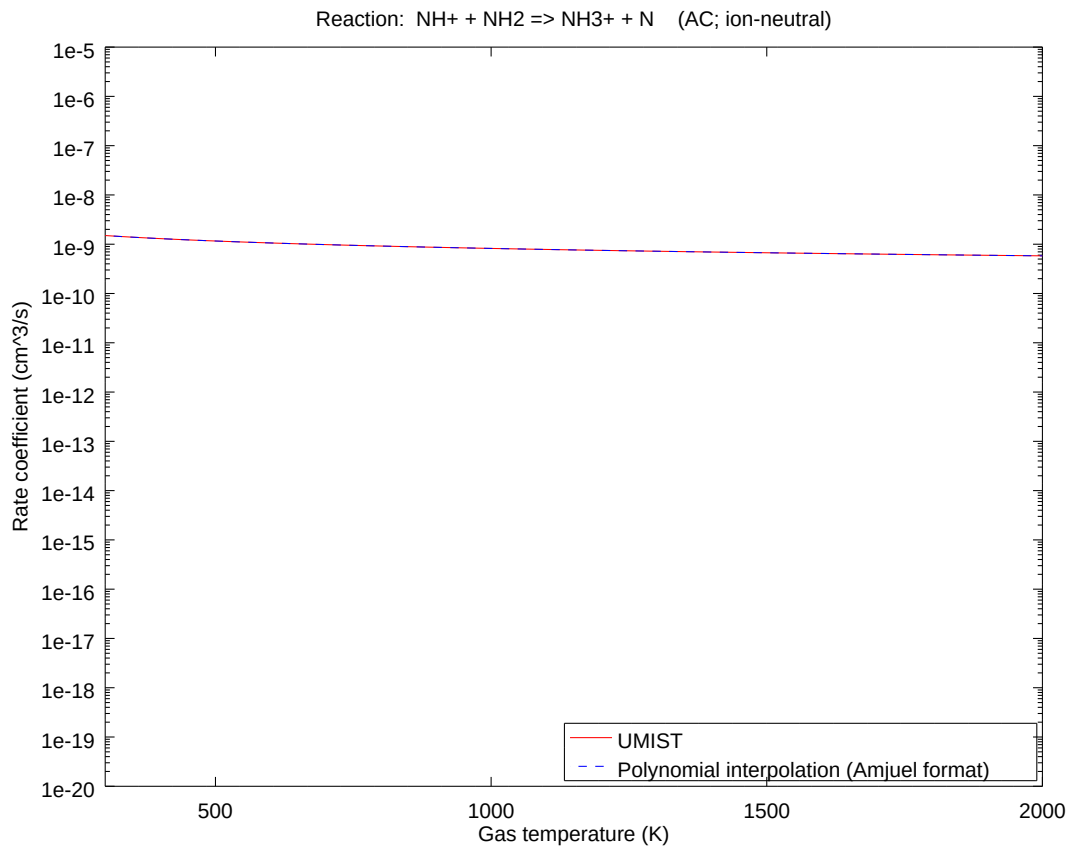
2.33 Reaction 48 $\text{NH}^+ + \text{NH}_3 = \text{NH}_4^+ + \text{N}$

b0	-2.123409146344D+01	b1	-8.573280849920D-09	b2	-1.166198676430D-08
b3	-8.971779203459D-09	b4	-4.269601390031D-09	b5	-1.287181187371D-09
b6	-2.401105548274D-10	b7	-2.534483430697D-11	b8	-1.159366359978D-12



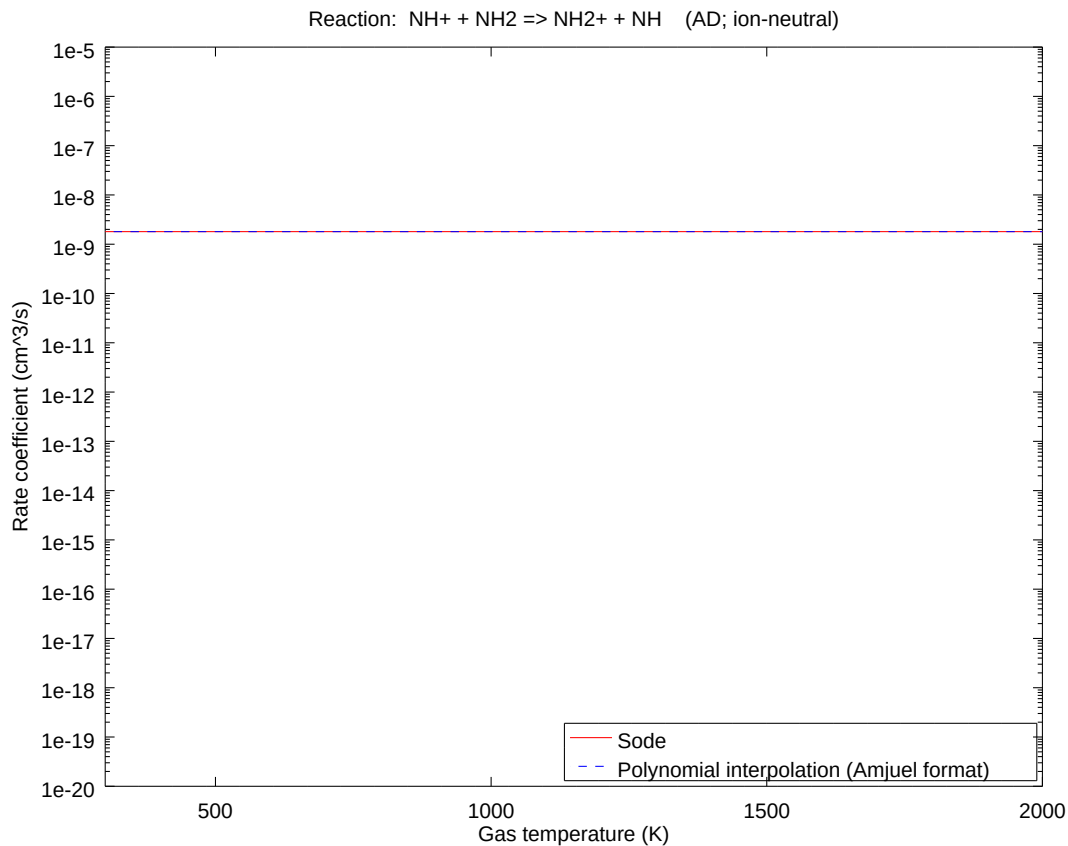
2.34 Reaction AC $\text{NH}^+ + \text{NH}_2 = \text{NH}_3^+ + \text{N}$

b0	-1.746590198948D+01	b1	-5.000091270135D-01	b2	4.850763255149D-06
b3	-1.470984691195D-06	b4	2.783830078130D-07	b5	-3.366788173866D-08
b6	2.541146799230D-09	b7	-1.094382601994D-10	b8	2.058984583179D-12



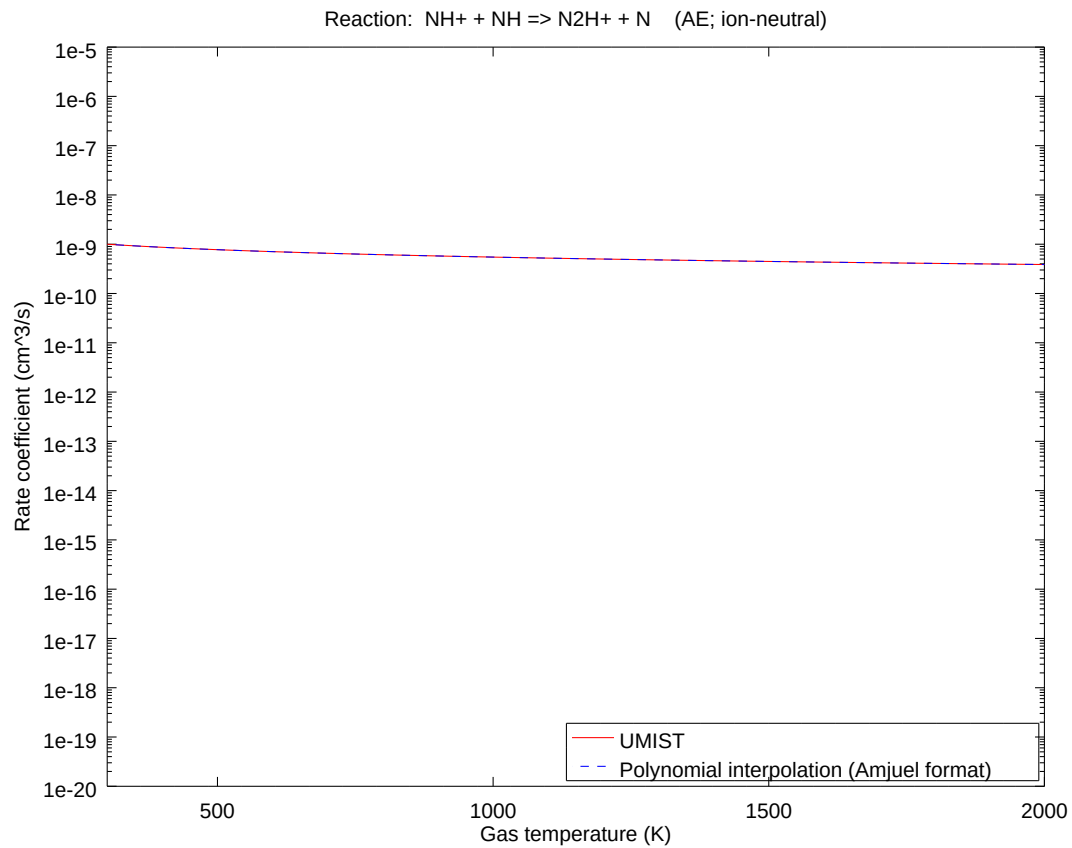
2.35 Reaction AD $\text{NH}^+ + \text{NH}_2 = \text{NH}_2^+ + \text{NH}$

b0	-2.013547917424D+01	b1	-6.805594076740D-09	b2	-9.136011450841D-09
b3	-6.926777430142D-09	b4	-3.243493823577D-09	b5	-9.603967624261D-10
b6	-1.756031873909D-10	b7	-1.812846346319D-11	b8	-8.090929294731D-13



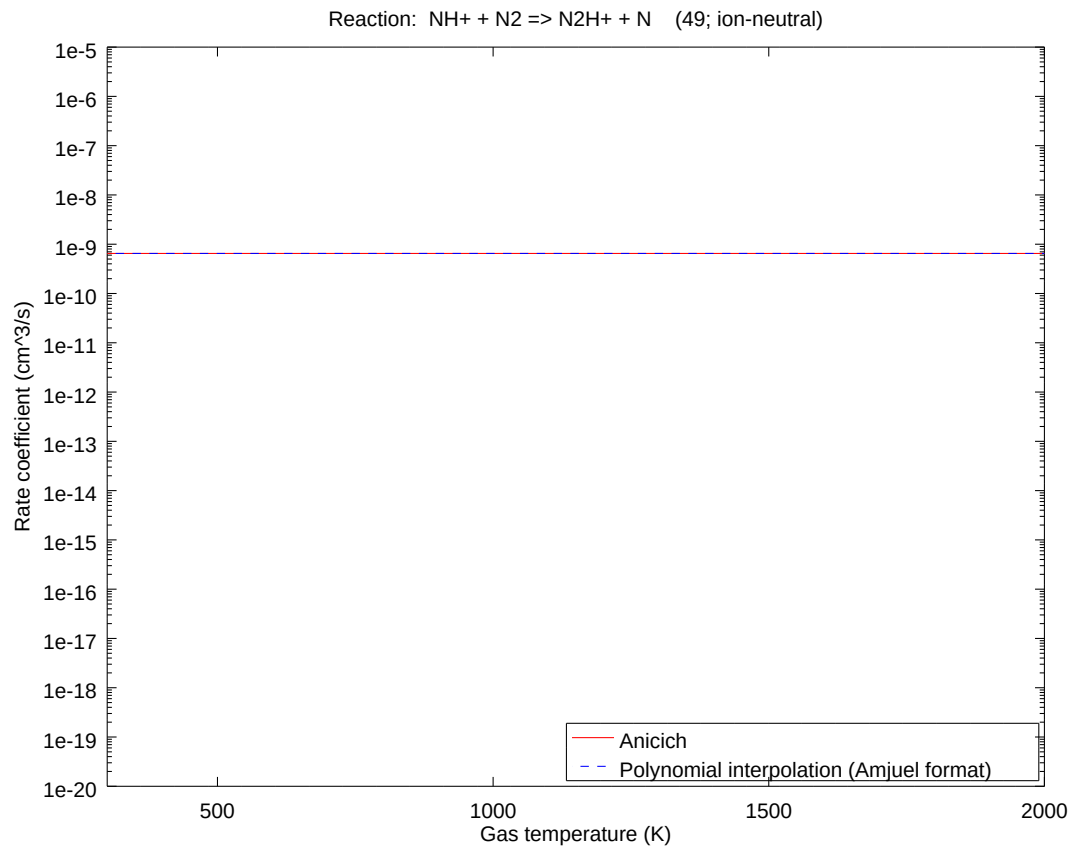
2.36 Reaction AE $\text{NH}^+ + \text{NH} = \text{N}_2\text{H}^+ + \text{N}$

b0	-1.787136826730D+01	b1	-5.000076950241D-01	b2	4.084561632382D-06
b3	-1.236959245409D-06	b4	2.337546976851D-07	b5	-2.822689445576D-08
b6	2.126996754399D-09	b7	-9.144435480296D-11	b8	1.717327548274D-12



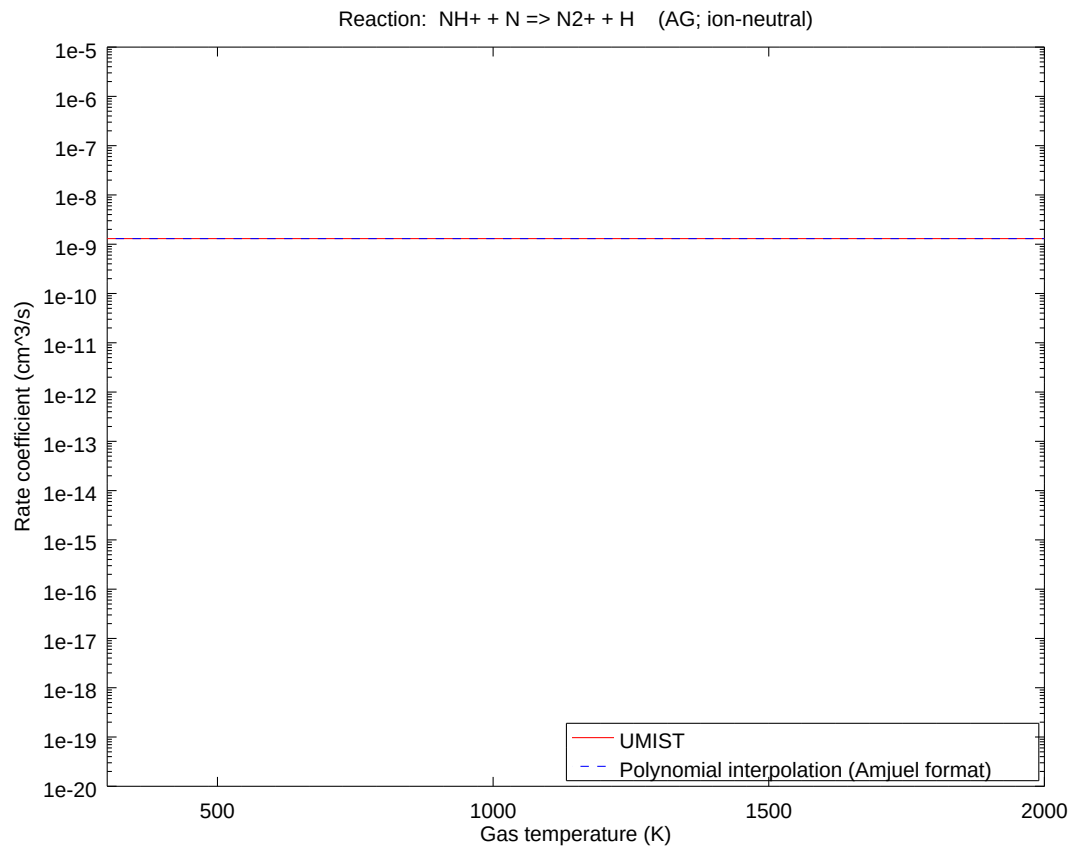
2.37 Reaction 49 $\text{NH}^+ + \text{N}_2 = \text{N}_2\text{H}^+ + \text{N}$

b0	-2.115404875565D+01	b1	-8.175551325954D-09	b2	-1.108959228976D-08
b3	-8.518585796004D-09	b4	-4.053165887750D-09	b5	-1.223274611361D-09
b6	-2.287208138927D-10	b7	-2.422633897627D-11	b8	-1.113179099929D-12



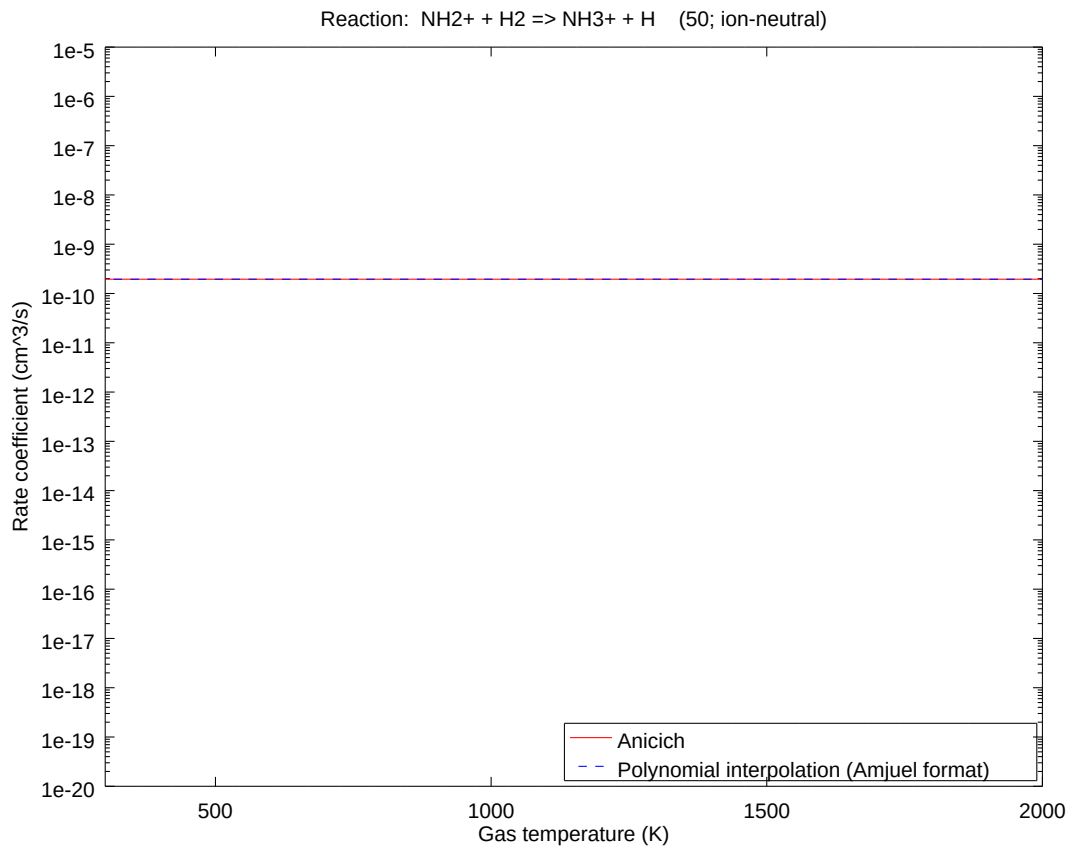
2.38 Reaction AG $\text{NH}^+ + \text{N} = \text{N}_2^+ + \text{H}$

b0	-2.046089640565D+01	b1	-6.309043749656D-06	b2	3.364679876218D-06
b3	-1.023652365486D-06	b4	1.943176986168D-07	b5	-2.356829800091D-08
b6	1.783632654019D-09	b7	-7.700722908794D-11	b8	1.452206891709D-12



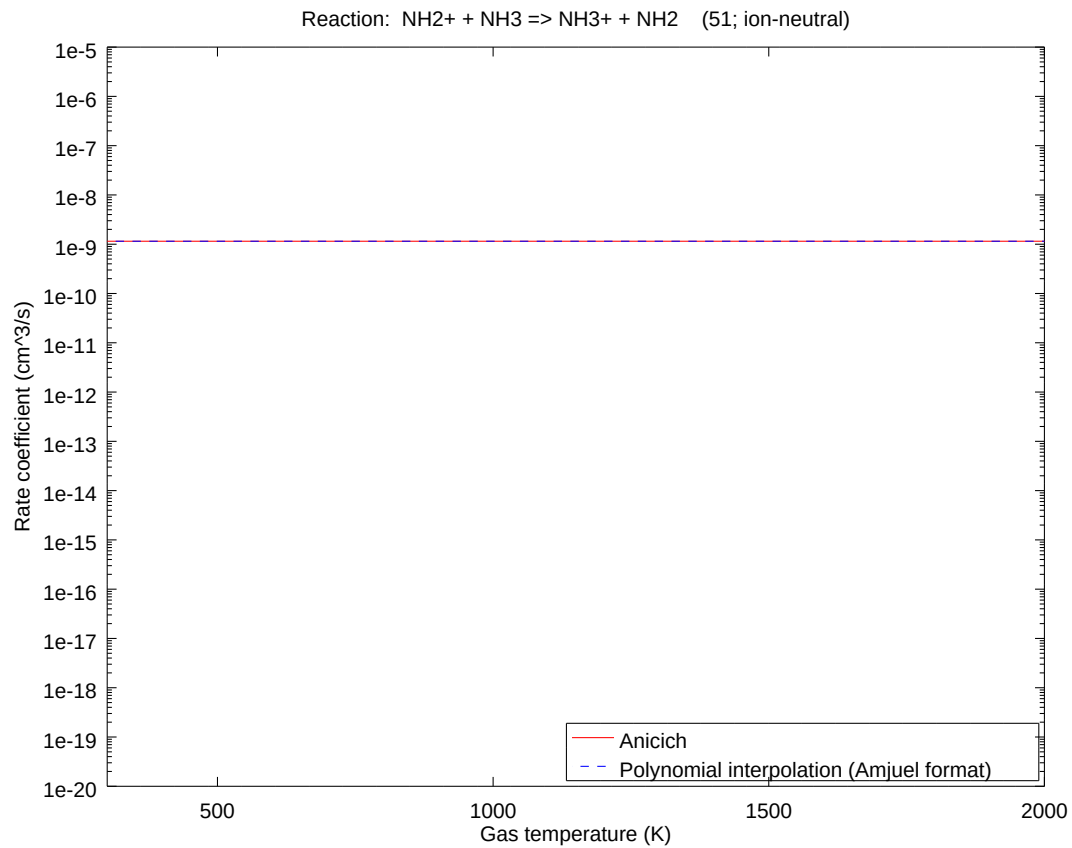
2.39 Reaction 50 $\text{NH}_2^+ + \text{H}_2 = \text{NH}_3^+ + \text{H}$

b0 -2.235802155903D+01	b1 -5.126291642220D-09	b2 -6.812959689968D-09
b3 -5.115130813147D-09	b4 -2.372142542164D-09	b5 -6.956403947333D-10
b6 -1.259589507198D-10	b7 -1.287480421967D-11	b8 -5.688106738779D-13



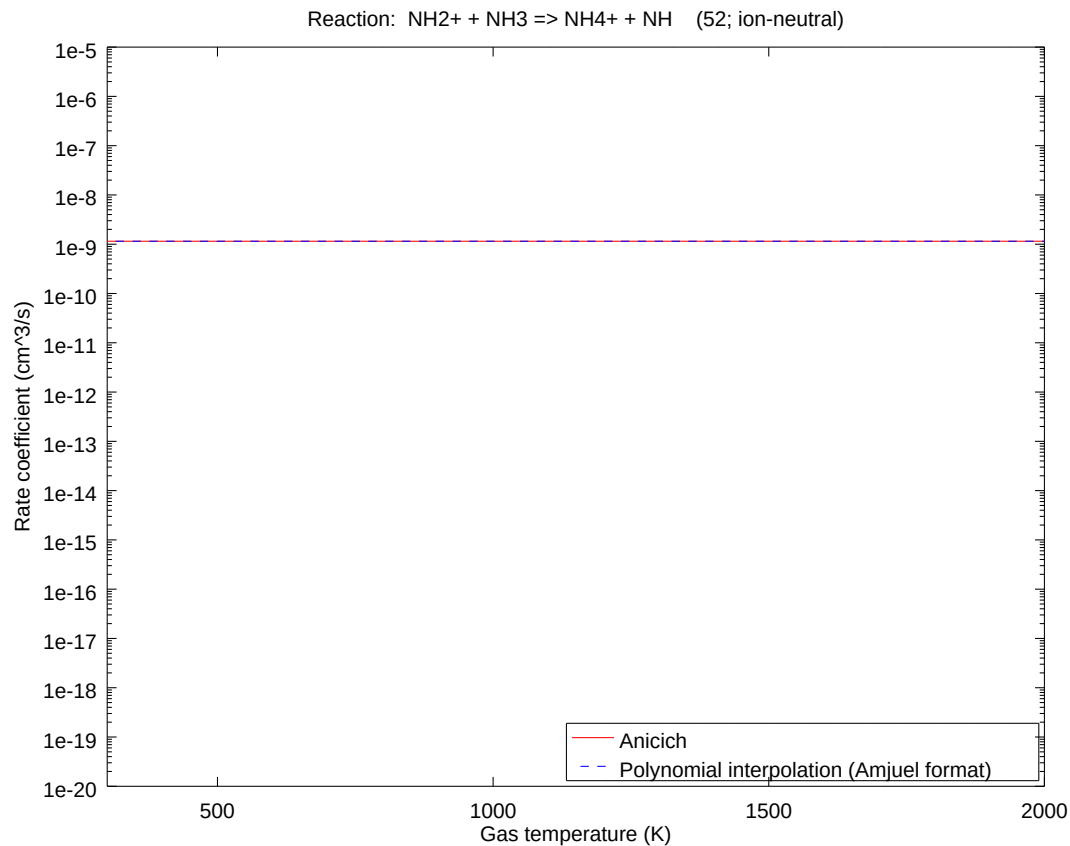
2.40 Reaction 51 $\text{NH}_2^+ + \text{NH}_3 = \text{NH}_3^+ + \text{NH}_2$

b0	-2.058350389575D+01	b1	-3.623757885018D-09	b2	-4.830435663478D-09
b3	-3.637883928480D-09	b4	-1.692890225051D-09	b5	-4.984780016176D-10
b6	-9.071303153841D-11	b7	-9.330157173760D-12	b8	-4.153866831075D-13



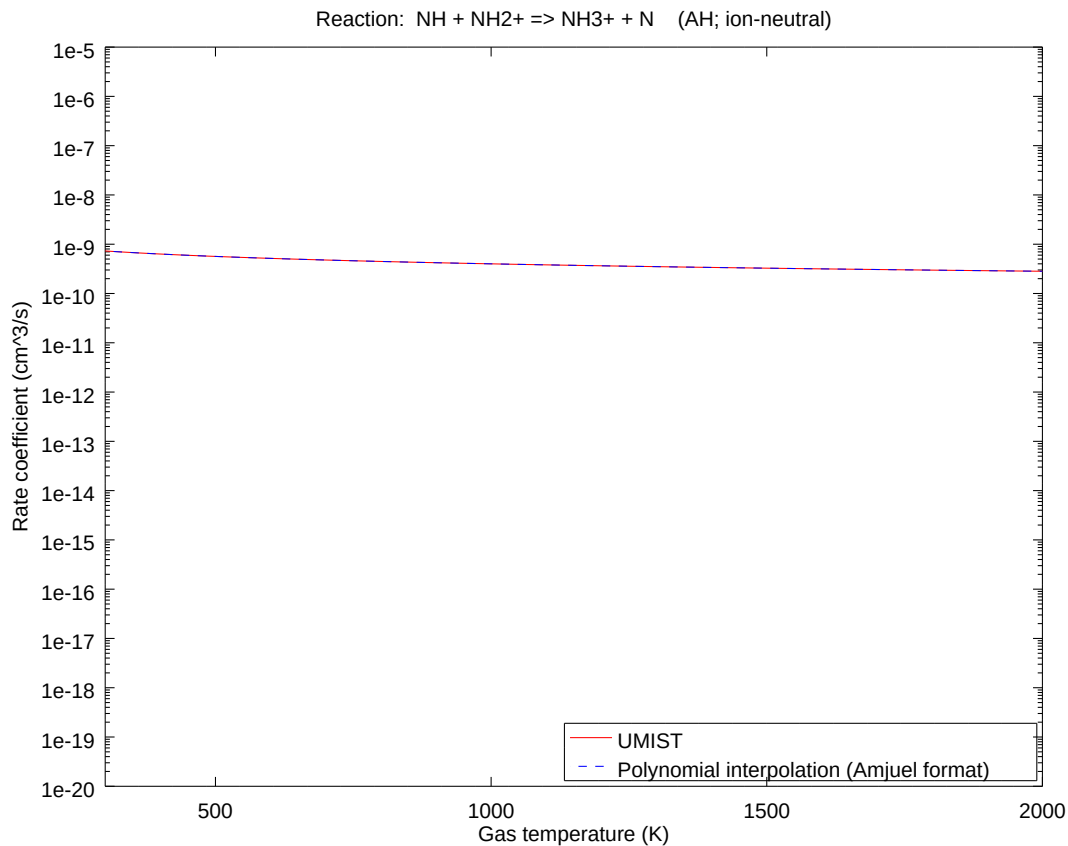
2.41 Reaction 52 $\text{NH}_2^+ + \text{NH}_3 = \text{NH}_4^+ + \text{NH}$

b0 -2.058350389575D+01	b1 -3.623757885018D-09	b2 -4.830435663478D-09
b3 -3.637883928480D-09	b4 -1.692890225051D-09	b5 -4.984780016176D-10
b6 -9.071303153841D-11	b7 -9.330157173760D-12	b8 -4.153866831075D-13



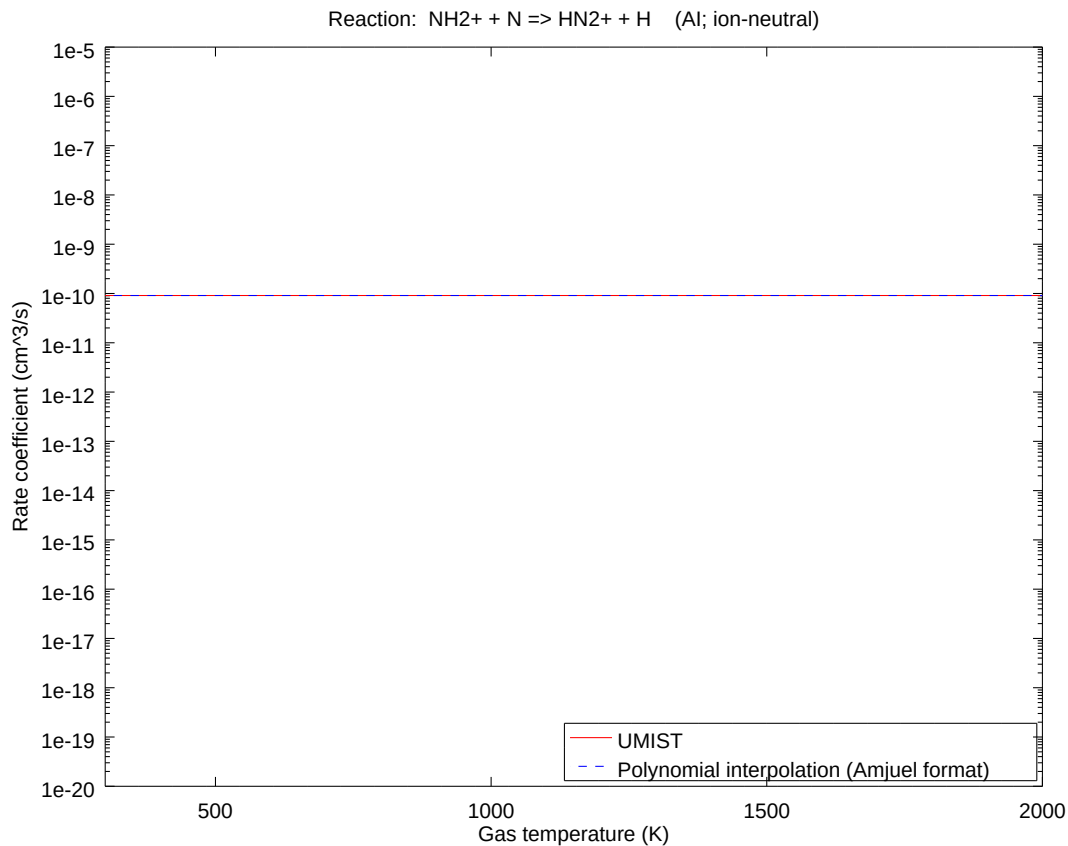
2.42 Reaction AH NH + NH2+ = NH3+ + N

b0	-1.818607887563D+01	b1	-5.000078456047D-01	b2	4.156349311216D-06
b3	-1.256230007379D-06	b4	2.369293567696D-07	b5	-2.855384218208D-08
b6	2.147385680129D-09	b7	-9.213857128273D-11	b8	1.726948689053D-12



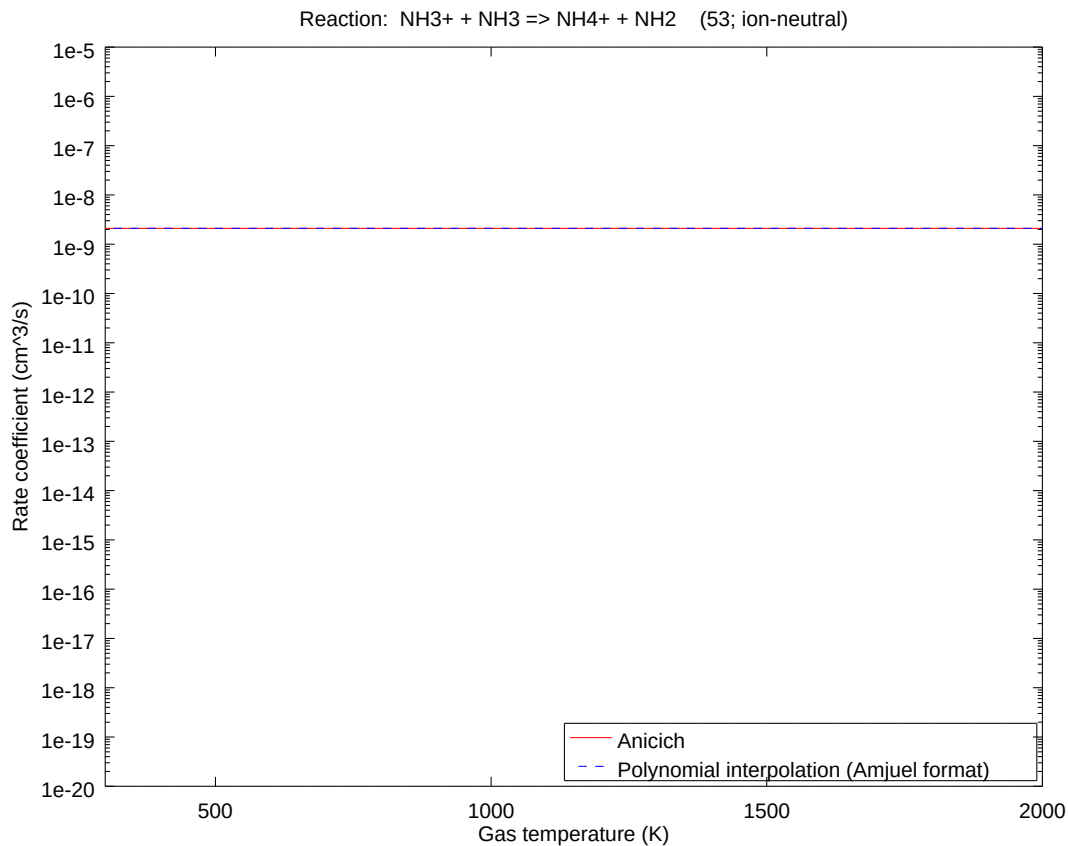
2.43 Reaction AI NH₂⁺ + N = HN₂⁺ + H

b0	-2.312015530949D+01	b1	-7.655206881455D-06	b2	4.063042893055D-06
b3	-1.230283525041D-06	b4	2.324553536999D-07	b5	-2.806454649117D-08
b6	2.114291047375D-09	b7	-9.087538100332D-11	b8	1.706174732568D-12



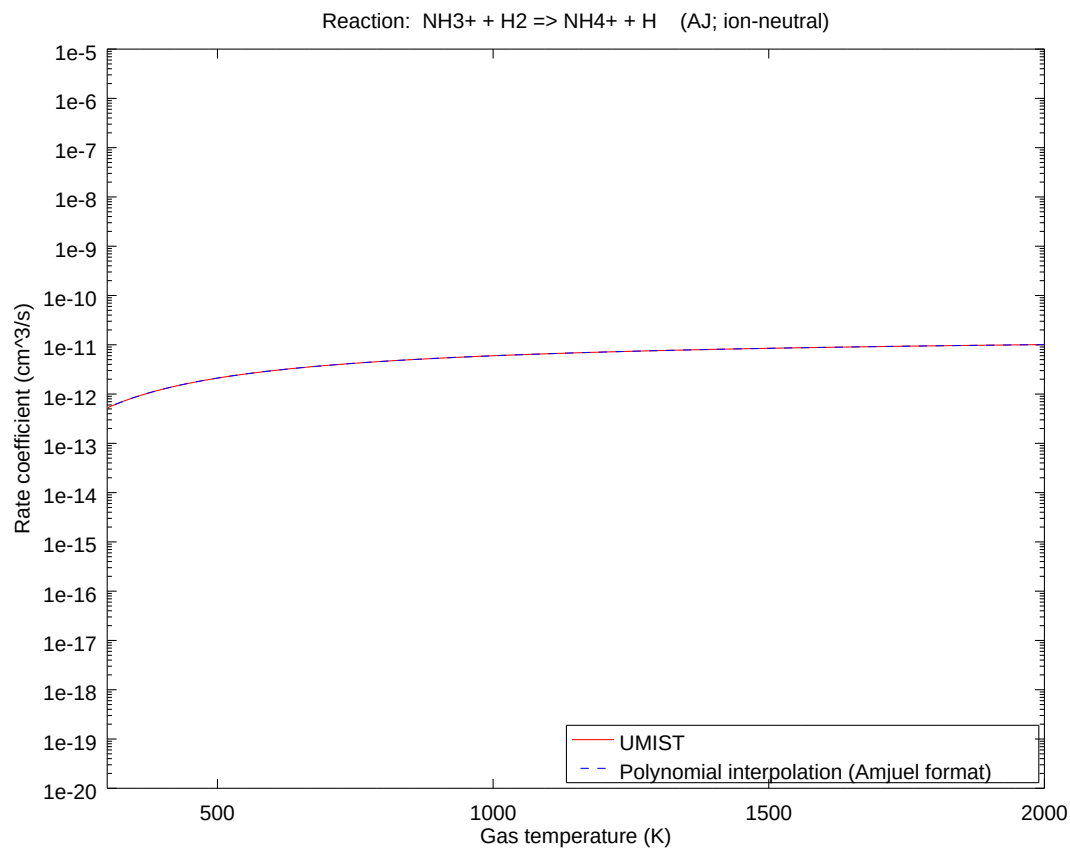
2.44 Reaction 53 $\text{NH}_3^+ + \text{NH}_3 = \text{NH}_4^+ + \text{NH}_2$

b0	-1.998132849358D+01	b1	-4.198256086301D-09	b2	-5.577663142080D-09
b3	-4.185441835682D-09	b4	-1.939584316088D-09	b5	-5.682572897106D-10
b6	-1.027727572340D-10	b7	-1.048955955390D-11	b8	-4.626069885568D-13



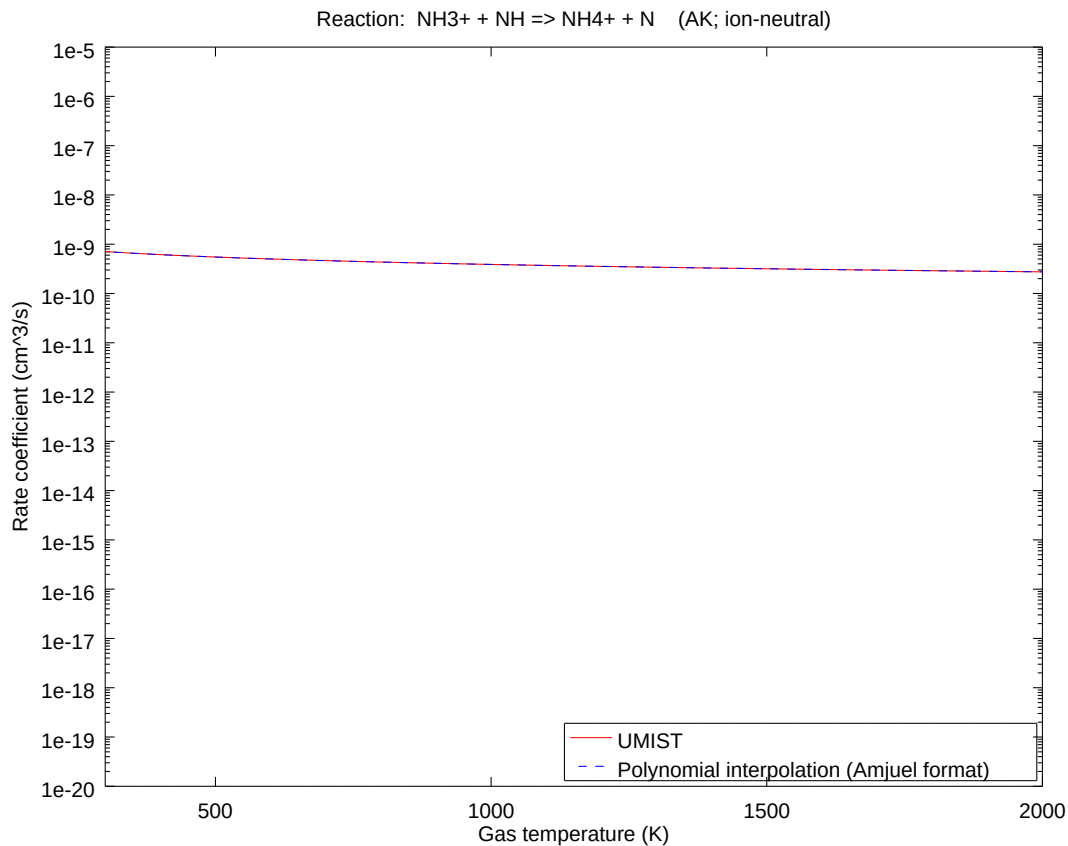
2.45 Reaction AJ $\text{NH}_3^+ + \text{H}_2 = \text{NH}_4^+ + \text{H}$

b0	-8.362035531499D+02	b1	6.834473349126D+02	b2	-2.644289375146D+02
b3	6.093998397467D+01	b4	-9.087919738986D+00	b5	8.926694453058D-01
b6	-5.611281142705D-02	b7	2.055034124102D-03	b8	-3.345597860355D-05



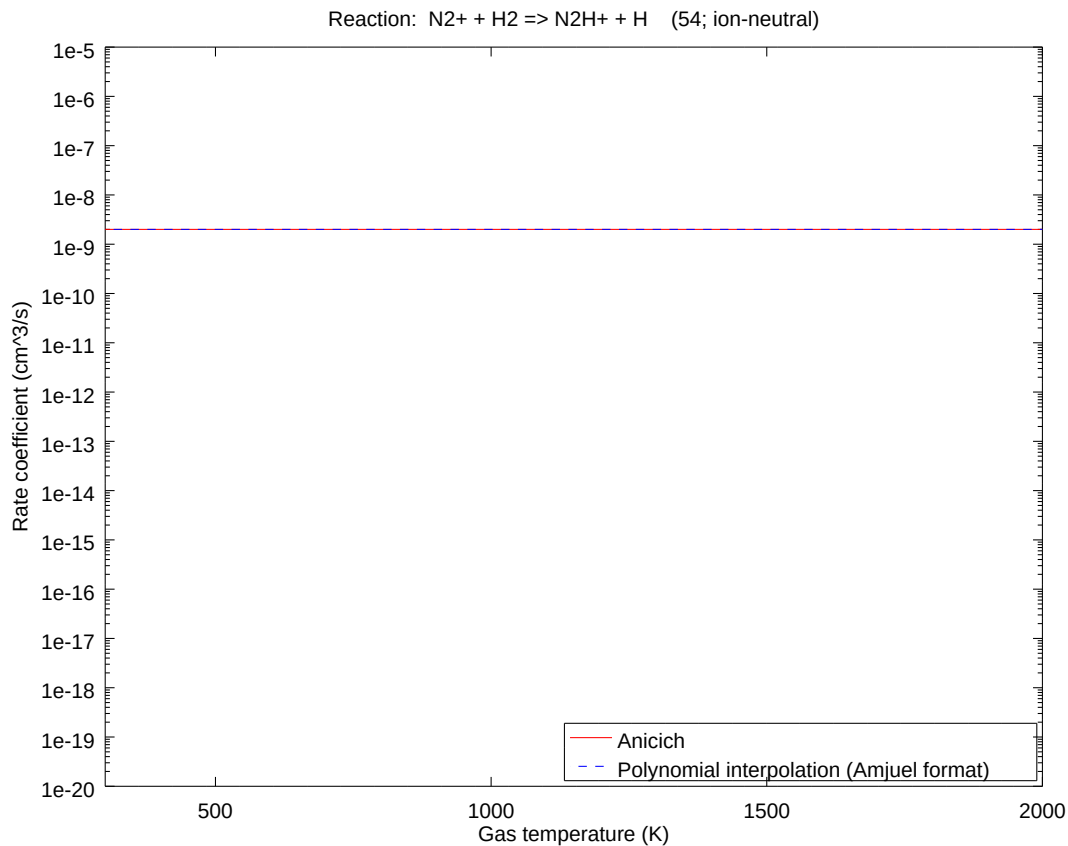
2.46 Reaction AK $\text{NH}_3^+ + \text{NH} = \text{NH}_4^+ + \text{N}$

b0	-1.821385782832D+01	b1	-5.000085852440D-01	b2	4.547278719427D-06
b3	-1.374140119126D-06	b4	2.591262218271D-07	b5	-3.122449013121D-08
b6	2.347936199197D-09	b7	-1.007325415014D-10	b8	1.887843793157D-12



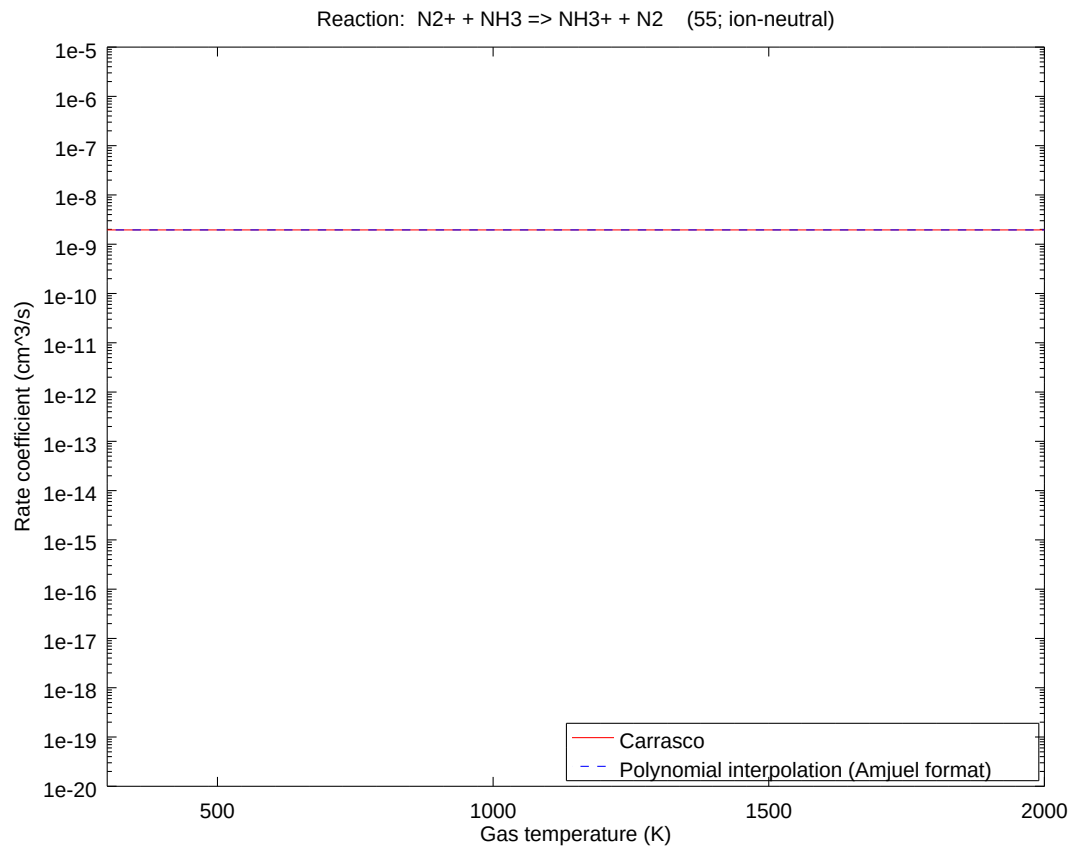
2.47 Reaction 54 $\text{N}_2^+ + \text{H}_2 = \text{N}_2\text{H}^+ + \text{H}$

b0	-2.003011865632D+01	b1	2.651530159769D-10	b2	4.417402249029D-10
b3	4.186177301543D-10	b4	2.453083470393D-10	b5	9.061934034478D-11
b6	2.054983068697D-11	b7	2.611571009143D-12	b8	1.423298839435D-13



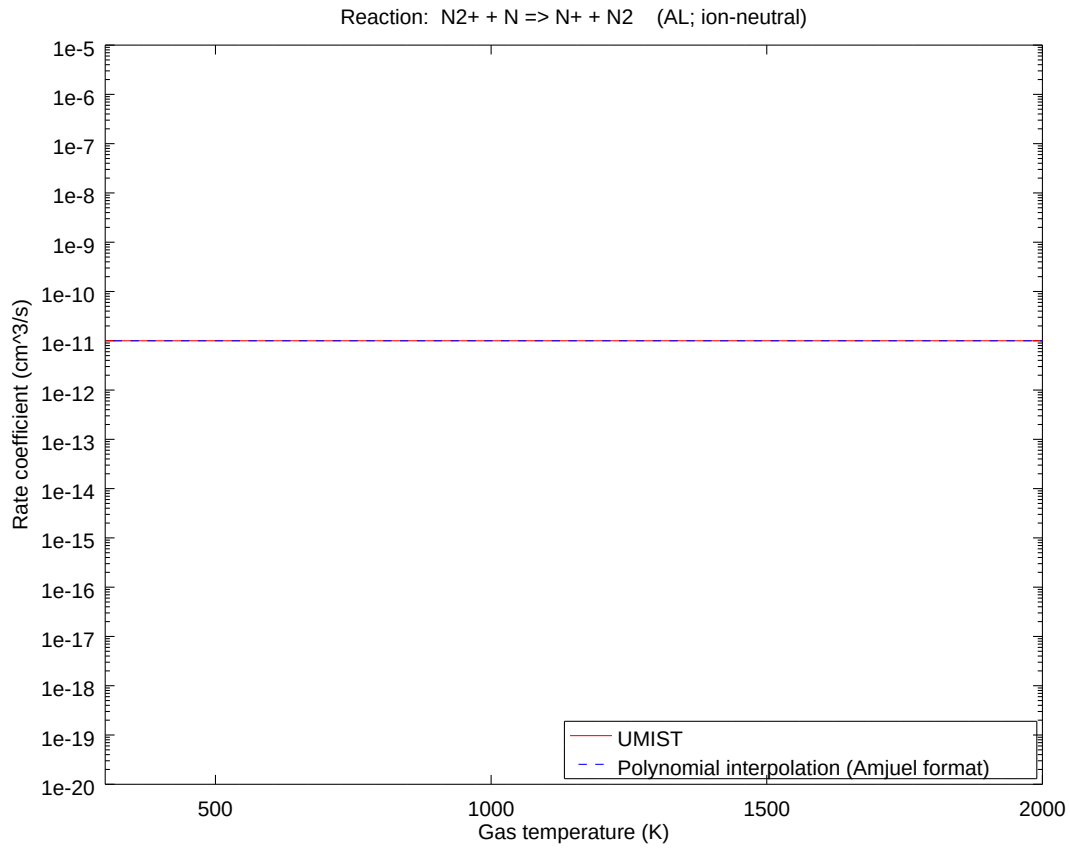
2.48 Reaction 55 $\text{N}_2^+ + \text{NH}_3 = \text{NH}_3^+ + \text{N}_2$

b0	-2.005543646628D+01	b1	-5.965942859480D-09	b2	-8.055431696629D-09
b3	-6.148792515450D-09	b4	-2.902225068604D-09	b5	-8.675064756898D-10
b6	-1.603987093760D-10	b7	-1.677659115819D-11	b8	-7.601828839840D-13



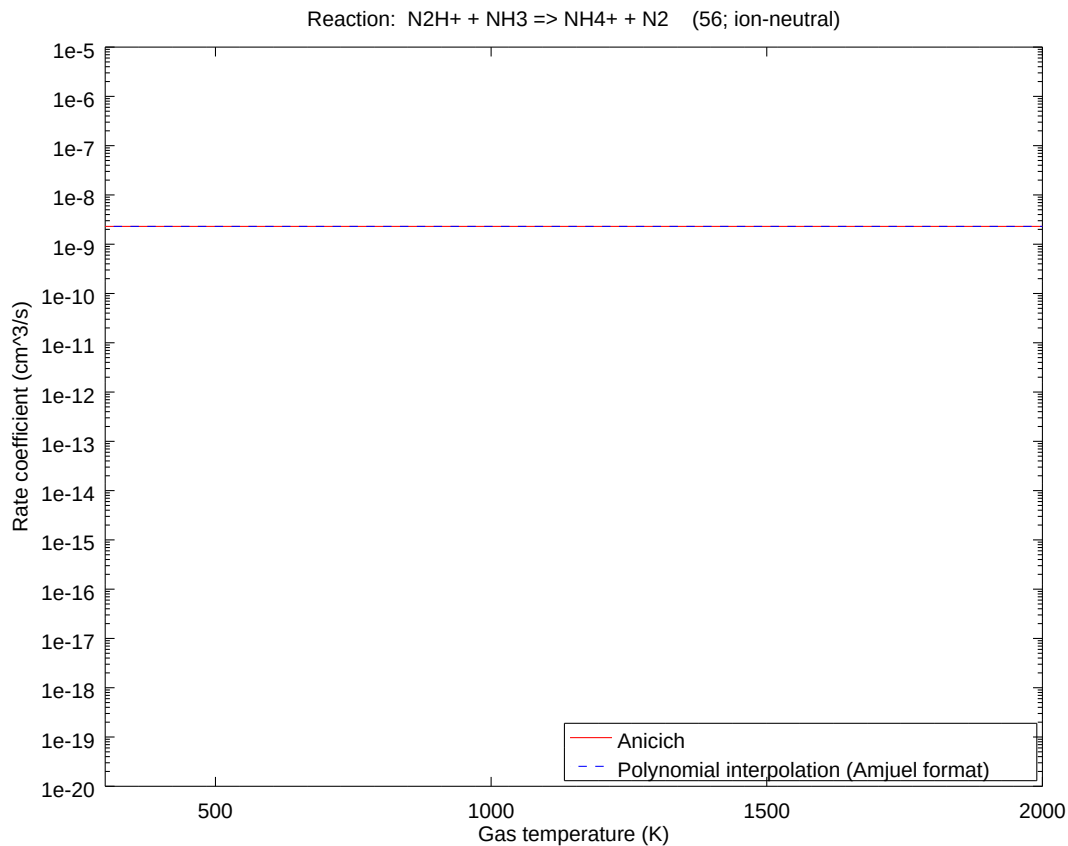
2.49 Reaction AL $\text{N}_2^+ + \text{N} = \text{N}^+ + \text{N}_2$

b0	-2.532843101849D+01	b1	-6.096669667990D-06	b2	3.244226613168D-06
b3	-9.849142194775D-07	b4	1.865845980529D-07	b5	-2.258639290203D-08
b6	1.706143264856D-09	b7	-7.353050543137D-11	b8	1.384277986480D-12



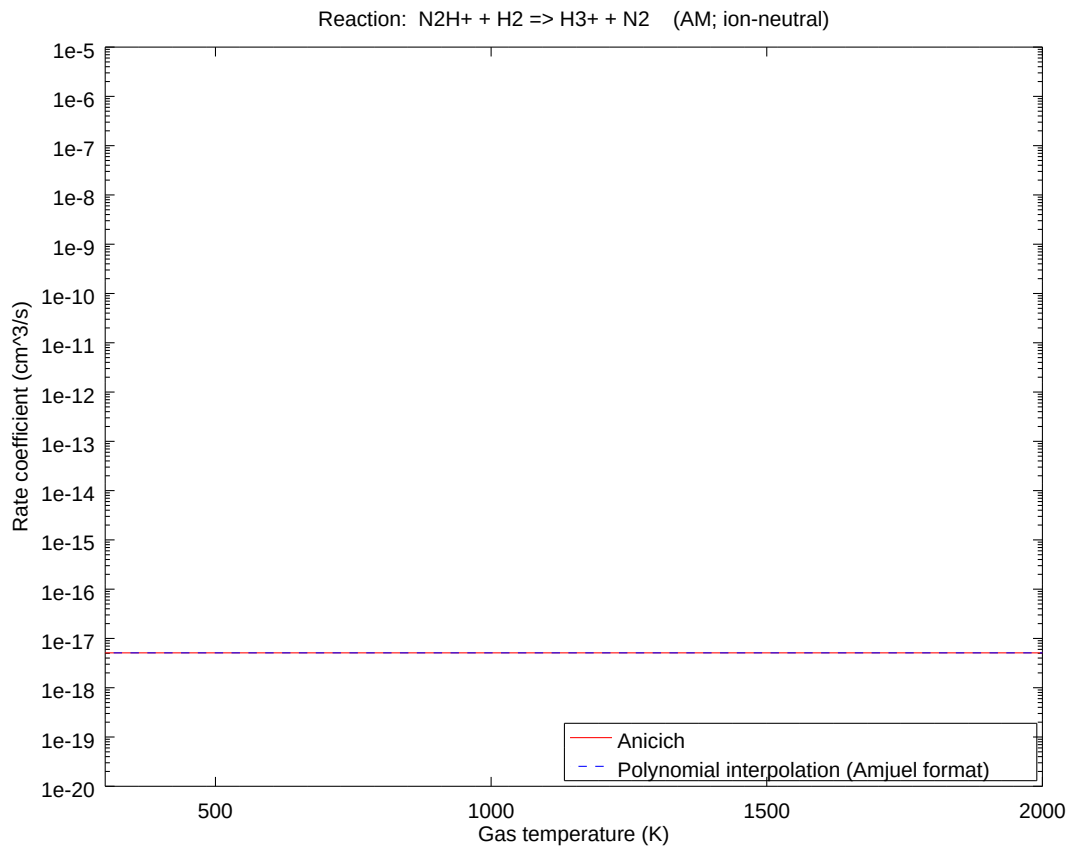
2.50 Reaction 56 $\text{N}_2\text{H}^+ + \text{NH}_3 = \text{NH}_4^+ + \text{N}_2$

b0	-1.989035671573D+01	b1	-5.347252488867D-09	b2	-7.199884370444D-09
b3	-5.478150766086D-09	b4	-2.575772829523D-09	b5	-7.663324140368D-10
b6	-1.408874458522D-10	b7	-1.463534994510D-11	b8	-6.578295424519D-13



2.51 Reaction AM $\text{N}_2\text{H}^+ + \text{H}_2 = \text{H}_3^+ + \text{N}_2$

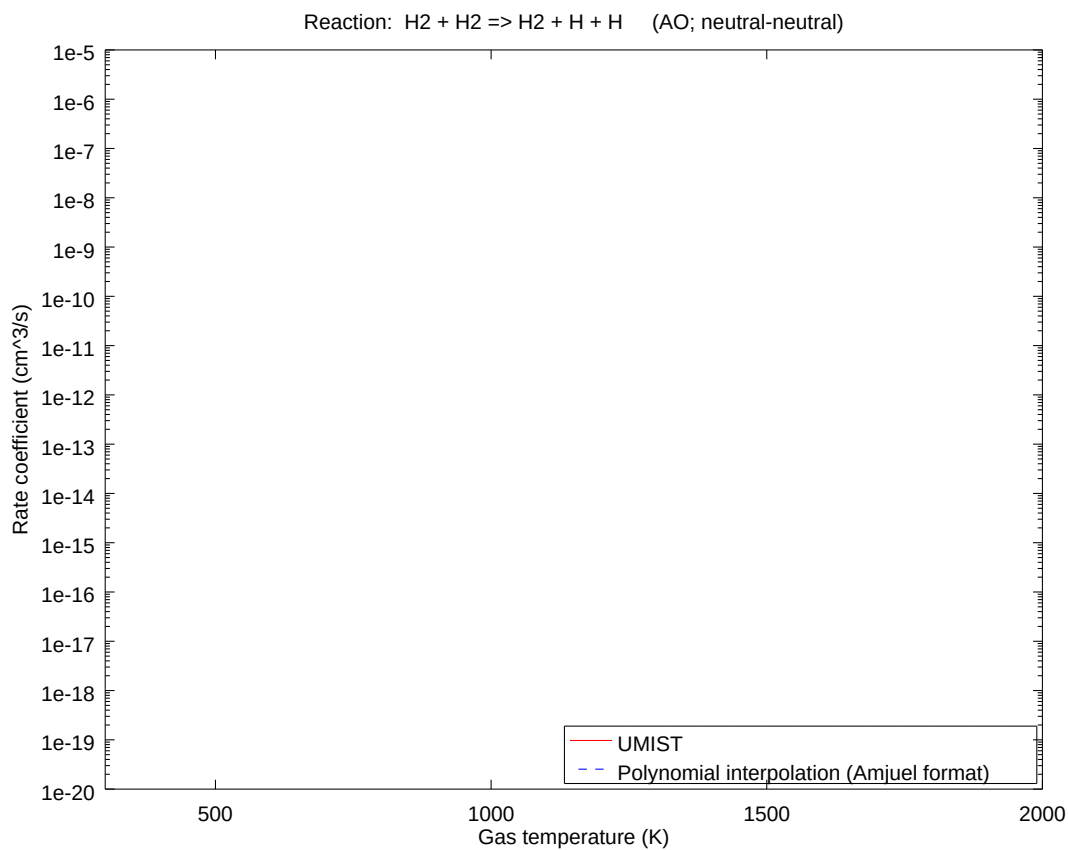
b0	-3.981729113612D+01	b1	-6.010135028810D-09	b2	-7.978031327521D-09
b3	-5.971762852958D-09	b4	-2.755089356818D-09	b5	-8.017297582443D-10
b6	-1.436200395246D-10	b7	-1.447105215130D-11	b8	-6.274632054775D-13



3 H.3 : Fits for neutral-neutral

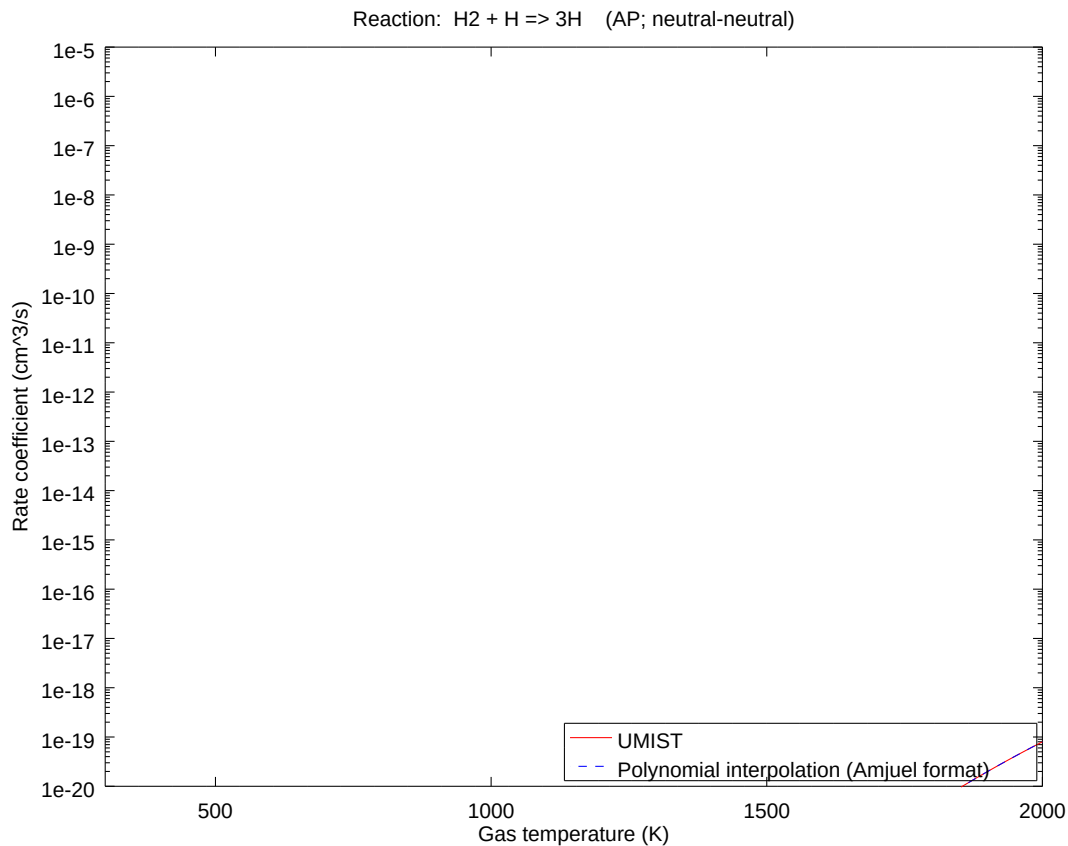
3.1 Reaction AO $\text{H}_2 + \text{H}_2 = \text{H}_2 + \text{H} + \text{H}$

b0	-6.538166197743D+04	b1	5.505548060307D+04	b2	-2.130122041540D+04
b3	4.909054400521D+03	b4	-7.320824490892D+02	b5	7.190948619296D+01
b6	-4.520198931743D+00	b7	1.655444255873D-01	b8	-2.695065131490D-03



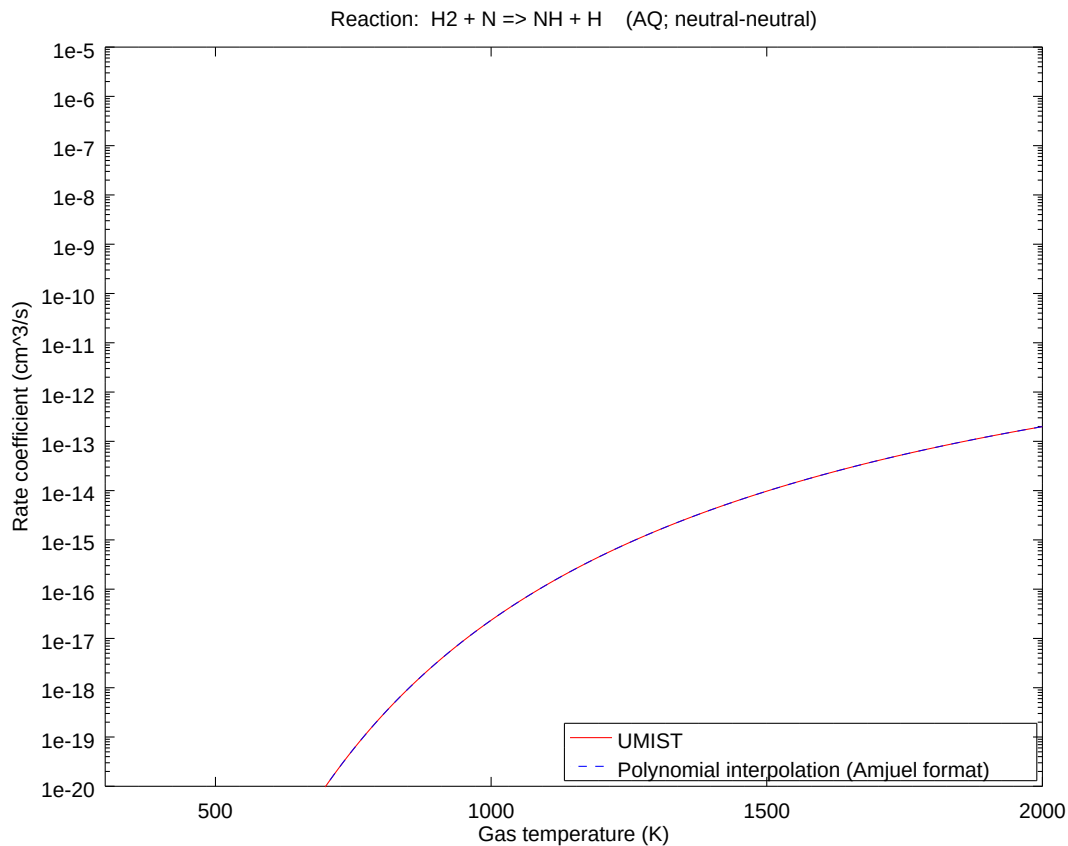
3.2 Reaction AP $\text{H}_2 + \text{H} = 3\text{H}$

b0	-4.275534486991D+04	b1	3.600436781375D+04	b2	-1.393064355320D+04
b3	3.210439857552D+03	b4	-4.787697348155D+02	b5	4.702760690064D+01
b6	-2.956134854018D+00	b7	1.082632985280D-01	b8	-1.762527731381D-03



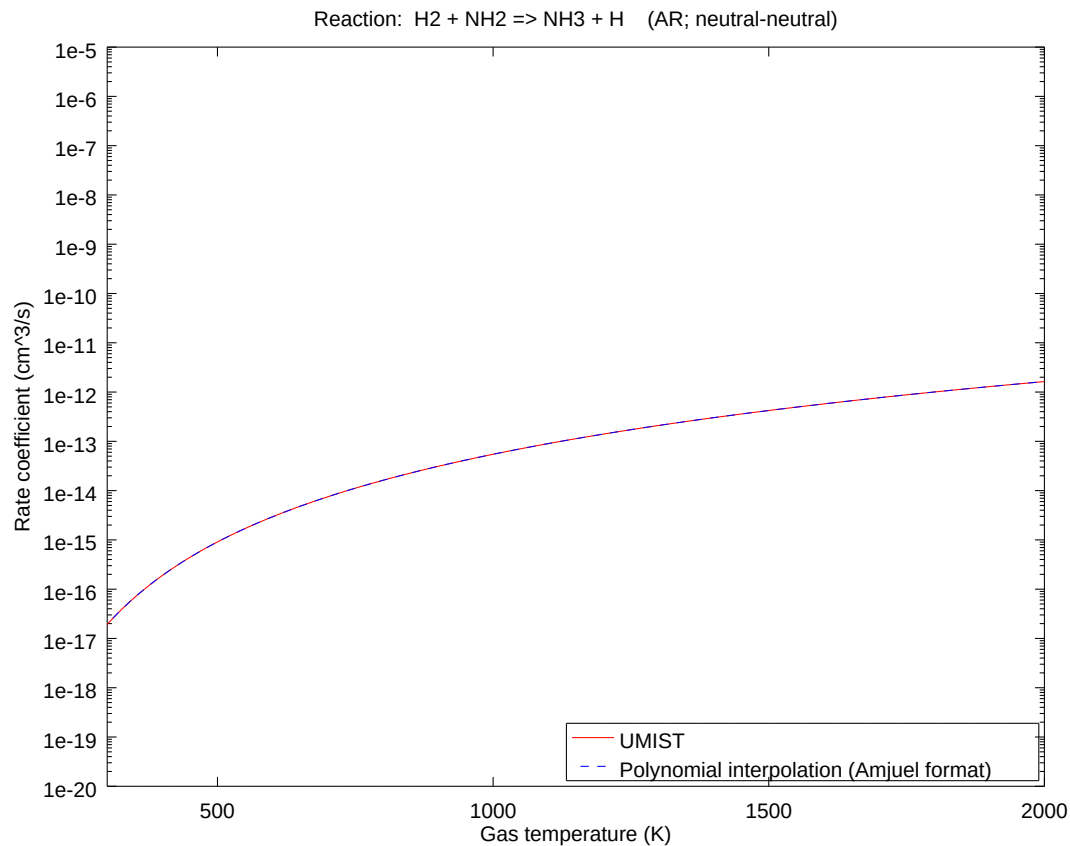
3.3 Reaction AQ $\text{H}_2 + \text{N} = \text{NH} + \text{H}$

b0	-1.408378772707D+04	b1	1.184576600502D+04	b2	-4.583181726009D+03
b3	1.056234712238D+03	b4	-1.575152425866D+02	b5	1.547208265028D+01
b6	-9.725683654795D-01	b7	3.561862515226D-02	b8	-5.798716224467D-04



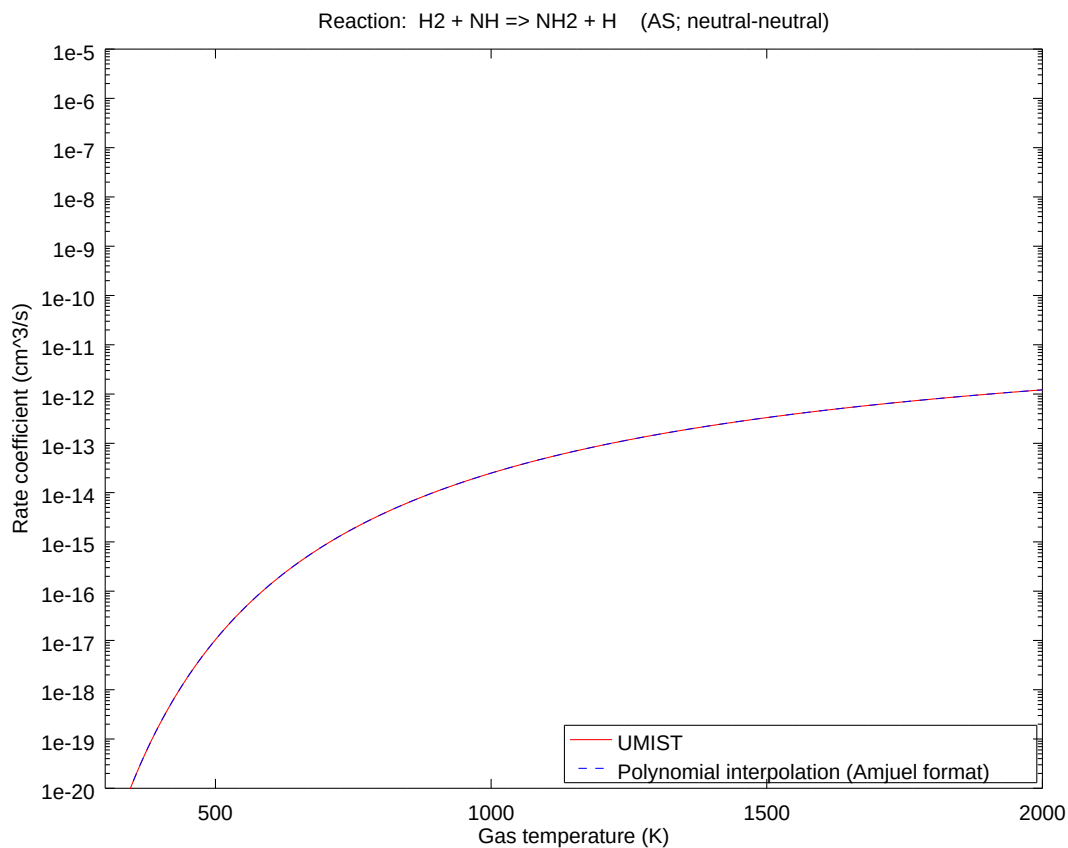
3.4 Reaction AR $\text{H}_2 + \text{NH}_2 = \text{NH}_3 + \text{H}$

b0	-1.144100645417D+03	b1	9.203902566862D+02	b2	-3.545981916263D+02
b3	8.172028489313D+01	b4	-1.218686552642D+01	b5	1.197066303154D+00
b6	-7.524706492596D-02	b7	2.755792878102D-03	b8	-4.486433896581D-05



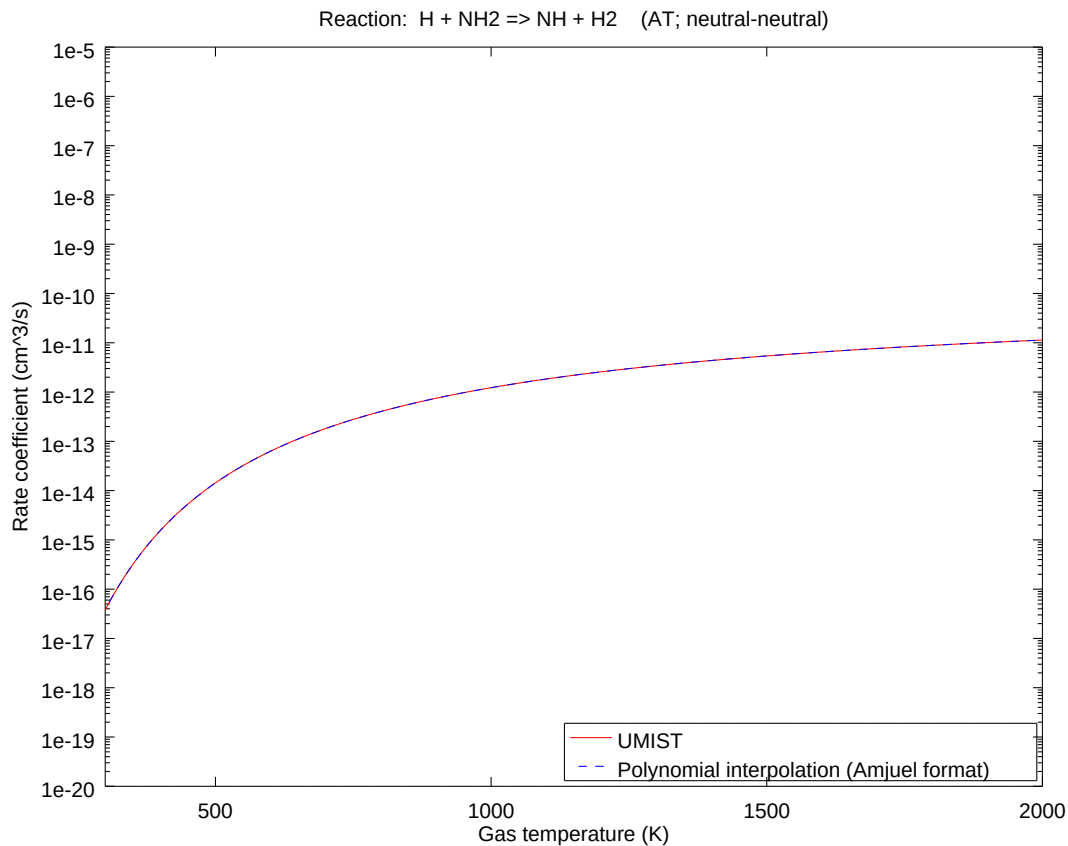
3.5 Reaction AS $\text{H}_2 + \text{NH} = \text{NH}_2 + \text{H}$

b0	-6.071780505248D+03	b1	5.094432219509D+03	b2	-1.971059418115D+03
b3	4.542480532851D+02	b4	-6.774156488412D+01	b5	6.653978835822D+00
b6	-4.182662066157D-01	b7	1.531827247512D-02	b8	-2.493816499829D-04



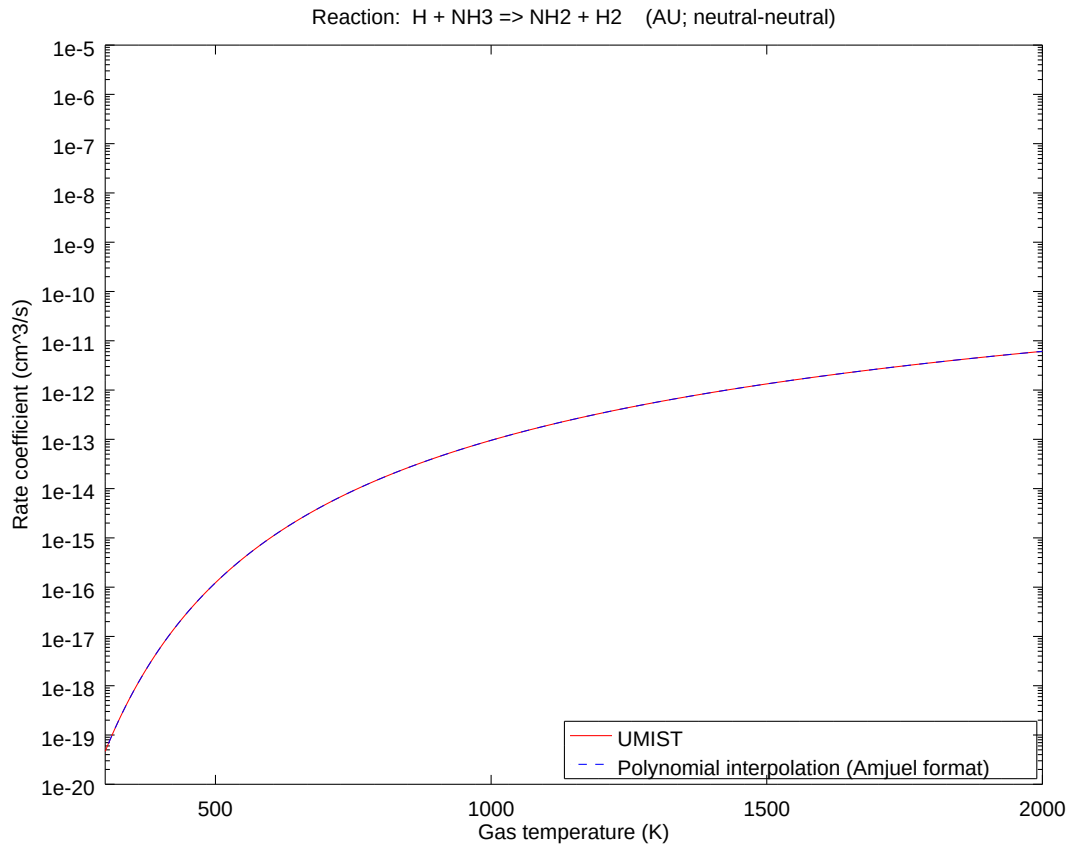
3.6 Reaction AT $\text{H} + \text{NH}_2 = \text{NH} + \text{H}_2$

b0	-3.481555219964D+03	b1	2.913161569274D+03	b2	-1.127115701218D+03
b3	2.597537689451D+02	b4	-3.873682374402D+01	b5	3.804960891206D+00
b6	-2.391781813258D-01	b7	8.759484963085D-03	b8	-1.426045145771D-04



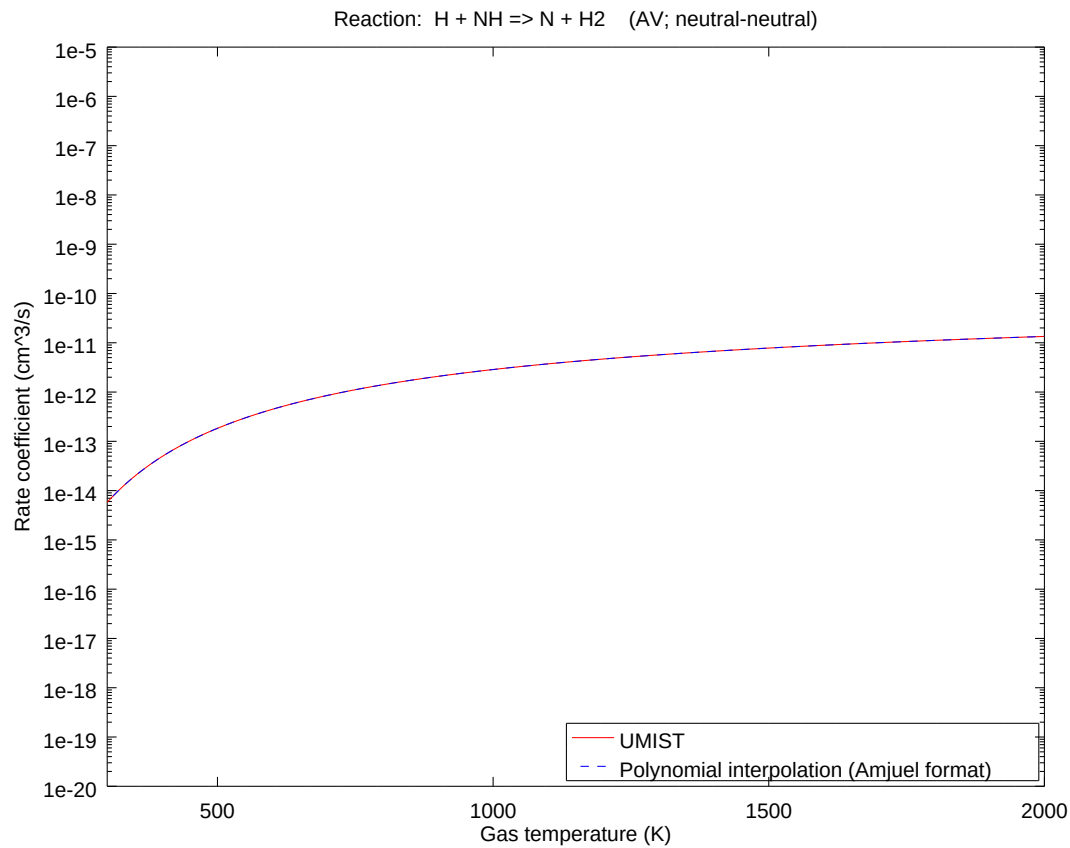
3.7 Reaction AU $\text{H} + \text{NH}_3 = \text{NH}_2 + \text{H}_2$

b0	-3.919839351098D+03	b1	3.269068816988D+03	b2	-1.263889292500D+03
b3	2.912744511884D+02	b4	-4.343747204971D+01	b5	4.266686486159D+00
b6	-2.682020507888D-01	b7	9.822433711395D-03	b8	-1.599093322809D-04



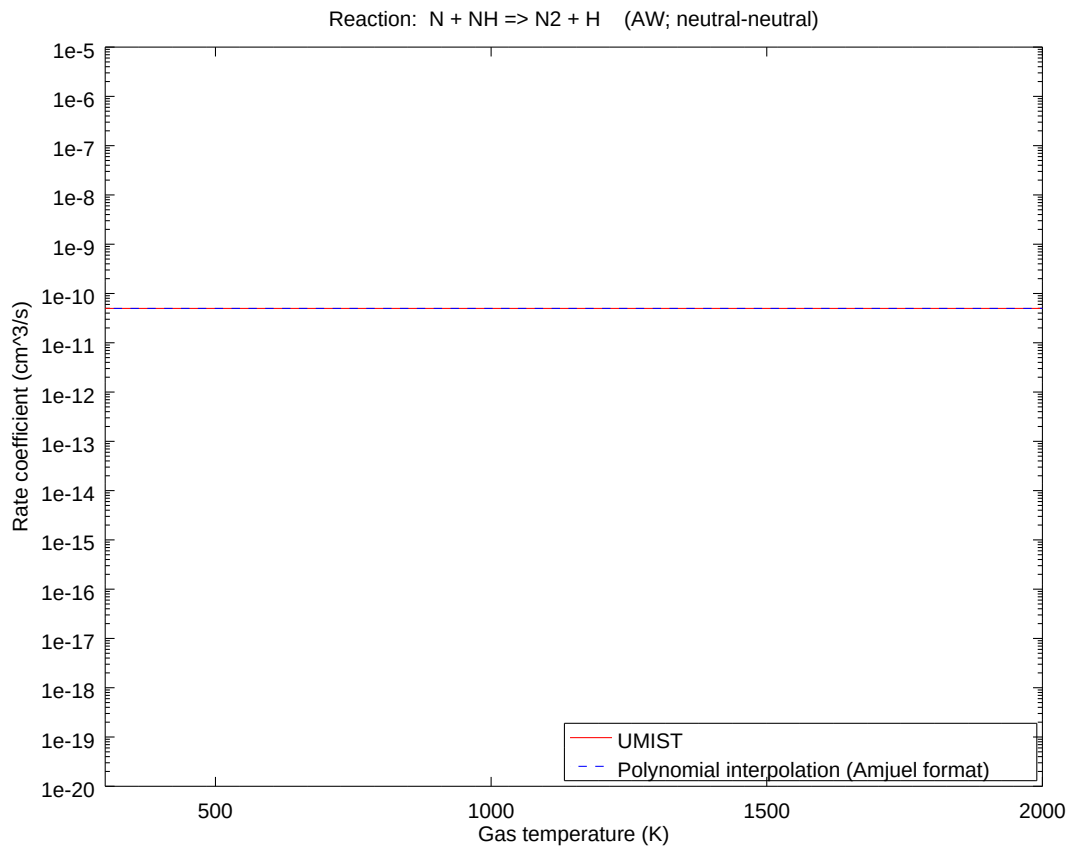
3.8 Reaction AV $\text{H} + \text{NH} = \text{N} + \text{H}_2$

b0	-1.892932785176D+03	b1	1.571643317114D+03	b2	-6.078826247516D+02
b3	1.400919201361D+02	b4	-2.089177007163D+01	b5	2.052113734444D+00
b6	-1.289949738414D-01	b7	4.724216593640D-03	b8	-7.691029969650D-05



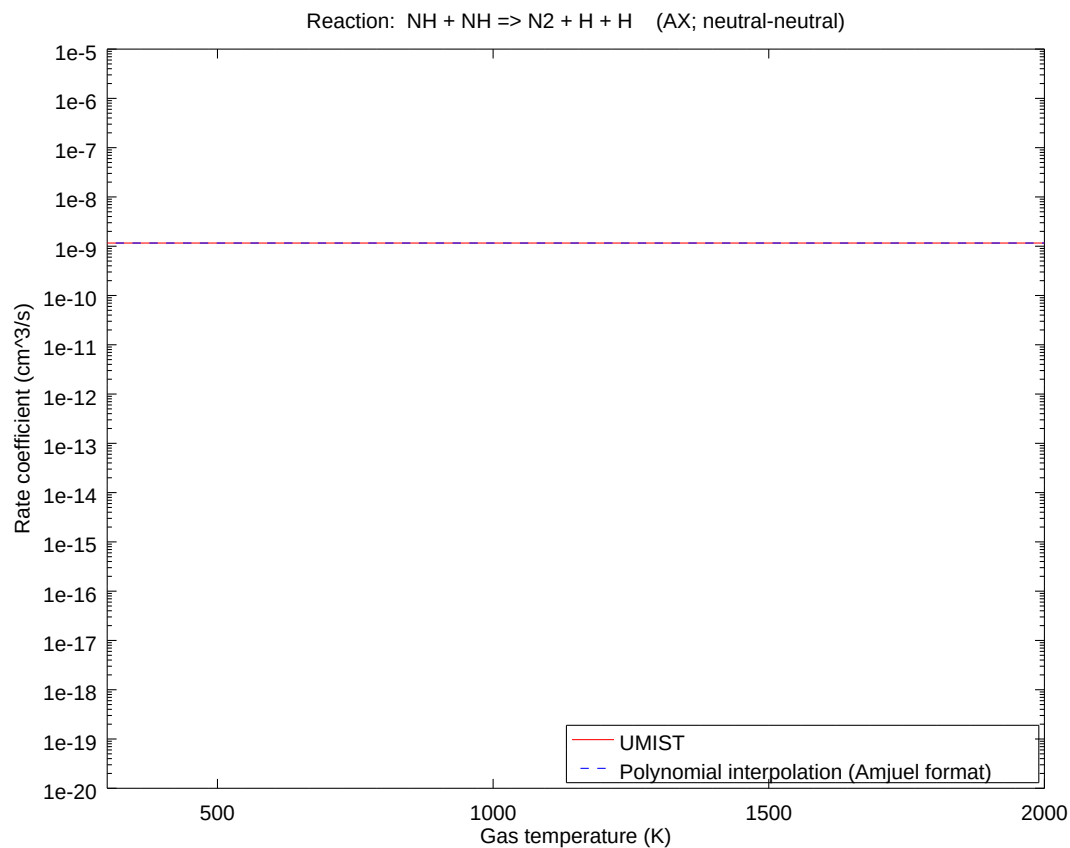
3.9 Reaction AW $\text{N} + \text{NH} = \text{N}_2 + \text{H}$

b0	-2.372299786395D+01	b1	-1.004784392574D-05	b2	5.334055515219D-06
b3	-1.615617532923D-06	b4	3.053778248118D-07	b5	-3.688571442980D-08
b6	2.780385400959D-09	b7	-1.195816380957D-10	b8	2.246765879181D-12



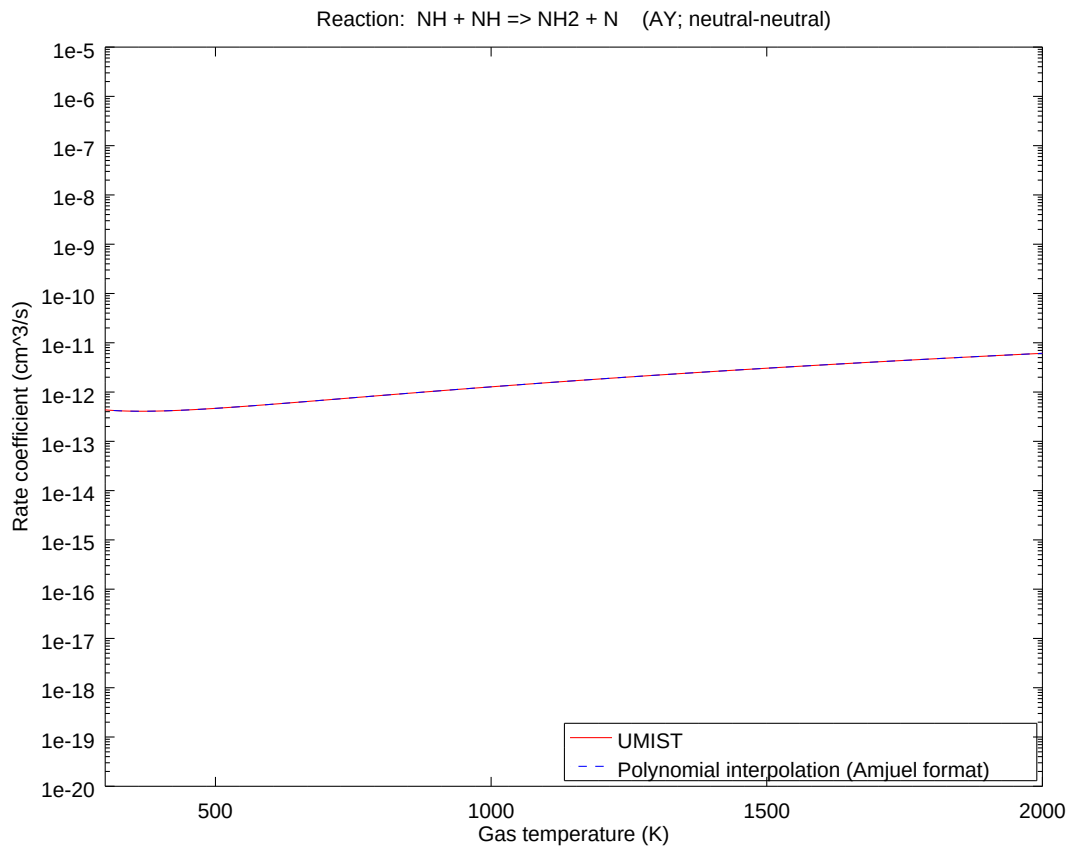
3.10 Reaction AX $\text{NH} + \text{NH} = \text{N}_2 + \text{H} + \text{H}$

b0	-2.057483893574D+01	b1	-8.386139979901D-06	b2	4.454723110308D-06
b3	-1.350091649278D-06	b4	2.553340718351D-07	b5	-3.085735939009D-08
b6	2.327101416926D-09	b7	-1.001297017297D-10	b8	1.882004895393D-12



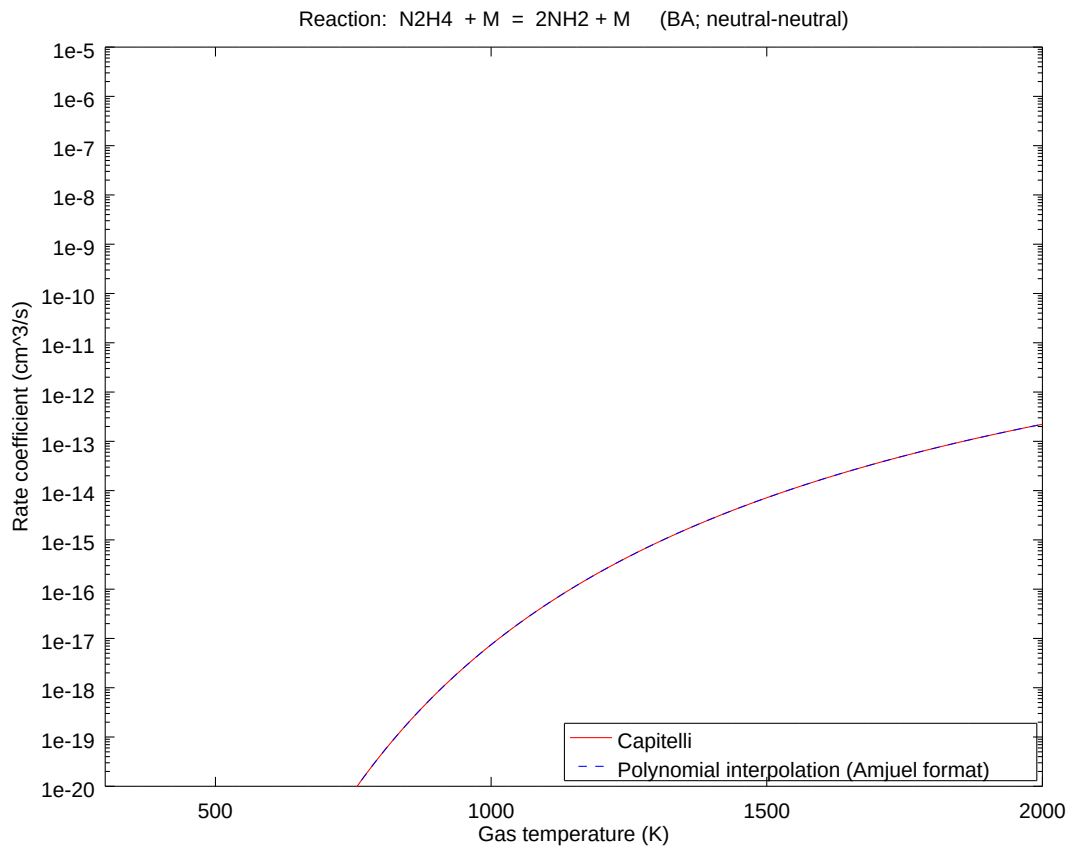
3.11 Reaction AY $\text{NH} + \text{NH} = \text{NH}_2 + \text{N}$

b0	8.230879970752D+02	b1	-7.320941588201D+02	b2	2.844384194694D+02
b3	-6.555134671043D+01	b4	9.775607868646D+00	b5	-9.602182732881D-01
b6	6.035890232345D-02	b7	-2.210539859342D-03	b8	3.598761441262D-05



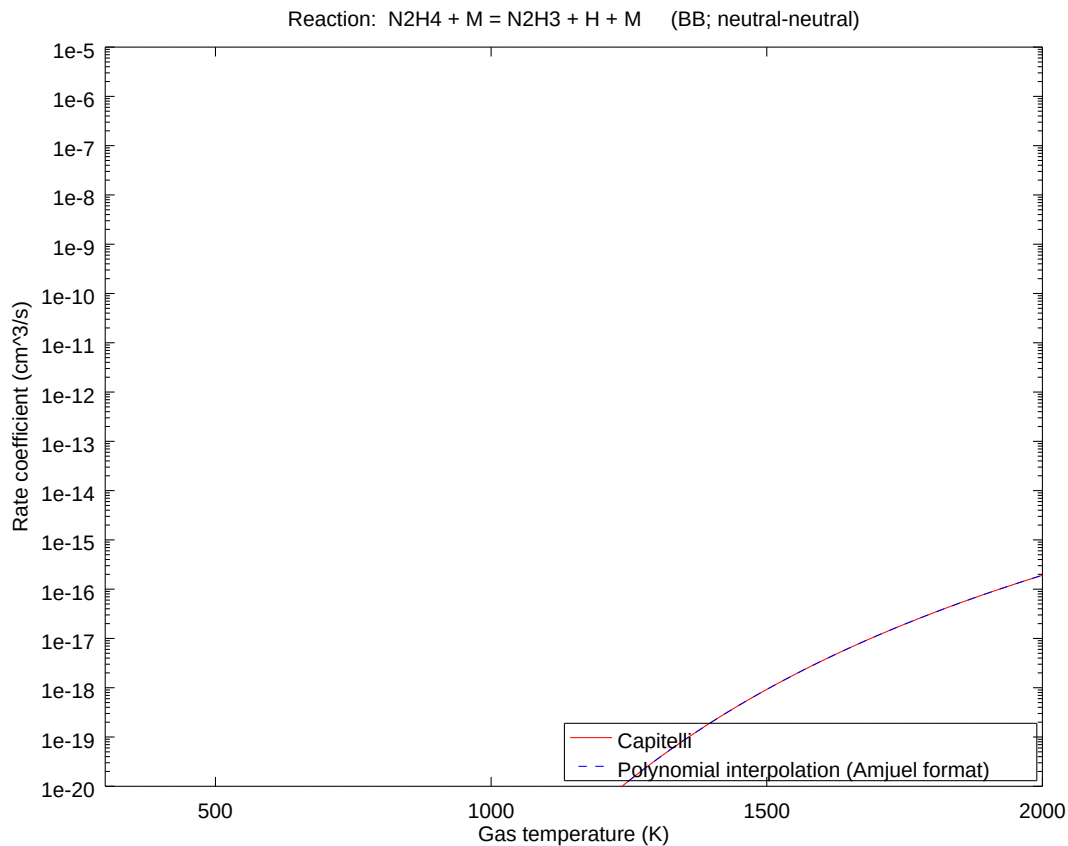
3.12 Reaction BA $\text{N}_2\text{H}_4 + \text{M} = 2\text{NH}_2 + \text{M}$

b0	-1.602933287178D+04	b1	1.348564684976D+04	b2	-5.217659219578D+03
b3	1.202455655087D+03	b4	-1.793210278272D+02	b5	1.761397638819D+01
b6	-1.107206871505D+00	b7	4.054952631122D-02	b8	-6.601467494319D-04



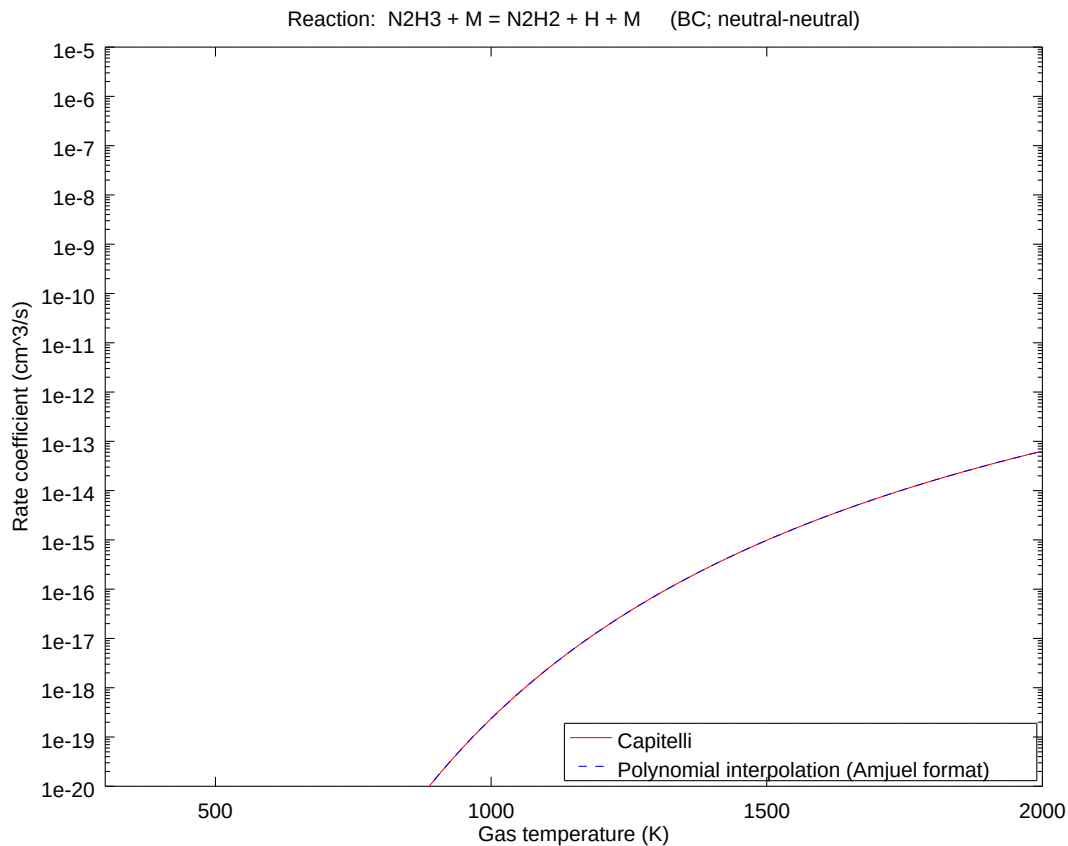
3.13 Reaction BB $\text{N}_2\text{H}_4 + \text{M} = \text{N}_2\text{H}_3 + \text{H} + \text{M}$

b0	-2.489086709460D+04	b1	2.094857764217D+04	b2	-8.105101706377D+03
b3	1.867892281575D+03	b4	-2.785569367746D+02	b5	2.736151676091D+01
b6	-1.719933007387D+00	b7	6.298955556807D-02	b8	-1.025470681223D-03



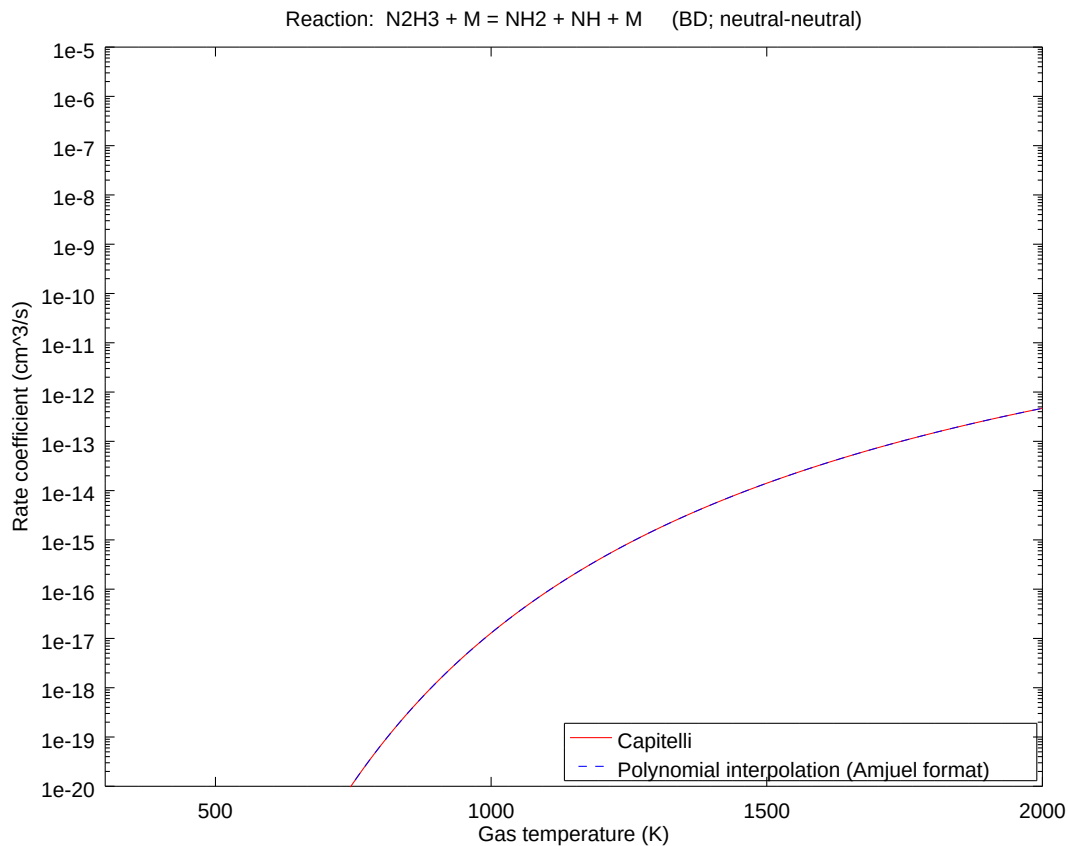
3.14 Reaction BC $\text{N}_2\text{H}_3 + \text{M} = \text{N}_2\text{H}_2 + \text{H} + \text{M}$

b0	-1.944810446444D+04	b1	1.636607627379D+04	b2	-6.332110703288D+03
b3	1.459290843532D+03	b4	-2.176226065836D+02	b5	2.137618493691D+01
b6	-1.343697659587D+00	b7	4.921059018364D-02	b8	-8.011489677674D-04



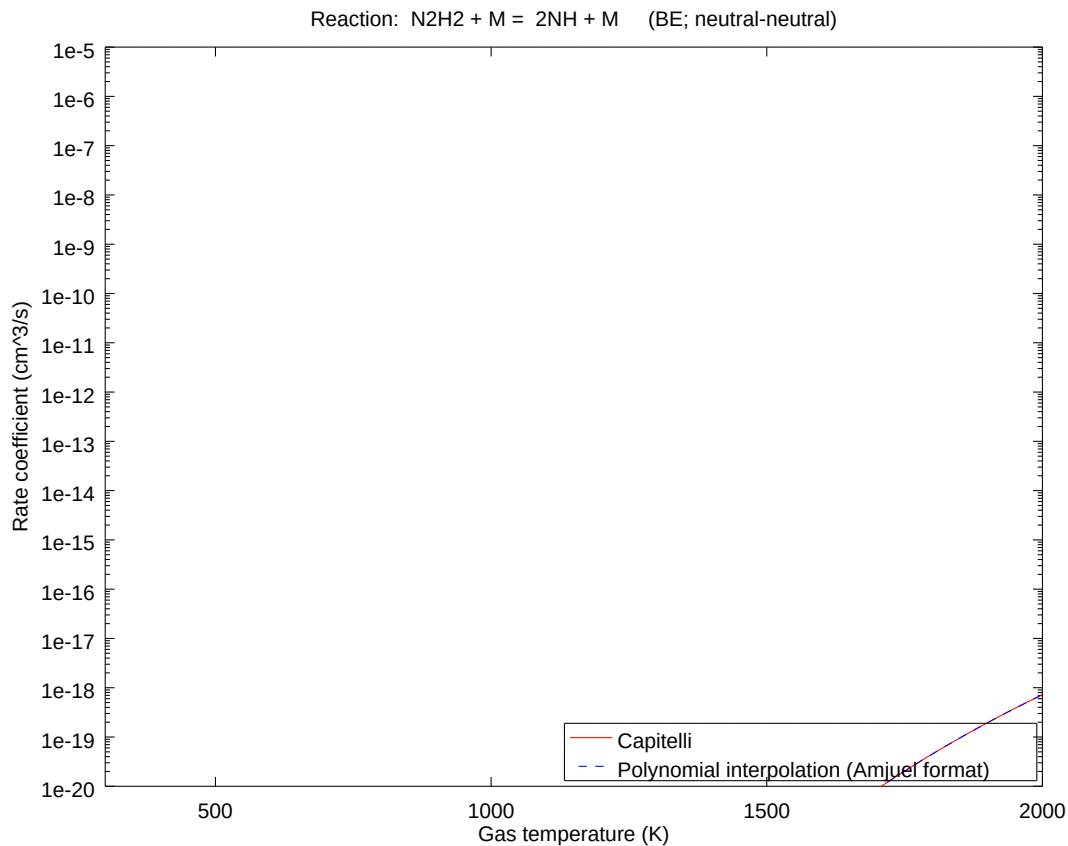
3.15 Reaction BD $\text{N}_2\text{H}_3 + \text{M} = \text{NH}_2 + \text{NH} + \text{M}$

b0	-1.633927015782D+04	b1	1.374750406914D+04	b2	-5.318972990327D+03
b3	1.225804308439D+03	b4	-1.828029895066D+02	b5	1.795599534421D+01
b6	-1.128706033846D+00	b7	4.133689574550D-02	b8	-6.729651327624D-04



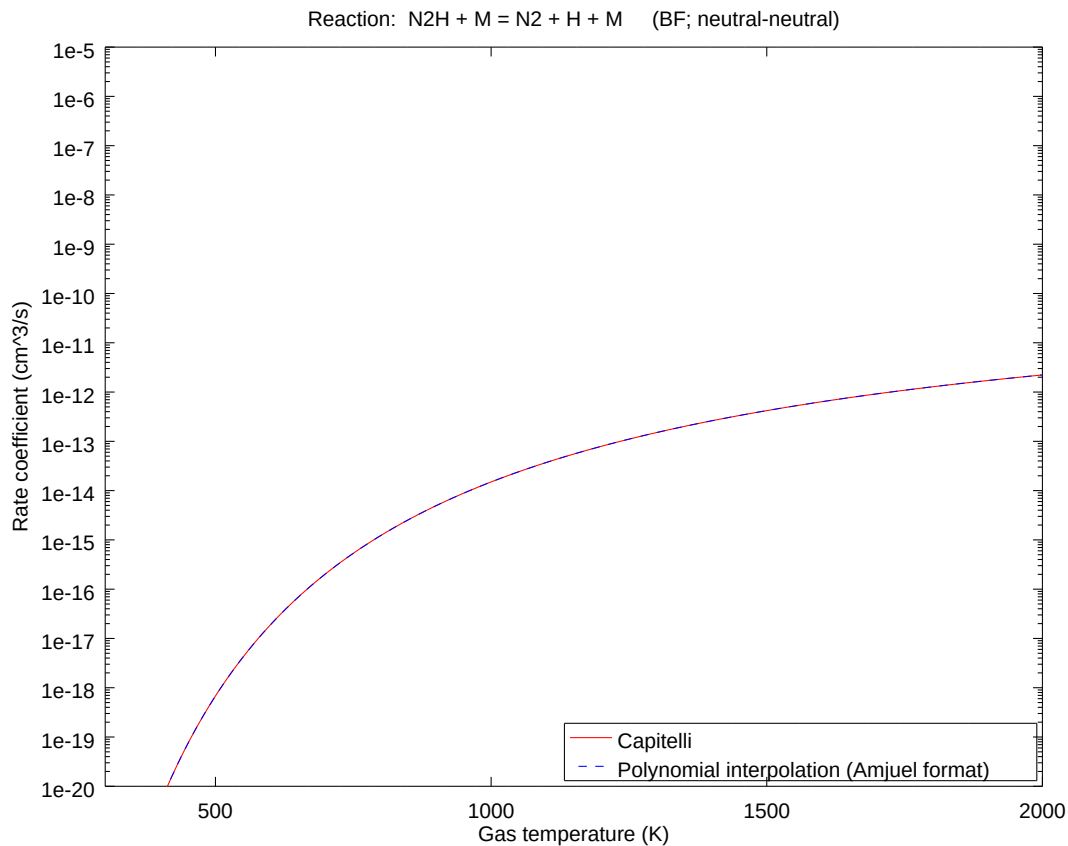
3.16 Reaction BE $\text{N}_2\text{H}_2 + \text{M} = 2\text{NH} + \text{M}$

b0	-3.887720086087D+04	b1	3.273215256544D+04	b2	-1.266422141596D+04
b3	2.918581689876D+03	b4	-4.352452136931D+02	b5	4.275236993669D+01
b6	-2.687395323863D+00	b7	9.842118056694D-02	b8	-1.602297939249D-03



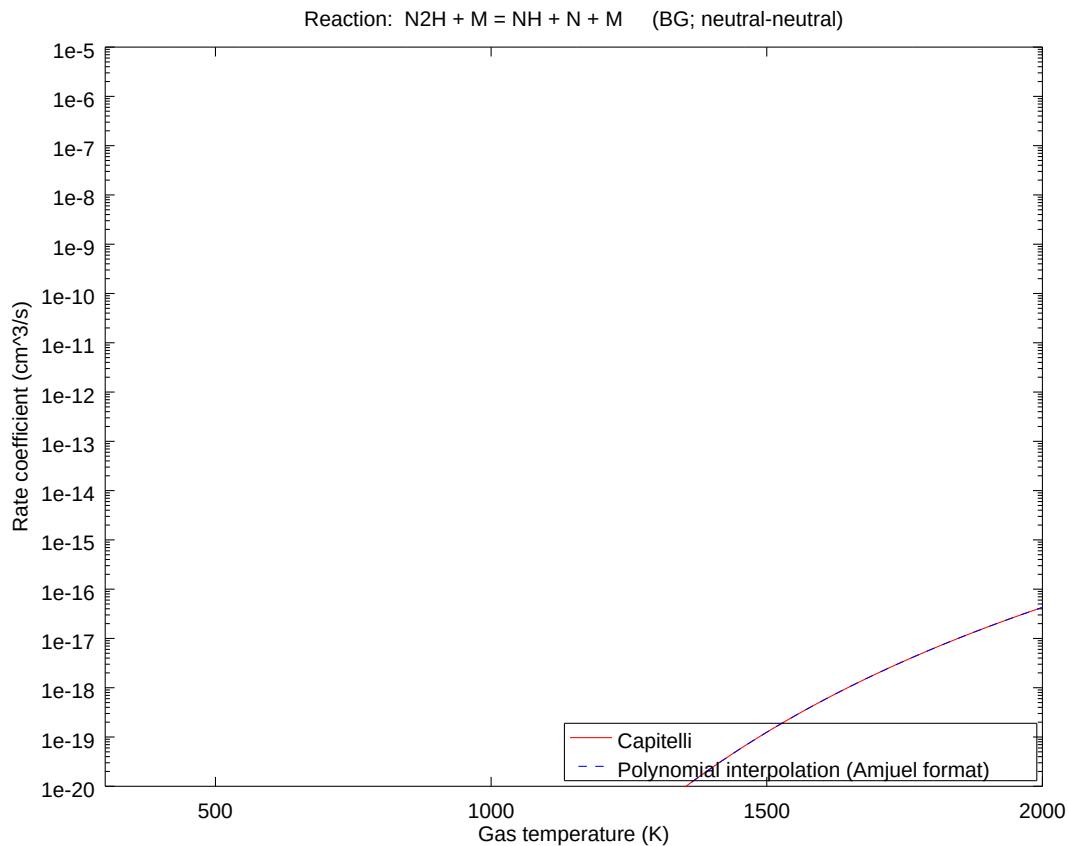
3.17 Reaction BF N2H + M = N2 + H + M

b0	-7.793917689605D+03	b1	6.546430505069D+03	b2	-2.532844278939D+03
b3	5.837163366888D+02	b4	-8.704904249575D+01	b5	8.550473958036D+00
b6	-5.374790625667D-01	b7	1.968423601863D-02	b8	-3.204595860716D-04



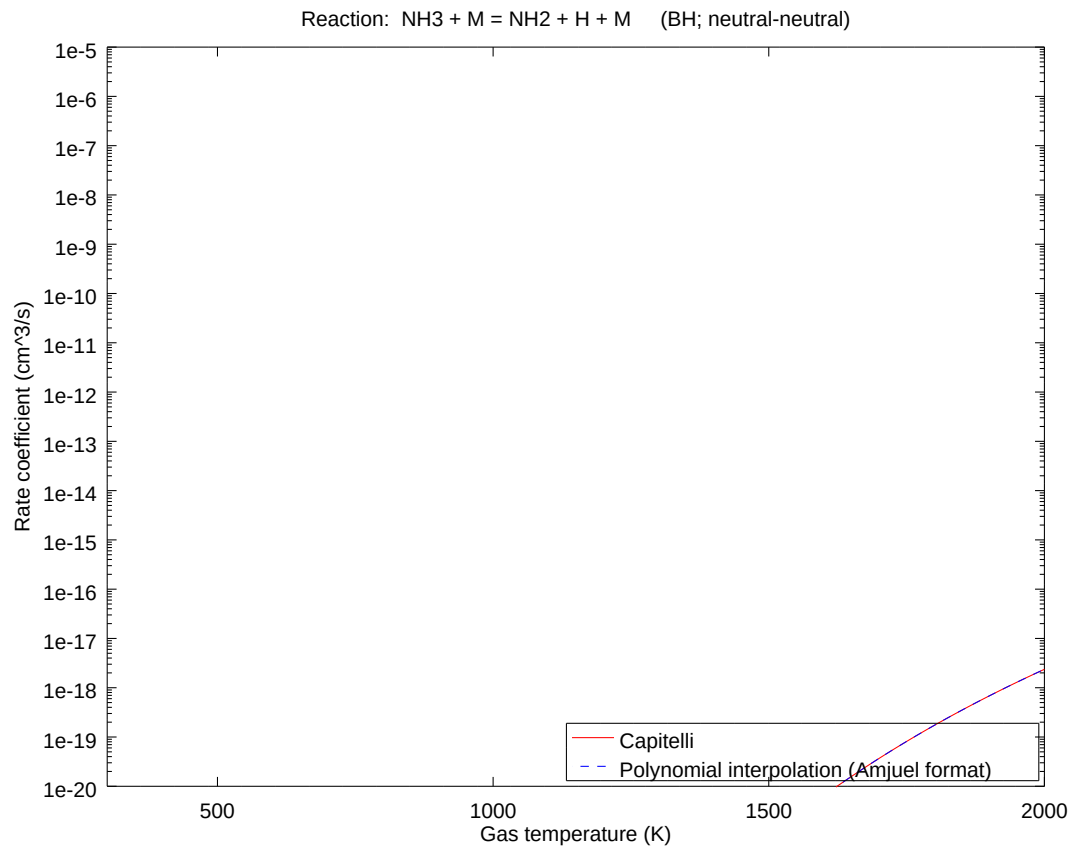
3.18 Reaction BG $\text{N}_2\text{H} + \text{M} = \text{NH} + \text{N} + \text{M}$

b0	-2.722249281401D+04	b1	2.291250678288D+04	b2	-8.864954984340D+03
b3	2.043007180855D+03	b4	-3.046716491980D+02	b5	2.992665890911D+01
b6	-1.881176723208D+00	b7	6.889482624701D-02	b8	-1.121608554668D-03



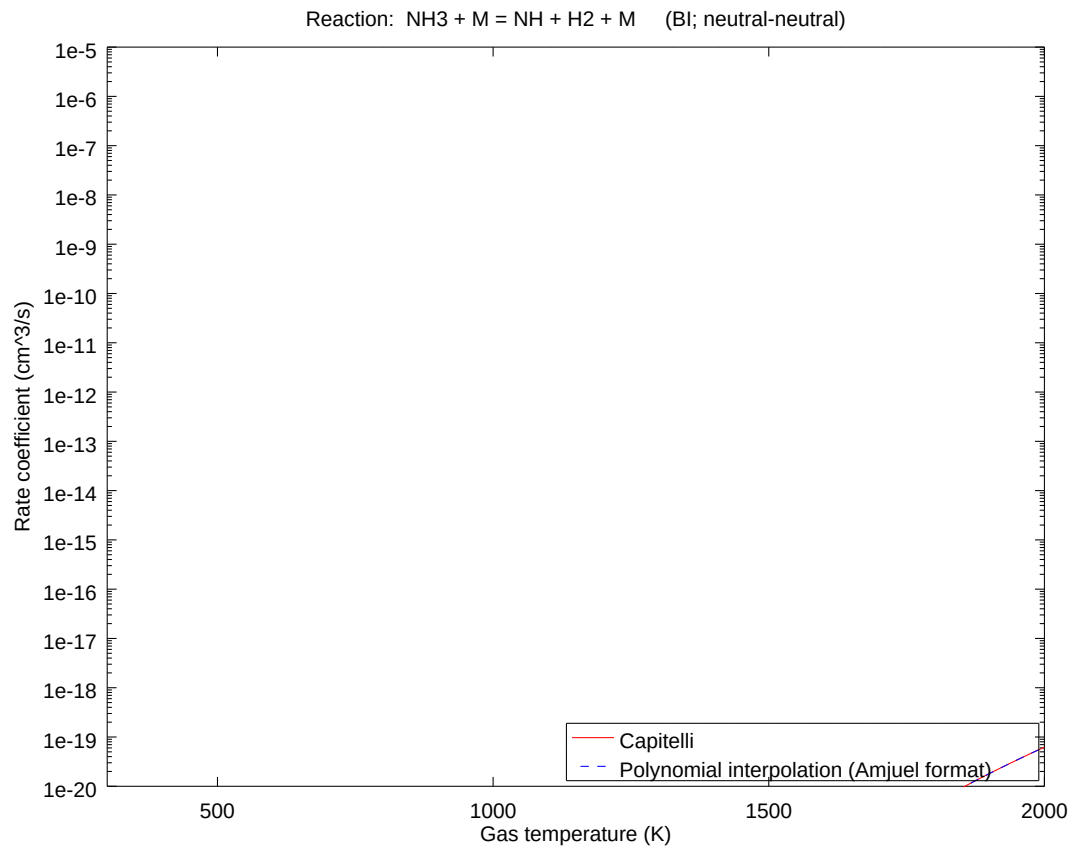
3.19 Reaction BH $\text{NH}_3 + \text{M} = \text{NH}_2 + \text{H} + \text{M}$

b0	-3.670123041776D+04	b1	3.089915201913D+04	b2	-1.195502501530D+04
b3	2.755141114840D+03	b4	-4.108714816522D+02	b5	4.035823721150D+01
b6	-2.536901185086D+00	b7	9.290959442828D-02	b8	-1.512569254158D-03



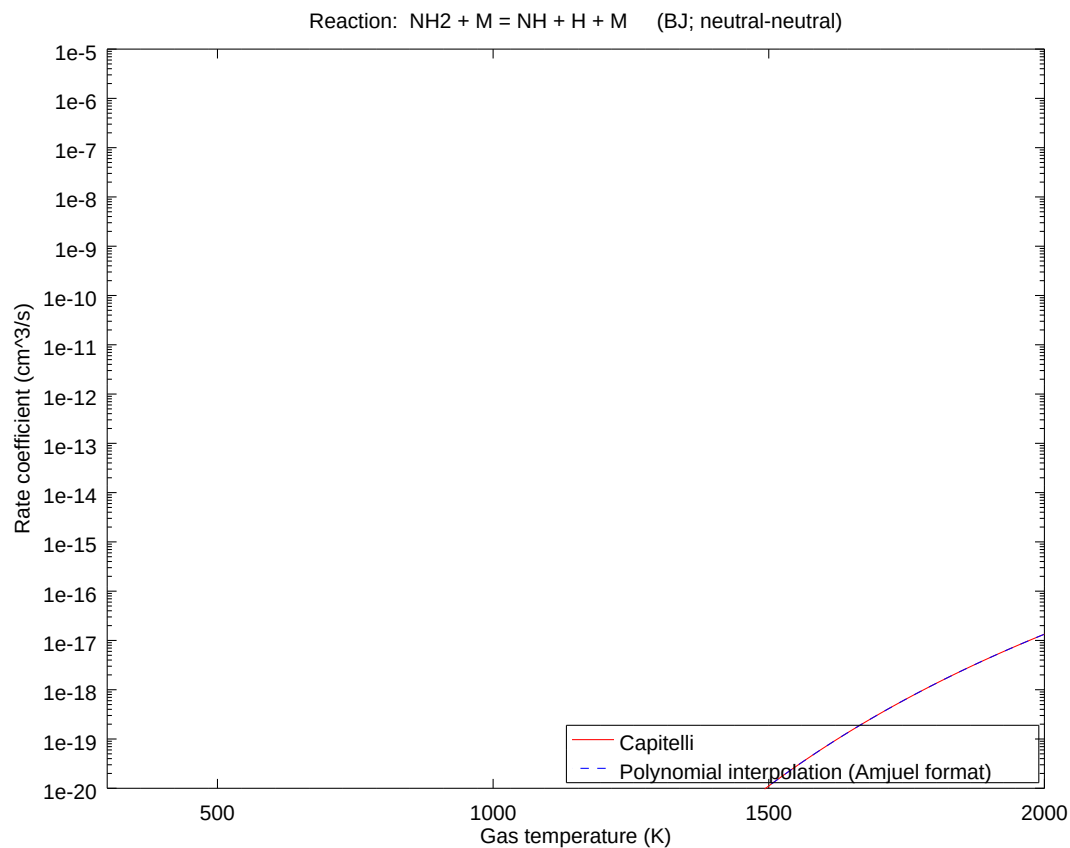
3.20 Reaction BI $\text{NH}_3 + \text{M} = \text{NH} + \text{H}_2 + \text{M}$

b0	-3.654952636521D+04	b1	3.076822340063D+04	b2	-1.190436812528D+04
b3	2.743466786770D+03	b4	-4.091305005509D+02	b5	4.018722770216D+01
b6	-2.526151601572D+00	b7	9.251590961120D-02	b8	-1.506160060630D-03



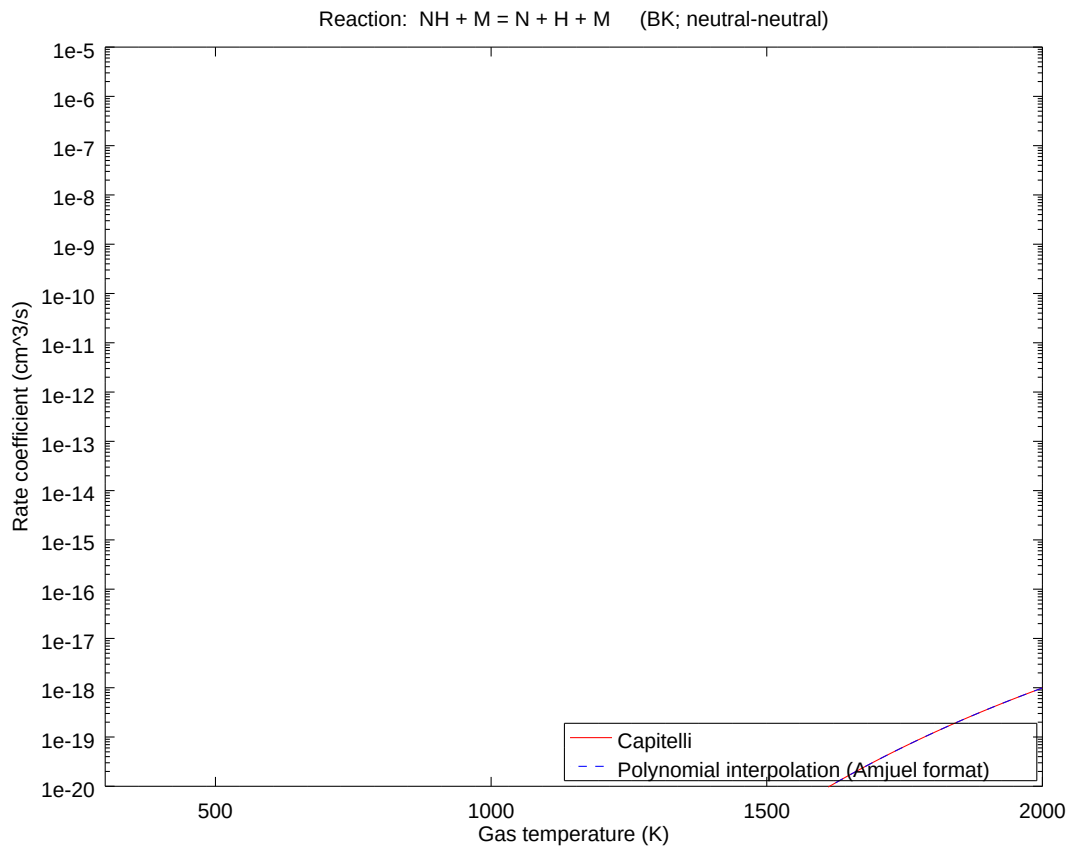
3.21 Reaction BJ $\text{NH}_2 + \text{M} = \text{NH} + \text{H} + \text{M}$

b0	-3.575224461767D+04	b1	3.011158035824D+04	b2	-1.165108370167D+04
b3	2.685095154392D+03	b4	-4.004255965440D+02	b5	3.933218033552D+01
b6	-2.472403697503D+00	b7	9.054748610296D-02	b8	-1.474114103774D-03



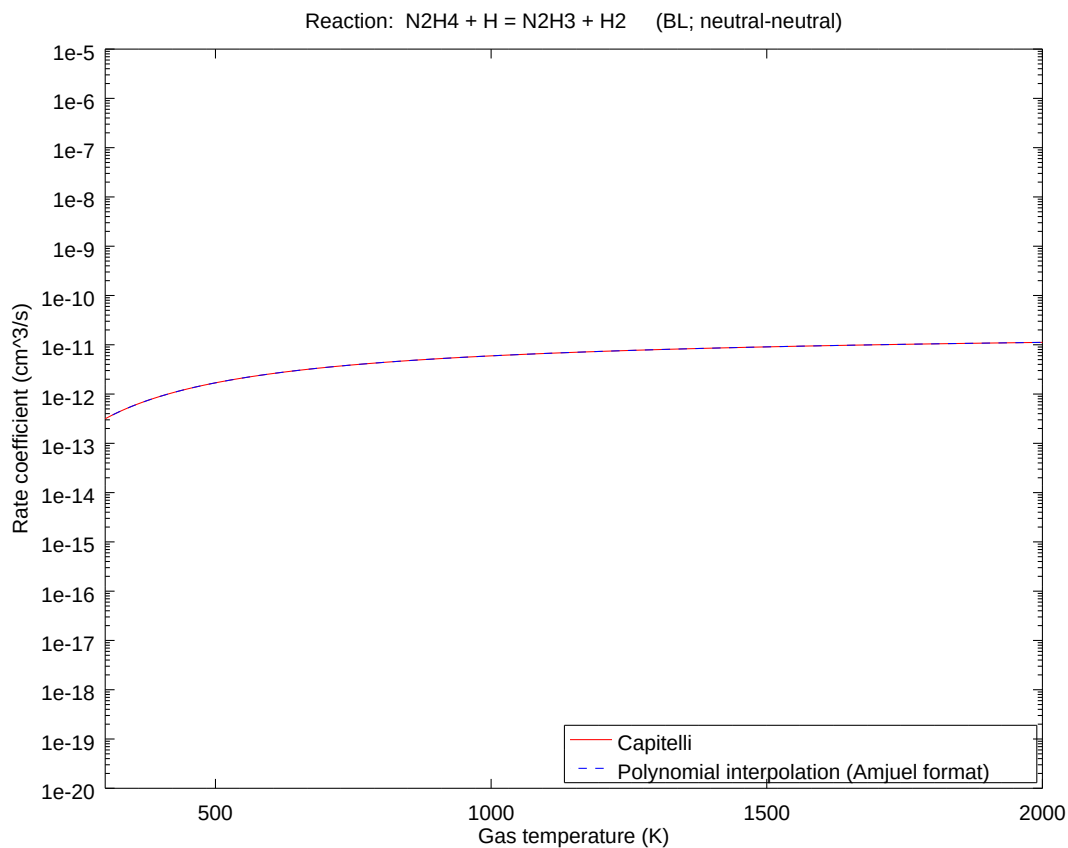
3.22 Reaction BK $\text{NH} + \text{M} = \text{N} + \text{H} + \text{M}$

b0	-3.264801548082D+04	b1	2.749300815308D+04	b2	-1.063794598843D+04
b3	2.451608619212D+03	b4	-3.656059794504D+02	b5	3.591199074075D+01
b6	-2.257412071602D+00	b7	8.267379165770D-02	b8	-1.345930268630D-03



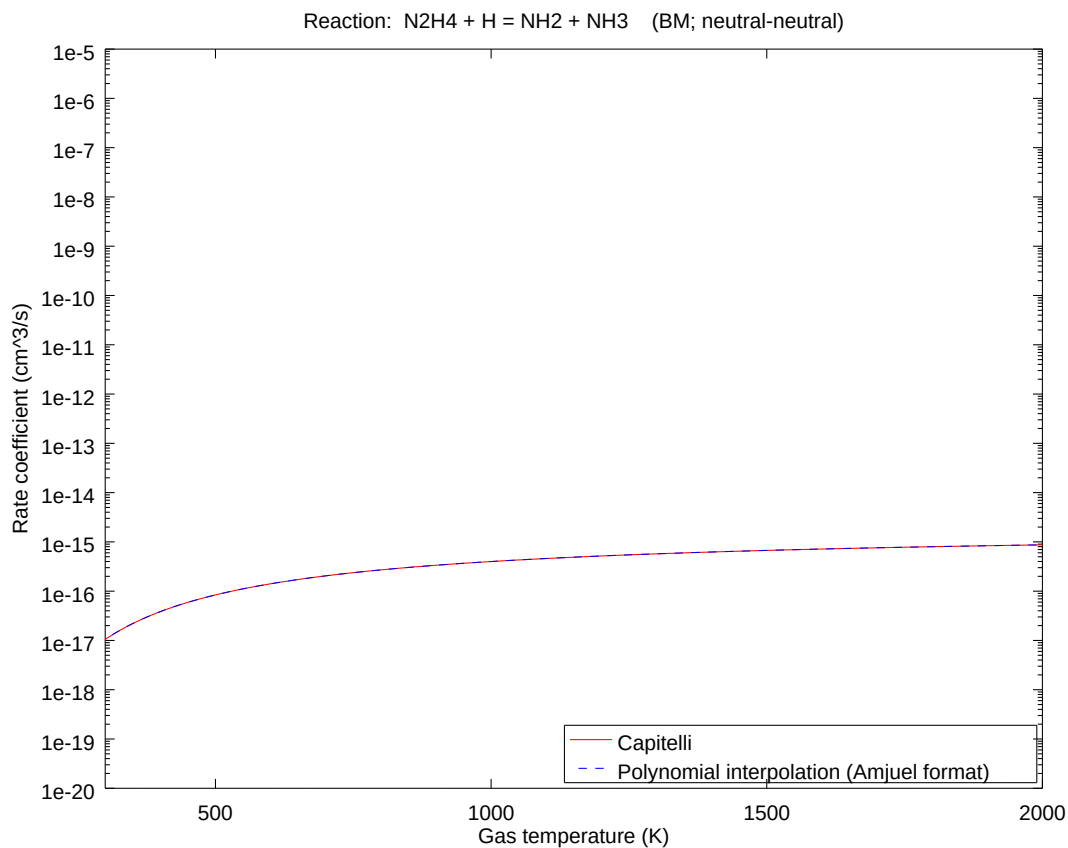
3.23 Reaction BL $\text{N}_2\text{H}_4 + \text{H} = \text{N}_2\text{H}_3 + \text{H}_2$

b0	-1.003869296880D+03	b1	8.248502342394D+02	b2	-3.191383741361D+02
b3	7.354825689922D+01	b4	-1.096817906537D+01	b5	1.077359683663D+00
b6	-6.772235923182D-02	b7	2.480213623904D-03	b8	-4.037790568747D-05



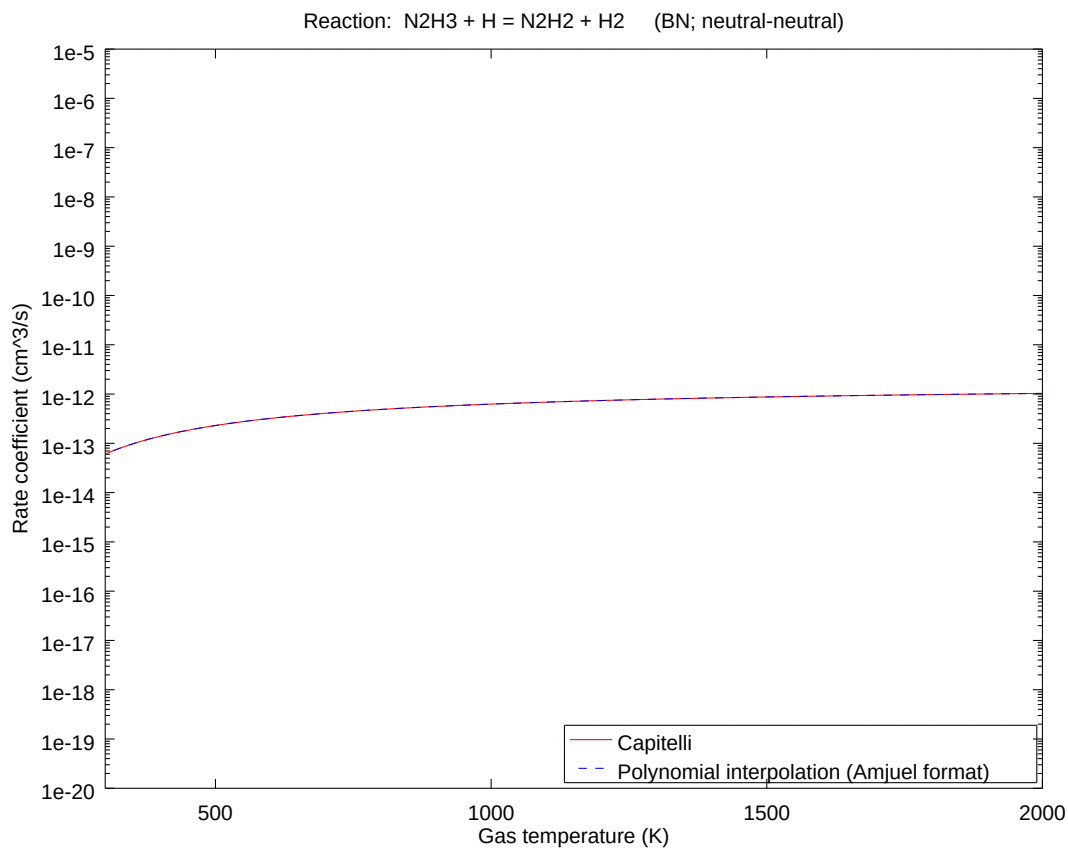
3.24 Reaction BM $\text{N}_2\text{H}_4 + \text{H} = \text{NH}_2 + \text{NH}_3$

b0	-1.246342291660D+03	b1	1.021243147172D+03	b2	-3.951237013637D+02
b3	9.105974666560D+01	b4	-1.357965027914D+01	b5	1.333873895269D+00
b6	-8.384673058920D-02	b7	3.070740682883D-03	b8	-4.999169287833D-05



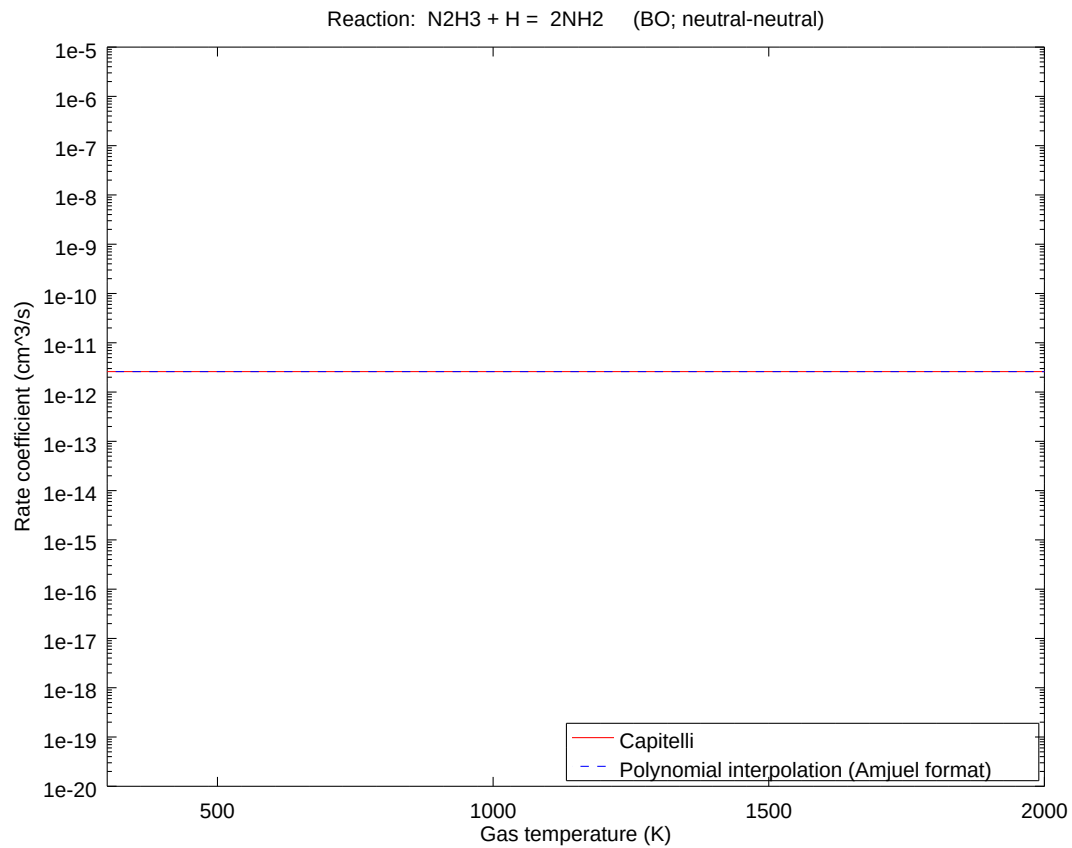
3.25 Reaction BN $\text{N}_2\text{H}_3 + \text{H} = \text{N}_2\text{H}_2 + \text{H}_2$

b0	-8.043089630003D+02	b1	6.546430432365D+02	b2	-2.532844240320D+02
b3	5.837163249848D+01	b4	-8.704904028223D+00	b5	8.550473690520D-01
b6	-5.374790423908D-02	b7	1.968423515043D-03	b8	-3.204595697514D-05



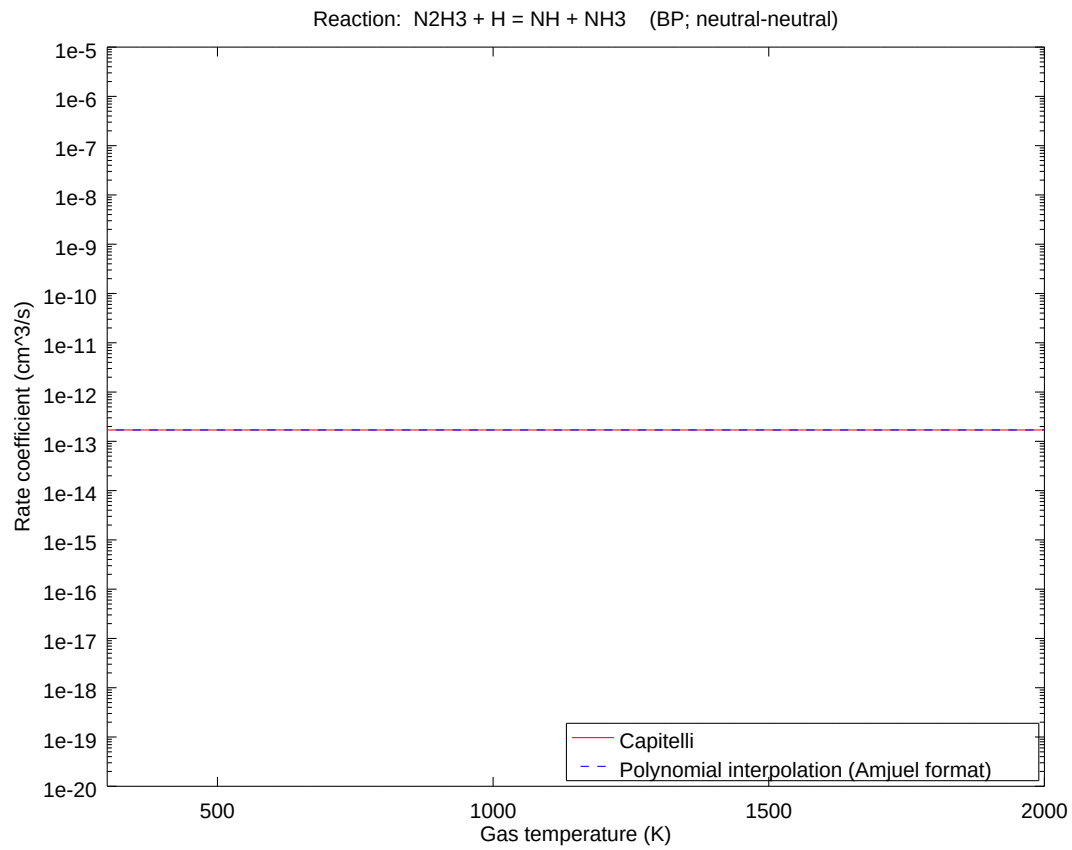
3.26 Reaction BO N2H3 + H = 2NH2

b0	-2.667550348071D+01	b1	-7.535507985259D-06	b2	4.007057470458D-06
b3	-1.215707287704D-06	b4	2.301669701002D-07	b5	-2.784645443528D-08
b6	2.102386708078D-09	b7	-9.056413171192D-11	b8	1.704199645492D-12



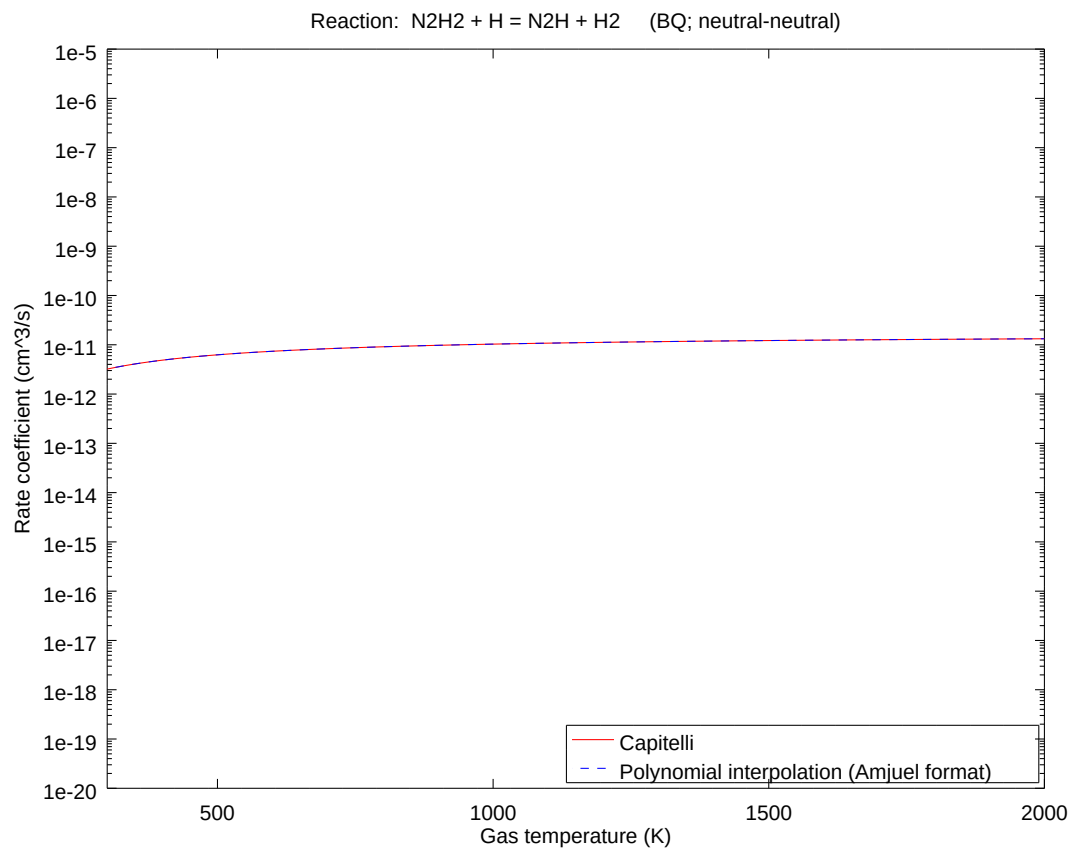
3.27 Reaction BP N2H3 + H = NH + NH3

b0	-2.940296853963D+01	b1	-1.148407191639D-05	b2	6.116559709289D-06
b3	-1.858617149687D-06	b4	3.524238327595D-07	b5	-4.270085866356D-08
b6	3.228551697330D-09	b7	-1.392722732081D-10	b8	2.624386667546D-12



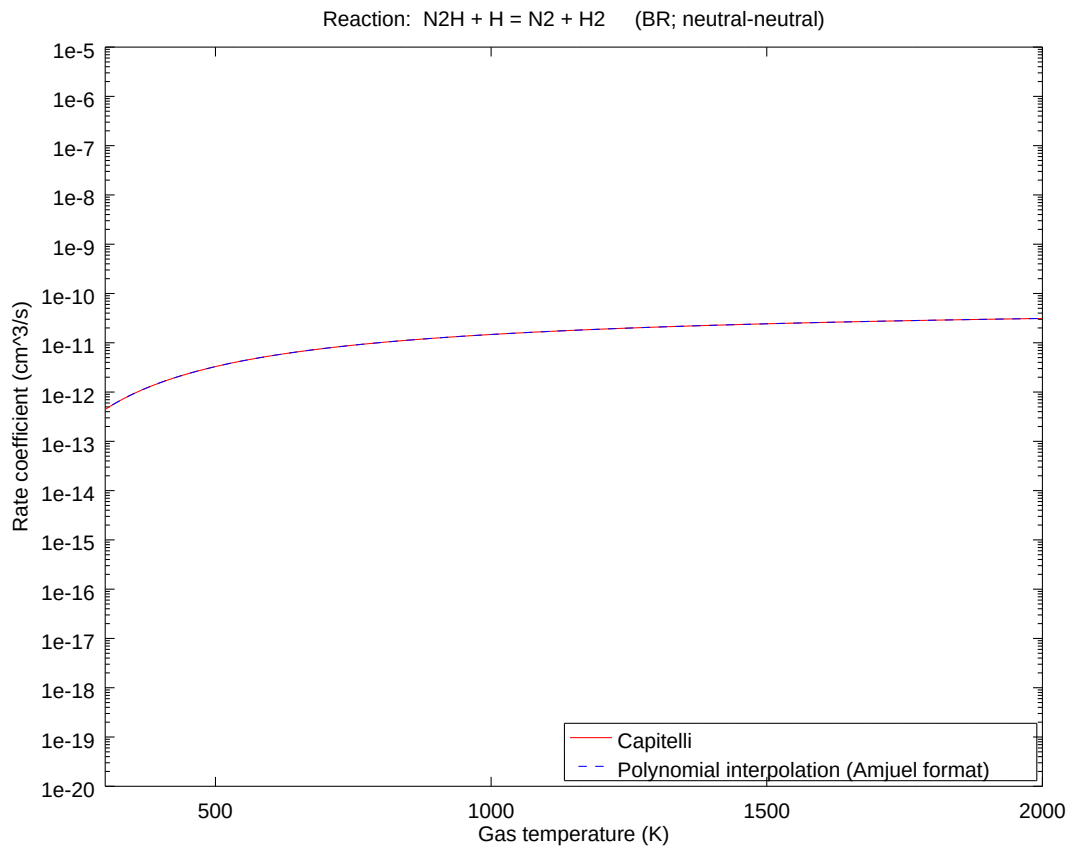
3.28 Reaction BQ $\text{N}_2\text{H}_2 + \text{H} = \text{N}_2\text{H} + \text{H}_2$

b0	-4.134020892313D+02	b1	3.273215172291D+02	b2	-1.266422096838D+02
b3	2.918581554216D+01	b4	-4.352451880335D+00	b5	4.275236683522D-01
b6	-2.687395089923D-02	b7	9.842117049880D-04	b8	-1.602297749962D-05



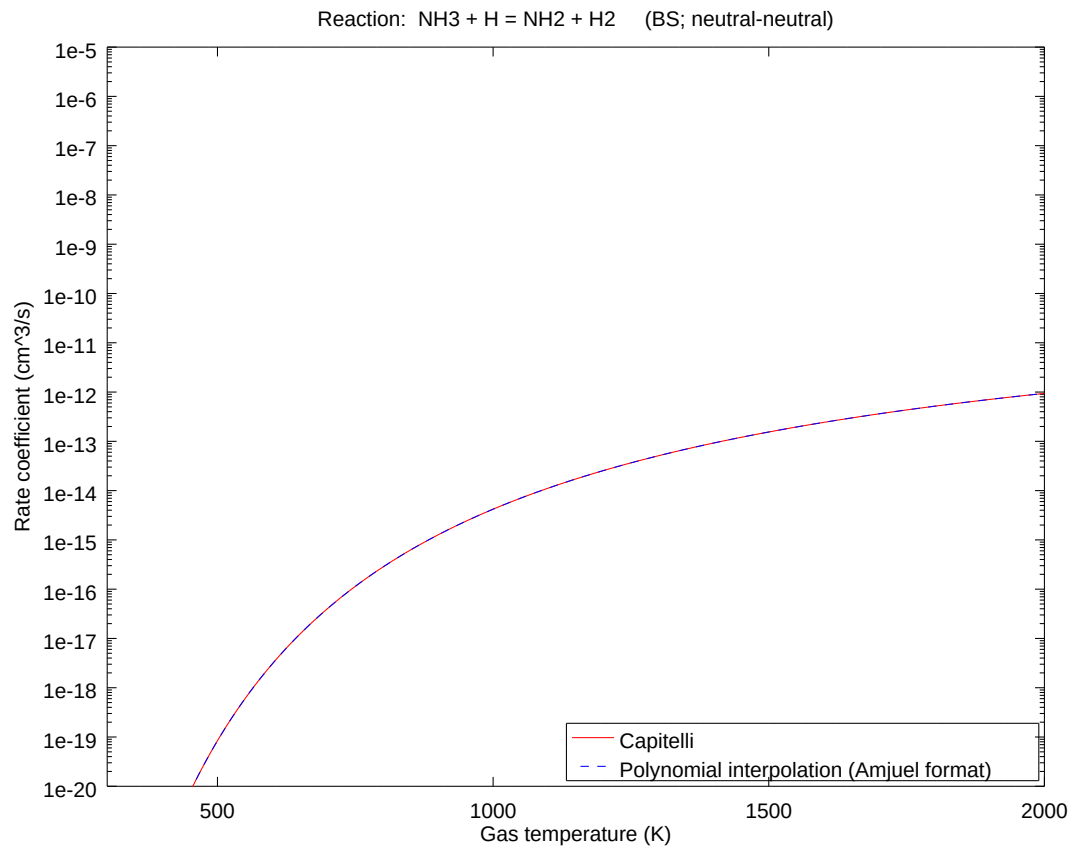
3.29 Reaction BR N2H + H = N2 + H2

b0	-1.189254224343D+03	b1	9.819645682089D+02	b2	-3.799266378243D+02
b3	8.755744928446D+01	b4	-1.305735614355D+01	b5	1.282571065774D+00
b6	-8.062185727567D-02	b7	2.952635311909D-03	b8	-4.806893620008D-05



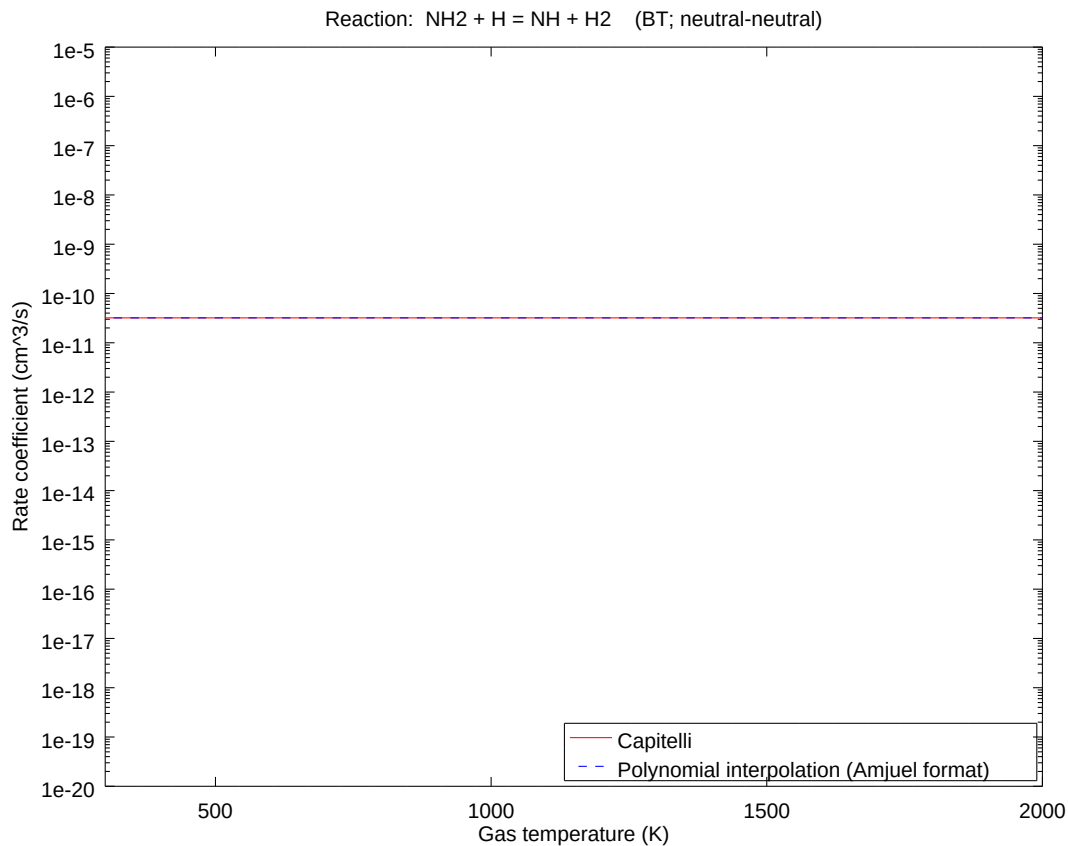
3.30 Reaction BS $\text{NH}_3 + \text{H} = \text{NH}_2 + \text{H}_2$

b0	-8.431680709526D+03	b1	7.083237809384D+03	b2	-2.740537511352D+03
b3	6.315810767637D+02	b4	-9.418706406853D+01	b5	9.251612833233D+00
b6	-5.815523464982D-01	b7	2.129834340656D-02	b8	-3.467372727748D-04



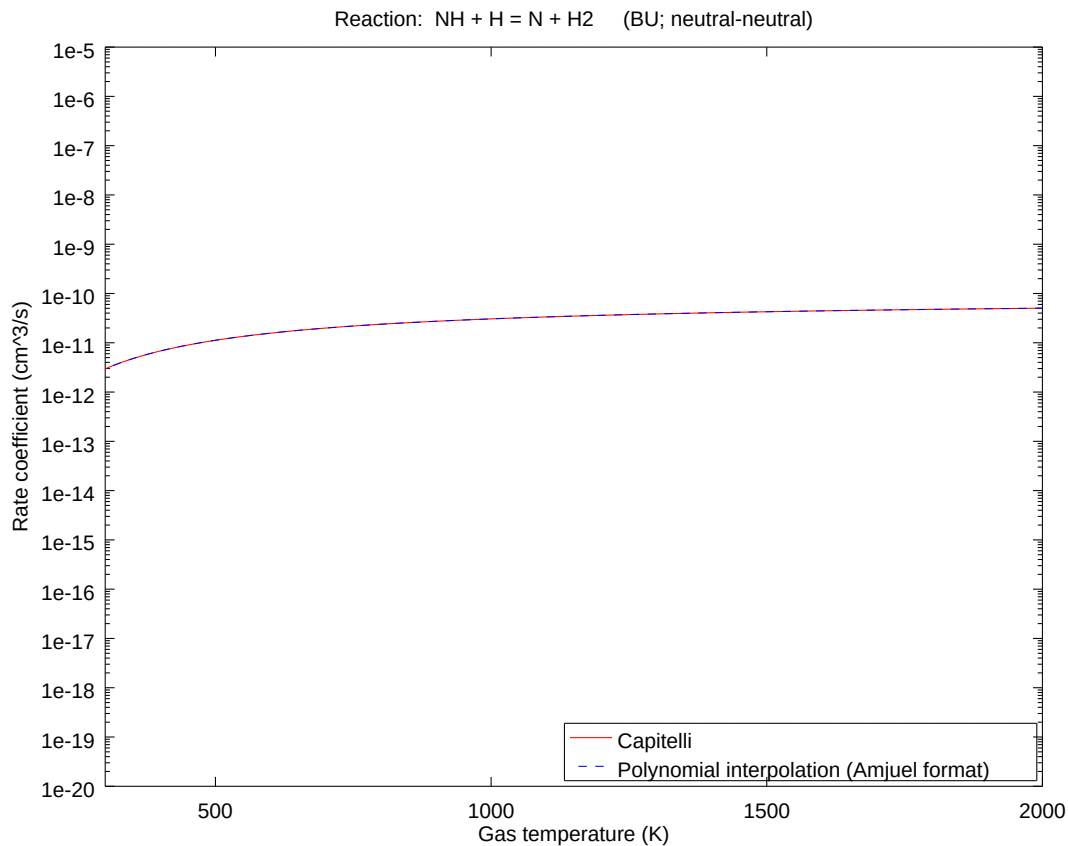
3.31 Reaction BT NH₂ + H = NH + H₂

b0	-2.416527628652D+01	b1	-1.084444335523D-05	b2	5.754794312628D-06
b3	-1.742367506413D-06	b4	3.291982926954D-07	b5	-3.974533062944D-08
b6	2.994531446955D-09	b7	-1.287276274913D-10	b8	2.417326629989D-12



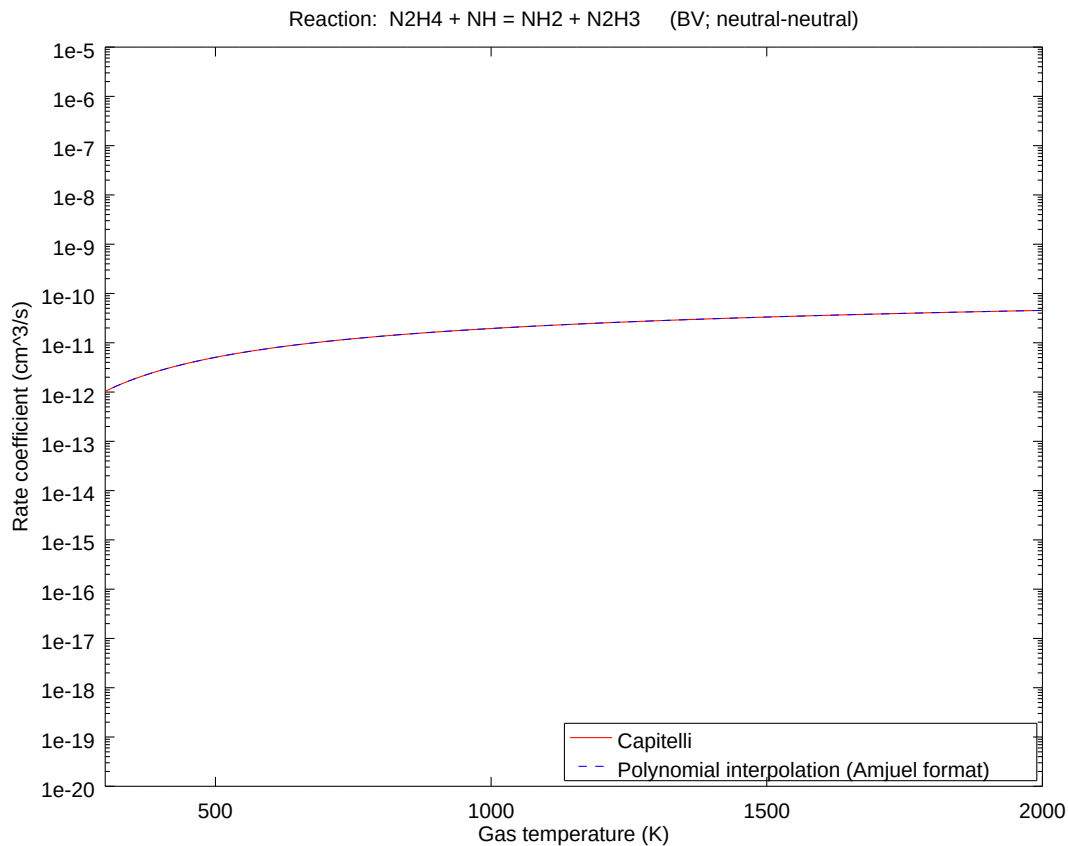
3.32 Reaction BU NH + H = N + H2

b0	-8.004207520405D+02	b1	6.546430449158D+02	b2	-2.532844249140D+02
b3	5.837163276286D+01	b4	-8.704904077691D+00	b5	8.550473749685D-01
b6	-5.374790468079D-02	b7	1.968423533864D-03	b8	-3.204595732555D-05



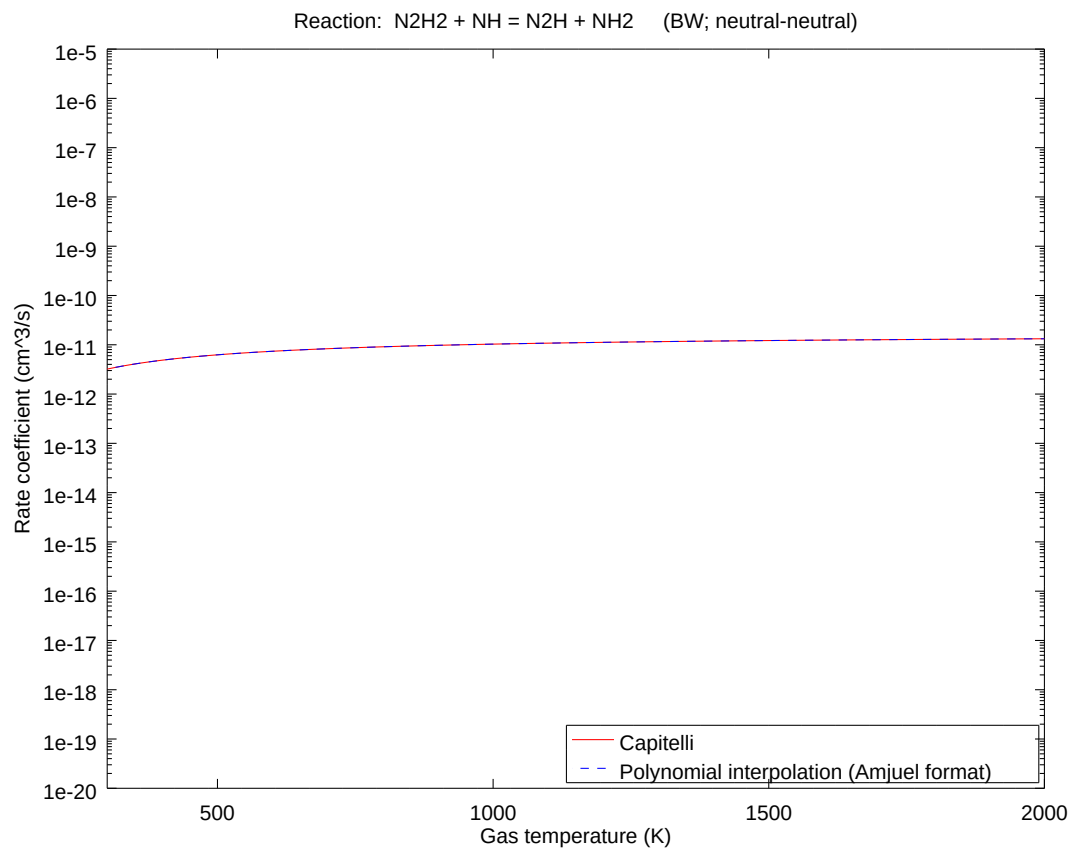
3.33 Reaction BV $\text{N}_2\text{H}_4 + \text{NH} = \text{NH}_2 + \text{N}_2\text{H}_3$

b0	-8.043241875986D+02	b1	6.551430443518D+02	b2	-2.532844246101D+02
b3	5.837163266942D+01	b4	-8.704904059764D+00	b5	8.550473727708D-01
b6	-5.374790451267D-02	b7	1.968423526526D-03	b8	-3.204595718567D-05



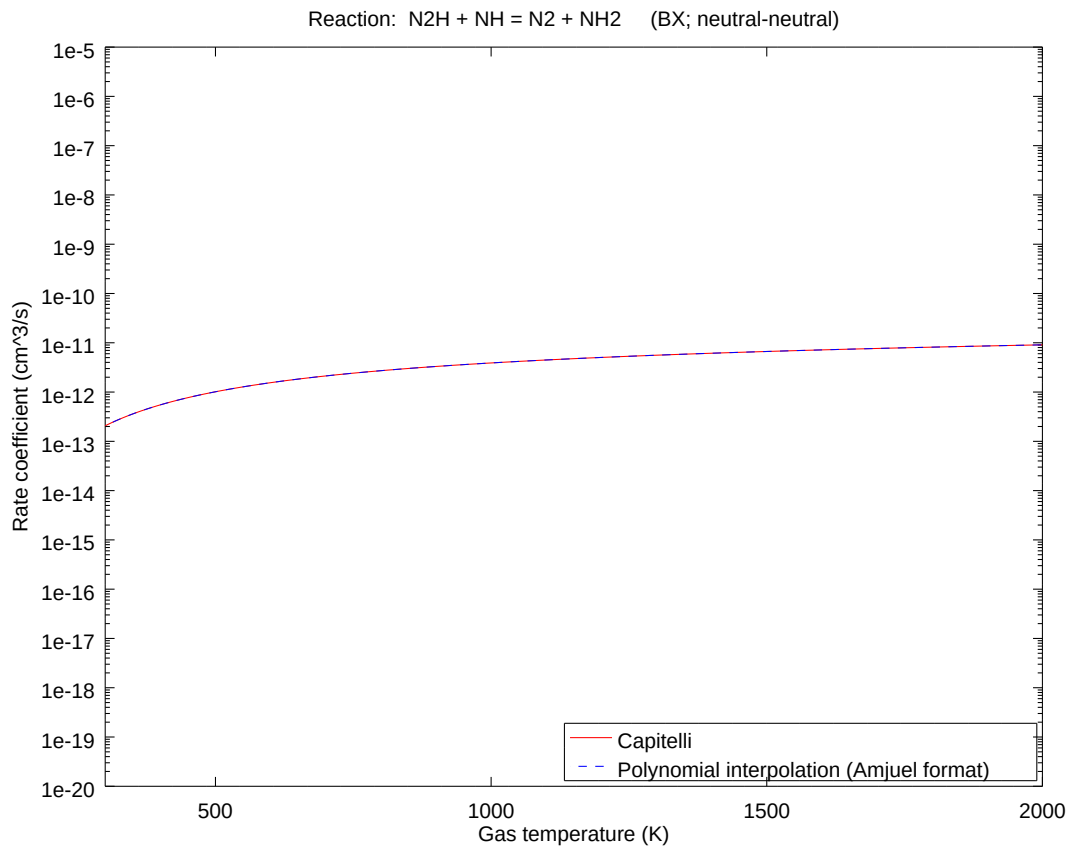
3.34 Reaction BW $\text{N}_2\text{H}_2 + \text{NH} = \text{N}_2\text{H} + \text{NH}_2$

b0	-4.134020892313D+02	b1	3.273215172291D+02	b2	-1.266422096838D+02
b3	2.918581554216D+01	b4	-4.352451880335D+00	b5	4.275236683522D-01
b6	-2.687395089923D-02	b7	9.842117049880D-04	b8	-1.602297749962D-05



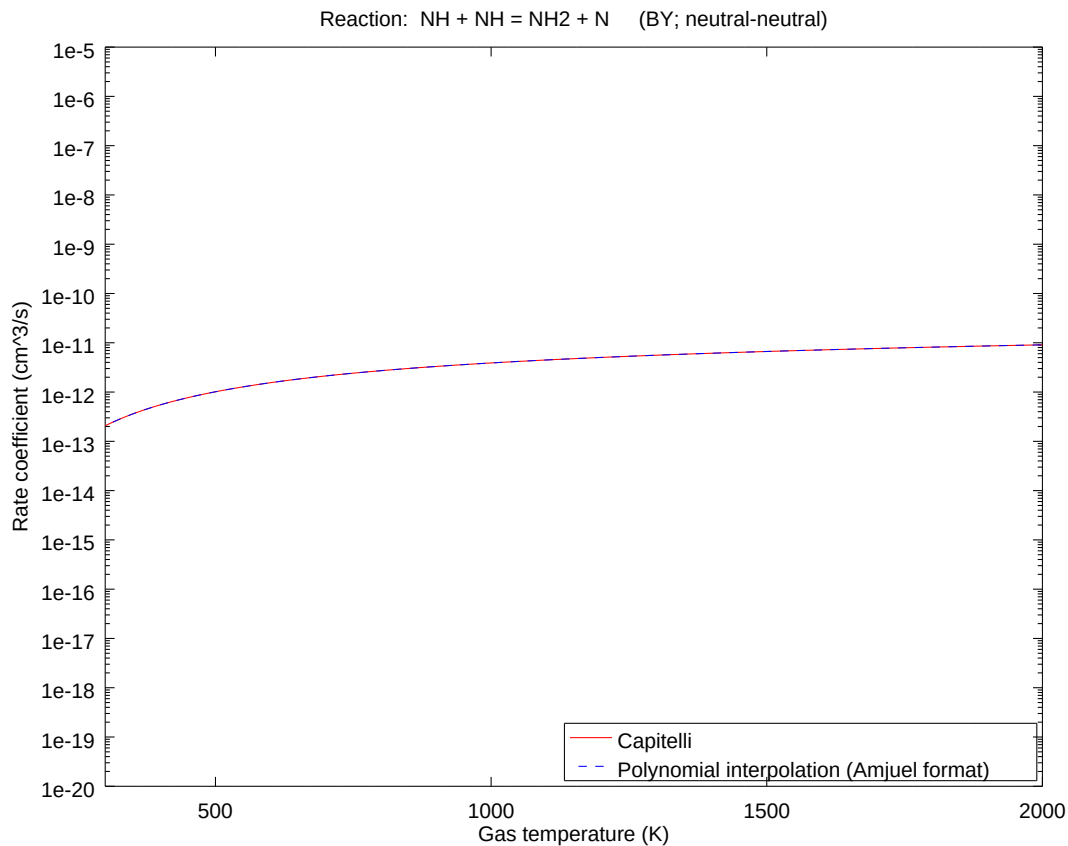
3.35 Reaction BX $\text{N}_2\text{H} + \text{NH} = \text{N}_2 + \text{NH}_2$

b0	-8.059336240690D+02	b1	6.551430426257D+02	b2	-2.532844237080D+02
b3	5.837163240060D+01	b4	-8.704904009803D+00	b5	8.550473668406D-01
b6	-5.374790407366D-02	b7	1.968423507994D-03	b8	-3.204595684411D-05



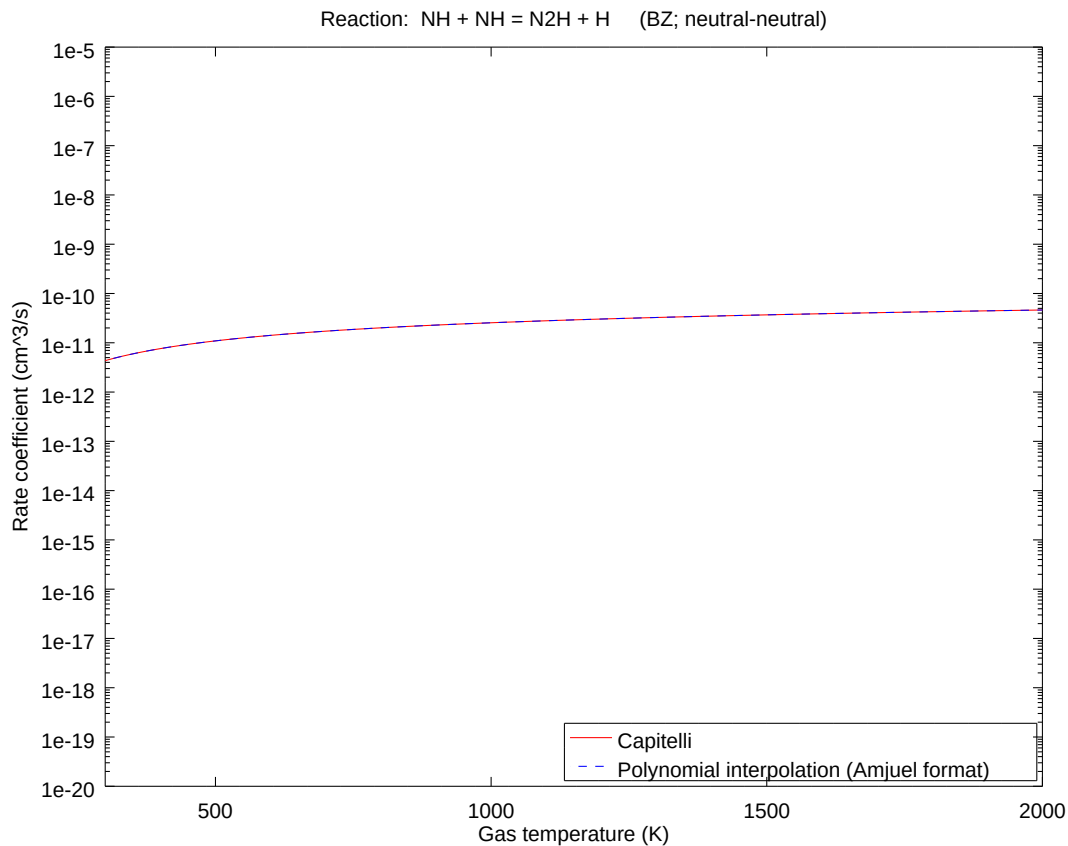
3.36 Reaction BY $\text{NH} + \text{NH} = \text{NH}_2 + \text{N}$

b0	-8.059336240690D+02	b1	6.551430426257D+02	b2	-2.532844237080D+02
b3	5.837163240060D+01	b4	-8.704904009803D+00	b5	8.550473668406D-01
b6	-5.374790407366D-02	b7	1.968423507994D-03	b8	-3.204595684411D-05



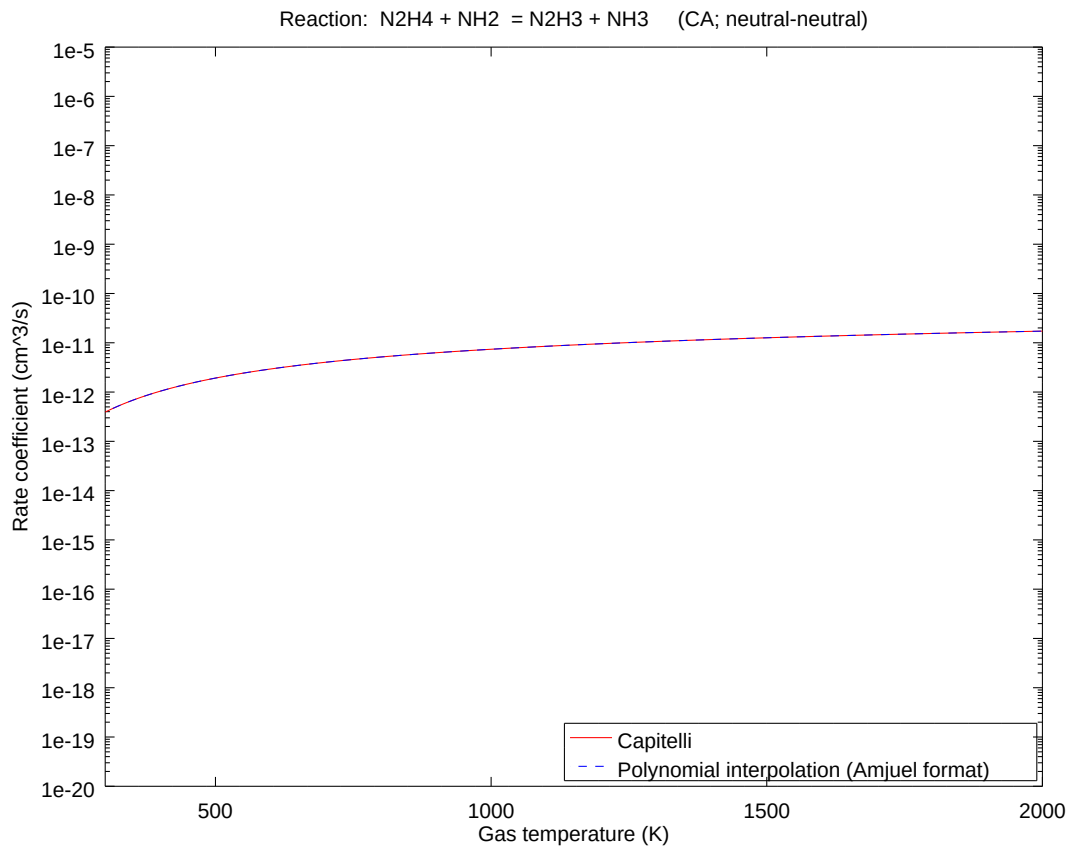
3.37 Reaction BZ $\text{NH} + \text{NH} = \text{N}_2\text{H} + \text{H}$

b0	-4.159517014478D+02	b1	3.278215194776D+02	b2	-1.266422108770D+02
b3	2.918581590350D+01	b4	-4.352451948630D+00	b5	4.275236766019D-01
b6	-2.687395152120D-02	b7	9.842117317469D-04	b8	-1.602297800261D-05



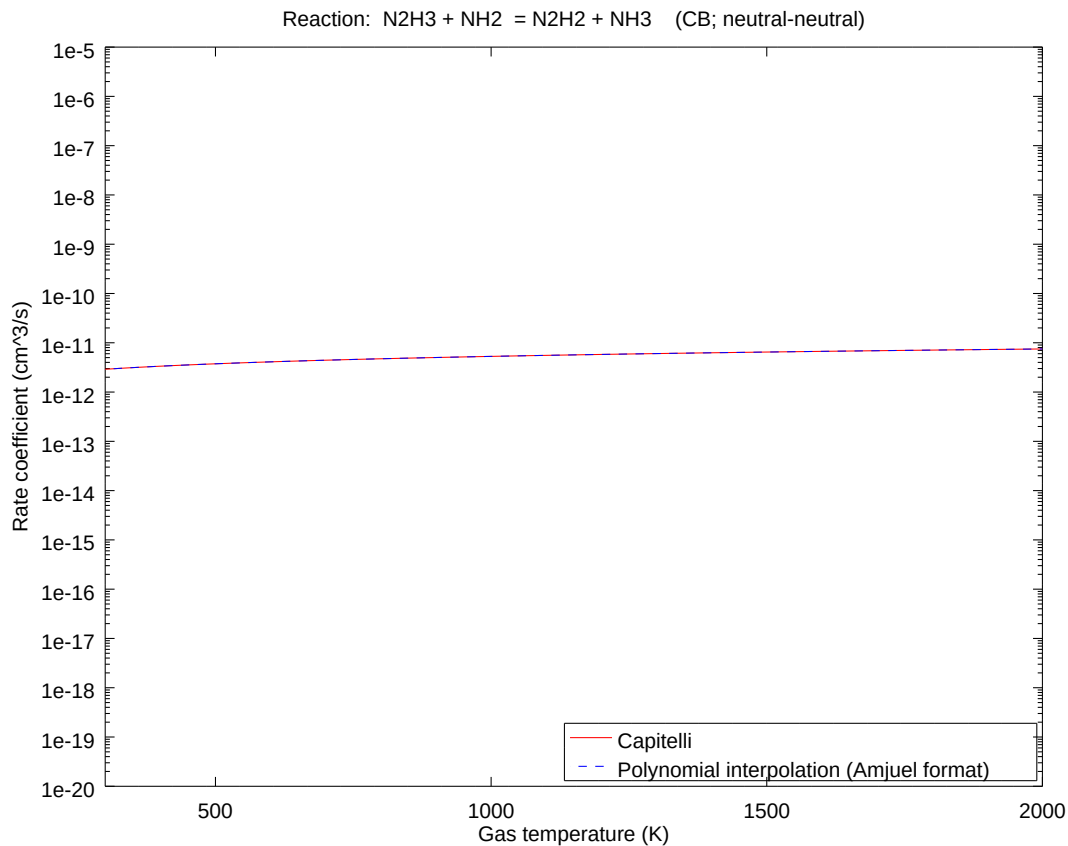
3.38 Reaction CA $\text{N}_2\text{H}_4 + \text{NH}_2 = \text{N}_2\text{H}_3 + \text{NH}_3$

b0	-8.052935865445D+02	b1	6.551430424173D+02	b2	-2.532844235951D+02
b3	5.837163236547D+01	b4	-8.704904002943D+00	b5	8.550473659801D-01
b6	-5.374790400602D-02	b7	1.968423504950D-03	b8	-3.204595678410D-05



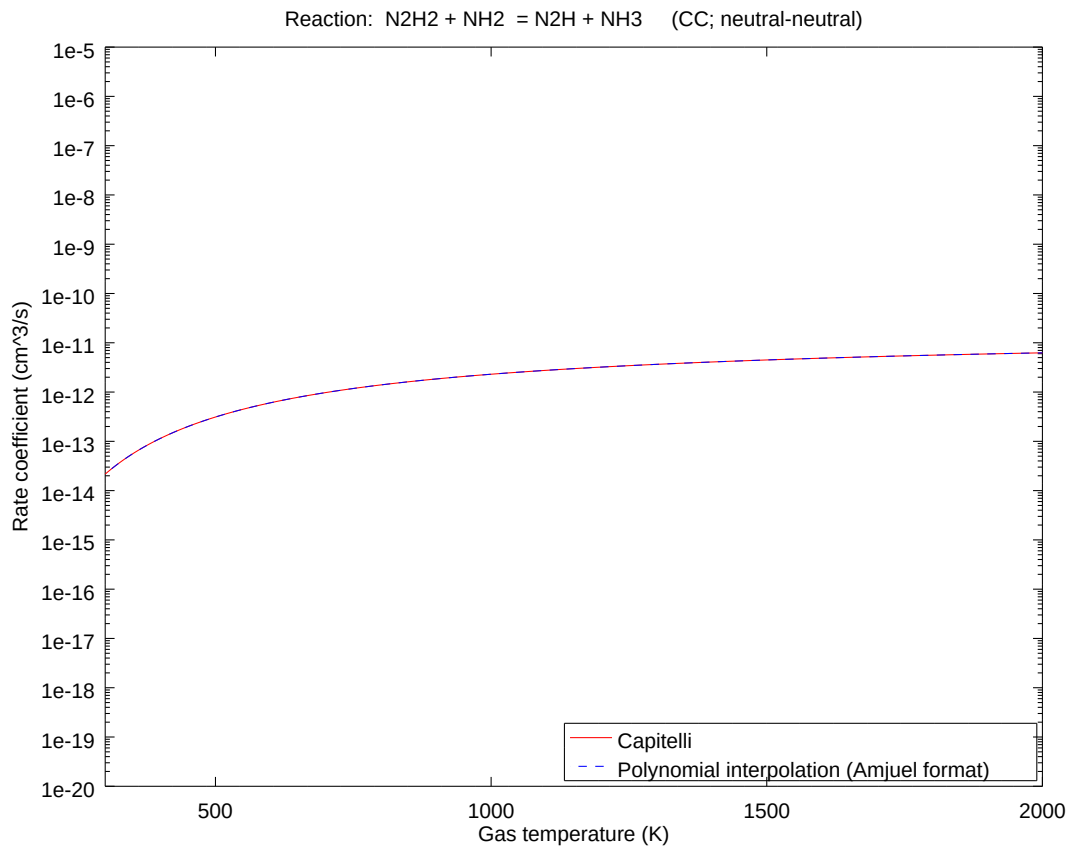
3.39 Reaction CB $\text{N}_2\text{H}_3 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{NH}_3$

b0	-2.941819276064D+01	b1	4.999892334092D-01	b2	5.717953260626D-06
b3	-1.732578702760D-06	b4	3.276094104984D-07	b5	-3.958532522557D-08
b6	2.984904934135D-09	b7	-1.284196713694D-10	b8	2.413561076059D-12



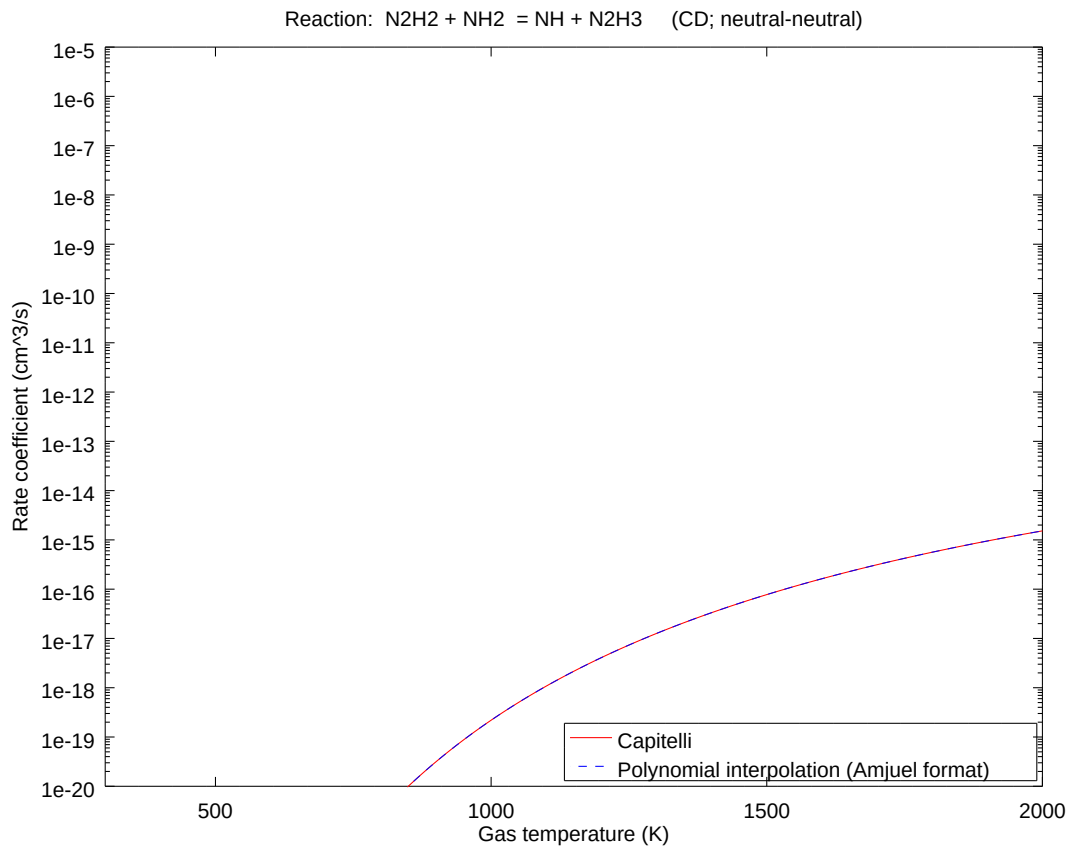
3.40 Reaction CC N2H2 + NH2 = N2H + NH3

b0	-1.579214953135D+03	b1	1.309286092655D+03	b2	-5.065688513420D+02
b3	1.167432659888D+02	b4	-1.740980824371D+01	b5	1.710094760698D+00
b6	-1.074958101793D-01	b7	3.936847103160D-03	b8	-6.409191532148D-05



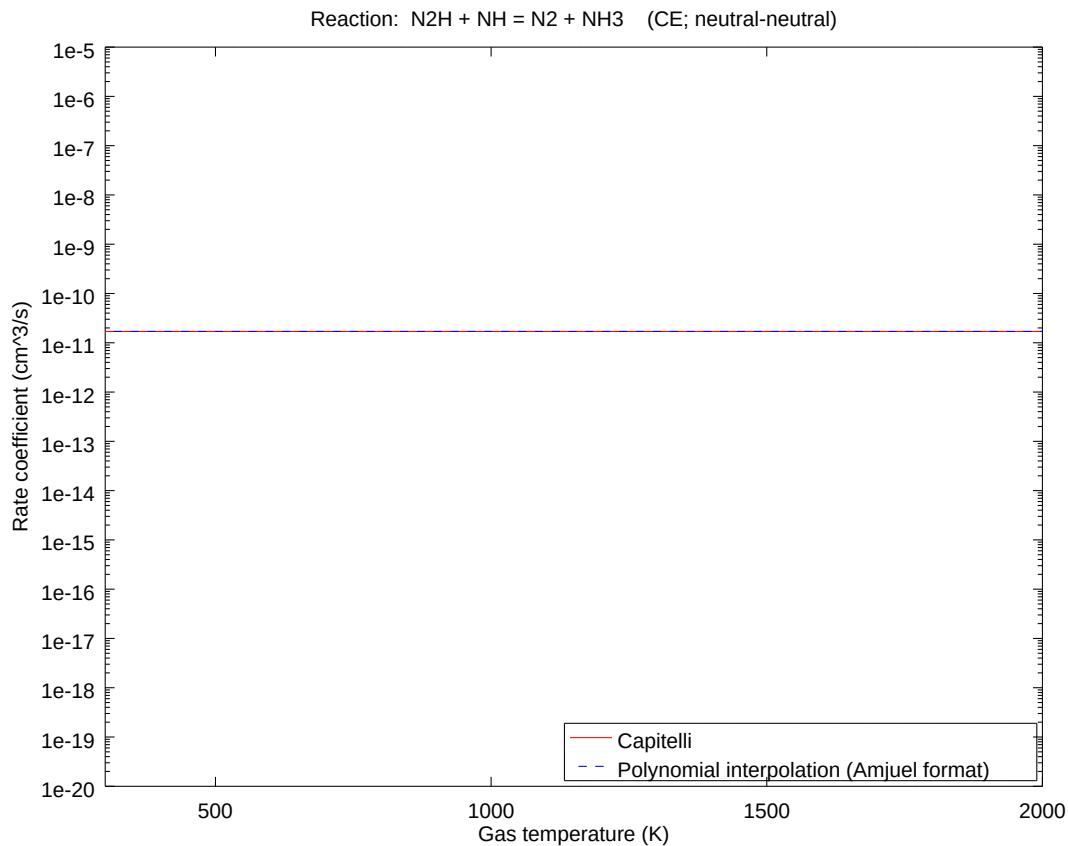
3.41 Reaction CD $\text{N}_2\text{H}_2 + \text{NH}_2 = \text{NH} + \text{N}_2\text{H}_3$

b0	-1.324196399778D+04	b1	1.112943186135D+04	b2	-4.305835275658D+03
b3	9.923177728176D+02	b4	-1.479833723279D+02	b5	1.453580573903D+01
b6	-9.137144071509D-01	b7	3.346320126579D-02	b8	-5.447812969673D-04



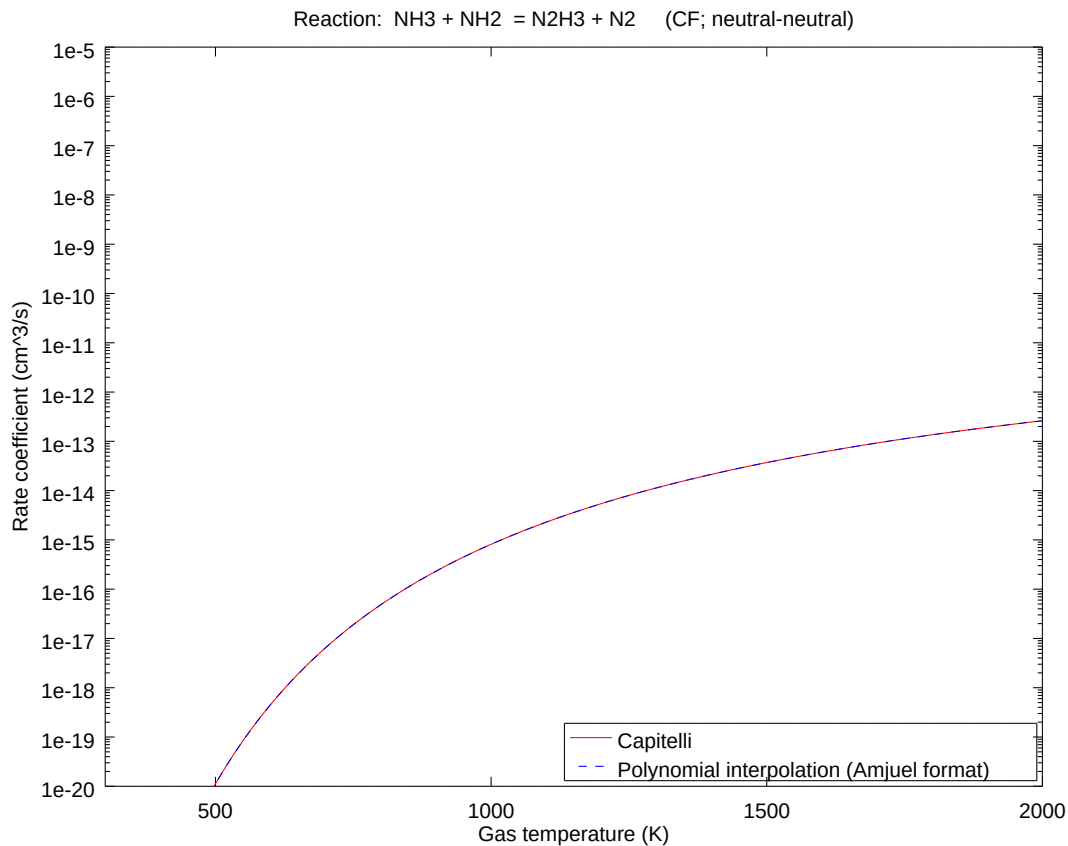
3.42 Reaction CE $\text{N}_2\text{H} + \text{NH} = \text{N}_2 + \text{NH}_3$

b0	-2.479779951964D+01	b1	-1.004140070841D-05	b2	5.337152282509D-06
b3	-1.618467303331D-06	b4	3.062659224950D-07	b5	-3.703373625214D-08
b6	2.794503657053D-09	b7	-1.203113524340D-10	b8	2.262686144425D-12



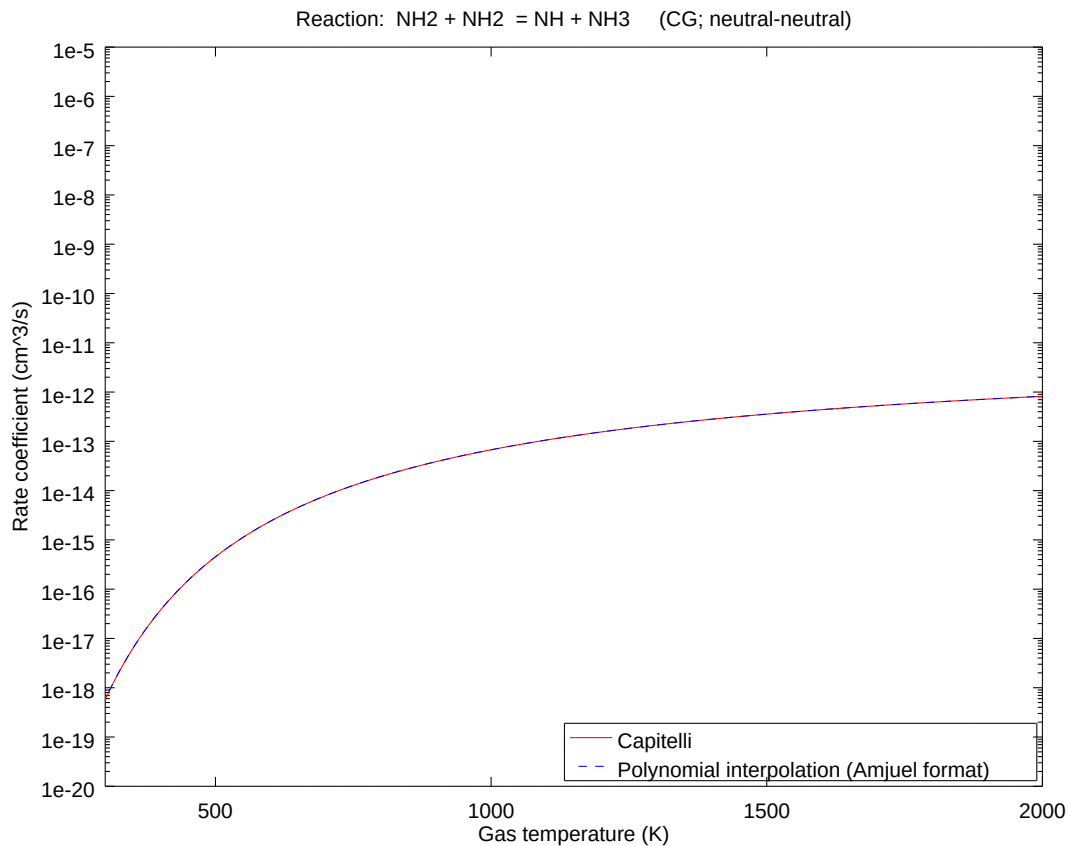
3.43 Reaction CF NH3 + NH2 = N2H3 + N2

b0	-8.460060472427D+03	b1	7.103377102199D+03	b2	-2.748136044882D+03
b3	6.333322259846D+02	b4	-9.444821123606D+01	b5	9.277264259965D+00
b6	-5.831647840537D-01	b7	2.135739613050D-02	b8	-3.476986518327D-04



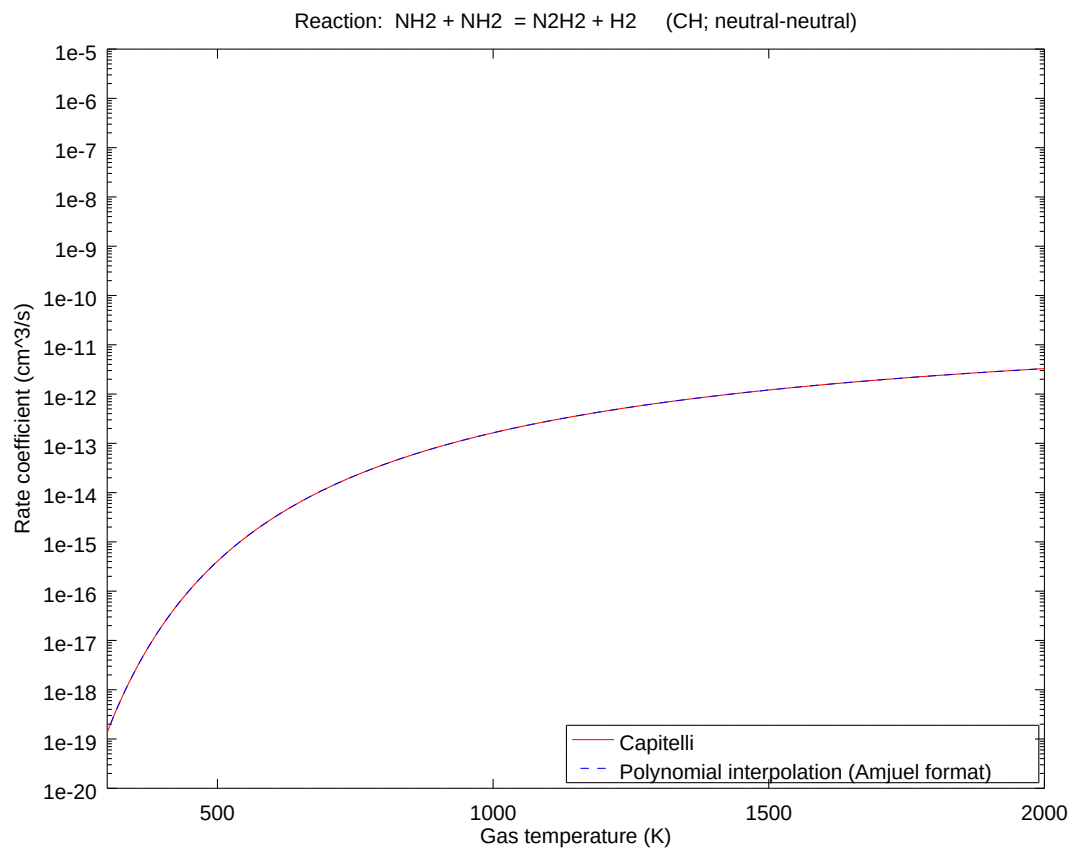
3.44 Reaction CG $\text{NH}_2 + \text{NH}_2 = \text{NH} + \text{NH}_3$

b0	-3.911371311308D+03	b1	3.273215246151D+03	b2	-1.266422136102D+03
b3	2.918581673305D+02	b4	-4.352452105736D+01	b5	4.275236956138D+00
b6	-2.687395295684D-01	b7	9.842117935957D-03	b8	-1.602297916648D-04



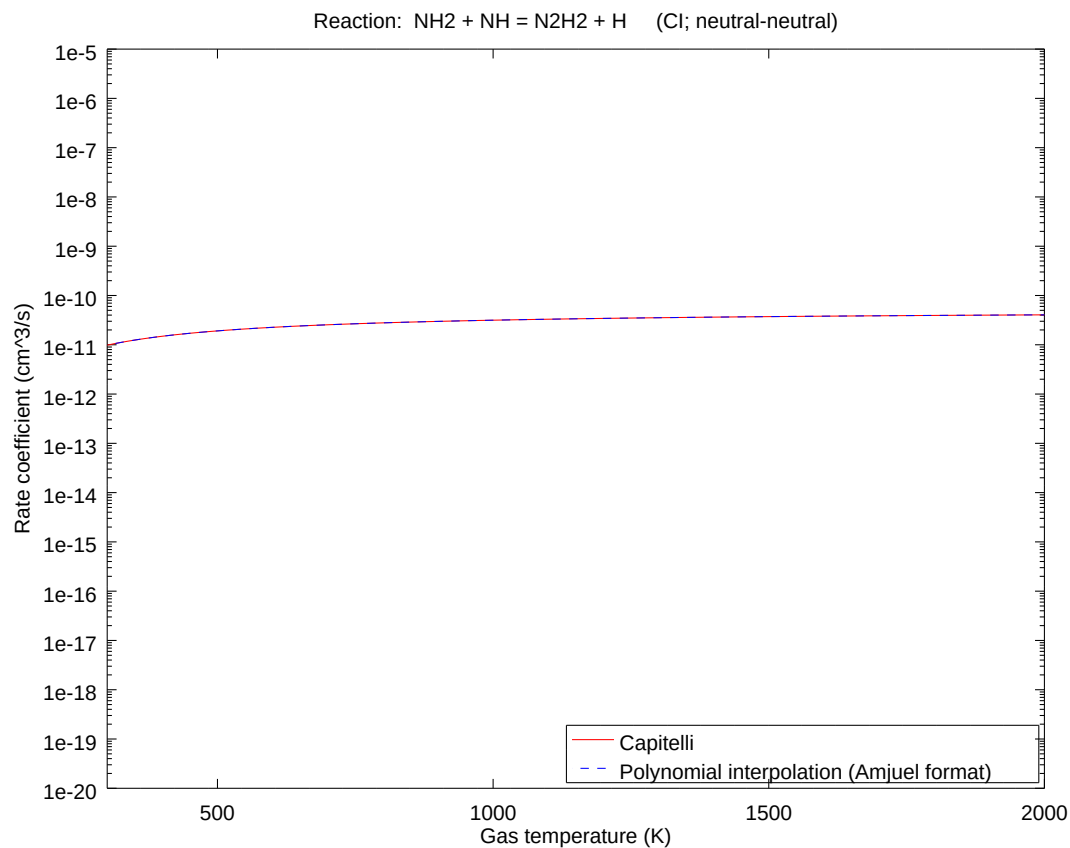
3.45 Reaction CH NH2 + NH2 = N2H2 + H2

b0	-4.686692819097D+03	b1	3.927858298238D+03	b2	-1.519706564820D+03
b3	3.502298012451D+02	b4	-5.222942535265D+01	b5	5.130284357378D+00
b6	-3.224874362286D-01	b7	1.181054155492D-02	b8	-1.922757505887D-04



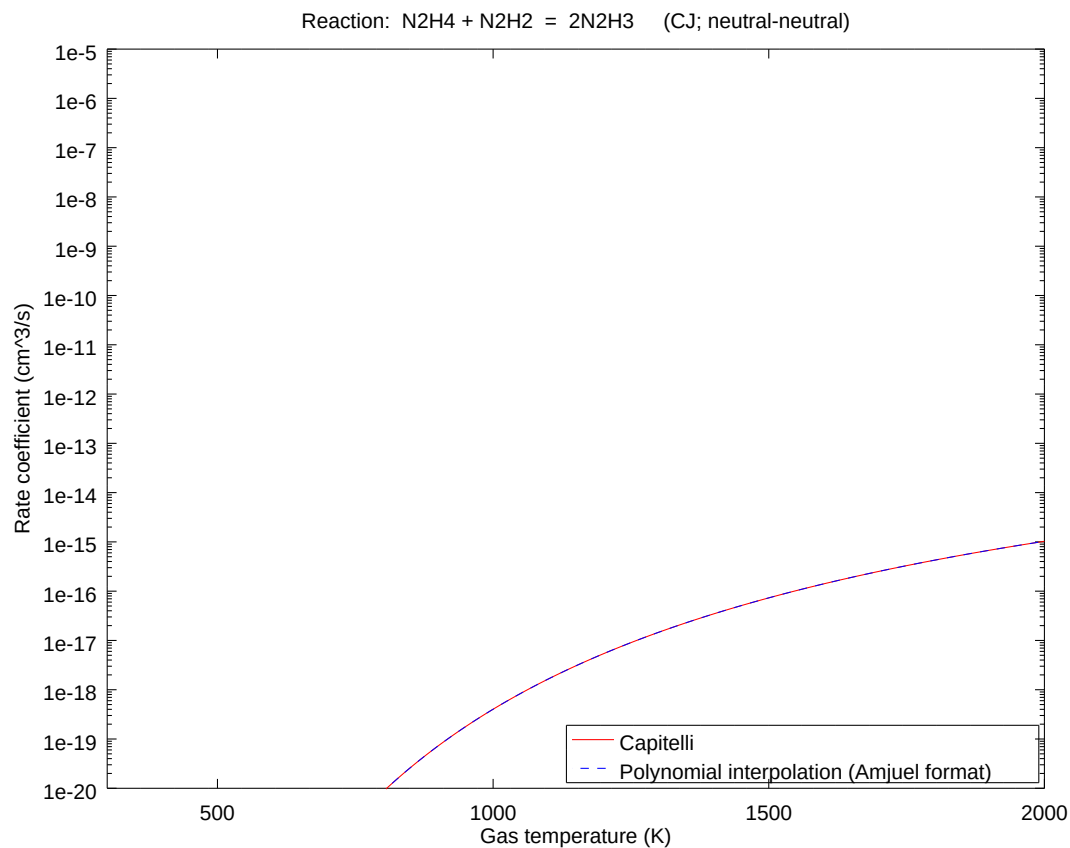
3.46 Reaction CI $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$

b0	-4.122840595613D+02	b1	3.273215180909D+02	b2	-1.266422101446D+02
b3	2.918581568282D+01	b4	-4.352451907143D+00	b5	4.275236716189D-01
b6	-2.687395114777D-02	b7	9.842117157823D-04	b8	-1.602297770451D-05



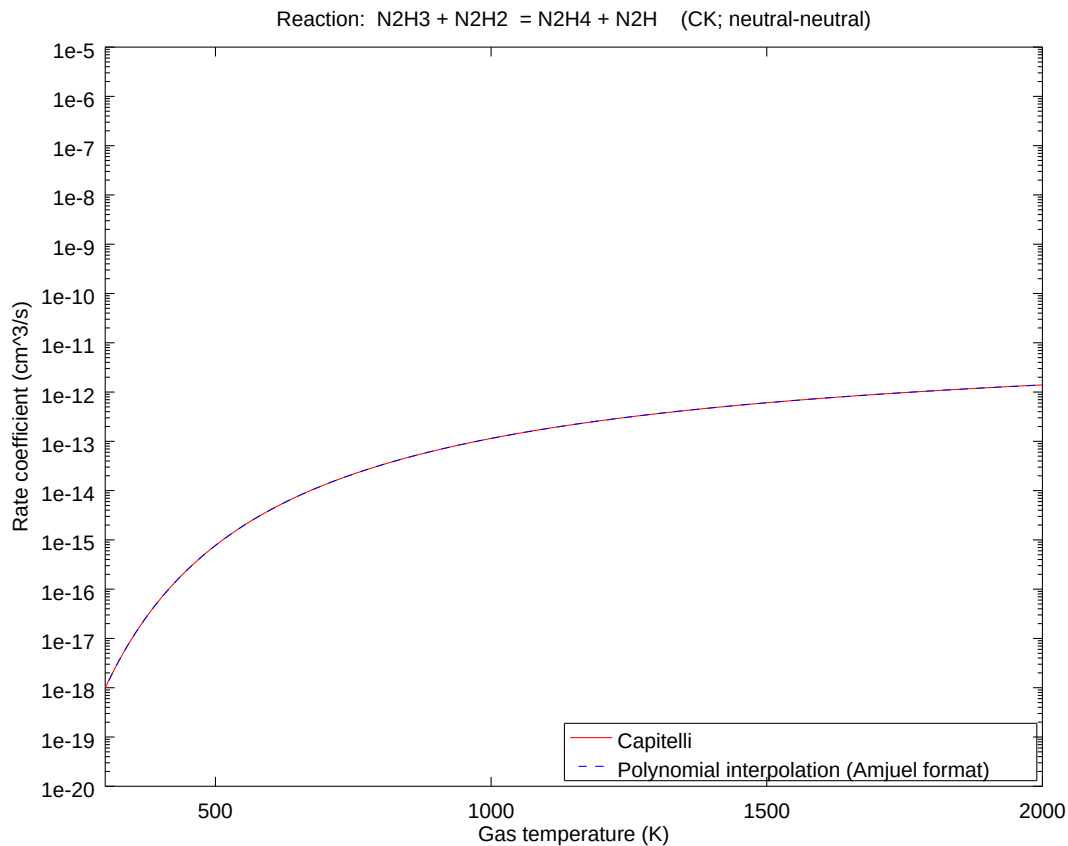
3.47 Reaction CJ $\text{N}_2\text{H}_4 + \text{N}_2\text{H}_2 = 2\text{N}_2\text{H}_3$

b0	-1.168894005641D+04	b1	9.820145755558D+03	b2	-3.799266417350D+03
b3	8.755745047211D+02	b4	-1.305735636862D+02	b5	1.282571093029D+01
b6	-8.062185933525D-01	b7	2.952635400707D-02	b8	-4.806893787246D-04



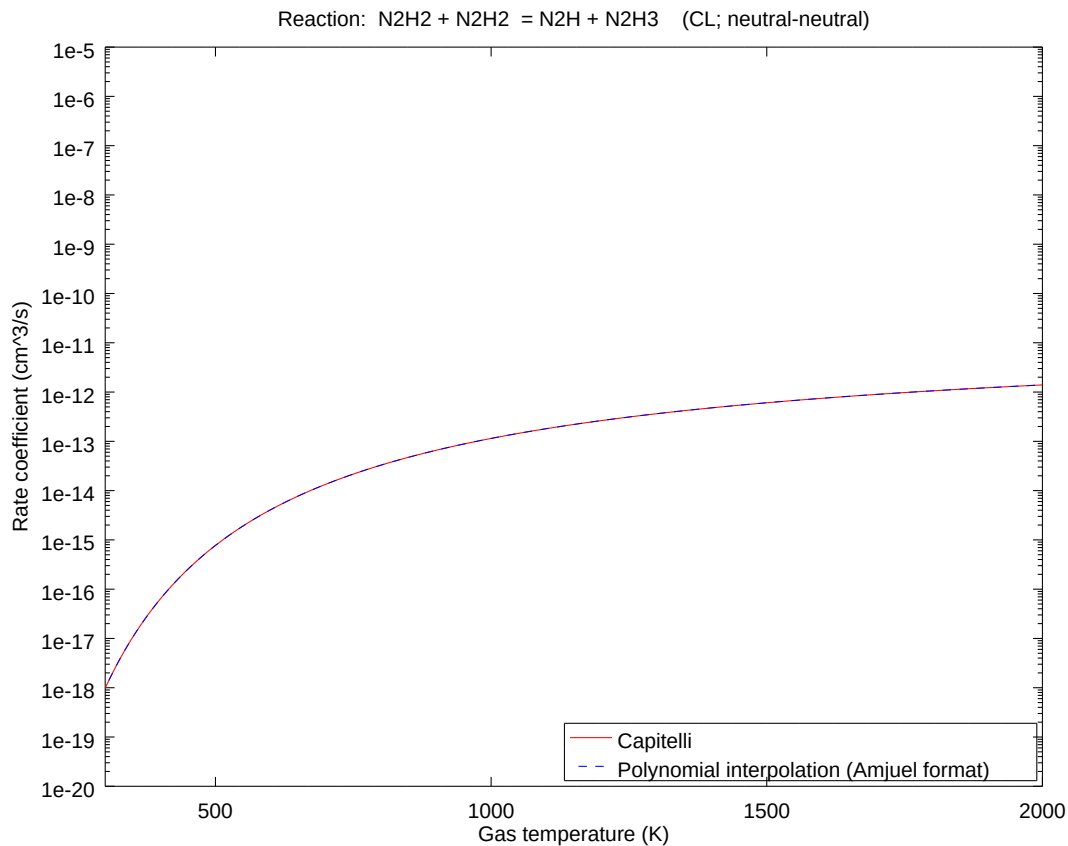
3.48 Reaction CK $\text{N}_2\text{H}_3 + \text{N}_2\text{H}_2 = \text{N}_2\text{H}_4 + \text{N}_2\text{H}$

b0	-3.910840685846D+03	b1	3.273215249482D+03	b2	-1.266422137839D+03
b3	2.918581678480D+02	b4	-4.352452115354D+01	b5	4.275236967563D+00
b6	-2.687395304155D-01	b7	9.842117971801D-03	b8	-1.602297923275D-04



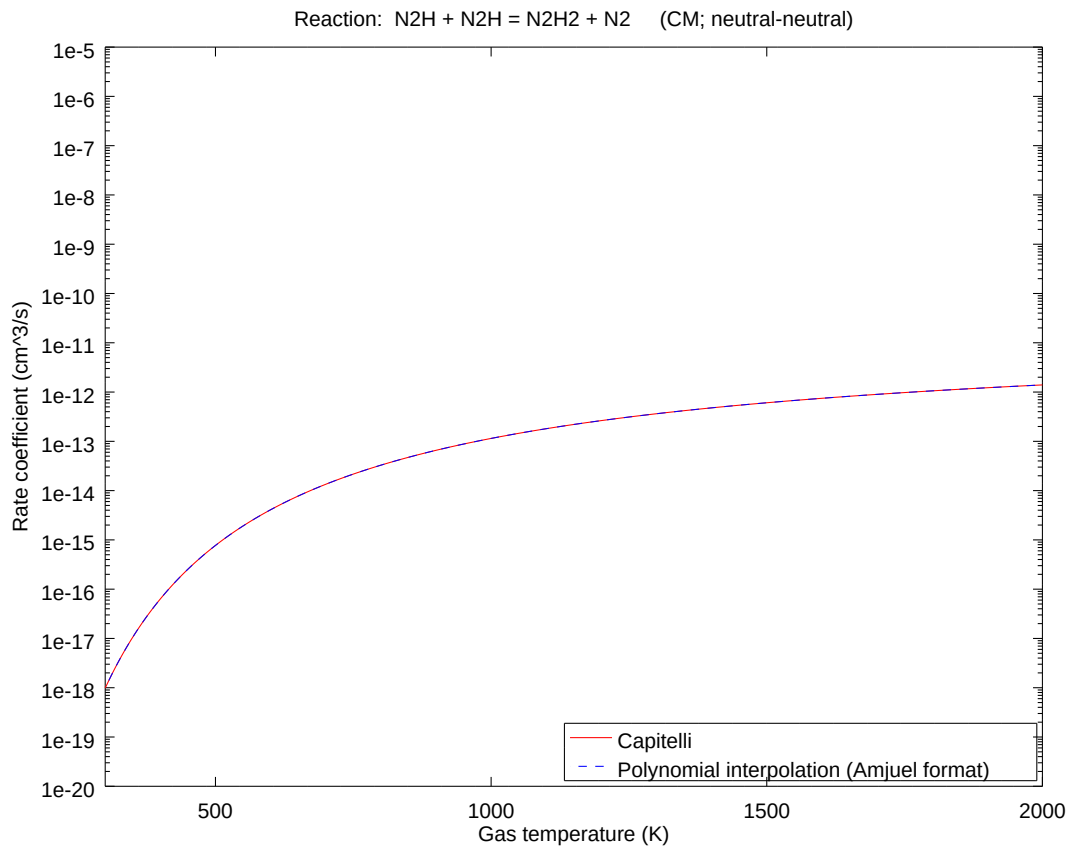
3.49 Reaction CL $\text{N}_2\text{H}_2 + \text{N}_2\text{H}_2 = \text{N}_2\text{H} + \text{N}_2\text{H}_3$

b0	-3.910840685846D+03	b1	3.273215249482D+03	b2	-1.266422137839D+03
b3	2.918581678480D+02	b4	-4.352452115354D+01	b5	4.275236967563D+00
b6	-2.687395304155D-01	b7	9.842117971801D-03	b8	-1.602297923275D-04



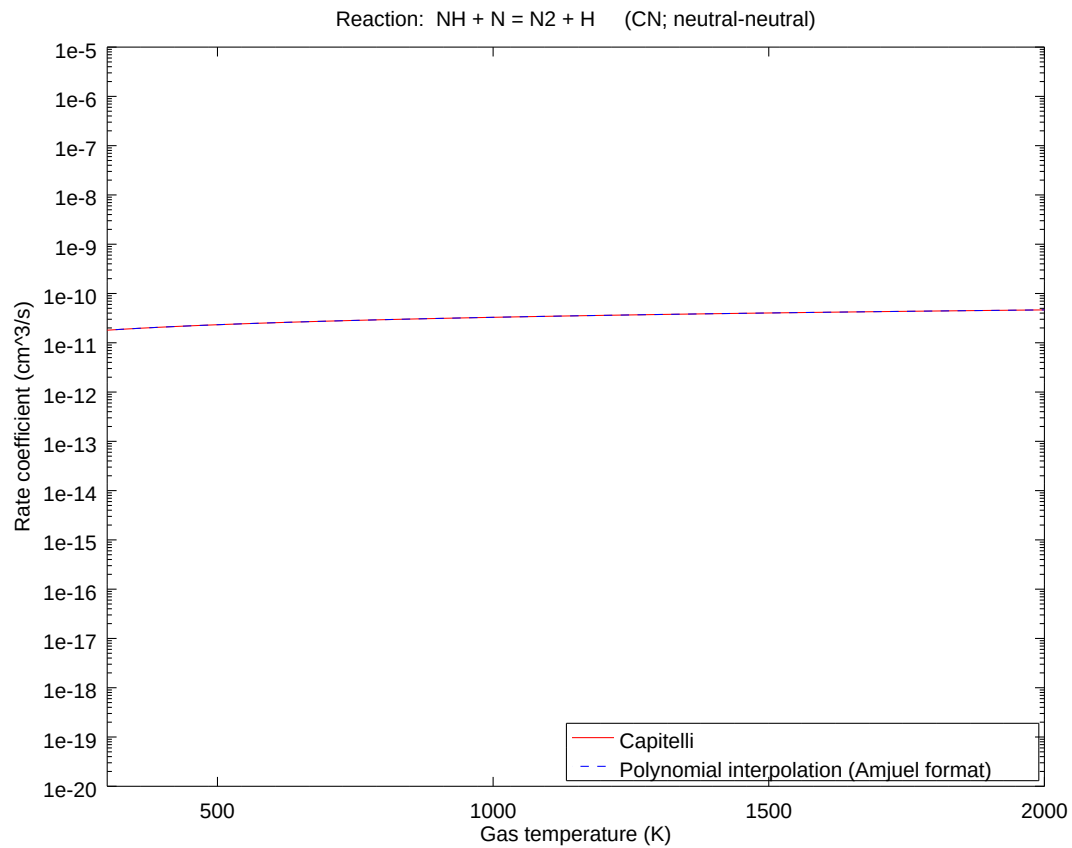
3.50 Reaction CM $\text{N}_2\text{H} + \text{N}_2\text{H} = \text{N}_2\text{H}_2 + \text{N}_2$

b0	-3.910840685846D+03	b1	3.273215249482D+03	b2	-1.266422137839D+03
b3	2.918581678480D+02	b4	-4.352452115354D+01	b5	4.275236967563D+00
b6	-2.687395304155D-01	b7	9.842117971801D-03	b8	-1.602297923275D-04



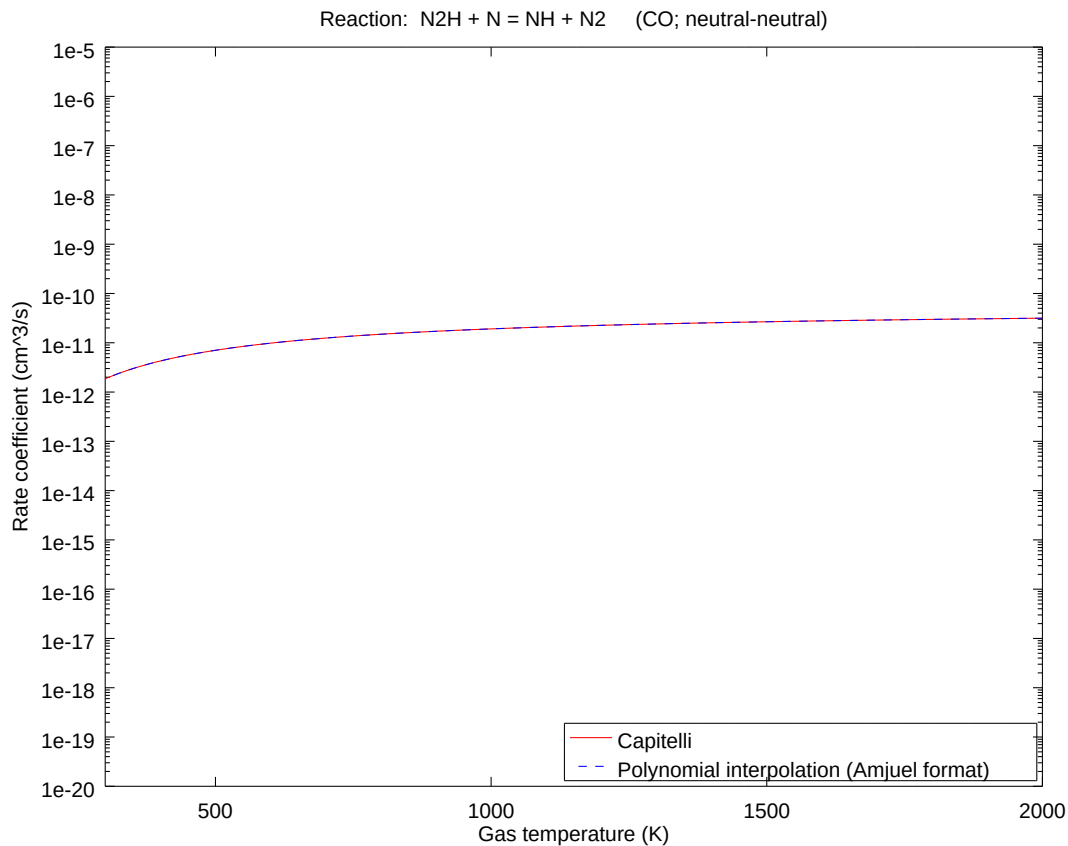
3.51 Reaction CN NH + N = N2 + H

b0	-2.759253321422D+01	b1	4.999910217002D-01	b2	4.770825976219D-06
b3	-1.446466709131D-06	b4	2.736904809845D-07	b5	-3.309409011558D-08
b6	2.497377141264D-09	b7	-1.075341209927D-10	b8	2.022818066678D-12



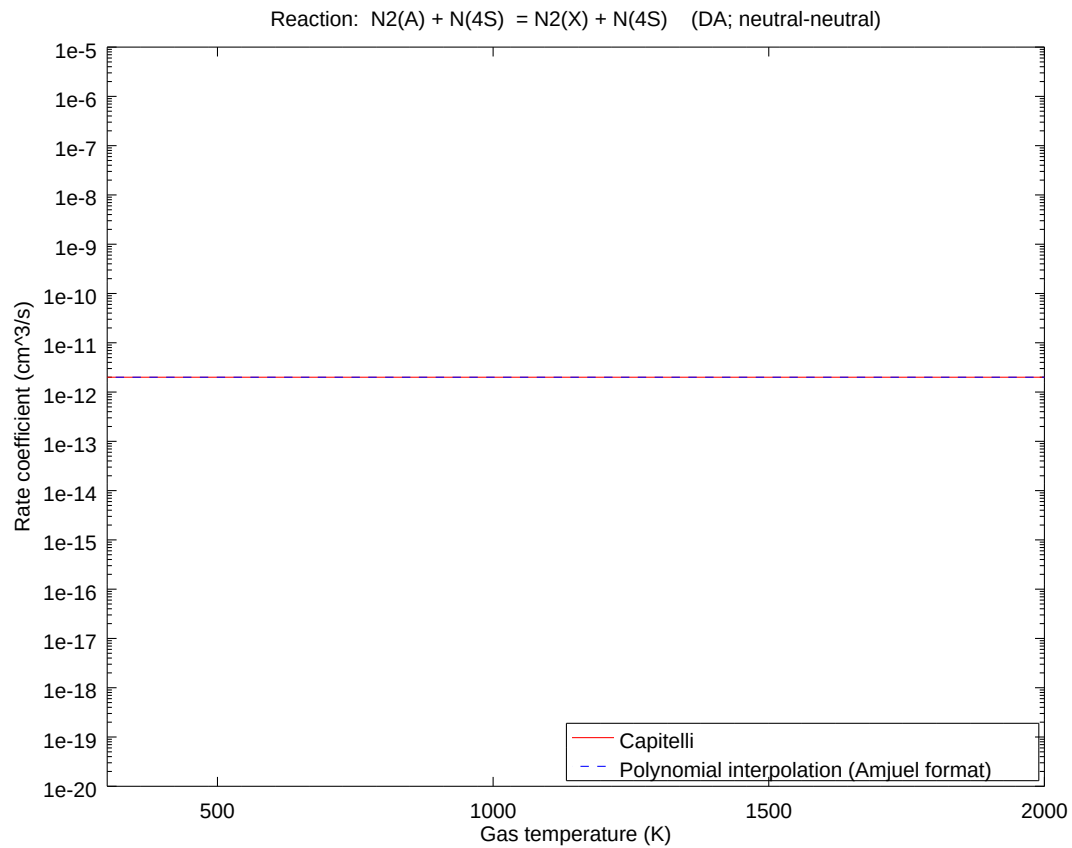
3.52 Reaction CO N2H + N = NH + N2

b0	-8.008883486442D+02	b1	6.546430445692D+02	b2	-2.532844247300D+02
b3	5.837163270704D+01	b4	-8.704904067114D+00	b5	8.550473736861D-01
b6	-5.374790458366D-02	b7	1.968423529663D-03	b8	-3.204595724609D-05



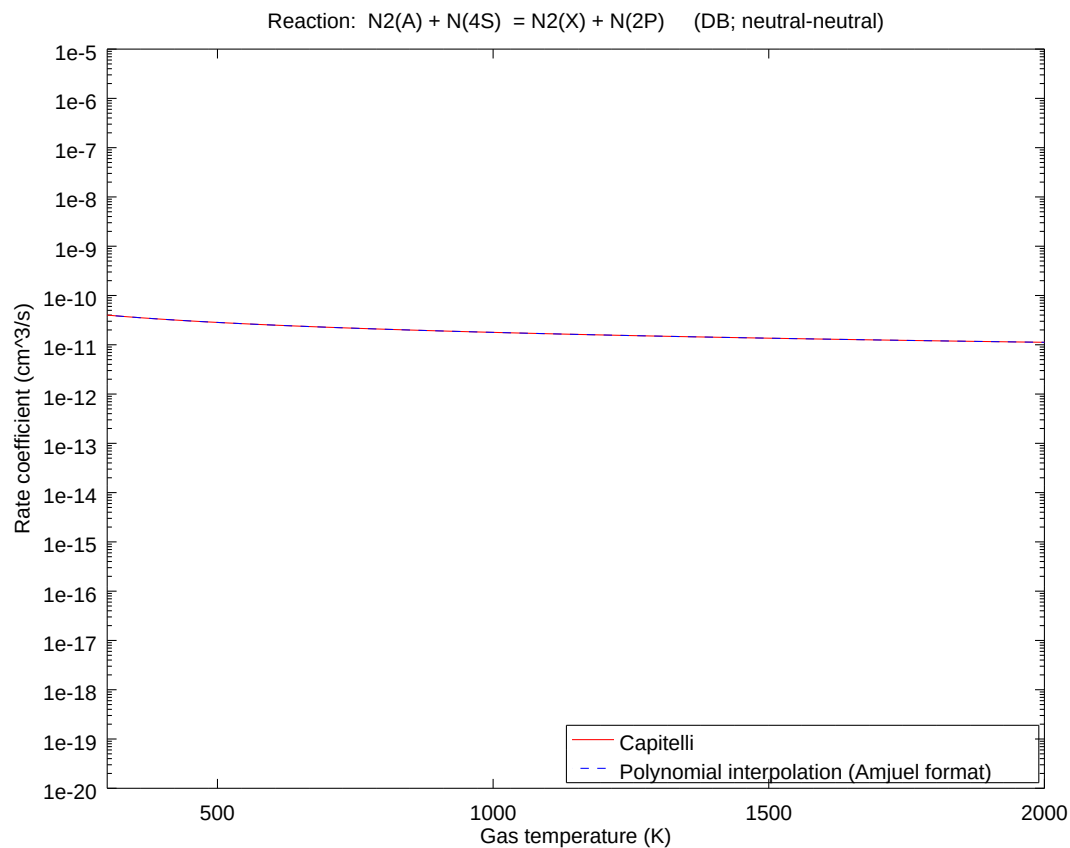
3.53 Reaction DA $\text{N}_2(\text{A}) + \text{N}(4\text{S}) = \text{N}_2(\text{X}) + \text{N}(4\text{S})$

b0	-2.693786625645D+01	b1	-9.307528833651D-06	b2	4.928219873907D-06
b3	-1.488854458533D-06	b4	2.806985700528D-07	b5	-3.381858275836D-08
b6	2.542726699222D-09	b7	-1.090829895082D-10	b8	2.044305905730D-12



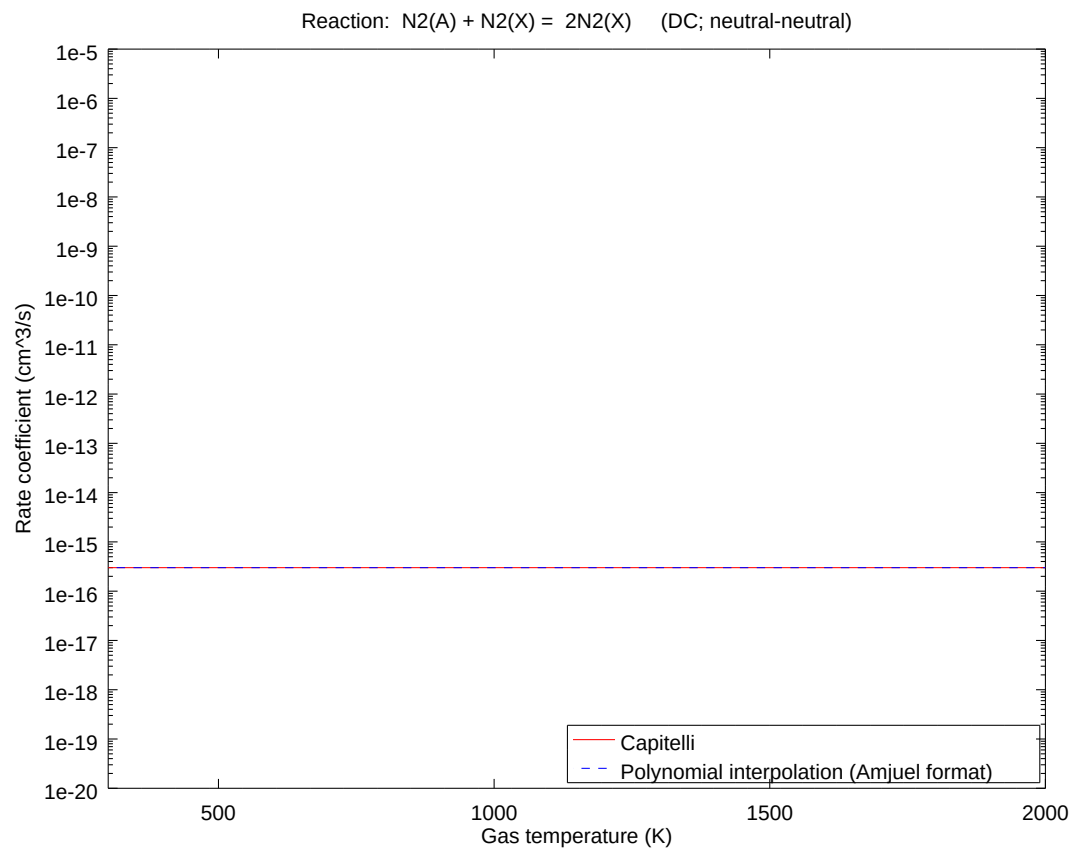
3.54 Reaction DB N2(A) + N(4S) = N2(X) + N(2P)

b0	-2.012060021704D+01	b1	-6.700087439511D-01	b2	4.647116229093D-06
b3	-1.409114699671D-06	b4	2.666349474316D-07	b5	-3.224028751636D-08
b6	2.432738947469D-09	b7	-1.047352301701D-10	b8	1.969750395445D-12



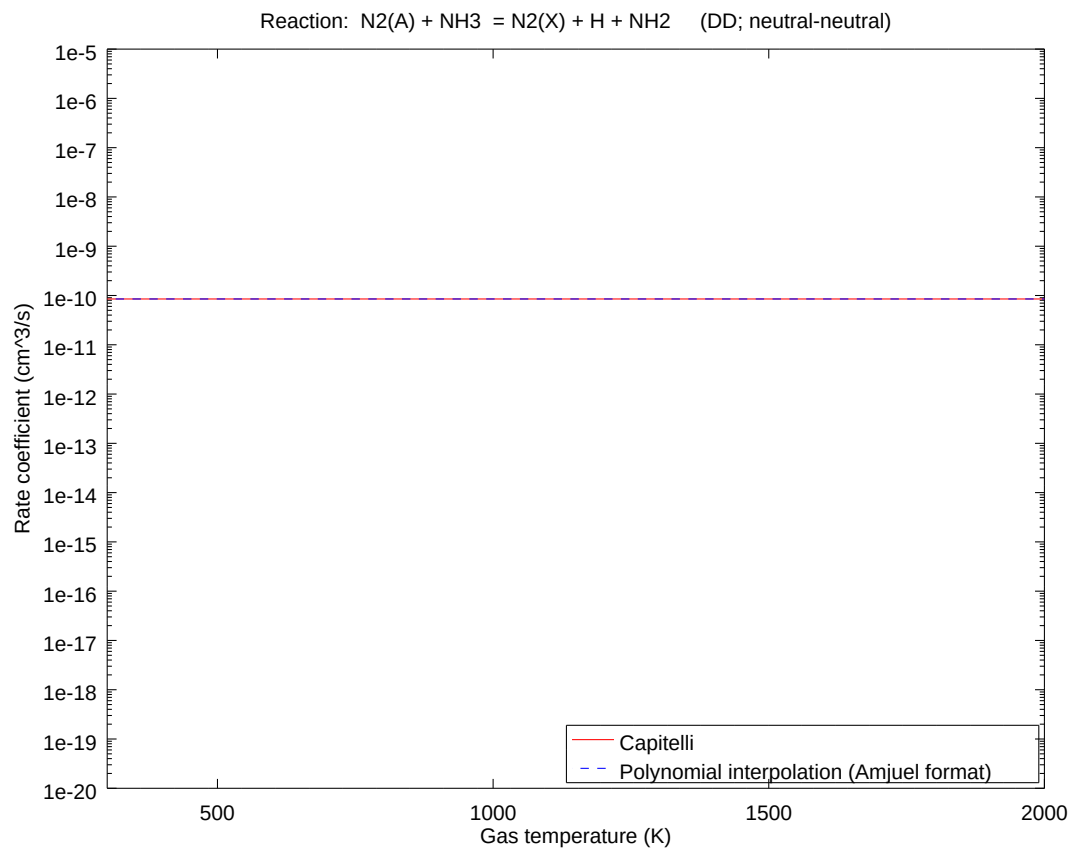
3.55 Reaction DC $\text{N}_2(\text{A}) + \text{N}_2(\text{X}) = 2\text{N}_2(\text{X})$

b0	-3.574274231973D+01	b1	-8.373338863656D-06	b2	4.451201130469D-06
b3	-1.349839977229D-06	b4	2.554098471469D-07	b5	-3.087799173966D-08
b6	2.329284293260D-09	b7	-1.002411393756D-10	b8	1.884264417516D-12



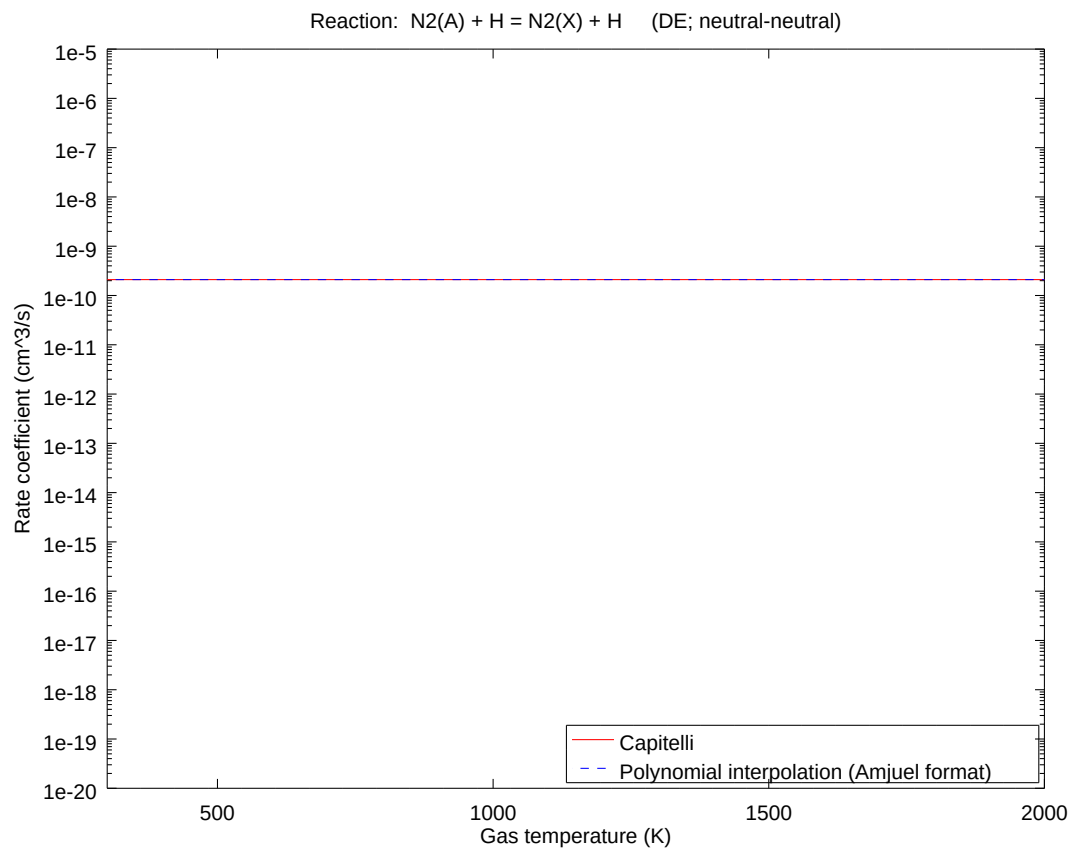
3.56 Reaction DD $\text{N}_2(\text{A}) + \text{NH}_3 = \text{N}_2(\text{X}) + \text{H} + \text{NH}_2$

b0	-2.318836146458D+01	b1	-1.017706561297D-05	b2	5.389198088252D-06
b3	-1.628187887750D-06	b4	3.069620049925D-07	b5	-3.698006822805D-08
b6	2.780081502789D-09	b7	-1.192449858295D-10	b8	2.234272311202D-12



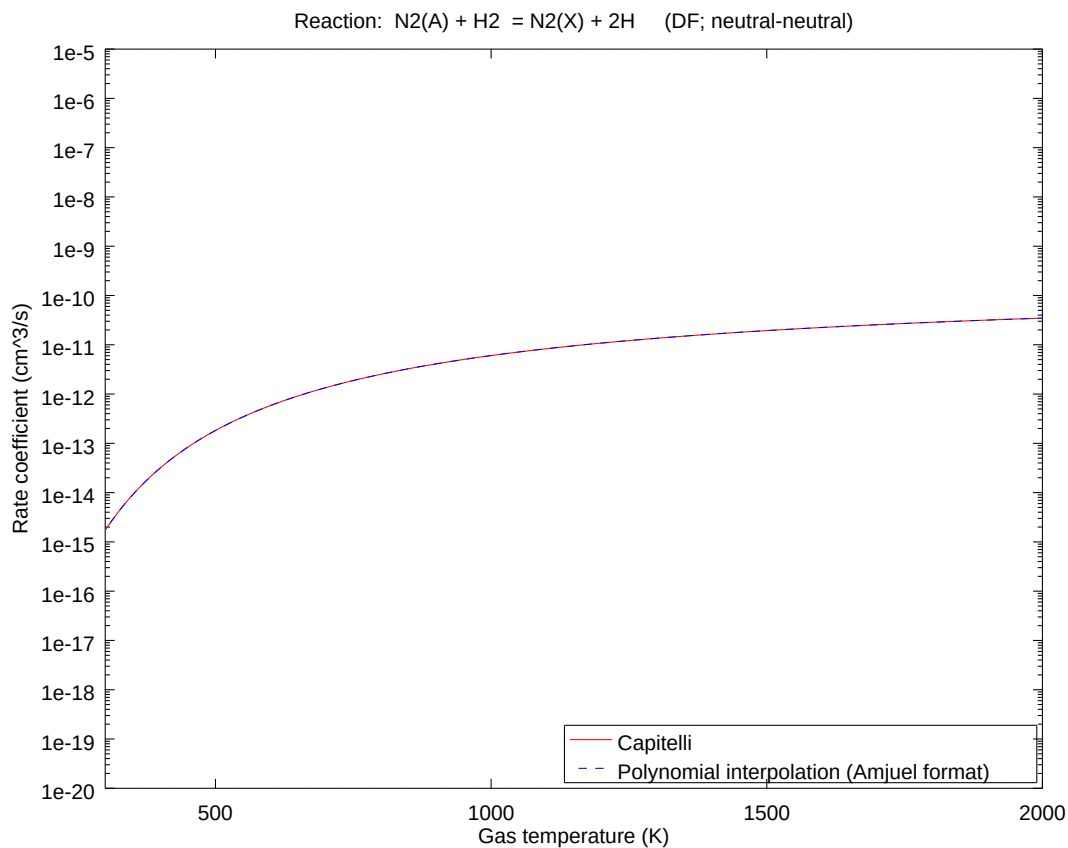
3.57 Reaction DE N2(A) + H = N2(X) + H

b0	-2.228390668558D+01	b1	-8.382509550888D-06	b2	4.448349574334D-06
b3	-1.346745957179D-06	b4	2.544218409556D-07	b5	-3.071195209043D-08
b6	2.313383909406D-09	b7	-9.941697559179D-11	b8	1.866232906590D-12



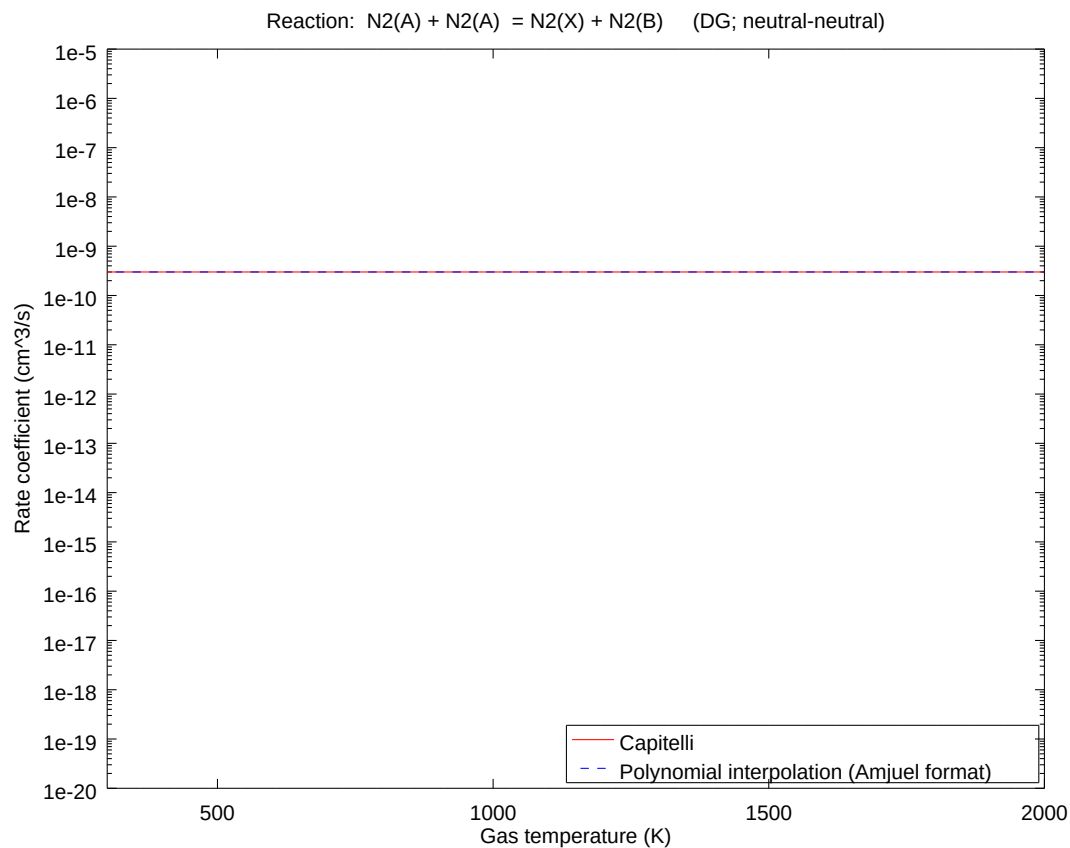
3.58 Reaction DF $\text{N}_2(\text{A}) + \text{H}_2 = \text{N}_2(\text{X}) + 2\text{H}$

b0	-2.742562716641D+03	b1	2.291250672485D+03	b2	-8.864954953386D+02
b3	2.043007171435D+02	b4	-3.046716474094D+01	b5	2.992665869213D+00
b6	-1.881176706783D-01	b7	6.889482553777D-03	b8	-1.121608541291D-04



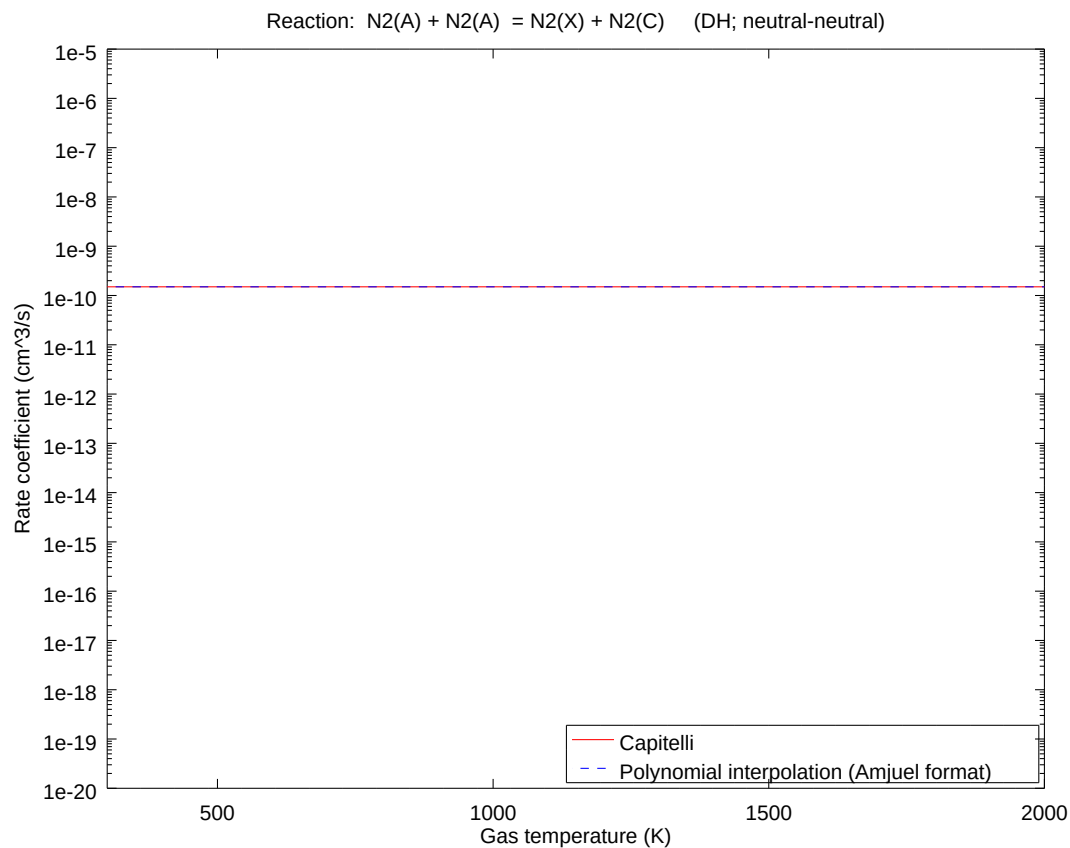
3.59 Reaction DG N2(A) + N2(A) = N2(X) + N2(B)

b0	-2.192723329785D+01	b1	-6.478664005524D-06	b2	3.430833170709D-06
b3	-1.036451744697D-06	b4	1.953690362165D-07	b5	-2.353009371820D-08
b6	1.768306073988D-09	b7	-7.581303820488D-11	b8	1.419725383988D-12



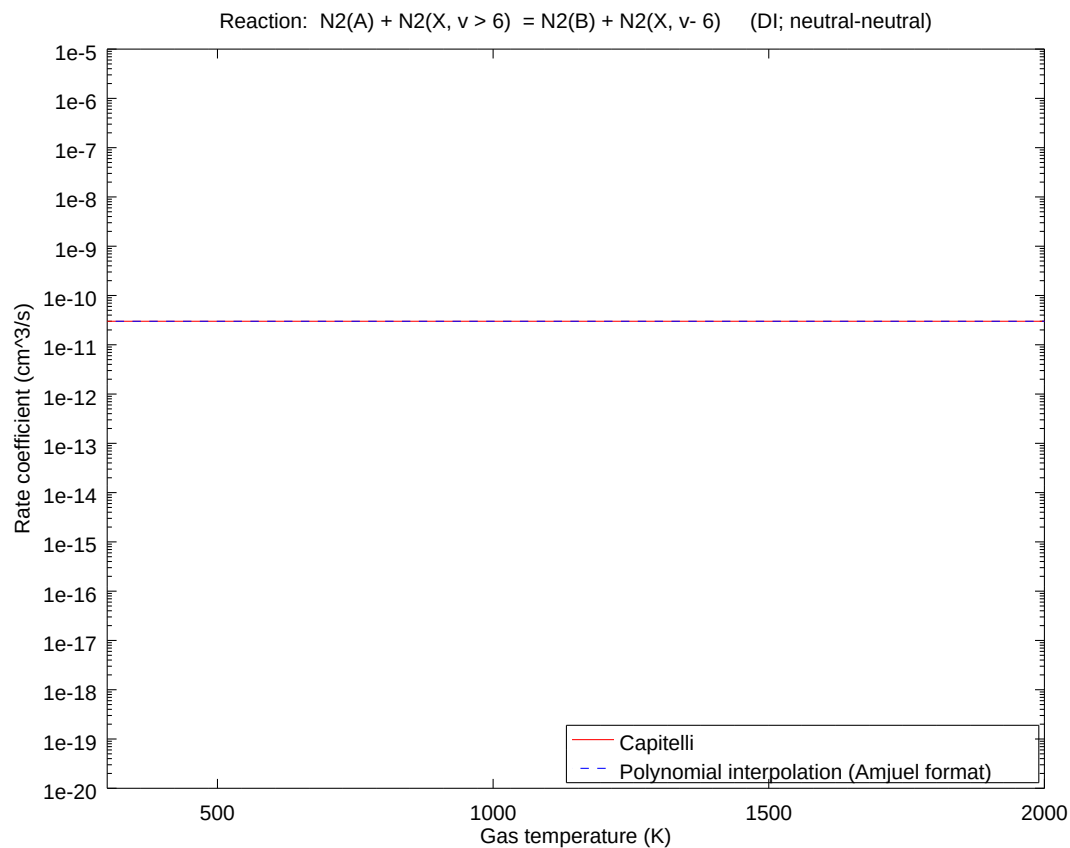
3.60 Reaction DH $\text{N}_2(\text{A}) + \text{N}_2(\text{A}) = \text{N}_2(\text{X}) + \text{N}_2(\text{C})$

b0	-2.262038083435D+01	b1	-6.080005149455D-06	b2	3.237347401322D-06
b3	-9.833866292555D-07	b4	1.863933013090D-07	b5	-2.257412860816D-08
b6	1.705966737246D-09	b7	-7.355216256471D-11	b8	1.385181383026D-12



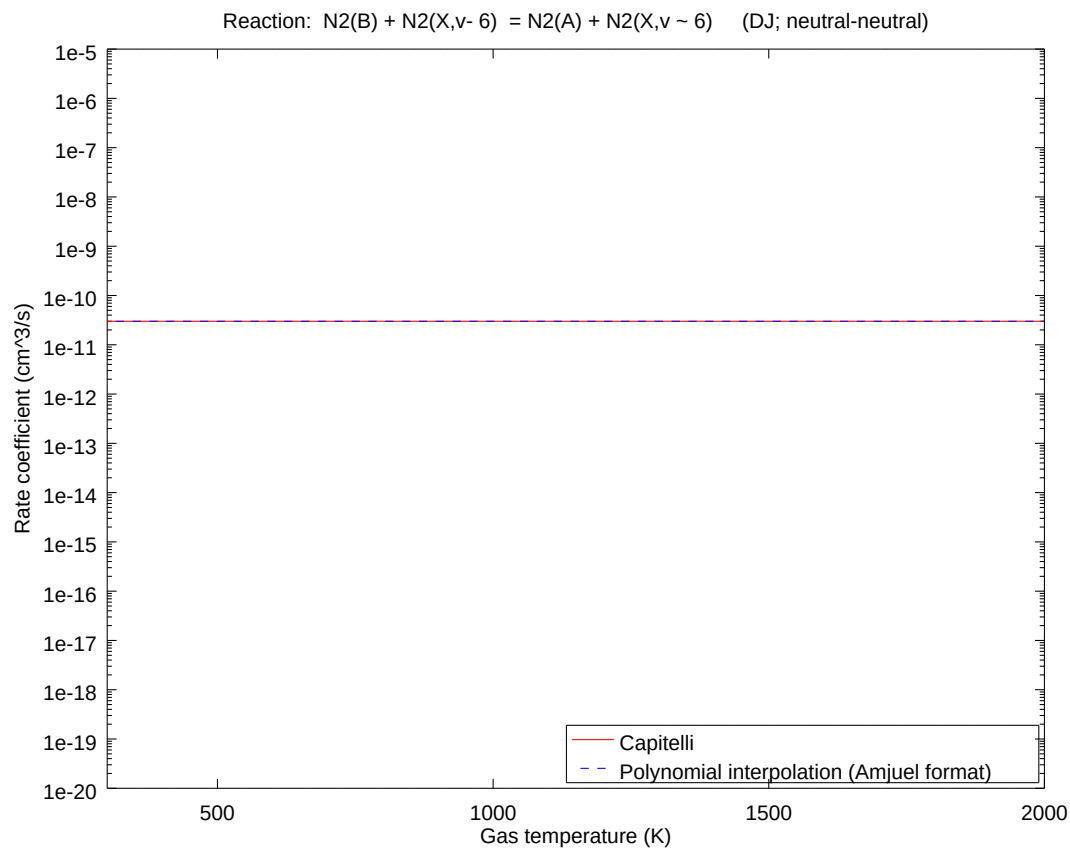
3.61 Reaction DI $\text{N}_2(\text{A}) + \text{N}_2(\text{X}, v \geq 6) = \text{N}_2(\text{B}) + \text{N}_2(\text{X}, v-6)$

b0	-2.422981695265D+01	b1	-8.252077863754D-06	b2	4.386056478354D-06
b3	-1.330002443172D-06	b4	2.516628213424D-07	b5	-3.042828271328D-08
b6	2.295786011921D-09	b7	-9.882504849018D-11	b8	1.858253498875D-12



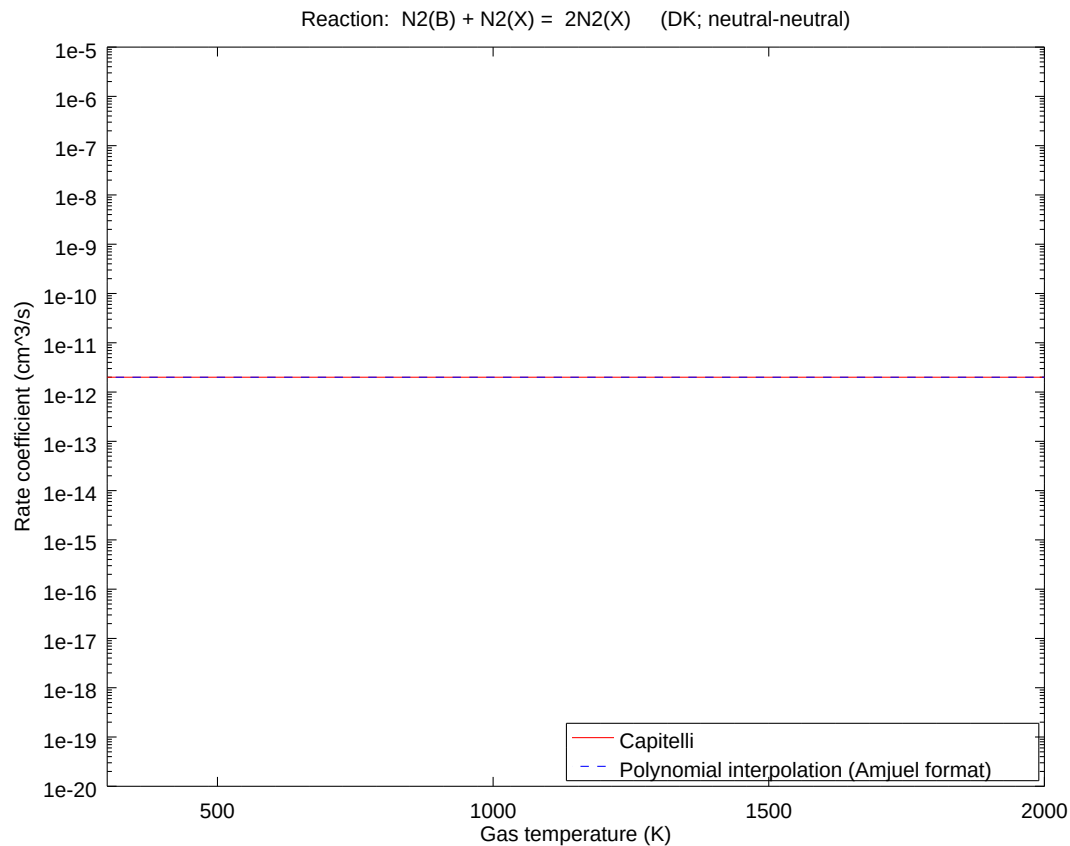
3.62 Reaction DJ $\text{N}_2(\text{B}) + \text{N}_2(\text{X}, v=6) = \text{N}_2(\text{A}) + \text{N}_2(\text{X}, v=6)$

b0	-2.422981695265D+01	b1	-8.252077863754D-06	b2	4.386056478354D-06
b3	-1.330002443172D-06	b4	2.516628213424D-07	b5	-3.042828271328D-08
b6	2.295786011921D-09	b7	-9.882504849018D-11	b8	1.858253498875D-12



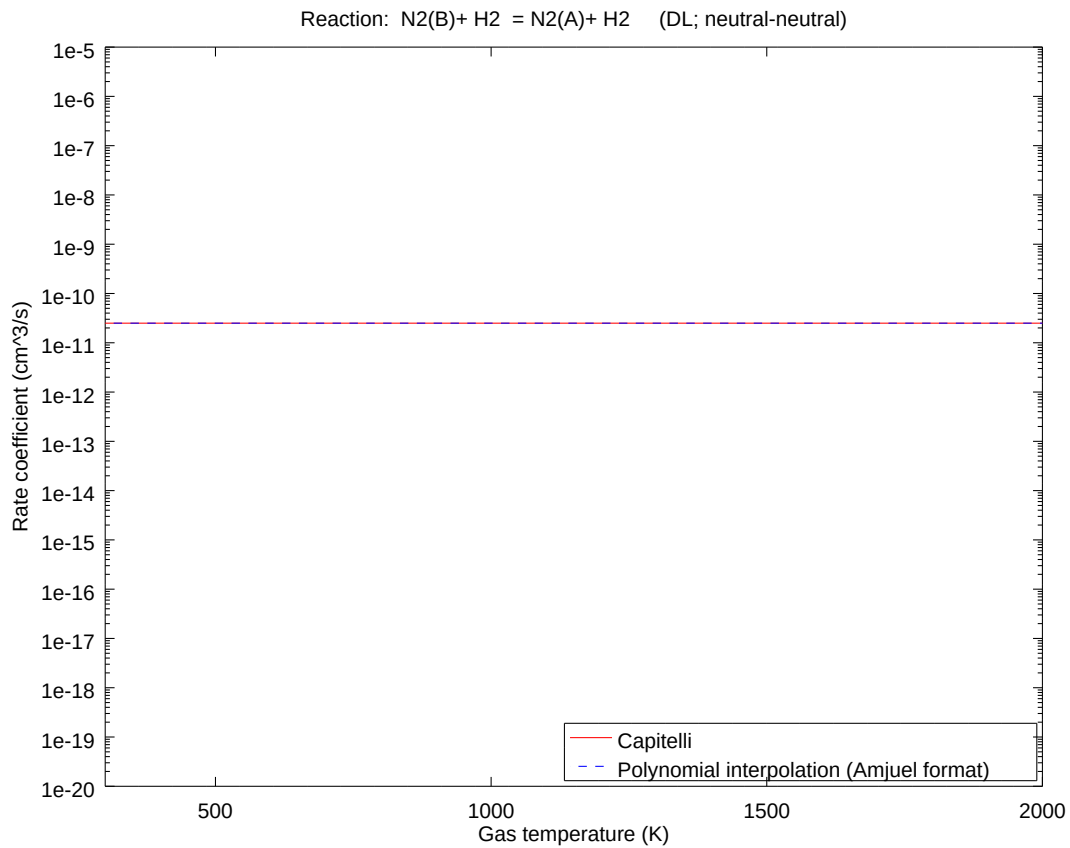
3.63 Reaction DK $\text{N}_2(\text{B}) + \text{N}_2(\text{X}) = 2\text{N}_2(\text{X})$

b0	-2.693786625645D+01	b1	-9.307528833651D-06	b2	4.928219873907D-06
b3	-1.488854458533D-06	b4	2.806985700528D-07	b5	-3.381858275836D-08
b6	2.542726699222D-09	b7	-1.090829895082D-10	b8	2.044305905730D-12



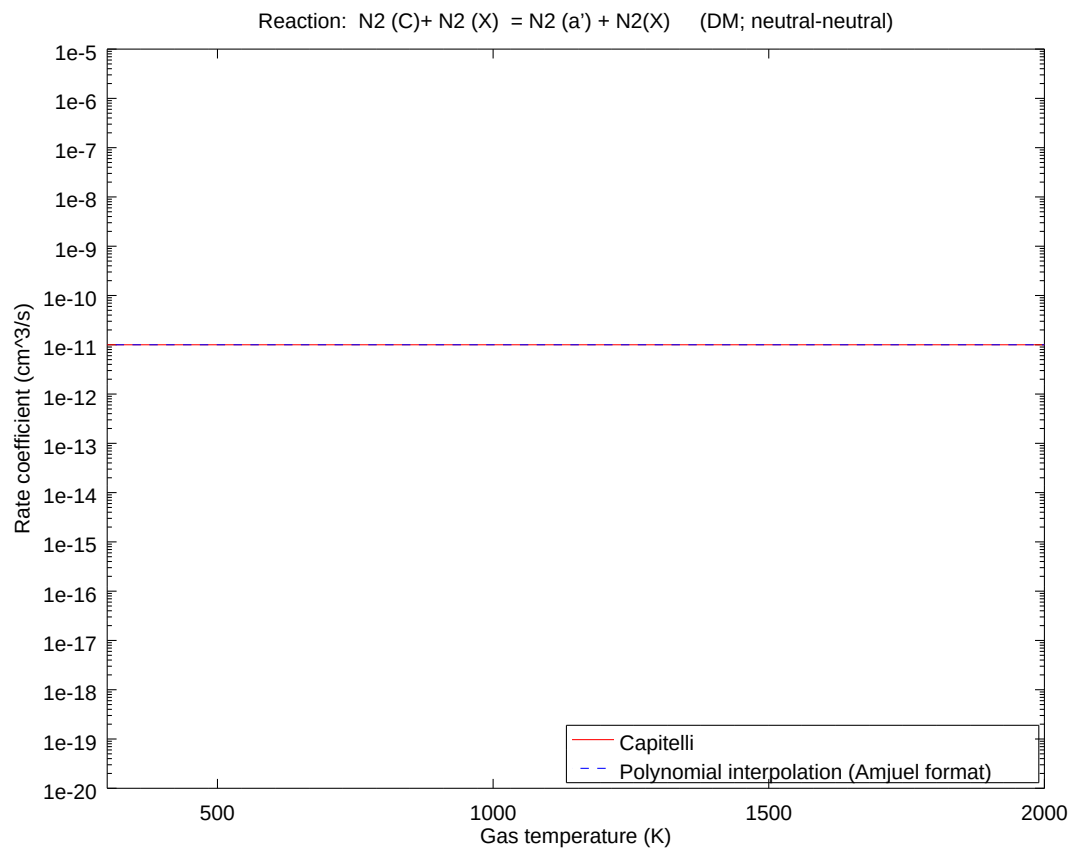
3.64 Reaction DL N2(B)+ H2 = N2(A)+ H2

b0	-2.441213466448D+01	b1	-1.289573010449D-05	b2	6.836460754101D-06
b3	-2.067926055892D-06	b4	3.903703978285D-07	b5	-4.709330311079D-08
b6	3.545557642161D-09	b7	-1.523133943080D-10	b8	2.858507948958D-12



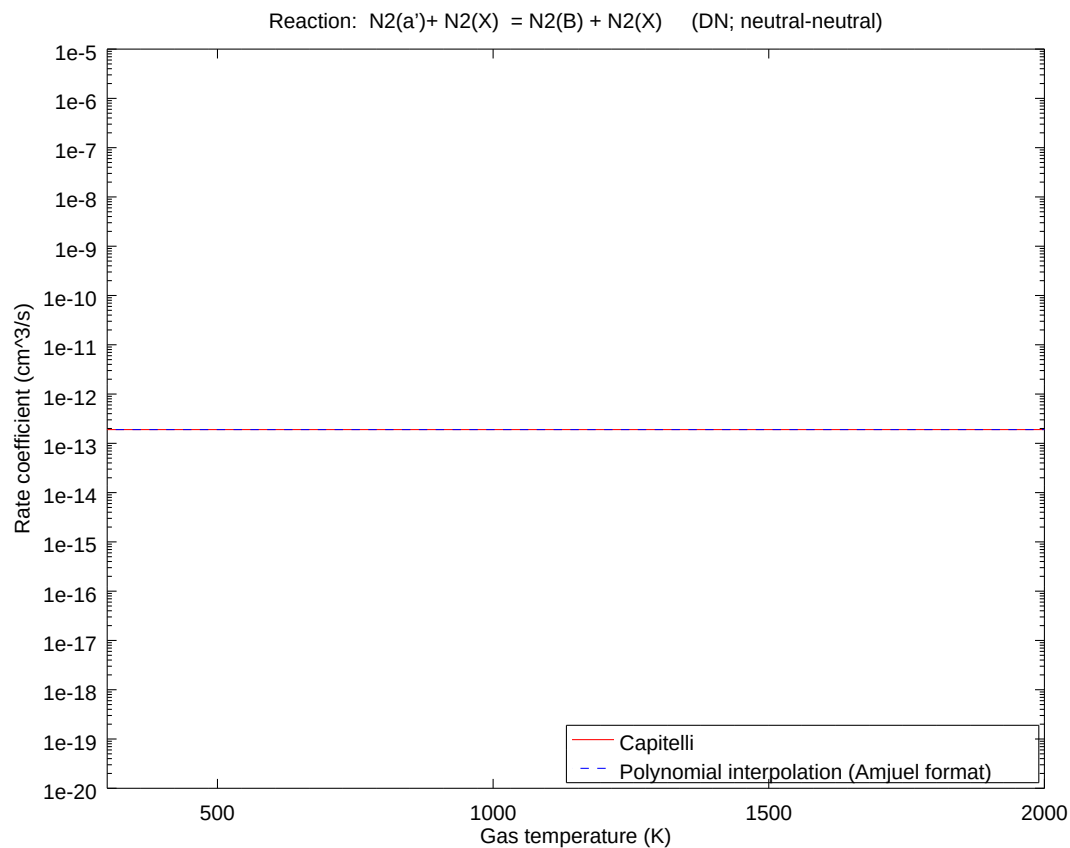
3.65 Reaction DM N2 (C)+ N2 (X) = N2 (a') + N2(X)

b0	-2.532843101849D+01	b1	-6.096669667990D-06	b2	3.244226613168D-06
b3	-9.849142194775D-07	b4	1.865845980529D-07	b5	-2.258639290203D-08
b6	1.706143264856D-09	b7	-7.353050543137D-11	b8	1.384277986480D-12



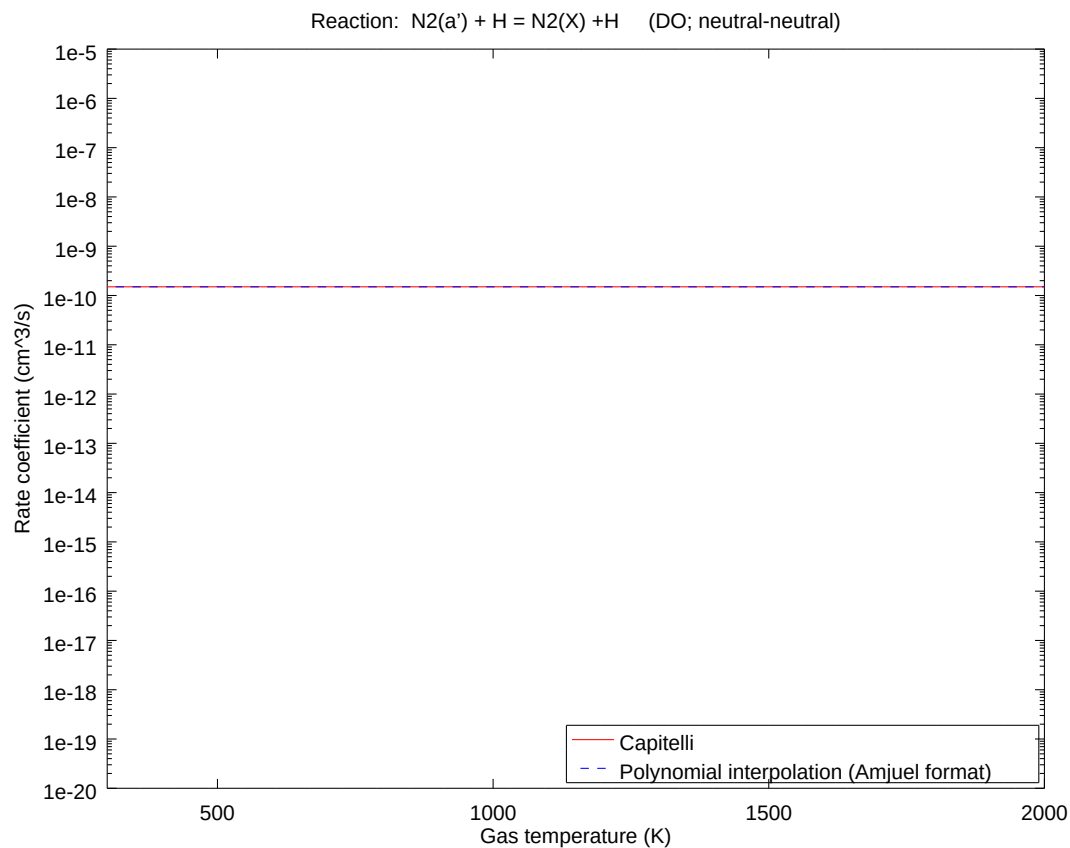
3.66 Reaction DN $\text{N}_2(\text{a}') + \text{N}_2(\text{X}) = \text{N}_2(\text{B}) + \text{N}_2(\text{X})$

b0	-2.929174193089D+01	b1	-1.261492545511D-05	b2	6.689492568324D-06
b3	-2.023970848880D-06	b4	3.821537629976D-07	b5	-4.611024598743D-08
b6	3.472044766407D-09	b7	-1.491719922589D-10	b8	2.799777333052D-12



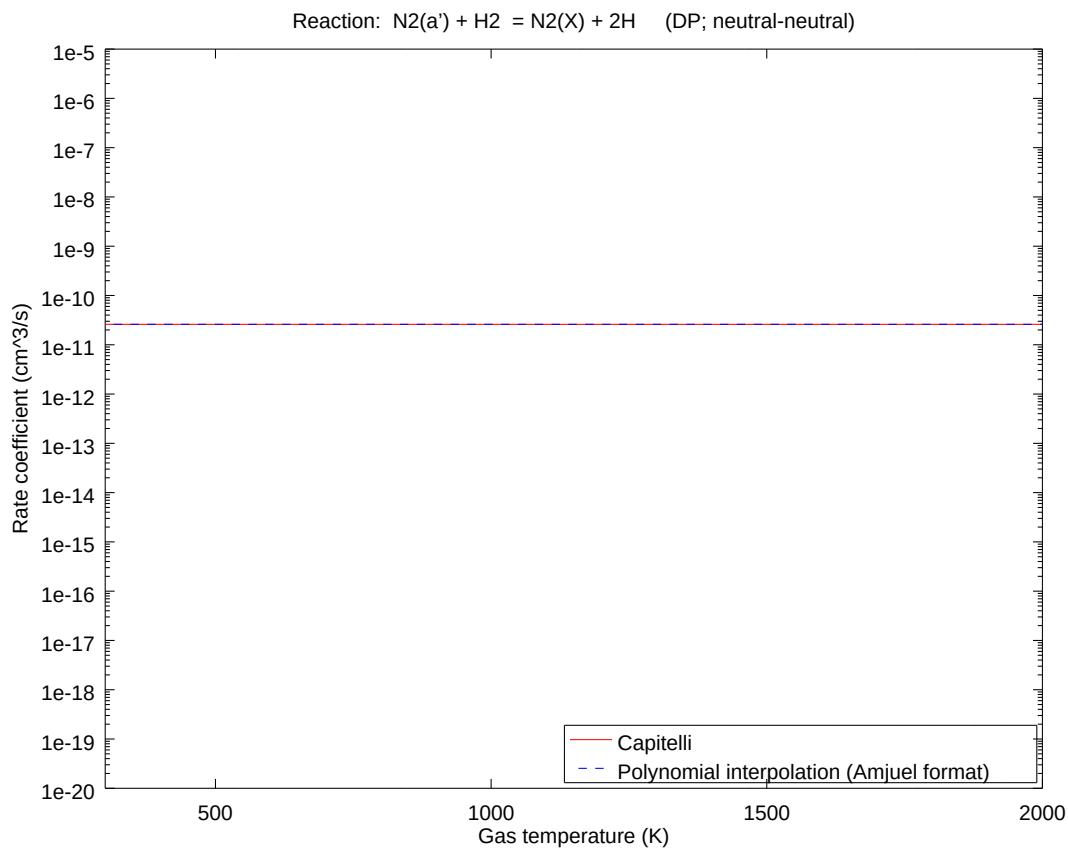
3.67 Reaction DO N2(a') + H = N2(X) +H

b0	-2.262038083435D+01	b1	-6.080005149455D-06	b2	3.237347401322D-06
b3	-9.833866292555D-07	b4	1.863933013090D-07	b5	-2.257412860816D-08
b6	1.705966737246D-09	b7	-7.355216256471D-11	b8	1.385181383026D-12



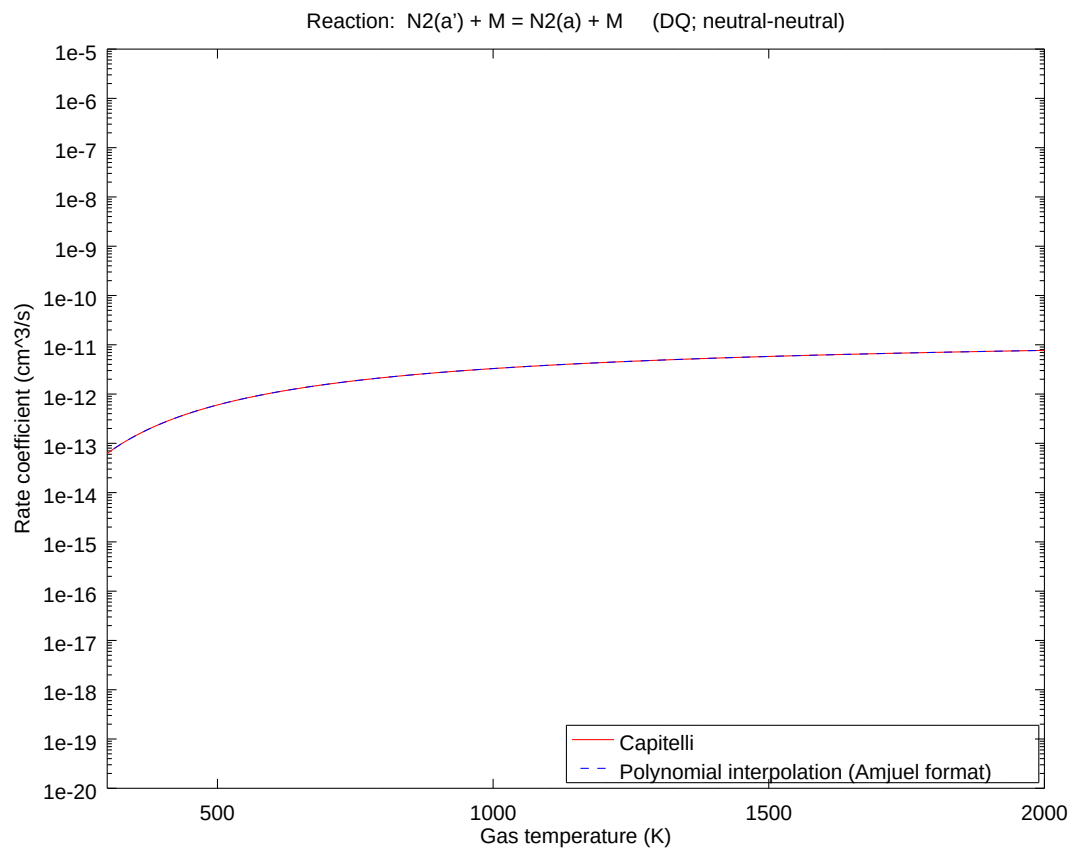
3.68 Reaction DP $\text{N}_2(\text{a}') + \text{H}_2 = \text{N}_2(\text{X}) + 2\text{H}$

b0	-2.437291829159D+01	b1	-7.630063239549D-06	b2	4.045093765858D-06
b3	-1.223448388807D-06	b4	2.308988365496D-07	b5	-2.784464154616D-08
b6	2.095321108064D-09	b7	-8.995725035280D-11	b8	1.687020945781D-12



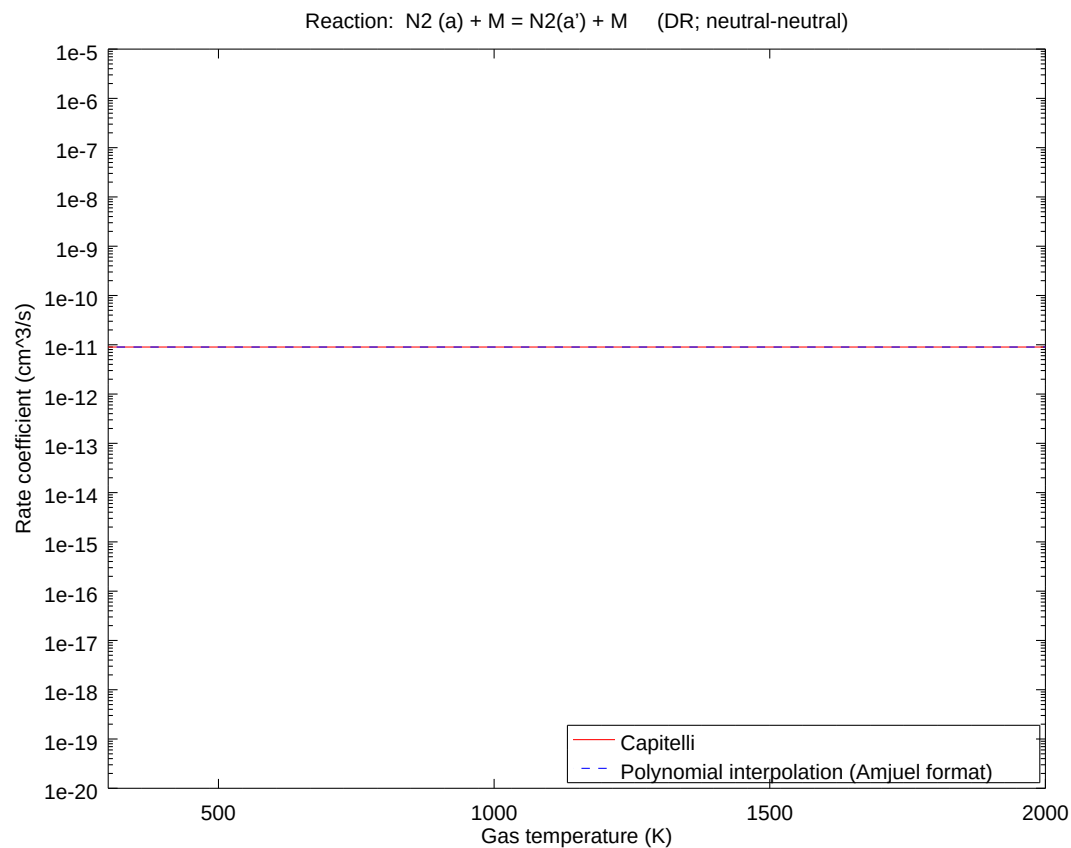
3.69 Reaction DQ $\text{N}_2(\text{a}') + \text{M} = \text{N}_2(\text{a}) + \text{M}$

b0	-1.345995221369D+03	b1	1.112893176897D+03	b2	-4.305835226649D+02
b3	9.923177579821D+01	b4	-1.479833695253D+01	b5	1.453580540068D+00
b6	-9.137143816591D-02	b7	3.346320016993D-03	b8	-5.447812763870D-05



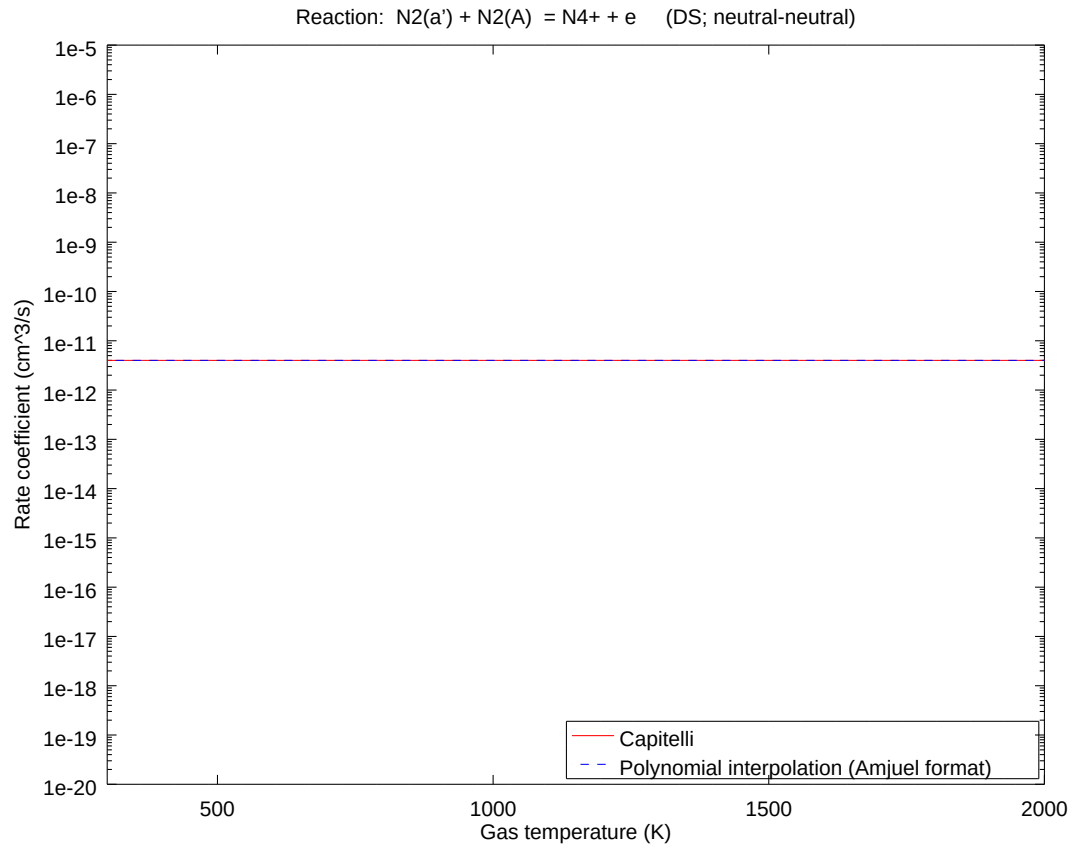
3.70 Reaction DR N2 (a) + M = N2(a') + M

b0	-2.543378870475D+01	b1	-9.505117712090D-06	b2	5.037998363738D-06
b3	-1.523568073038D-06	b4	2.875338576498D-07	b5	-3.467701281804D-08
b6	2.609892969512D-09	b7	-1.120771573916D-10	b8	2.102541263165D-12



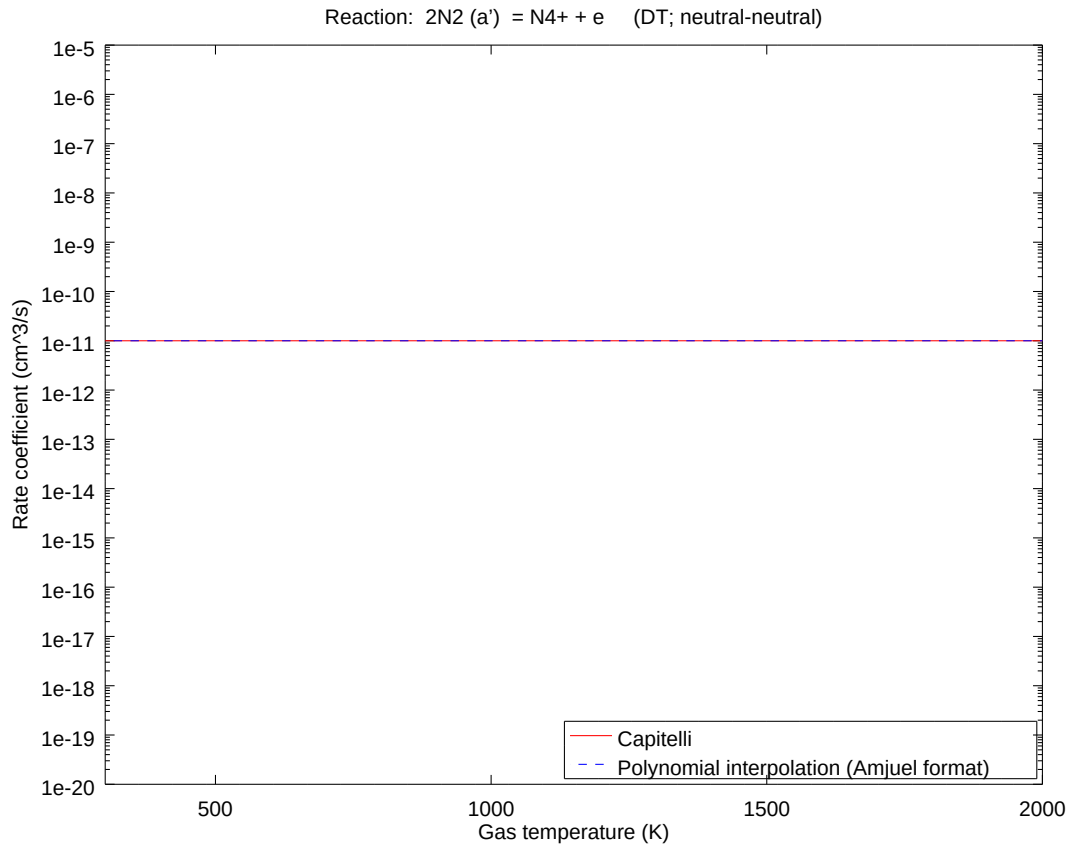
3.71 Reaction DS $\text{N}_2(\text{a}') + \text{N}_2(\text{A}) = \text{N}_4^+ + \text{e}$

b0	-2.624471937392D+01	b1	-8.961512942463D-06	b2	4.753386914248D-06
b3	-1.438662079166D-06	b4	2.717485942561D-07	b5	-3.280423982553D-08
b6	2.471431061480D-09	b7	-1.062445765709D-10	b8	1.995368756981D-12



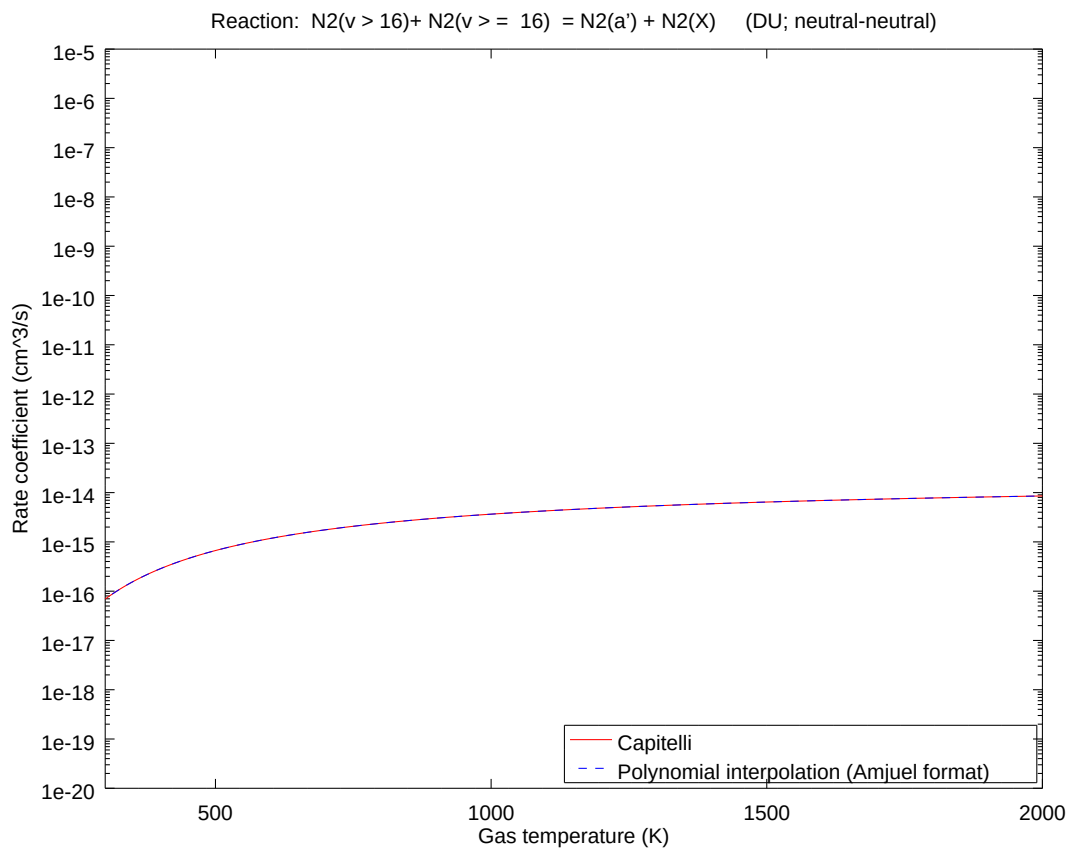
3.72 Reaction DT $2\text{N}_2(\text{a}') = \text{N}_4^+ + \text{e}$

b0	-2.532843101849D+01	b1	-6.096669667990D-06	b2	3.244226613168D-06
b3	-9.849142194775D-07	b4	1.865845980529D-07	b5	-2.258639290203D-08
b6	1.706143264856D-09	b7	-7.353050543137D-11	b8	1.384277986480D-12



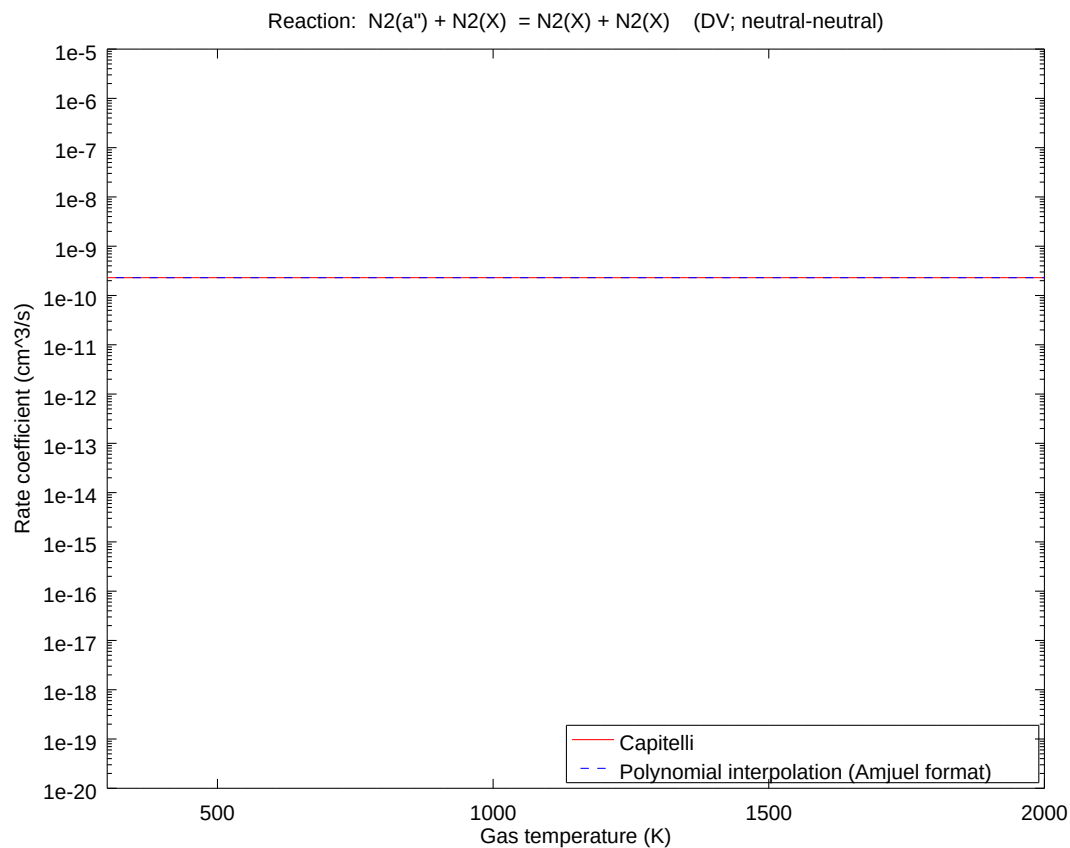
3.73 Reaction DU $\text{N}_2(v=16) + \text{N}_2(v=16) = \text{N}_2(a') + \text{N}_2(X)$

b0	-1.352797615527D+03	b1	1.112893176159D+03	b2	-4.305835222711D+02
b3	9.923177567848D+01	b4	-1.479833692982D+01	b5	1.453580537317D+00
b6	-9.137143795799D-02	b7	3.346320008028D-03	b8	-5.447812746987D-05



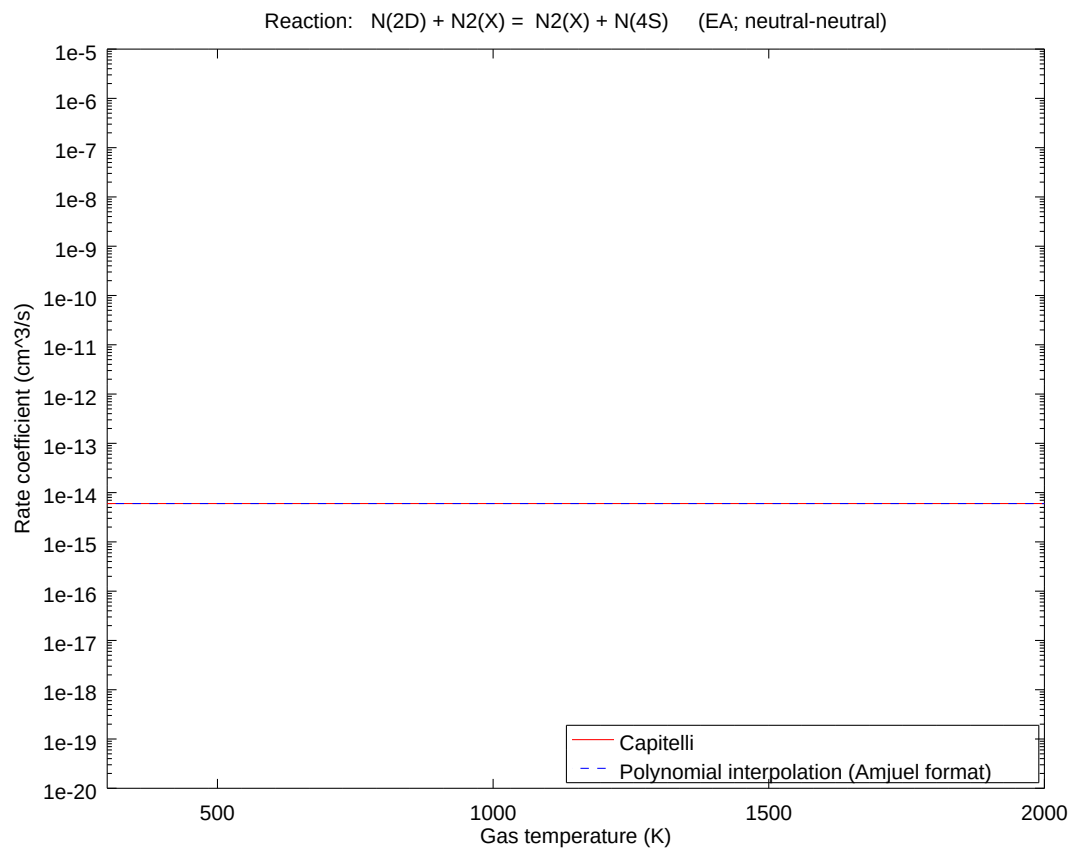
3.74 Reaction DV $\text{N}_2(\text{a}'') + \text{N}_2(\text{X}) = \text{N}_2(\text{X}) + \text{N}_2(\text{X})$

b0	-2.219293540650D+01	b1	-7.762925255744D-06	b2	4.112906109191D-06
b3	-1.243283629654D-06	b4	2.345361915936D-07	b5	-2.827289486648D-08
b6	2.126936279091D-09	b7	-9.129517295128D-11	b8	1.711867751830D-12



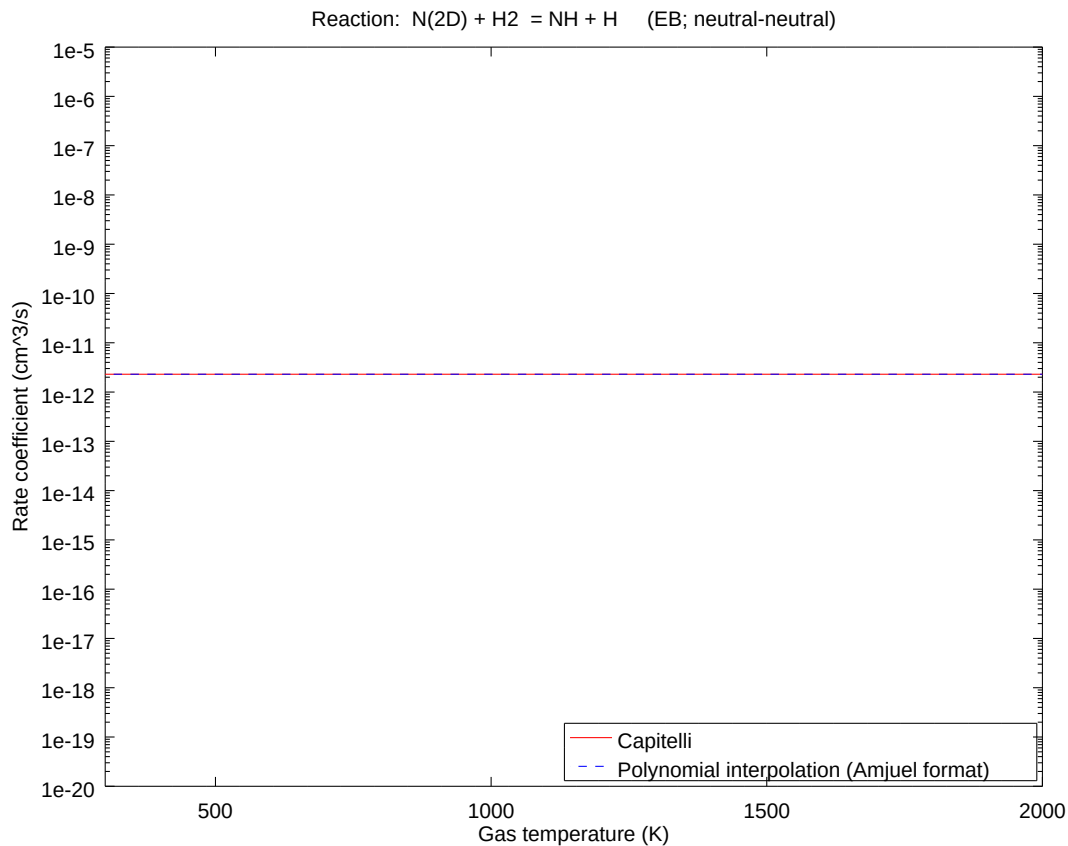
3.75 Reaction EA $\text{N}(2\text{D}) + \text{N}_2(\text{X}) = \text{N}_2(\text{X}) + \text{N}(4\text{S})$

b0	-3.274700639971D+01	b1	-1.280116255233D-05	b2	6.800615336835D-06
b3	-2.061317454552D-06	b4	3.899064191179D-07	b5	-4.713006052324D-08
b6	3.555178936547D-09	b7	-1.530159747808D-10	b8	2.877030079357D-12



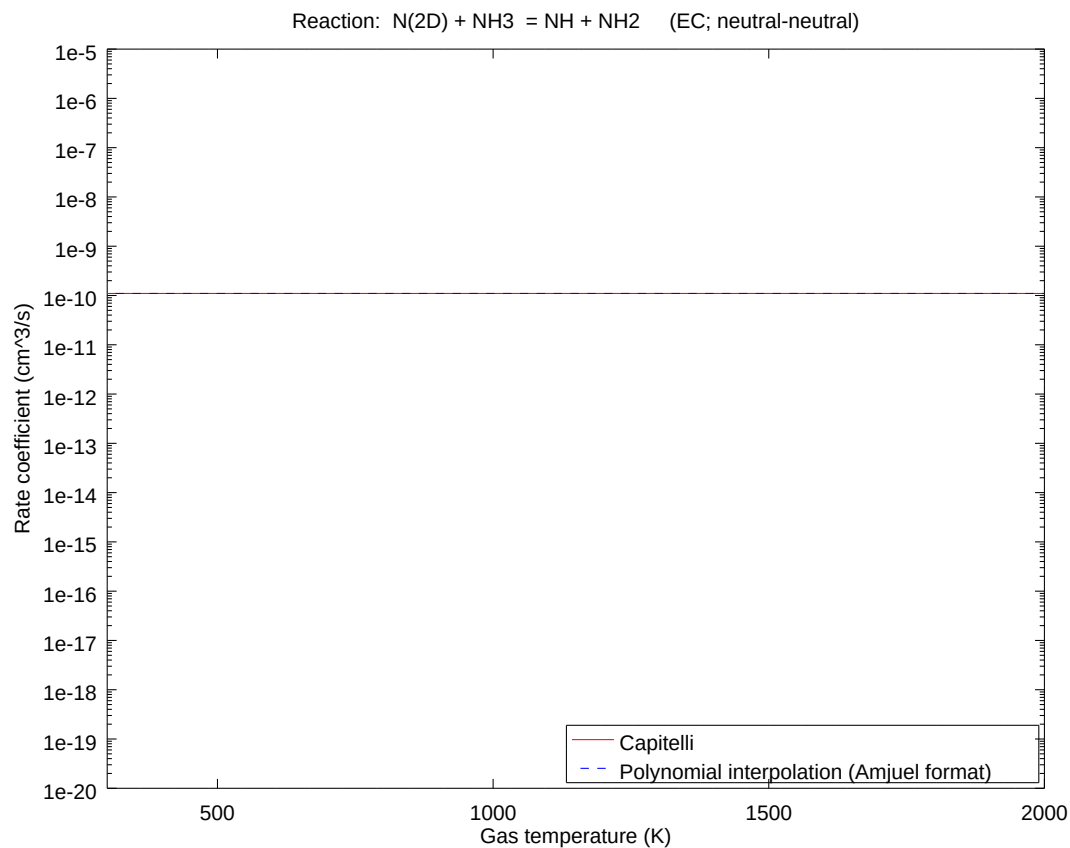
3.76 Reaction EB $\text{N}(2\text{D}) + \text{H}_2 = \text{NH} + \text{H}$

b0	-2.679810510561D+01	b1	-8.368030496002D-06	b2	4.441064197952D-06
b3	-1.344720059874D-06	b4	2.540848203749D-07	b5	-3.067827423487D-08
b6	2.311492231699D-09	b7	-9.936868667999D-11	b8	1.866049462814D-12



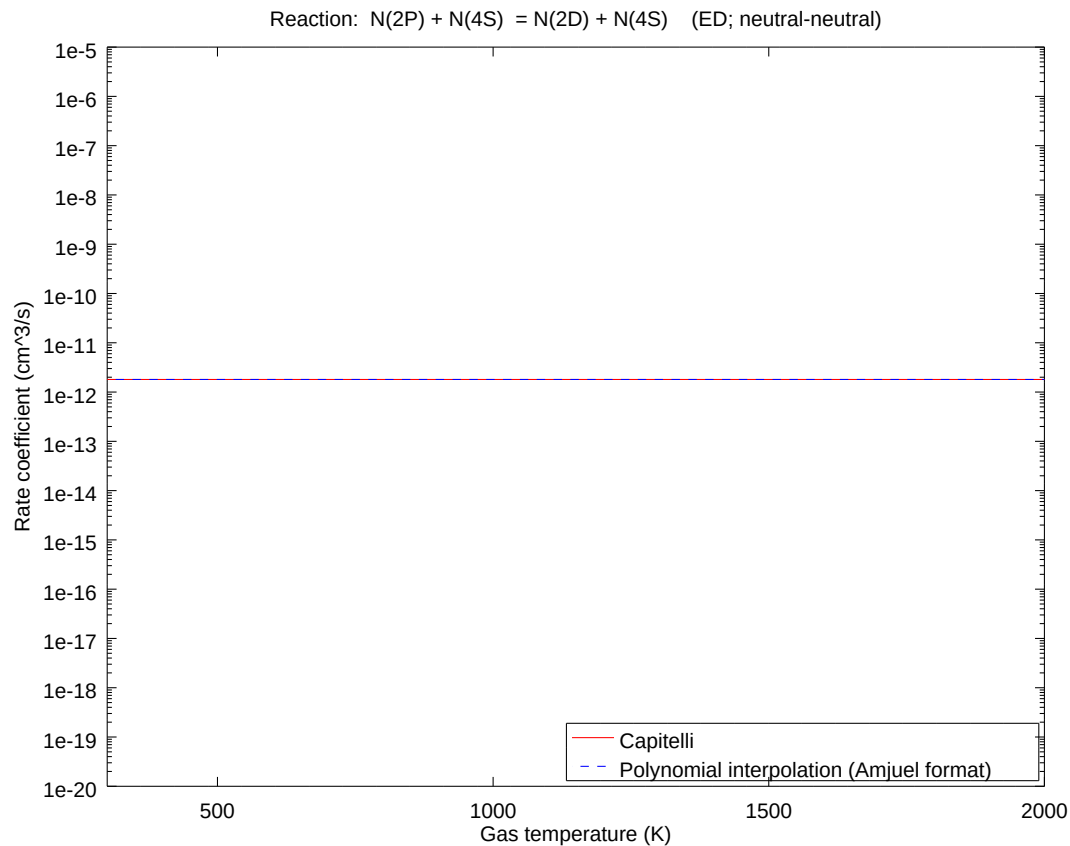
3.77 Reaction EC $\text{N}(2\text{D}) + \text{NH}_3 = \text{NH} + \text{NH}_2$

b0	-2.293053368413D+01	b1	-8.575421428320D-06	b2	4.546510936730D-06
b3	-1.375396253777D-06	b4	2.596714199524D-07	b5	-3.133064394724D-08
b6	2.359206214301D-09	b7	-1.013674863060D-10	b8	1.902772651883D-12



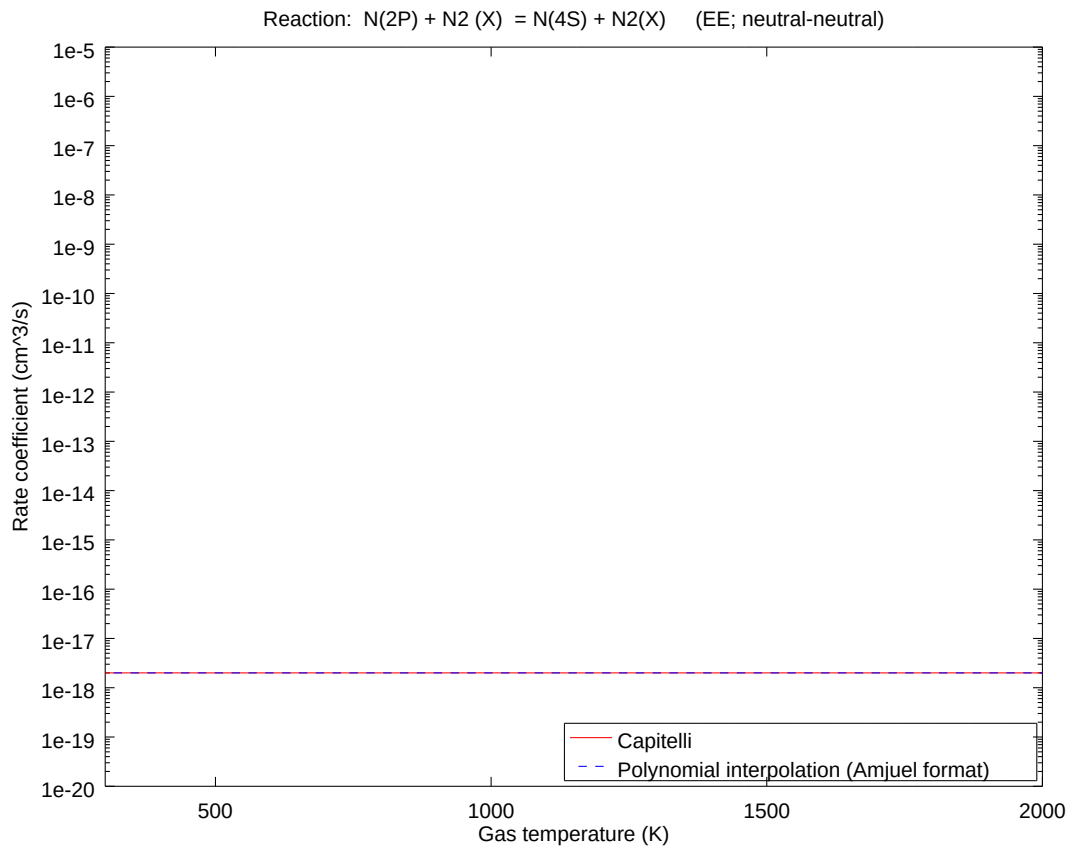
3.78 Reaction ED $\text{N}(2\text{P}) + \text{N}(4\text{S}) = \text{N}(2\text{D}) + \text{N}(4\text{S})$

b0	-2.704322522549D+01	b1	-1.121668826051D-05	b2	5.957309063578D-06
b3	-1.805239152863D-06	b4	3.413814147645D-07	b5	-4.125429107056D-08
b6	3.111189865388D-09	b7	-1.338746126320D-10	b8	2.516546155127D-12



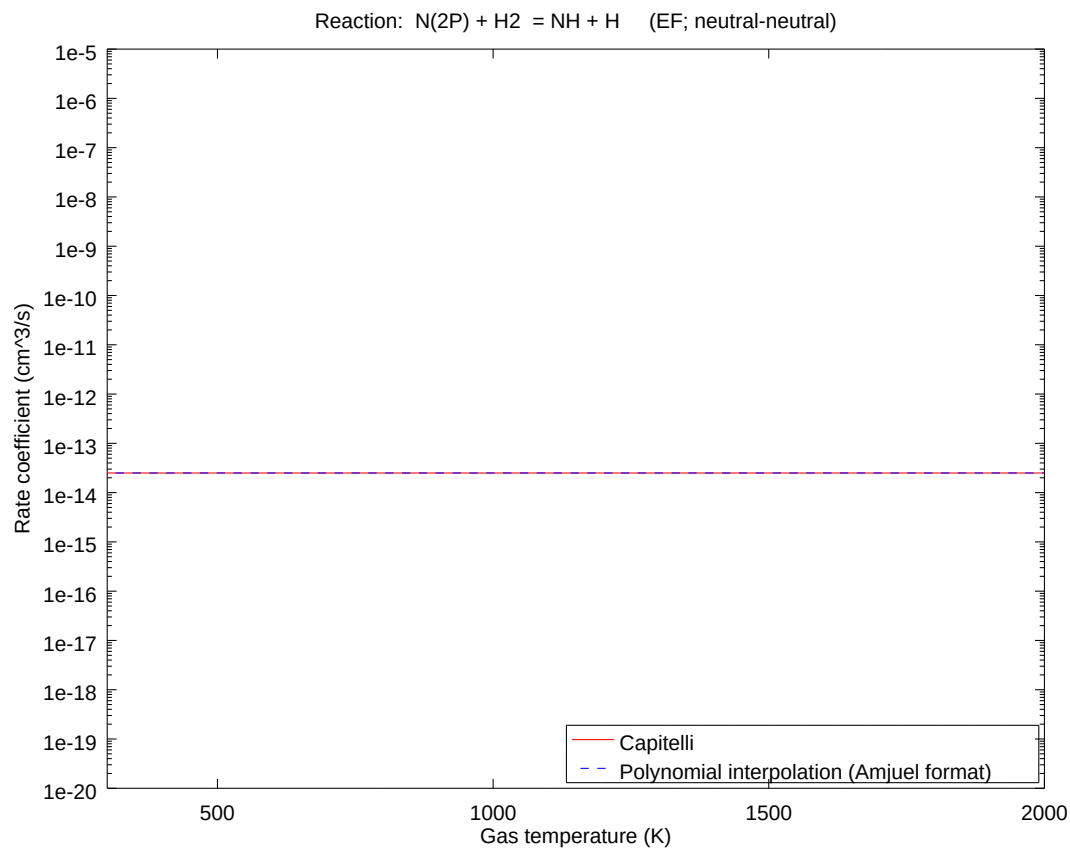
3.79 Reaction EE $\text{N}(2\text{P}) + \text{N}_2(\text{X}) = \text{N}(4\text{S}) + \text{N}_2(\text{X})$

b0	-4.075337354279D+01	b1	-1.328254674853D-05	b2	7.037855976896D-06
b3	-2.127642347491D-06	b4	4.013991549009D-07	b5	-4.839233143345D-08
b6	3.640844307679D-09	b7	-1.562926846721D-10	b8	2.930935484390D-12



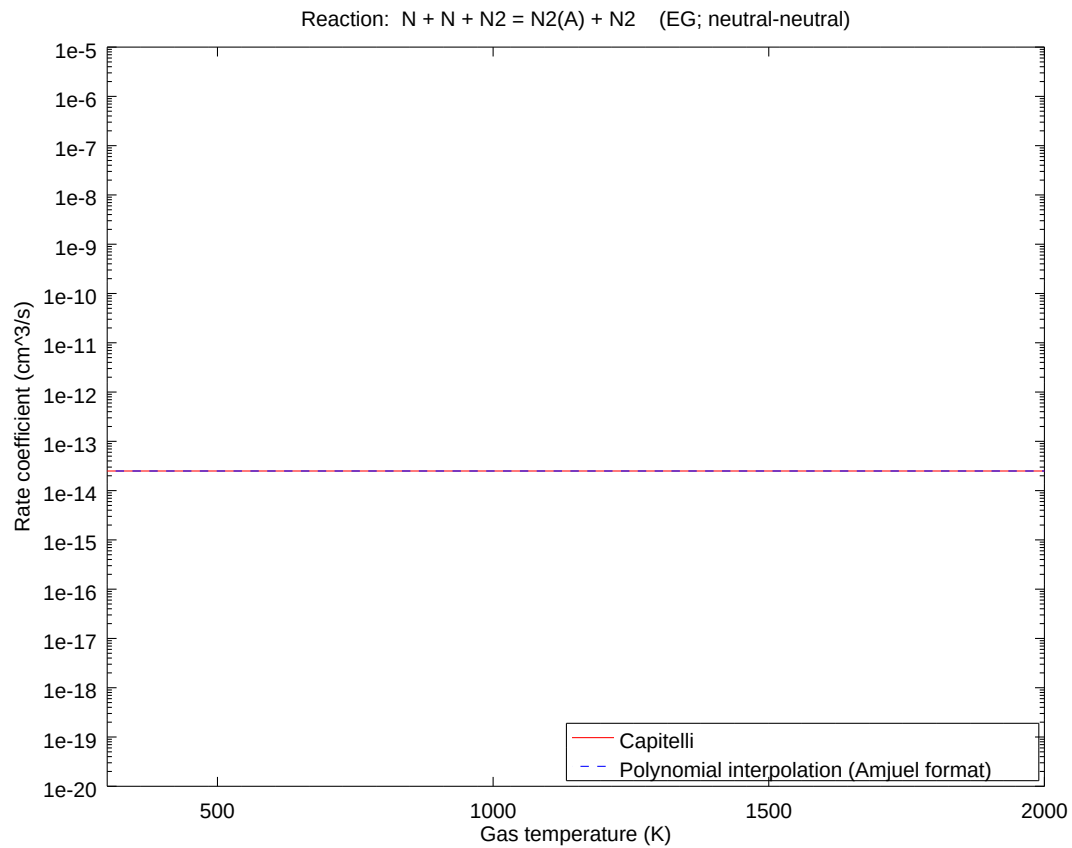
3.80 Reaction EF N(2P) + H2 = NH + H

b0	-3.131989179131D+01	b1	-1.065147739860D-05	b2	5.645285531895D-06
b3	-1.707046087806D-06	b4	3.221125994149D-07	b5	-3.883966973715D-08
b6	2.922479364271D-09	b7	-1.254644176994D-10	b8	2.352895128306D-12



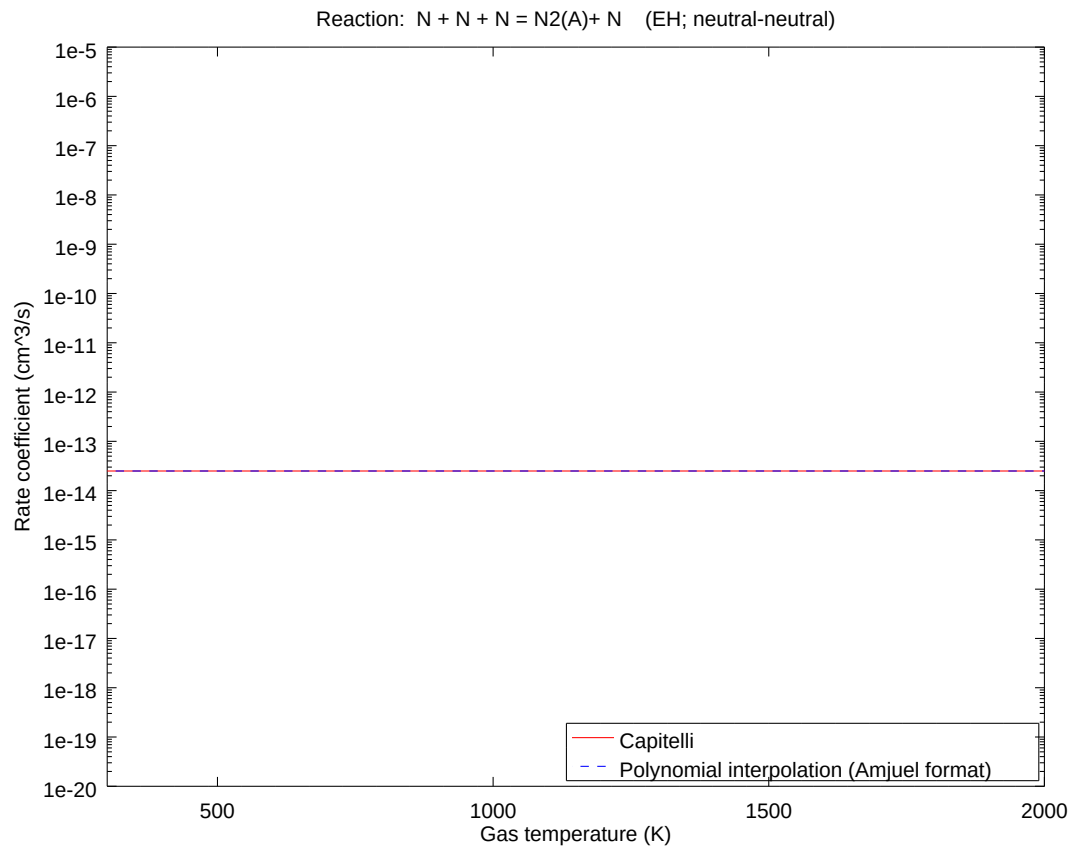
3.81 Reaction EG $\text{N} + \text{N} + \text{N}_2 = \text{N}_2(\text{A}) + \text{N}_2$

b0	-7.545465906650D+01	b1	-2.526496083530D-05	b2	1.343618552354D-05
b3	-4.076645263800D-06	b4	7.718284935539D-07	b5	-9.337558916624D-08
b6	7.049279984618D-09	b7	-3.036277607256D-10	b8	5.712740075695D-12



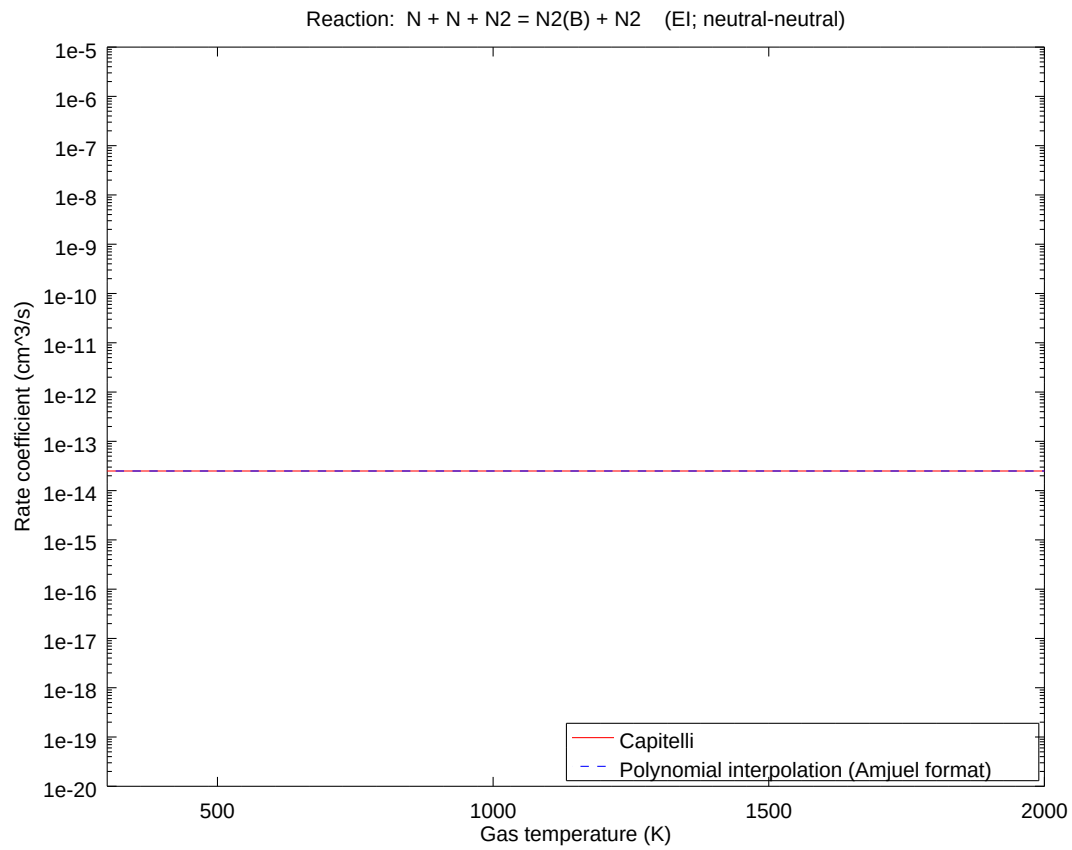
3.82 Reaction EH $\text{N} + \text{N} + \text{N} = \text{N}_2(\text{A}) + \text{N}$

b0	-7.368270230759D+01	b1	-2.512454980844D-05	b2	1.334007142274D-05
b3	-4.040818386514D-06	b4	7.637532554277D-07	b5	-9.223864696612D-08
b6	6.951089605116D-09	b7	-2.988541031300D-10	b8	5.612452815179D-12



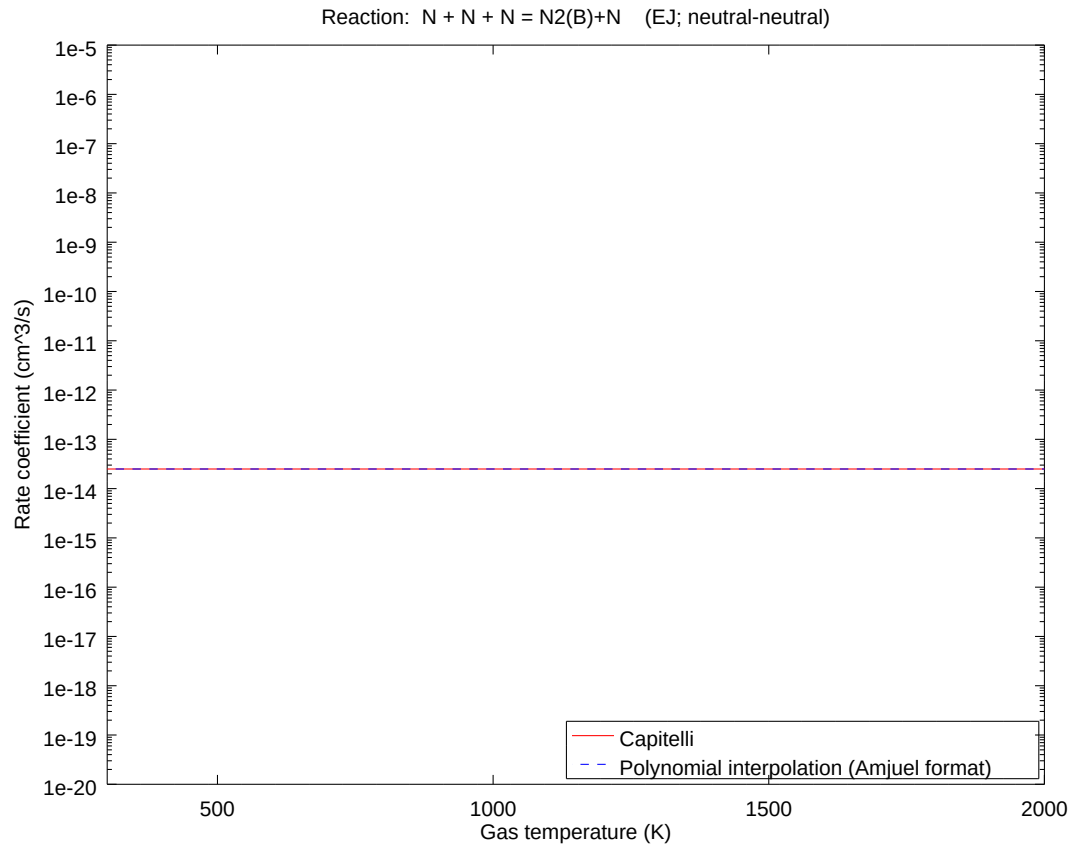
3.83 Reaction EI $\text{N} + \text{N} + \text{N}_2 = \text{N}_2(\text{B}) + \text{N}_2$

b0	-7.510981923005D+01	b1	-2.437617153819D-05	b2	1.291337735692D-05
b3	-3.903330449722D-06	b4	7.363299477103D-07	b5	-8.876752128535D-08
b6	6.678569971547D-09	b7	-2.867121788249D-10	b8	5.377272986844D-12



3.84 Reaction EJ $\text{N} + \text{N} + \text{N} = \text{N}_2(\text{B}) + \text{N}$

b0	-7.334622083937E+01	b1	-3.624488351015D-05	b2	1.919274950342D-05
b3	-5.798640470706D-06	b4	1.093287203987D-06	b5	-1.317237978984D-07
b6	9.904198102672D-09	b7	-4.248981963834D-10	b8	7.963075365350D-12



4 Appendix

References

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